FINITELY GENERATED UNIT LAURENT CRYSTALS

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ABSTRACT. Let p be an odd prime, let k be a perfect field of characteristic p, and let X be a smooth qcqs p-adic formal scheme over $\mathrm{W}(k)$ equipped with the log structure $\mathcal{M}$ induced by a horizontal normal crossings divisor D. We show that there is an equivalence of categories between constructible p-Kummer-étale $\mathbf{Z}_p$-sheaves on the log adic generic fiber $(\mathrm{X}, \mathcal{M})_\eta$ for the logarithmic stratification induced by D and certain finitely generated unit Laurent crystals on the log prismatic site. We verify p^+-perverse t-exactness, give a local description of finitely generated unit crystals on a framed log affine in terms of (φ, Γ)-modules, and identify at the level of hearts the objects corresponding to ordinary constructible sheaves on the adic generic fiber of X.

CONTENTS

1. Introduction

Let k be a perfect field of characteristic p. For smooth k-schemes, Emerton and Kisin gave a contravariant Riemann-Hilbert correspondence describing constructible $\mathbf{F}_p$ sheaves using finitely generated unit Frobenius modules in [EK04b]. Their result is compatible with Katz's correspondence in [Kat73], which describes the full subcategory of étale local systems.

In mixed characteristic, Katz's theorem plays an important role in describing local systems on the adic generic fiber of a p-adic formal scheme using (φ, Γ) modules; importantly, this can also be done over a deperfected base ring. This is originally due to Fontaine in [Fon90], and later generalized in [AI08]. Due to Kedlaya-Liu in [KL15], this theory also admits a logarithmic variant describing what are now called p-Kummer-étale local systems on the log adic generic fiber (see Definition 4.19). Unlike the situation in positive characteristic, there is not a known constructible variant of these theories of (φ, Γ)-modules.

In this article, we will show such a result in logarithmic setting using a new notion of finitely generated unit (φ, Γ) modules over a deperfected base. Our correspondence allows one to study p-Kummer-étale constructible sheaves on a fixed normal crossings stratification on the log adic generic fiber. We use a log formal scheme $(\mathrm{X}, \mathcal{M})$ where $\mathrm{X}/\operatorname{Spf} \mathrm{W}(k)$ is smooth and qcqs and $\mathcal{M}$ is the log structure attached to a horizontal normal crossings divisor D. By this we mean that $\mathrm{D} \cup \mathrm{X}_s$ is a normal crossings divisor and no irreducible component of D is contained in the special fiber; equivalently, strict-étale locally the pair has the form of Definition 3.2.[1] Since any description using (φ, Γ)-modules is necessarily local and coordinate dependent, similarly to [BS23] and [IKY26] we globalize local results about (φ, Γ)-modules using crystals on the absolute log prismatic site $(\mathrm{X}, \mathcal{M})_{\Delta}$. This is done in Definition 4.3, where we introduce the category $\mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)$ of finitely generated unit Laurent crystals.

We now explain the target of the functor in more detail. We write $\mathrm{D}^{(b)}_{\mathrm{cons,D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$ for the category of Kummer-étale sheaves which are constructible for the stratification induced by D on the log adic generic fiber $(\mathrm{X}, \mathcal{M})_\eta$. For a geometric point $\bar{x}$ of X_η, put

$$\rho(\bar{x}) = \mathrm{rk}_{\mathbf{Z}}\, \overline{\mathcal{M}}^{\mathrm{gp}}_{\bar{x}}.$$

For $a \geq 0$, let $\mathrm{X}^{(a)}_\eta$ be the locus where $\rho = a$. The logarithmic strata are the connected components of the locally closed loci $\mathrm{X}^{(a)}_\eta$. Strict-étale locally, after labeling the branches

[1] Note this is not the same as simply asking D to have no vertical components.

$\mathrm{D}_1, \ldots, \mathrm{D}_r$, they are the connected components of the coordinate strata

$$\mathrm{X}_{\eta,\Lambda} = \left(\bigcap_{i\in\Lambda}(\mathrm{D}_i)_\eta\right) \setminus \left(\bigcup_{j\notin\Lambda}(\mathrm{D}_j)_\eta\right) \text{ for } \Lambda \subseteq \{1,\ldots,r\}.$$

As with the correspondence of [IKY26] for Kummer-étale local systems, there is a condition on objects $\mathscr{F}$ in the essential image which they call *p-Kummer* (see Definition 4.19). In a normal crossings setting, at a log geometric point $\bar{x}$ the log inertia group is given by $\mathrm{I}^{\log}_{\bar{x}} = \mathrm{Hom}(\overline{\mathcal{M}}^{\mathrm{gp}}_{\bar{x}}, \widehat{\mathbf{Z}}(1))$ which will look like $\widehat{\mathbf{Z}}(1)^r$. The p-Kummer condition simply asks that the prime-to-p component $\mathrm{I}^{\log,(p')}_{\overline{x}}$ acts trivially on $\mathscr{F}_{\overline{x}}$ for all $\overline{x}$. This condition is not so restrictive, as it still allows all ordinary constructible sheaves.

The main result of this article is then the following. We let $(\mathrm{X},\mathcal{M})_{\Delta}$ denote the category of log prisms, opposite to the log prismatic site of [Kos22], and $\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p$ the coefficient system assigning a log prism $(\mathrm{A},\mathrm{I},\mathrm{M}_{\mathrm{Spf}\,\mathrm{A}})^a$ to $\mathrm{A}[1/\mathrm{I}]^{\wedge}_p$ as in [IKY26]. We assume p is odd in everything that follows.

Theorem A (Theorem 5.25, Corollary 5.29)**.** *Let* $\mathrm{X}/\operatorname{Spf}\mathrm{W}(k)$ *be smooth and qcqs, and let* $\mathcal{M} = \mathcal{M}_{\mathrm{D}}$ *be induced by a horizontal normal crossings divisor. There is a fully faithful symmetric monoidal functor*

$$\mathrm{T}_{\text{ét}} : \mathrm{D}_{\mathrm{fgu}}((\mathrm{X},\mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)^{\mathrm{op}} \longrightarrow \mathrm{D}^{(b)}_{\mathrm{cons,D}}((\mathrm{X},\mathcal{M})_\eta, \mathbf{Z}_p)$$

with essential image the p-Kummer complexes of Definition 4.19. If X_η *is pure of dimension d, then* $\mathrm{T}_{\text{ét}}[d]$ *identifies the standard t-structure on the source with the* p^+*-perverse t-structure on its essential image.*

We emphasize that it is not too difficult to check the finitely generated unit condition for a Laurent crystal. A Laurent crystal assigns to every log prism (A,I) a complex of $\mathrm{A}[1/\mathrm{I}]^{\wedge}_p$-modules, compatibly with pullback. It is unit if it is equipped with an isomorphism $\varphi^*\mathcal{E} \simeq \mathcal{E}$. A unit Laurent crystal is further *finitely generated unit* when for any log prism $(\mathrm{A},\mathrm{I},\mathrm{M}_{\mathrm{Spf}\,\mathrm{A}})^a \in (\mathrm{X},\mathcal{M})_{\Delta}$ one has $\mathcal{E}(\mathrm{A},\mathrm{I},\mathrm{M}) \in \mathsf{Perf}(\mathrm{A}[1/\mathrm{I}]^{\wedge}_p\langle\mathrm{F}\rangle)$ (here we mean the thick subcategory generated by the unit). The noncommutative ring $\mathrm{A}[1/\mathrm{I}]^{\wedge}_p\langle\mathrm{F}\rangle$ is obtained by adjoining F with $\mathrm{F}a = \varphi(a)\mathrm{F}$.[2] One does not need to check all prisms to verify the condition. It in fact suffices to check on a covering family of logarithmic q-de Rham prisms, as in Remark 4.12.

Because the equivalence is derived, it also immediately implies a cohomological comparison. In Corollary 5.29 we explain how straightforward modifications allow one to deduce all results over the base $\mathrm{W}(k)[\zeta_{p^\ell}]$ for $\ell \geq 0$. We primarily work with $\mathrm{W}(k)[\zeta_p]$ in the paper for expository reasons, and later descend the result to $\mathrm{W}(k)$. It is also worth pointing out that one can treat ordinary constructible sheaves within this correspondence. There is a morphism of sites

$$\epsilon : (\mathrm{X},\mathcal{M})_{\eta,\text{qpkét}} \longrightarrow \mathrm{X}_{\eta,\text{proét}}$$

[2]Strictly speaking we must do this all in a solid setting, but we ignore this for the introduction.

whose pullback ϵ^* is fully faithful on constructible sheaves and lands in the p-Kummer sheaves in the heart of $\mathrm{D}^{(b)}_{\mathrm{cons,D}}((\mathrm{X},\mathcal{M})_\eta, \mathbf{Z}_p)$. Thus ordinary constructible sheaves form a full subcategory at the level of hearts. See Remark 5.28 for the corresponding condition on the crystal side, which is easily checked locally.

As mentioned earlier, finitely generated unit Laurent crystals admit an explicit local description in terms of (φ,Γ)-modules. We use this to give an explicit example in Example 5.31. We state the variant over $\mathrm{W}(k)$. In what follows a framed log affine is a $\mathrm{W}(k)$-algebra R that is p-completely étale over $\mathrm{W}(k)\langle \mathrm{T}_1,\dots,\mathrm{T}_r,\mathrm{T}^{\pm}_{r+1},\dots,\mathrm{T}^{\pm}_n\rangle$ with log structure $\mathcal{M}$ induced by $\mathrm{D}=\mathrm{V}(\mathrm{T}_1\dots\mathrm{T}_r)$.

Theorem B (Variant of Theorem 4.6)**.** *Let* $(\mathrm{Spf}\,\mathrm{R},\mathcal{M})$ *be a framed log affine, with framing denoted by* $\square$*. Evaluation on the cyclotomic log q-de Rham prism* $(\mathrm{R}^\square[\![q-1]\!],[p]_q,\mathbf{N}^r)^a$ *over* R *induces an equivalence*

$$\mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf}\,\mathrm{R},\mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^\wedge_p) \simeq \mathrm{D}_{\mathrm{fgu}}(\mathrm{R}^\square(\!(q-1)\!)^\wedge_p)^{h\Gamma},$$

where $(a,b_1,\dots,b_n)\in\Gamma=\mathbf{Z}_p^\times\ltimes\mathbf{Z}_p(1)^n$ *acts by* $q\mapsto q^a$ *and* $\mathrm{T}_i\mapsto q^{pb_i}\mathrm{T}_i$*, and "fgu" means that the complex has a unit Frobenius and is perfect over* $\mathrm{R}^\square(\!(q-1)\!)^\wedge_p\langle\mathrm{F}\rangle$*.*

We also use this to make $\mathrm{T}_{\text{ét}}$ completely explicit locally in Lemma 4.26 (although in principle the definition in Definition 4.17 is already computable).

Our third main result is pushforward compatibility. The required functor is a shift and twist of the right adjoint to pullback on Laurent crystals.

Theorem C (Theorem 6.11)**.** *Let* $f:(\mathrm{X},\mathcal{M})\to(\mathrm{Y},\mathcal{N})$ *be a strict log smooth proper morphism of pure relative dimension d between smooth qcqs p-adic formal schemes over* $\mathrm{W}(k)[\zeta_{p^\ell}]$ *for some* $\ell\geq 0$*, and let* $\mathrm{D_X}$ *and* $\mathrm{D_Y}$ *be horizontal normal crossings divisors inducing* $\mathcal{M}$ *and* $\mathcal{N}$*. Suppose the adic generic fibers have pure dimensions* d_X *and* d_Y*. Thus* $d=d_\mathrm{X}-d_\mathrm{Y}$*. There is a commutative diagram*

$$\begin{array}{ccc}
\mathrm{D}_{\mathrm{fgu}}((\mathrm{X},\mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^\wedge_p)^{\mathrm{op}} & \xrightarrow{\mathrm{T}_{\text{ét},\mathrm{X}}[d_\mathrm{X}]} & \mathrm{D}^{(b)}_{\mathrm{cons,D_X}}((\mathrm{X},\mathcal{M})_\eta,\mathbf{Z}_p) \\
\Big\downarrow{\scriptstyle \mathrm{R}f_*\{d\}[d]} & & \Big\downarrow{\scriptstyle \mathrm{R}f_{\eta,*}} \\
\mathrm{D}_{\mathrm{fgu}}((\mathrm{Y},\mathcal{N})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^\wedge_p)^{\mathrm{op}} & \xrightarrow{\mathrm{T}_{\text{ét},\mathrm{Y}}[d_\mathrm{Y}]} & \mathrm{D}^{(b)}_{\mathrm{cons,D_Y}}((\mathrm{Y},\mathcal{N})_\eta,\mathbf{Z}_p).
\end{array}$$

The shift and twist arise from a logarithmic Poincaré duality result for perfect Laurent crystals proved in Corollary 6.7.

It is worth pointing out here that actually for the Laurent F-crystal equivalences in either [BS23] or [IKY26], the analogous pushforward compatibility theorem does not seem to appear in the literature (to the best of the author's knowledge). The problem is that it is not obvious that the right adjoint $\mathrm{R}f_*$ on Laurent crystals preserves $\mathsf{Perf}(\mathrm{X}_{\mathbb{\Delta}},\mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^\wedge_p)^{\varphi=1}$ (once this is known the claim is more or less immediate). For this reason existing pushforward compatibility results (such as the result stated in [GL25]) assume that the Laurent F-crystal arises from a prismatic F-crystal, as in that situation perfectness can be deduced from the Hodge-Tate comparison. So it appears that even in the lisse case, the pushforward compatibility result is new.

Relation to the work of Bhatt and Lurie. There is a related Riemann-Hilbert correspondence due to Bhatt-Lurie in [Bha25]. Let $X^{\Delta,\mathrm{pfd}}$ denote the perfected prismatic stack attached to a p-adic formal scheme X as defined in [Bha25]. In *loc. cit.*, Bhatt and Lurie construct a fully faithful p-adic Riemann-Hilbert functor

$$\mathrm{RH}_{\Delta} : \mathrm{D}((\mathrm{X}_\eta)_{\text{ét}}, \mathbf{Z}_p)^{\mathrm{oc}} \to \mathrm{D}(\mathrm{X}^{\Delta,\mathrm{pfd}})^{a,\varphi=1}$$

which is compatible with their mod p Riemann-Hilbert correspondence in positive characteristic (constructed in [BL19]; note they also prove their functor is dual to the Emerton-Kisin functor in a precise sense). When X arises as a formal completion of a finite type $\mathcal{O}_K$ scheme this admits a variant without using almost mathematics. The functor also has good properties when applied to Zariski-constructible sheaves on X_η, for example compatibility with the perverse t-structure constructed in [BH22]; they also prove pushforward compatibility results.

Our correspondence is much less general, but there is a reasonable tradeoff which makes both correspondences valuable. First, our crystals are deperfected. This allows us to locally formulate the crystal side of the correspondence as a generalization of (φ, Γ) modules over a deperfected base, and also allows us to write them down explicitly. The second is that the functor is an *equivalence* onto the full subcategory of p-Kummer constructible sheaves – in [Bha25] there is not a clear (non-tautological) condition describing constructible sheaves.

Overview of the proofs. All arguments work first over $\mathrm{W}(k)[\zeta_p]$, and then we extend to other bases. The first main result we work towards is strict-étale descent for finitely generated unit Laurent crystals, since most arguments reduce to a framed log affine. We work in the ordinary solid category $\mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})$ with derived p-complete coefficient rings, as in Definition A.11. The appendix proves that the logarithmic Breuil-Kisin atlases are completely descendable in the sense of [Kip26b, Definition 2.4.1.7]. Consequently, ordinary solid Laurent crystals are computed by the Čech nerve of the log q-de Rham prism. We also show strict-étale descent using similar methods.

The next step is to show that the finitely generated unit condition itself is strict-étale local. We do this by using *roots*. A coherent root of a unit Frobenius module M is a coherent submodule $\mathrm{M}_0 \subset \mathrm{M}$ equipped with an injection $\mathrm{M}_0 \to \varphi^*\mathrm{M}_0$ so $\mathrm{M} \simeq \operatorname{colim} \varphi^{*n}\mathrm{M}_0$; it is minimal if it contains no proper root. Modulo p, [Bli08] gives a unique minimal coherent root in the sense of Definition 4.5 for a finitely generated unit module over a regular F-finite $\mathbf{F}_p$-algebra, for example $(\mathrm{R}_0^{\square}/p)(\!(q-1)\!)$. The main point is to promote these minimal roots *as modules* from individual log q-de Rham evaluations to minimal roots in *coherent Laurent crystals* on framed log affines. Once this is done, their uniqueness as subcrystals makes them glue and proves the required descent (since ordinary coherent Laurent crystals do form a stack, using [Mat22]).

The section on coherent Laurent crystals provides the technical ingredients to carry this out. The group $\Gamma = (1 + p\mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)^n$ acts on the log q-de Rham prism by $q \mapsto q^a$ and $\mathrm{T}_i \mapsto q^{pb_i}\mathrm{T}_i$. Proposition 3.12 shows that evaluation on this prism is fully faithful. On objects with p-completely nuclear evaluation, this admits the usual interpretation in terms of continuous semilinear Γ-modules; this includes coherent and finitely generated unit crystals. Theorem 3.5 carries a Γ-equivariant root over $\mathrm{R}_0^{\square}((q-1))_p^{\wedge}$ to a coherent Laurent crystal. Together they give the local (φ, Γ)-module description in Theorem B. Applying Theorem 3.5 to the distinguished minimal roots of the algebraic evaluations produces distinguished minimal roots as crystals. In the process of strict-étale descent we also characterize finitely generated unit Laurent crystals in terms of admitting a global coherent root modulo p.

Once strict-étale descent is known the arguments are more elementary and module-theoretic, as we can localize to a situation where we deal only with finitely generated unit (φ, Γ) modules as in Theorem B. The main equivalence in Theorem A is built first on the open stratum $\mathrm{U}_\eta = \mathrm{X}_\eta \setminus \mathrm{D}_\eta$. Theorem B reduces the open-stratum equivalence locally to a statement about (φ, Γ)-modules. For essential surjectivity on the open stratum, one uses Lütkebohmert's theorem to extend a local system on U_η to a Kummer-étale local system $\mathbb{L}$ on X_η. Letting $j : \mathrm{U}_\eta \to \mathrm{X}_\eta$ be the inclusion, we construct a crystal $\mathcal{O}_{\Delta}(*\mathrm{D})$ corresponding to $j_!\mathbf{Z}_p$. We show that after a finite prime-to-p Kummer cover $(\mathrm{X}_\mathrm{N})_\eta$ we may make $\mathbb{L}$ a p-Kummer local system, and thus [IKY26] provides a logarithmic Laurent crystal $\mathcal{E}$. Keeping track of the descent data in the form of a μ_N-action, one may descend $\mathcal{E}(*\mathrm{D})$ to construct a preimage. Full faithfulness uses as its main component a more delicate decompletion statement that base change to the perfection is fully faithful; Remark 5.13 explains why this does not follow from Frobenius alone. Closed strata are far easier and are handled by a variant of Kashiwara's lemma, which reduces to the corresponding result of [EK04b].

A dévissage argument using Kashiwara's lemma reduces the main theorem to the open and closed comparisons on strata, along with some (derived) full faithfulness results about maps between strata. The essential image is identified because extensions by zero of p-Kummer local systems on the strata generate the p-Kummer constructible subcategory. One must also verify that $\mathrm{T}_{\text{ét}}$ is constructible for the stratification along D; to show this strict-étale descent allows one to use the local formulation in terms of finitely generated unit (φ, Γ) modules where a Fitting ideal argument controls the possible roots (and proves constructibility along D). Once all results are shown over $\mathrm{W}(k)[\zeta_p]$, with trivial modifications we can work over $\mathrm{W}(k)[\zeta_{p^\ell}]$.

Finally, Theorem C first uses results in [Kip26b] and [Kip26a] to construct a stacky approach to these solid Laurent crystals and equip them with a suitable six functor formalism. The existence of this six functor formalism helps greatly in proving pushforward compatibility (for example by showing $\mathrm{R}f_* \simeq \mathrm{R}f_!$ is compatible with colimits, or showing strict log smooth proper maps preserve perfect Laurent crystals) but does not make the conclusion automatic. The idea is to show that the pushforward preserves the finitely generated unit condition by producing a variant of a root (using a perfect complex) and then formally deducing pushforward compatibility by adjunction.

Notation. Unless otherwise stated, all categories are ∞-categories, $(\mathrm{X}, \mathcal{M})$ denotes a smooth qcqs log p-adic formal scheme over $\mathrm{Spf}\ \mathrm{W}(k)[\zeta_p]$ equipped with the log structure coming from a horizontal normal crossings divisor D. We use the logarithmic stratification by the rank of $\overline{\mathcal{M}}^{\mathrm{gp}}$. On a framed strict-étale chart, $\mathrm{X}_{\eta,\Lambda}$ denotes the coordinate stratum on which exactly the boundary coordinates indexed by Λ vanish. We write $\mathrm{D}^{(b)}_{\mathrm{cons,D}}((\mathrm{X},\mathcal{M})_\eta, \mathbf{Z}_p)$ for the quasi-pro-Kummer-étale complexes which are derived lisse on each logarithmic stratum. The p-Kummer sheaves form a full subcategory; see Definition 4.19.

We use $(\mathrm{X},\mathcal{M})^{\Delta}$ for Olsson's log prismatic stack (see Appendix A). We reserve $(\mathrm{X},\mathcal{M})^{\Delta,\blacksquare}$ and $(\mathrm{X},\mathcal{M})^{\mathcal{L},\blacksquare}$ for the prismatic and Laurent analytic stacks derived from the construction of [Kip26b] and used only for perfect-coefficient duality.

On a framed log affine $(\mathrm{Spf}\,\mathrm{R},\mathcal{M})$ with framing $\square$ and boundary coordinates $\mathrm{T}_1,\dots,\mathrm{T}_r$, let R_0 be the $\mathrm{W}(k)$-algebra in the framing, so that $\mathrm{R}\simeq \mathrm{R}_0\widehat{\otimes}_{\mathrm{W}(k)}\mathrm{W}(k)[\zeta_p]$. The attached log q-de Rham prism is $(\mathrm{R}_0^{\square}[\![q-1]\!],[p]_q,\mathbf{N}^r)^a$, where $\mathrm{R}_0^{\square}[\![q-1]\!]=\mathrm{R}_0[\![q-1]\!]$. Its quotient by $[p]_q$ is R. The associated coefficient ring is $\mathrm{R}_0^{\square}(\!(q-1)\!)^{\wedge}_p = \mathrm{R}_0^{\square}[\![q-1]\!][1/[p]_q]^{\wedge}_p$. The superscript $\square$ records the framing. The group $\Gamma=(1+p\mathbf{Z}_p)\ltimes\mathbf{Z}_p(1)^n$ acts on the prism by $q\mapsto q^a$ and $\mathrm{T}_i\mapsto q^{pb_i}\mathrm{T}_i$. We write A for the coefficient ring when the framing is fixed, and $\overline{\mathrm{A}}$ for its reduction modulo p.

We write $\mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)$ for the category of Laurent crystals with ordinary solid coefficients and derived p-complete coefficient rings. We write $\mathrm{D}_{\mathrm{fgu}}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)$ for its subcategory of finitely generated unit Laurent crystals as in Definition 4.3; their evaluations are derived p-complete. The functor $\mathrm{T}_{\text{ét}}$ is the (Kummer)-étale realization functor, while RSol denotes the crystalline Emerton-Kisin solution functor in characteristic p.

We write $\mathrm{RHom}(-,-)$ for mapping complexes and $\underline{\mathrm{RHom}}(-,-)$ for internal derived Hom. Underived internal Hom is denoted by $\underline{\mathrm{Hom}}(-,-)$.

Acknowledgements. The author would like to thank Mark Kisin for suggesting this problem to the author and for his guidance throughout the project. The author would also like to thank Oakley Edens, Daishi Kiyohara, and Kush Singhal for helpful conversations.

AI Use Disclosure. AI tools assisted in technical details in Lemma 3.11 and the helper lemmas for Proposition 5.9. The author's first counterexample in Remark 5.13 was also simplified (by finding a less complicated Artin-Schreier class).

2. The Emerton-Kisin correspondence

In this section, we review the Emerton-Kisin correspondence from [EK04b].

The story naturally begins with Katz's correspondence, which the Emerton-Kisin correspondence generalizes to constructible sheaves. The original statement of this result is [Kat73, Proposition 4.1.1], allowing the generality of a regular $\mathbf{F}_p$-algebra R, but for our purposes we will be most interested in the case where R is a perfect $\mathbf{F}_p$-algebra. In this setting the theorem appears in [KL15, Proposition 3.2.7] and [KL15, Theorem 8.5.3] (once we pass to the limit). However we will also need a derived variant, which appears to not have a direct reference but is a formal extension of existing results. We also take the opportunity to introduce a dévissage method for idempotent complete categories which we will use frequently in what follows.

We let $\mathrm{D}^{(b)}_{\mathrm{lisse}}(\operatorname{Spec}\mathrm{R}, \mathbf{Z}_p)$ denote locally bounded derived p-complete complexes in $\mathrm{D}((\operatorname{Spec}\mathrm{R})_{\text{proét}}, \mathbf{Z}_p)$ such that each cohomology sheaf of the derived mod p reduction is lisse in the usual sense.

Theorem 2.1 (Katz). *Let* R *be a perfect* $\mathbf{F}_p$*-algebra. Then there is an equivalence of categories*

$$\mathsf{Perf}(\mathrm{W}(\mathrm{R}))^{\varphi=1} \simeq \mathrm{D}^{(b)}_{\mathrm{lisse}}(\operatorname{Spec}\mathrm{R}, \mathbf{Z}_p),$$

induced by the solution functor

$$\mathrm{M} \longmapsto \left(\mathrm{M} \otimes_{\mathrm{W}(\mathrm{R})} \mathrm{W}(\mathcal{O}_{\operatorname{Spec}\mathrm{R},\text{proét}})\right)^{\varphi=1}.$$

Proof. We apply Lemma 2.3. We first verify the hypotheses. The solution functor is easily checked to be exact and $\mathsf{Perf}(\mathbf{Z}_p)$-linear, as well as compatible with derived reduction modulo p^n (using perfectness). Both categories are idempotent complete, since their defining conditions are preserved under retracts. The source is presented as $\lim_n \mathsf{Perf}(\mathrm{W}_n(\mathrm{R}))^{\varphi=1}$ and the target as a limit of categories of lisse complexes whose stalks have finite Tor-dimension over $\mathbf{Z}/p^n$, i.e. $\mathrm{D}^{(b)}_{\mathrm{lisse}}(\operatorname{Spec}\mathrm{R}, \mathbf{Z}/p^n)$. The mod p functor is the equivalence of [BS23, Proposition 3.4], so Lemma 2.3 proves the result. □

Remark 2.2. The Emerton-Kisin correspondence is contravariant. Since Perf is self-dual, the previous correspondence also admits a dual or contravariant version. The solution functor is $\mathrm{M} \mapsto \mathrm{RHom}_{\mathsf{Perf}^{\varphi}(\mathrm{W}(\mathrm{R})_{\text{proét}})}(\mathrm{M}_{\text{proét}}, \mathrm{W}(\mathrm{R})_{\text{proét}})$ (regarded as a proétale sheaf). Similarly other correspondences reducing to Katz's theorem, such as [BS23, Corollary 3.7] or [IKY26] admit contravariant versions.

The dévissage method we have just applied is a categorified version of Nakayama's lemma. For an object X of a $\mathsf{Perf}(\mathbf{Z}_p)$-linear stable ∞-category, write $\mathrm{X}/p^n := \operatorname{cofib}(p^n : \mathrm{X} \to \mathrm{X})$. Recall a stable ∞-category $\mathcal{C}$ is idempotent complete when $\mathcal{C}$ is closed under retracts in $\mathsf{Ind}(\mathcal{C})$.

Setup. Let $\mathcal{C}$ be an idempotent complete $\mathsf{Perf}(\mathbf{Z}_p)$-linear stable ∞-category. All tensor products of categories below are taken among idempotent complete stable ∞-categories. Put

$$\mathcal{C}_n := \mathcal{C} \otimes_{\mathsf{Perf}(\mathbf{Z}_p)} \mathsf{Perf}(\mathbf{Z}/p^n).$$

We say that $\mathcal{C}$ is derived p-complete if the natural functor $\mathcal{C} \to \varprojlim_n \mathcal{C}_n$ is an equivalence. Write $\mathrm{red}_n : \mathcal{C} \to \mathcal{C}_n$ for reduction and $\mathrm{X}_n = \mathrm{red}_n\mathrm{X}$. Since $\mathsf{Perf}(\mathbf{Z}/p^n)$ is thickly generated by its unit, $\mathcal{C}_n$ is thickly generated by the objects X_n, and

$$\mathrm{Map}_{\mathcal{C}_n}(\mathrm{X}_n, \mathrm{Y}_n) \simeq \mathrm{Map}_{\mathcal{C}}(\mathrm{X}, \mathrm{Y}/p^n).$$

We ask that functors between derived p-complete categories $\mathrm{F} : \mathcal{C} \to \mathcal{D}$ in this setup induce compatible functors $\mathrm{F}_n : \mathcal{C}_n \to \mathcal{D}_n$.

Lemma 2.3. *Let $\mathcal{C}$ and $\mathcal{D}$ be derived p-complete idempotent complete $\mathsf{Perf}(\mathbf{Z}_p)$-linear stable ∞-categories. Let $\mathrm{F} : \mathcal{C} \to \mathcal{D}$ be an exact $\mathsf{Perf}(\mathbf{Z}_p)$-linear functor in the setup above. If F_1 is an equivalence, then F is an equivalence.*

Proof. Full faithfulness is clear, using derived Nakayama on mapping complexes. Indeed,

$$\mathrm{Map}_{\mathcal{C}_1}(\mathrm{X}_1, \mathrm{Y}_1) \simeq \mathrm{Map}_{\mathcal{C}}(\mathrm{X}, \mathrm{Y}/p) \simeq \mathrm{Map}_{\mathcal{C}}(\mathrm{X}, \mathrm{Y}) \otimes^{\mathrm{L}}_{\mathbf{Z}_p} \mathbf{F}_p.$$

Note that by our setup $\mathrm{Map}_{\mathcal{C}}(\mathrm{X}, \mathrm{Y})$ is derived p-complete. Note that similarly one gets full faithfulness of F_n as we will need later in the argument.

For essential surjectivity, it suffices to show the F_n are essentially surjective; we know this for $n = 1$. Let $\mathrm{Y}_n \in \mathcal{D}_n$, and consider the fiber sequence

$$\mathrm{Y}_n/p \longrightarrow \mathrm{Y}_n/p^n \longrightarrow \mathrm{Y}_n/p^{n-1}.$$

This shows Y_n/p^n belongs to the thick subcategory generated by Y_n/p. One can show Y_n/p is in the essential image of F_n because F_1 is essentially surjective, and the essential image is a thick subcategory. There is a minor subtlety here, which is that $\mathrm{Y}_n/p \in \mathcal{D}_n$ and not $\mathcal{D}_1$ despite the notation. However since $\mathcal{D}_n$ is thickly generated by reductions it suffices to consider the case when Y_n is the reduction of $\mathrm{Y} \in \mathcal{D}$. We may choose a preimage for $\mathrm{red}_1 \mathrm{Y} \in \mathcal{D}_1$, and applying restriction of scalars to regard this preimage in $\mathcal{C}$ one obtains a preimage for Y/p; reducing we get the claim.

Thus Y_n/p^n is in the essential image of F_n, so denote a choice of preimage by X_n.

Because we take the derived quotient by p^n, since $p^n = 0$ in $\mathcal{D}_n$ one has $\mathrm{Y}_n/p^n = \mathrm{cofib}(0 : \mathrm{Y}_n \to \mathrm{Y}_n) = \mathrm{Y}_n \oplus \mathrm{Y}_n[1]$. So in particular Y_n is a retract of Y_n/p^n. But then idempotent completeness implies that Y_n itself is in the essential image of F_n. Indeed we have

$$\mathrm{Y}_n \xrightarrow{\ i\ } \mathrm{Y}_n/p^n \xrightarrow{\ r\ } \mathrm{Y}_n$$

where ri is the identity; in particular ir is an idempotent in $\mathrm{End}_{\mathcal{D}_n}(\mathrm{F}_n(\mathrm{X}_n))$. By derived full faithfulness, it is an image of an idempotent e in $\mathrm{End}_{\mathcal{C}_n}(\mathrm{X}_n)$ which splits as

$$\mathrm{X}'_n \xrightarrow{\ i'\ } \mathrm{X}_n \xrightarrow{\ r'\ } \mathrm{X}'_n$$

where $r'i' = \mathrm{id}_{\mathrm{X}'_n}$ and $i'r' = e$. Then necessarily $\mathrm{F}_n(\mathrm{X}'_n) \simeq \mathrm{Y}_n$, and the F_n are all essentially surjective. Thus every F_n is an equivalence, and derived p-completeness implies that F is an equivalence. ◻

Katz's theorem is crucial in proving the equivalence

$$\mathsf{Perf}(\mathrm{X}_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)^{\varphi=1} \simeq \mathrm{D}^{(b)}_{\mathrm{lisse}}(\mathrm{X}^{\diamond}_\eta)$$

for bounded p-adic formal schemes X in [BS23]. Precisely, this means that when $\mathrm{X} = \mathrm{Spf}\, \mathrm{R}$ is perfectoid, the result reduces to Katz's equivalence; their theorem is deduced by a reduction to this case. Our result for finitely generated unit Laurent crystals will be compatible in a similar way with the Emerton-Kisin correspondence for constructible sheaves.

We now extend the Emerton-Kisin correspondence of [EK04b] to $\mathbf{Z}_p$ coefficients, deduced from the mod p theorem by Lemma 2.3. We fix a perfect field k of characteristic p. If $\mathscr{X} / \operatorname{Spf} \mathrm{W}(k)$ is a smooth formal scheme, we let $\mathrm{D}_{\mathscr{X}}$ denote the sheaf of infinitesimal differential operators. Locally, if $\mathrm{X} \to \mathbf{A}^n_k$ is étale, this can be pulled back from the sheaf associated to $\mathrm{D}_{\operatorname{Spf} \mathrm{W}(k)\langle \mathrm{T}_1, \ldots, \mathrm{T}_n\rangle} = \mathrm{W}(k)\langle \mathrm{T}_1, \ldots, \mathrm{T}_n, \partial_1^{[i_1]}, \ldots, \partial_n^{[i_n]}\rangle$, where all ∂_i are given full divided powers. We put

$$\mathrm{D}_{\mathrm{F},\mathscr{X}} := \widehat{\bigoplus}_{i \geq 0} \mathrm{F}^{*i}\mathrm{D}_{\mathscr{X}}.$$

Modules over this noncommutative sheaf are infinitesimal crystals $\mathcal{E}$ with semilinear Frobenius $\varphi_{\mathcal{E}}$. A certain Taylor formula comparing Frobenius lifts makes $\mathrm{D}_{\mathrm{F},\mathscr{X}}$ canonical and global; see [EK04a, §§6.1-6.2].

The Emerton-Kisin correspondence is compatible with a perverse t-structure on the étale side, which we now introduce.

Definition 2.4. Write $({}^p\mathrm{D}^{\leq 0}, {}^p\mathrm{D}^{\geq 0})$ for Gabber's perverse t-structure on $\mathrm{D}^{(b)}_{\mathrm{cons}}(\mathrm{X}, \mathbf{Z}_p)$ [Gab04], and write ${}^p\mathrm{H}^i$ for perverse cohomology. Its dual tilt is the p^+-perverse t-structure

$${}^{p^+}\mathrm{D}^{\leq 0} = \{\mathrm{K} \in {}^p\mathrm{D}^{\leq 1} \mid {}^p\mathrm{H}^1(\mathrm{K}) \text{ is locally bounded } p\text{-power torsion}\},$$

$${}^{p^+}\mathrm{D}^{\geq 0} = \{\mathrm{K} \mid \mathrm{K}/p \in {}^p\mathrm{D}^{\geq 0}(\mathrm{X}, \mathbf{F}_p)\}.$$

Here locally bounded means that the torsion is killed by a fixed power of p after restriction to any quasicompact open. For every $n \geq 1$, derived reduction modulo p^n is right t-exact for p-perversity and left t-exact for p^+-perversity. Moreover, $\mathrm{K} \in {}^p\mathrm{D}^{\leq 0}$ if and only if $\mathrm{K}/p \in {}^p\mathrm{D}^{\leq 0}(\mathrm{X}, \mathbf{F}_p)$, while $\mathrm{K} \in {}^{p^+}\mathrm{D}^{\geq 0}$ if and only if $\mathrm{K}/p \in {}^p\mathrm{D}^{\geq 0}(\mathrm{X}, \mathbf{F}_p)$. This is similar in spirit to [BH22, Construction 4.3 and Proposition 4.6].

One defines $\mathrm{D}^b_{\mathrm{fgu}}(\mathrm{D}_{\mathrm{F},\mathscr{X}})$ as the full subcategory of $\mathrm{D}_{\mathrm{unit}}(\mathrm{D}_{\mathrm{F},\mathscr{X}})$ (i.e. $\mathrm{F}^*\mathrm{M} \simeq \mathrm{M}$) which are locally bounded and have cohomology sheaves finitely generated over $\mathrm{D}_{\mathrm{F},\mathscr{X}}$. We will assume X is quasicompact in this section for convenience (it is not strictly needed), so this amounts to being bounded.

Theorem 2.5. *Let $\mathscr{X} / \operatorname{Spf} \mathrm{W}(k)$ be smooth and quasicompact. Then there is an equivalence*

$$\mathrm{RSol} : \mathrm{D}^b_{\mathrm{fgu}}(\mathrm{D}_{\mathrm{F},\mathscr{X}})^{\mathrm{op}} \simeq \mathrm{D}^b_{\mathrm{cons}}(\mathrm{X}, \mathbf{Z}_p),$$

where X/k is the smooth special fiber of $\mathscr{X}$. Explicitly, the functor is given by $\mathrm{RSol}_{\mathrm{X}}(\mathrm{M}) = \mathrm{RHom}_{\mathrm{D}(\mathrm{D}_{\mathrm{F},\mathscr{X}_{\mathrm{proét}}})}(\mathrm{M}_{\mathrm{proét}}, \mathcal{O}_{\mathscr{X},\mathrm{proét}}) \in \mathrm{D}(\mathrm{X}_{\mathrm{proét}}, \mathbf{Z}_p)$. If X is pure of dimension d, then $\mathrm{RSol}_{\mathrm{X}}[d]$ identifies the standard t-structure on the source with the p^+-perverse t-structure on the target.

Proof. We first note that one has

$$\mathrm{D}^b_{\mathrm{fgu}}(\mathrm{D}_{\mathrm{F},\mathscr{X}}) \simeq \varprojlim_n \mathrm{D}^b_{\mathrm{fgu,tf}}(\mathrm{D}_{\mathrm{F},\mathscr{X}_{p^n=0}})$$

and similarly

$$\mathrm{D}^b_{\mathrm{cons}}(\mathrm{X}, \mathbf{Z}_p) \simeq \varprojlim_n \mathrm{D}^b_{\mathrm{ctf}}(\mathrm{X}_{\mathrm{pro\acute{e}t}}, \mathbf{Z}/p^n).$$

Here tf and ctf mean finite Tor dimension and constructible finite Tor dimension, respectively. These categories are closed under retracts in their ambient idempotent complete categories. They therefore satisfy the hypotheses of Lemma 2.3. Moreover, [EK04b, Proposition 15.4.1 and Equation (16.1.3)] give

$$\mathrm{RSol}_{\mathrm{X}}(\mathrm{M})/p^n \simeq \mathrm{RSol}_{\mathrm{X},\mathbf{Z}/p^n}(\mathrm{M}/p^n).$$

Thus the solution functor induces the required compatible functors on reductions, and Lemma 2.3 reduces the assertion to the case modulo p.

Since we have defined RSol using proétale sheaves to handle $\mathbf{Z}_p$ coefficients well, after reducing modulo p in the target category we must use the equivalence

$$\mathrm{D}^b_{\mathrm{cons}}(\mathrm{X}_{\mathrm{pro\acute{e}t}}, \mathbf{F}_p) \simeq \mathrm{D}^b_{\mathrm{cons}}(\mathrm{X}_{\mathrm{\acute{e}t}}, \mathbf{F}_p).$$

Let $\nu : \mathrm{X}_{\mathrm{pro\acute{e}t}} \to \mathrm{X}_{\mathrm{\acute{e}t}}$ be the natural morphism of sites. Since X is smooth, it is locally Noetherian, and for each n pullback along ν identifies $\mathrm{D}^b_{\mathrm{ctf}}(\mathrm{X}_{\mathrm{\acute{e}t}}, \mathbf{Z}/p^n)$ with $\mathrm{D}^b_{\mathrm{ctf}}(\mathrm{X}_{\mathrm{pro\acute{e}t}}, \mathbf{Z}/p^n)$ by the torsion comparison of [BS15, Corollary 5.1.6 and Proposition 5.2.6], applied on quasi-compact opens. Combining this with [HRS23, Theorem 1.6(1)] gives the desired result.

Modulo p, [EK04b, Proposition 15.4.3] identifies finitely generated unit $\mathrm{D}_{\mathrm{F},\mathscr{X}}$-complexes with the usual finitely generated unit $\mathcal{O}_{\mathrm{F,X}}$-complexes, and [EK04b, §16.1.5] identifies their solution functors. Theorem 11.4.2 of [EK04b] therefore proves the required mod-p equivalence, while Theorem 11.5.4 gives the compatibility of its standard heart with the middle-perverse heart. This proves the equivalence.

Finally we discuss the compatibility with t-structures. Assume X has pure dimension d, let M lie in the source category's heart, and put $\mathrm{K} = \mathrm{RSol}_{\mathrm{X}}(\mathrm{M})[d]$. If M is p-torsionfree, then the derived reduction M/p lies in the mod-p heart, so K/p is perverse. The two reduction criteria in Definition 2.4 give $\mathrm{K} \in {}^p\mathrm{D}^{\leq 0} \subset {}^{p^+}\mathrm{D}^{\leq 0}$ and $\mathrm{K} \in {}^{p^+}\mathrm{D}^{\geq 0}$. Hence K lies in the p^+-perverse heart. If $p\mathrm{M} = 0$, one obtains

$$\mathrm{RSol}_{\mathrm{X},\mathbf{Z}_p}(\mathrm{M})[d] \simeq \mathrm{RSol}_{\mathrm{X},\mathbf{F}_p}(\mathrm{M})[d][-1],$$

which is a torsion perverse object shifted by $[-1]$, hence is p^+-perverse. The p-power torsion in a finitely generated unit module is locally bounded, so dévissage and the exact sequence

$$0 \to \mathrm{M}[p^\infty] \to \mathrm{M} \to \mathrm{M}/[p^\infty] \to 0$$

prove the claim for every M in the heart.

Conversely, the p^+-perverse heart is generated under extensions by torsionfree p-perverse sheaves and by $\mathrm{L}[-1]$ for locally bounded torsion p-perverse objects L. For the first kind, derived reduction modulo p is perverse; its inverse image under the stable equivalence has reduction in the mod-p source heart, so derived Nakayama places it in the source heart and makes it p-torsion-free. The formula $\mathrm{RSol}_{\mathrm{X},\mathbf{Z}_p}(\mathrm{M})[d] \simeq \mathrm{RSol}_{\mathrm{X},\mathbf{F}_p}(\mathrm{M})[d][-1]$ and dévissage lift the second kind. Thus $\mathrm{RSol}_{\mathrm{X}}[d]$ identifies the two hearts. □

We also record a variant for perfect rings, which we will use (in a mild way – we really only use existence of the functor and full faithfulness). In this setting we use the definition from [BL19] to make sense of the finitely generated unit condition: one asks for modules to be in $\mathrm{Thick}(\mathrm{W}(\mathrm{R})\langle F\rangle_p)$ and to be unit.

Theorem 2.6. *Let* R *be a perfect* $\mathbf{F}_p$*-algebra. Then there is an equivalence*

$$\mathrm{RSol} : \mathrm{D}_{\mathrm{fgu}}(\mathrm{W}(\mathrm{R}))^{\mathrm{op}} \simeq \mathrm{D}^b_{\mathrm{cons}}(\mathrm{Spec}\,\mathrm{R}, \mathbf{Z}_p)$$

given by $\mathrm{RHom}_{\mathrm{D}^{\varphi}(\mathrm{W}(\mathrm{R}))}(\mathrm{M}_{\text{proét}}, \mathrm{W}(\mathrm{R})_{\text{proét}}) \in \mathrm{D}((\mathrm{Spec}\,\mathrm{R})_{\text{proét}}, \mathbf{Z}_p)$. *The equivalence is moreover symmetric monoidal.*

Proof. The argument is similar to the previous case. Both sides are stable subcategories of their ambient derived categories closed under retracts, hence are idempotent complete and $\mathsf{Perf}(\mathbf{Z}_p)$-linear; the dual action of $\mathsf{Perf}(\mathbf{Z}_p)$ on the source makes RSol linear. Their reductions (in the sense of the setting of Lemma 2.3) are the perfect unit categories over $\mathrm{W}_n(\mathrm{R})$ and the constructible complexes whose stalks have finite Tor-dimension over $\mathbf{Z}/p^n$.

Applying lemma 2.3 now reduces the claim to [BL19, Theorem 11.4.4], together with [BL19, Theorems 12.5.4 and 12.6.1].

For the final claim, one may essentially copy the argument in [BL19, Theorem 8.4.1]. The formula for RSol gives a lax symmetric monoidal structure

$$\mathrm{RSol}(\mathrm{M}) \otimes^{\mathrm{L}}_{\mathbf{Z}_p} \mathrm{RSol}(\mathrm{N}) \longrightarrow \mathrm{RSol}(\mathrm{M} \otimes^{\mathrm{L}}_{\mathrm{W}(\mathrm{R})} \mathrm{N}).$$

Both sides are derived p-complete, so derived Nakayama reduces the assertion that this map is an equivalence to the case modulo p, where it may be checked on geometric stalks. Over an algebraically closed perfect field κ, use the dual or contravariant form of Katz's theorem described in the remark following Theorem 2.1. The fact that this is symmetric monoidal can be checked in the same way as [BL19, Theorem 8.4.1]: the category is equivalent to $\mathsf{Perf}(\mathbf{F}_p)$, and reducing to the heart (which is fine since it thickly generates) one can decompose objects as direct summands of the unit where the claim is obvious.

To check tensor closure, Remark 4.4 reduces us to characteristic p, where [BL19, Theorems 12.4.1 and 12.5.4, Proposition 12.3.8] express every finitely generated unit complex as $\mathrm{D}(\mathrm{P})$ for a Frobenius complex P perfect over R, where D denotes a *weak dual*. By [BL19, Construction 11.2.7 and the proof of Proposition 12.5.1], $\mathrm{D}(\mathrm{P}) \simeq \mathrm{colim}_n\, \varphi^{n*}(\mathrm{P}^\vee)$, so cofinality of the diagonal $\mathbf{N} \to \mathbf{N}^2$ gives $\mathrm{D}(\mathrm{P}) \otimes^{\mathrm{L}}_{\mathrm{R}} \mathrm{D}(\mathrm{Q}) \simeq \mathrm{D}(\mathrm{P} \otimes^{\mathrm{L}}_{\mathrm{R}} \mathrm{Q})$. Since $\mathrm{P} \otimes^{\mathrm{L}}_{\mathrm{R}} \mathrm{Q}$ is again R-perfect, its weak dual is finitely generated unit by [BL19, Proposition 12.3.11]. □

3. Coherent Laurent crystals

In this section, we prove several important results about coherent Laurent crystals on the framed log affines of Definition 3.2. These will be used to understand the local structure of finitely generated unit crystals in terms of (φ, Γ)-modules.

Assumption 3.1. Let $\mathrm{X}/\mathcal{O}_K$ be smooth and qcqs, let D be a horizontal normal crossings divisor, and let $\mathcal{M} = \mathcal{M}_{\mathrm{D}}$ be the log structure induced by D.

Fix a compatible system $(\zeta_{p^m})_{m\geq 1}$ extending ζ_p, and use it to identify $\mathbf{Z}_p(1)$ with $\mathbf{Z}_p$.

In this section we assume p is odd and work over $\mathrm{W}(k)[\zeta_p]$ (mostly for expository reasons). At the end of this section in Corollary 3.17, we explain how to make modifications to obtain the result over $\mathrm{W}(k)$ (and also with more roots of unity).

Definition 3.2. A framing of $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ consists of a p-completely étale $\mathrm{W}(k)$-algebra R_0 over $\mathrm{W}(k)\langle \mathrm{T}_1, \ldots, \mathrm{T}_r, \mathrm{T}_{r+1}^{\pm}, \ldots, \mathrm{T}_n^{\pm}\rangle$ and an identification $\mathrm{R} \simeq \mathrm{R}_0 \widehat{\otimes}_{\mathrm{W}(k)} \mathrm{W}(k)[\zeta_p]$ under which $\mathcal{M}$ is the pullback of the log structure with chart $\mathbf{N}^r$ whose generators map to $\mathrm{T}_1, \ldots, \mathrm{T}_r$. We denote the resulting log formal scheme $\mathrm{Spf}\,\mathrm{W}(k)[\zeta_p]\langle \mathrm{T}_1, \ldots, \mathrm{T}_r, \mathrm{T}_{r+1}^{\pm}, \ldots, \mathrm{T}_n^{\pm}\rangle$ by $\mathbb{T}_r^n$, the induced strict morphism to it by $\square$, and call $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ a framed log affine.

There is also a variant over $\mathcal{O}_K$ for general $K/\mathrm{W}(k)[1/p]$, obtained by replacing $\mathrm{W}(k)[\zeta_p]$ with $\mathcal{O}_K$ and requiring $\mathrm{R} \simeq \mathrm{R}_0 \widehat{\otimes}_{\mathrm{W}(k)} \mathcal{O}_K$.

Remark 3.3. For a framed log affine, we use n and r as the corresponding integers in Definition 3.2. We call the étale lifts of $\mathrm{T}_1, \ldots, \mathrm{T}_r$ "boundary coordinates".

Once we have later shown descent results, the following lemma will frequently be used to reduce to the situation of a framed log affine.

Lemma 3.4. *Let $(\mathrm{X}, \mathcal{M})$ satisfy Assumption 3.1 over $\mathcal{O}_K$. Then $(\mathrm{X}, \mathcal{M})$ admits a finite strict-étale cover by framed log affines over $\mathcal{O}_K$.*

Proof. By the normal crossings hypothesis, a strict-étale cover separates the local branches of D and makes them a simple normal crossings divisor. Fix $x \in \mathrm{X}$. Let $\mathrm{D}_1, \ldots, \mathrm{D}_r$ be the branches of D through x, and pick a small affine Zariski neighborhood Spf R around x so that no other component of D meets the chosen neighborhood. Since D is simple normal crossings relative to $\mathrm{Spf}\,\mathcal{O}_K$ and $\mathrm{X}/\,\mathrm{Spf}\,\mathcal{O}_K$ is smooth, there are functions $t_1, \ldots, t_r, s_{r+1}, \ldots, s_n \in \mathrm{R}$ such that $\mathrm{D}_i = \mathrm{V}(t_i)$ and

$$\mathrm{d}t_1, \ldots, \mathrm{d}t_r, \mathrm{d}s_{r+1}, \ldots, \mathrm{d}s_n$$

is a basis of $\widehat{\Omega}^1_{\mathrm{R}/\mathcal{O}_K}$. Choose constants $c_j \in \mathcal{O}_K$ such that $u_j = s_j + c_j$ has nonzero image in $\kappa(x)$; thus after Zariski localizing once more, each u_j is a unit. The functions t_i and u_j define a morphism

$$\square : \mathrm{Spf}\,\mathrm{R} \to \mathbb{T}_r^n.$$

It is induced by the ring map sending T_i to t_i for $i \leq r$ and T_j to u_j for $j > r$ (the latter are invertible, which is why we localized to make u_j units).

The pullback of $\mathrm{V}(\mathrm{T}_1 \cdots \mathrm{T}_r)$ is $\mathrm{V}(t_1 \cdots t_r) = \mathrm{D} \cap \operatorname{Spf} \mathrm{R}$, hence $\square$ is strict as a morphism of log formal schemes. The Jacobian criterion shows that the underlying morphism is p-completely étale.

Let π be a uniformizer of $\mathcal{O}_K$ with Eisenstein polynomial $\mathrm{E}(u) \in \mathrm{W}(k)[u]$ of degree e. Since $\mathcal{O}_K/p \simeq k[\pi]/(\pi^e)$, invariance of étale algebras under nilpotent thickenings descends R/p uniquely to an étale algebra $\overline{\mathrm{R}}_0$ over the reduction modulo (p, π) of the target coordinate ring. Its unique p-completely étale lift R_0 over the corresponding $\mathrm{W}(k)$-algebra satisfies $\mathrm{R} \simeq \mathrm{R}_0 \widehat{\otimes}_{\mathrm{W}(k)} \mathcal{O}_K$, compatibly with the framing. $\square$

Now let $(\operatorname{Spf} \mathrm{R}, \mathcal{M})$ be a framed log affine with framing $\square$. Let R_0 be the algebra in the framing, and put $\mathrm{A}^+ = \mathrm{R}_0^\square[\![q-1]\!] := \mathrm{R}_0[\![q-1]\!]$ and $\mathrm{A} = \mathrm{R}_0^\square(\!(q-1)\!)_p^\wedge := (\mathrm{A}^+[1/[p]_q])_p^\wedge$. The identification $\mathrm{A}^+/([p]_q) \simeq \mathrm{R}$ sends q to ζ_p. The log q-de Rham prism $(\mathrm{A}^+, [p]_q, \mathbf{N}^r)^a$ (in the sense of [Kos22]) uses this to witness it as an object in $(\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\Delta}$, and its log chart sends e_i to T_i for $i \leq r$. We write $\overline{\mathrm{R}} = \mathrm{R}_0/p$, $\overline{\mathrm{A}}^+ = \mathrm{A}^+/p$, and $\overline{\mathrm{A}} = \mathrm{A}/p$.

Put $\Gamma = (1 + p\mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)^n$. For $\gamma = (a, b_1, \ldots, b_n) \in \Gamma$, regard b_i as an element of $\mathbf{Z}_p$ through the identification fixed above. The formulas $q \mapsto q^a$ and $\mathrm{T}_i \mapsto q^{pb_i}\mathrm{T}_i$ for $i \leq n$ reduce to the identity modulo $[p]_q$. They therefore lift uniquely along the framing to log prism automorphisms over $(\operatorname{Spf} \mathrm{R}, \mathcal{M})$; see Lemma A.17. Thus the crystal property gives an action of Γ on every evaluation in $\mathrm{D}_\blacksquare(\mathrm{A})$ for a solid Laurent crystal as defined in Definition A.11. We use the symbol $\square$ to indicate that the Γ action on the prism depends on the framing.

Our first goal is to prove that evaluation on the log q-de Rham prism induces a t-exact conservative functor

$$\rho_{\mathrm{A}}^* : \mathrm{D}_\blacksquare((\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge) \longrightarrow \mathrm{D}_\blacksquare(\mathrm{A})^{h\Gamma},$$

which is fully faithful. On objects with p-completely nuclear evaluation (p-complete with derived mod p reduction nuclear), the target admits an interpretation in terms of continuous semilinear Γ-modules.

We also establish a result about roots. This will be crucial in giving a workable local description of finitely generated unit crystals. Its proof uses the full faithfulness of Proposition 3.12 for coherent Laurent crystals.

Theorem 3.5. *Let $(\operatorname{Spf} \mathrm{R}, \mathcal{M})$ be a framed log affine and put $\overline{\mathrm{A}} = \mathrm{A}/p$. Let M be a finite $\overline{\mathrm{A}}$-module equipped with a continuous semilinear Γ-action as above. Suppose there is a Γ-equivariant injection $\mathrm{M} \hookrightarrow \varphi^*\mathrm{M}$. Then there is a coherent Laurent crystal $\mathcal{E}$ killed by p, with evaluation M, and an injection $\mathcal{E} \hookrightarrow \varphi^*\mathcal{E}$ whose evaluation is the given map.*

3.1. **Laurent crystals and Γ-modules.** Fix a framed log affine $(\operatorname{Spf} \mathrm{R}, \mathcal{M})$ over $\operatorname{Spf} \mathrm{W}(k)[\zeta_p]$. We let $\mathrm{A}^+ = \mathrm{R}_0^\square[\![q-1]\!]$ denote the q-de Rham prism attached to the framing $\square$. In this subsection we will define a functor

$$\rho_{\mathrm{A}}^{\Gamma,*} : \mathrm{D}_{\blacksquare}((\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}) \longrightarrow \mathrm{D}_{\blacksquare}(\mathrm{A})^{h\Gamma}$$

and check that it is fully faithful. Here $(-)^{h\Gamma}$ denotes the continuous bar construction; only its p-completely nuclear subcategory is identified with continuous semilinear representations.

We explain some of the content of Appendix A.3 here, as we will be working in this formalism. Throughout we work with analytic rings as defined by Clausen and Scholze in [CS23] – see also [RC24] and [Kip26b] for more background material. If $(\mathrm{B}, (d))$ is a prism, we give $\mathrm{B}[1/d]_p^{\wedge}$ the stack-level p-complete analytic structure defined by the tower of Tate-Huber pairs $((\mathrm{B}/p^n)[1/d], \mathrm{B}/p^n)$ from [Kip26a, Example 2.1.5 and Remark 2.1.7]. For dealing with semilinear Γ-modules we will need $\mathrm{C}(\Gamma, -)$ as well, with its natural induced analytic ring structure (as a topological ring one uses the compact-open topology). The precise definition of $\mathrm{D}_{\blacksquare}((\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})$ is given in Definition A.11 using these analytic structures, as a limit over all log prisms A in $(\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\Delta}$.

We will show that actually there is a natural stacky interpretation of this functor, using the results in A and A.3. For the reader not wishing to wade through the technical details needed to properly set up this theory, the essential point is that $(\operatorname{Spf} \mathrm{R}, \mathcal{M})^{\mathcal{L},\blacksquare}$ is an analytic stack such that $\mathrm{D}_{\blacksquare}((\operatorname{Spf} \mathrm{R}, \mathcal{M})^{\mathcal{L},\blacksquare}) \simeq \mathrm{D}_{\blacksquare}((\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})$. We postpone explaining details about this Laurent stack until §6.1 as we will only use relatively formal properties of it here, where it is explained in detail – we use the stack $\mathrm{X}^{\Delta,\blacksquare}$ from [Kip26a].

It is possible to carry the full faithfulness argument out entirely site-theoretically (indeed this was the first proof the author obtained) but the strategy is far easier to understand at a high level from a stacky perspective. We will realize $\rho_{\mathrm{A}}^{\Gamma,*}$ as a pullback functor along

$$\rho_{\mathrm{A}}^{\Gamma} : [\operatorname{AnSpec} \mathrm{A}/\Gamma] \to (\operatorname{Spf} \mathrm{R}, \mathcal{M})^{\mathcal{L},\blacksquare}.$$

The stacky interpretation then gives a natural strategy to attack the problem. In such a situation to verify full faithfulness what one needs to do is to check that the morphism of stacks is cohomologically proper, and moreover that the canonical map $\mathcal{O} \to \mathrm{R}\rho_{\mathrm{A},*}^{\Gamma}\mathcal{O}$ is an isomorphism. Once this is established full faithfulness is a purely formal consequence of the projection formula.

The proof will be a reduction the cohomology calculation in Lemma 3.11 (used to establish that $\mathcal{O} \simeq \mathrm{R}\rho_{\mathrm{A},*}^{\Gamma}\mathcal{O}$), for which we will need a very explicit description of $\mathrm{A}^{(1)}$ and how it sits inside the algebra of continuous functions $\mathrm{C}(\Gamma, \mathrm{A})$. The following lemma computes $\mathrm{A}^{+,(1)}$ as a log prismatic envelope.

Lemma 3.6. *Let* $\mathrm{A}^+ = \mathrm{R}_0^{\square}[\![q-1]\!]$, *and let* $(\mathrm{A}^+)^{(1)}$ *be the ring of the self-coproduct of* $(\mathrm{A}^+, [p]_q, \mathcal{M})^a$ *in* $(\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\Delta}$. *Write* q_0, q_1 *for the two copies of* q, *put* $u_0 = q_1/q_0$, *and let* $u_i = \mathrm{T}_{i,1}/\mathrm{T}_{i,0}$ *for* $1 \leq i \leq n$. *Then*

$$(\mathrm{A}^+)^{(1)} \simeq \mathrm{R}_0^{\square}[\![q_0 - 1]\!][u_0^{\pm 1}, u_1^{\pm 1}, \ldots, u_n^{\pm 1}] \left\{ \frac{u_0 - 1}{[p]_{q_0}}, \ldots, \frac{u_n - 1}{[p]_{q_0}} \right\}^{\wedge}_{\delta, (p, [p]_{q_0})}.$$

Here $\delta(u_j) = 0$ *for* $0 \leq j \leq n$.

Proof. This follows by verifying universal properties. For simplicity write $(\mathrm{A}^+)^{(1)}$ for $\mathrm{R}_0^\square[\![q_0 - 1]\!][u_0^{\pm 1}, u_1^{\pm 1}, \ldots, u_n^{\pm 1}]\left\{\frac{u_0-1}{[p]_{q_0}}, \ldots, \frac{u_n-1}{[p]_{q_0}}\right\}^\wedge_{\delta,(p,[p]_{q_0})}$ with its natural log prism structure (with prismatic ideal $[p]_{q_0}$ and log structure pulled back from $\mathrm{Spf}\ \mathrm{A}^+$); we will verify that it has the required universal property.

Let us recall what this universal property should be for the self-coproduct of the log q-de Rham prism. We ask, for a test log prism $(\mathrm{B}, [p]_{q_0}, \mathcal{N})^a \in (\mathrm{Spf}\ \mathrm{R}, \mathcal{M})_{\Delta}$ that in the diagram

$$\begin{tikzcd}
(\mathrm{B}, [p]_{q_0}, \mathcal{N})^a \arrow[drr, bend left, "f_0"] \arrow[dr, dashed, "g"] \arrow[ddr, bend right, "f_1"'] & & \\
& (\mathrm{A}^{(1)}, [p]_{q_0}, \mathcal{M}^{(1)})^a \arrow[r, "p_0"] \arrow[d, "p_1"] & (\mathrm{Spf}\ \mathrm{A}_0^+, ([p]_{q_0}), \mathcal{M}_0)^a \arrow[d] \\
& (\mathrm{Spf}\ \mathrm{A}_1^+, ([p]_{q_1}), \mathcal{M}_1)^a \arrow[r] & *
\end{tikzcd}$$

there is a unique dashed arrow g (we can assume the ideal is $[p]_{q_0}$ by rigidity). Here the diagram is as sheaves in the log prismatic site, where the prisms are regarded this way via the Yoneda embedding.

First, we can replace the log structure on B with a common strict log structure to test such universal properties. One can use the same method as [DLMS24, Corollary 2.2 and proof of Lemma 2.10].

A map g into the self-coproduct (when it is represented by a log prism) then amounts to the data of strict maps of prisms f_0 and f_1. After fixing $f_0 : \mathrm{A}_0^+ \to \mathrm{B}$, the choices of f_1 are

$$\left\{(v_j)_{j=0}^n \in (\mathrm{B}^\times)^{n+1} : \delta(v_j) = 0 \text{ and } v_j \equiv 1 \pmod{[p]_{q_0}}\right\}.$$

By the universal property of log prismatic envelopes [Kos22, Propositions 3.6 and 3.7], after fixing f_0 the maps g are classified by the same data, through $g(u_j) = v_j$ on the underlying rings. Indeed, the divided generators map to $(v_j - 1)/[p]_{q_0}$, while the second chart maps e_i to (e_i, v_i) for $i \leq r$. Conversely, these assignments give the unique map g, and its composite with p_1 is f_1 by the unique étale lift along the framing. Hence the two universal properties agree. □

Our next task will be to understand $\mathrm{A}^{(1)} := \mathrm{A}^{+,(1)}[1/[p]_q]^\wedge$ as a subring of $\mathrm{C}(\Gamma, \mathrm{A})$.

Once this is defined, as in Definition A.11 one sets

$$\mathrm{D}_{\blacksquare}(\mathrm{B}[1/d]_p^\wedge) := \varprojlim_n \mathsf{Mod}_{(\mathrm{B}/^{\mathrm{L}}p^n)[1/d]}\big(\mathrm{D}_{\blacksquare}(\mathrm{Spa}(\mathrm{B},\mathrm{B}))\big),$$

with transition functors given by derived reduction.

Definition 3.7. For $\gamma = (a, b_1, \dots, b_n) \in \Gamma$, put $x_0(\gamma) = (a-1)/p$ and $x_i(\gamma) = b_i$ for $1 \le i \le n$. Set $\eta_{0,m}(x_0) = q^{m+p\binom{m}{2}}(q-1)^m\binom{x_0}{m}_{q^p}$ and $\eta_{i,m}(x_i) = q^{p\binom{m}{2}}(q-1)^m\binom{x_i}{m}_{q^p}$ for $i > 0$. Write η_α for the products of these functions in the $(n+1)$ variables and set

$$\mathrm{C}_q(\Gamma, \mathrm{A}) = \left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} \mathrm{A}^+\eta_\alpha\right)^\wedge_{(p,q-1)}[1/[p]_q]_p^\wedge.$$

We define $\mathrm{C}_q(\Gamma^s, \mathrm{A})$ similarly, over indices $\alpha \in \mathbf{N}^{s(n+1)}$.

There is also an alternative formula which will be of use. Pullback along (id, γ) then sends $u_i \in \mathrm{A}^{+,(1)}$ to $(q^p)^{x_i(\gamma)}$ for $0 \le i \le n$. Set

$$[m]_{q^p}!\,\gamma_{m,q^p}(x) = \prod_{j=0}^{m-1}((q^p)^x - q^{pj}),$$

where $[m]_{q^p}! = \prod_{j=1}^m [j]_{q^p}$. Then $\eta_{0,m}(x_0) = q^m[p]_q^{-m}\gamma_{m,q^p}(x_0)$ and $\eta_{i,m}(x_i) = [p]_q^{-m}\gamma_{m,q^p}(x_i)$ for $i > 0$.

The formula for η_α in Definition 3.7 shows that these functions are uniformly bounded. Consequently, an expansion whose coefficients tend to zero $(p, q-1)$-adically converges uniformly, and evaluation defines a continuous map $\mathrm{C}_q(\Gamma^s, \mathrm{A}) \to \mathrm{C}(\Gamma^s, \mathrm{A})$, where the target carries the compact-open topology. Here we give $\mathrm{C}_q(\Gamma^s, \mathrm{A})$ the natural topology induced by the completion in Definition 3.7, followed by Laurentization and p-adic completion. Lemma 3.8 shows that this map is injective, but its topology on the source is not the subspace topology.

For $s \ge 0$, let $(\mathrm{A}^+)^{(s)}$ be the sth term of the Čech nerve of the log q-de Rham prism in $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}$, and put $\mathrm{A}^{(s)} = ((\mathrm{A}^+)^{(s)}[1/[p]_q])_p^\wedge$.

We will show evaluation on the log q-de Rham prism induces a functor

$$\rho_{\mathrm{A}}^{\Gamma,*} : \mathrm{D}_{\blacksquare}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge) \longrightarrow \mathrm{D}_{\blacksquare}(\mathrm{A})^{h\Gamma}.$$

The crystal property does give an abstract Γ action, but does not obviously ensure continuity and higher coherences which is why we make the following more delicate construction. Lemma A.17 writes the source as the limit

$$\mathrm{D}_{\blacksquare}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge) \simeq \lim\Big(\ \mathrm{D}_{\blacksquare}(\mathrm{A}) \rightrightarrows \mathrm{D}_{\blacksquare}(\mathrm{A}^{(1)}) \substack{\rightarrow\\\rightarrow\\\rightarrow} \mathrm{D}_{\blacksquare}(\mathrm{A}^{(2)}) \substack{\rightarrow\\\rightarrow\\\rightarrow\\\rightarrow} \cdots\ .\Big)$$

The target can be thought of as solid A-modules with continuous semilinear Γ-action, although Lemma 3.9 tells us this is only reasonable under a nuclearity hypothesis (which will be satisfied in all applications). Modulo p^n, the rings in the preceding diagram and in the continuous bar construction carry their natural analytic structures $\mathrm{Spa}(\mathrm{B}, \mathrm{B})$. The Γ-action on $\mathrm{A}^+ = \mathrm{R}_0^{\square}[\![q-1]\!]$, after p-completely inverting $[p]_q$ induces maps

$$\theta_s : \mathrm{A}^{(s)} \longrightarrow \mathrm{C}(\Gamma^s, \mathrm{A}).$$

A priori these land only in all functions, but we will show they land in continuous functions. This then induces $\rho_{\mathrm{A}}^{\Gamma,*}$ as

$$\mathrm{D}_{\blacksquare}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}) \xrightarrow{\widetilde{\rho}_{\mathrm{A}}^{*}} \lim_{[s]\in\Delta} \mathrm{D}_{\blacksquare}(\mathrm{C}(\Gamma^s, \mathrm{A})) =: \mathrm{D}_{\blacksquare}(\mathrm{A})^{h\Gamma}.$$

We point out that because A is induced from a topological ring that is an inverse limit of Banach rings, one can write $\mathrm{C}(\Gamma^s, \mathrm{A}) = \lim_n \mathrm{C}(\Gamma^s, \mathrm{A}/p^n)$. We also note that in the situation the quotient topology on A/p^n and the Tate topology given by $(\mathrm{A}^+/p^n)[1/(q-1)]$ agree, so there is no ambiguity (indeed this holds for any prism). Finally we also note that $\mathrm{C}(\Gamma^s, \mathrm{B})[1/d]_p^{\wedge} \simeq \mathrm{C}(\Gamma^s, \mathrm{B}[1/d]_p^{\wedge})$ for any prism $(\mathrm{B}, (d))$ because Γ^s is profinite (combined with the previous fact about inverse limits). Thus we freely exchange these in what follows.

These maps θ_s are in fact injective as ordinary ring maps, and we show the previously defined $\mathrm{C}_q(\Gamma^s, \mathrm{A})$ are their images. In what follows we give $\mathrm{A}^{(s)}$ the analytic ring structure induced from the log prism $\mathrm{A}^{+,(s)}$ in the logarithmic Čech nerve, using $\lim_n((\mathrm{A}^{+(s)}/p^n)[1/d], (\mathrm{A}^{+(s)}/p^n))$ with their Huber pair analytic ring structures (as in the convention set in the discussion after Lemma 3.6). For $\mathrm{C}_q(\Gamma^s, \mathrm{A})$ we make a similar definition, using the completed direct sum presentation as we did for its topological ring structure.

Lemma 3.8. *The map θ_s identifies $\mathrm{A}^{(s)}$ with the subring $\mathrm{C}_q(\Gamma^s, \mathrm{A}) \subset \mathrm{C}(\Gamma^s, \mathrm{A})$:*

$$\mathrm{A}^{(s)} \xrightarrow{\sim} \mathrm{C}_q(\Gamma^s, \mathrm{A}).$$

This is further an isomorphism of analytic rings.

The map $\mathrm{C}_q(\Gamma^s, \mathrm{A}) \to \mathrm{C}(\Gamma^s, \mathrm{A})$ is continuous as topological rings where the target is equipped with the compact-open topology; the same assertion holds when regarded as solid modules.

Proof. As above, we use $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}$. We first treat $s = 1$. We actually will show a stronger claim, which is that there is a natural isomorphism of adic rings

$$\theta_1 : \mathrm{A}^{+,(1)} \simeq \left(\bigoplus_{\alpha\in\mathbf{N}^{n+1}} \mathrm{A}^+\eta_\alpha\right)^{\wedge}_{(p,q-1)}.$$

After p-completely inverting $[p]_q$ or equivalently $q-1$, this isomorphism of ordinary adic rings implies the desired isomorphism of analytic rings (as in this setting passing from topological rings to analytic rings is fully faithful).

We do this by using the explicit log prismatic envelope presentation of $\mathrm{A}^{+,(1)}$. Recall $(\mathrm{A}^+)^{(1)}$ from Lemma 3.6 is the $(p, q-1)$-adic (equivalently $(p, [p]_q)$) completion of

$$\mathrm{A}^+[u_0^{\pm 1}, \dots, u_n^{\pm 1}] \left\{ \frac{u_0 - 1}{[p]_{q_0}}, \dots, \frac{u_n - 1}{[p]_{q_0}} \right\}_\delta .$$

We used $u_0 = q_1/q_0$ and $u_i = \mathrm{T}_{i,1}/\mathrm{T}_{i,0}$ as shorthand. The map θ_1 can be made explicit integrally. For $g \in \Gamma$, using the coordinates $g = (x_0, \dots, x_n)$ of Definition 3.7 write the evaluation of $\theta_1(a) \in \mathrm{Map}(\Gamma, \mathrm{A}^+)$[3] for $a \in \mathrm{A}^+$ at g as the pullback map determined by $u_i \mapsto (q^p)^{x_i(g)}$ for $0 \le i \le n$.

Using [GLSQ23, Theorem 4.4], the cyclotomic factor $\mathrm{A}^+[u_0^{\pm 1}] \left\{ \frac{u_0 - 1}{[p]_{q_0}} \right\}_\delta^\wedge$ is, up to a change of variables, the $(p, q-1)$-completion of the free A^+-module on their Δ-divided powers $\omega_{0,\Delta}^{\{n\}}$ (see [GLSQ23, Definition 1.7]) as a $(p, q-1)$-adic ring. Indeed from the discussion preceding Theorem 4.4. in *loc. cit.* they identify $\widehat{\bigoplus}_{n \ge 0} \mathrm{A}^+ \omega_{0,\Delta}^{\{n\}}$ with the prismatic envelope of the δ-pair $(\mathrm{A}^+[\![\xi]\!], (\xi))$, which checking universal properties is indeed our ring (one has $\xi \mapsto q_0(u_0 - 1)$).

Now we see where θ_1 sends this basis. Using [GLSQ23, Remark 1.8(5)] provides a "blowing up" map Bl, yielding

$$\mathrm{A}^+\langle \xi \rangle_{q_0^p} \xrightarrow{\mathrm{Bl}} \widehat{\bigoplus}_{n \ge 0} \mathrm{A}^+ \omega_{0,\Delta}^{\{n\}} \simeq \mathrm{A}^+[u_0^{\pm 1}] \left\{ \frac{u_0 - 1}{[p]_{q_0}} \right\}_\delta^\wedge \xrightarrow{\theta_1} \mathrm{Map}(\Gamma, \mathrm{A}^+)$$

such that the q-divided power $\xi^{[m]_{q_0^p}}$ is sent to $[p]_{q_0}^m \omega_{0,\Delta}^{\{m\}}$ under Bl. This is helpful, because we know where $\xi^{[m]_{q_0^p}}$ is sent to under the composite map $\theta_1 \circ \mathrm{Bl}$: ξ is sent by θ_1 to

$$q((q^p)^{x_0} - 1) \in \mathrm{Map}(\Gamma, \mathrm{A}^+).$$

Thus we may deduce

$$[p]_{q_0}^m \theta_1(\omega_{0,\Delta}^{\{m\}}) = q^m \gamma_{m,q^p}(x_0),$$

so in particular using the alternative formula after Definition 3.7 we see $\theta_1(\omega_{0,\Delta}^{\{m\}}) = \eta_{0,m}$ using $[p]_{q_0}$-torsionfreeness.

Next, we handle the remaining factors corresponding to u_i for $i > 0$. The argument is more or less the same. Using [GLSQ22, Definition 1.6 and Corollary 5.6], the corresponding factor $\mathrm{A}^+[u_i^{\pm 1}] \left\{ \frac{u_i - 1}{[p]_{q_0}} \right\}_\delta^\wedge$ is the completion of the free A^+-module on its negative-level divided powers $\omega_{i,\Delta}^{\{m\}}$. The analogous map in [GLSQ22, Proposition 1.7] let us deduce the formula

$$[p]_q^m \theta_1(\omega_{i,\Delta}^{\{m\}}) = \gamma_{m,q^p}(x_i).$$

[3] We haven't yet checked the output is a continuous function.

Similarly we find $\theta_1(\omega_{i,\mathbb{A}}^{\{m\}}) = \eta_{i,m}$. Thus θ_1 induces a surjection onto

$$\left(\bigoplus_{\alpha\in\mathbf{N}^{n+1}} \mathrm{A}^+\eta_\alpha\right)^\wedge_{(p,q-1)}.$$

It is injective: evaluation at nonnegative integral values of each x_i is triangular, with $\eta_{i,m}(j) = 0$ for $m > j$ and $\eta_{i,m}(m)$ a unit times $(q-1)^m$. Only finitely many terms contribute at each such value, and A^+ is $(q-1)$-torsion-free. Hence the integral map is, as desired, an isomorphism onto $(\bigoplus_\alpha \mathrm{A}^+\eta_\alpha)^\wedge_{(p,q-1)}$; the isomorphism is of $(p,q-1)$-adic rings, so the claim about analytic ring structures also follows. One sees θ_1 lands in $\mathrm{C}(\Gamma, \mathrm{A})$ by the discussion after Definition 3.7.

For general s, $\mathrm{A}^{+,(s)} = (\mathrm{A}^{+(1)})^{\widehat{\otimes}^s_{\mathrm{A}^+}}$. Using this one may check

$$(\mathrm{A}^+)^{(s)} \simeq \left(\bigoplus_{\alpha\in\mathbf{N}^{s(n+1)}} \mathrm{A}^+\eta_\alpha\right)^\wedge_{(p,q-1)},$$

so by the same argument the general claim follows. □

By definition, $\mathrm{D}_\blacksquare(\mathrm{A})^{h\Gamma} = \lim_{[s]\in\Delta} \mathrm{D}_\blacksquare(\mathrm{C}(\Gamma^s, \mathrm{A}))$. The next lemma identifies its p-completely nuclear subcategory with continuous semilinear representations.

We first fix notation. Let G be a profinite group and let B be a derived p-complete analytic ring with continuous G-action. Write $\mathsf{Rep}_{\mathrm{B}^t_\blacksquare}(\mathrm{G})$ for the category of solid modules over the twisted solid group algebra $\mathrm{B}^t_\blacksquare[\mathrm{G}]$; equivalently, this is the category of solid B-modules with continuous semilinear G-action. Here $\mathbf{Z}_\blacksquare[\mathrm{G}]$ denotes the solid group algebra. We set

$$\mathrm{R\Gamma}_\blacksquare(\mathrm{G}, \mathrm{M}) := \mathrm{RHom}_{\mathbf{Z}_\blacksquare[\mathrm{G}]}(\mathbf{Z}_\blacksquare, \mathrm{M}) \simeq \mathrm{RHom}_{\mathrm{B}^t_\blacksquare[\mathrm{G}]}(\mathrm{B}, \mathrm{M}).$$

For $\mathrm{M} \in \mathrm{D}_\blacksquare(\mathrm{B})$, we write

$$\mathrm{C}(\mathrm{G}, \mathrm{M}) := \underline{\mathrm{RHom}}_{\mathrm{B}}(\mathrm{B}[\underline{\mathrm{G}}], \mathrm{M}).$$

Recall that $\mathrm{D}_\blacksquare(\mathrm{B})^{\mathrm{nuc}}$ consists of the objects M for which, for every compact $\mathrm{P} \in \mathrm{D}_\blacksquare(\mathrm{B})^\omega$, the canonical map

$$\mathrm{P}^\vee \otimes^{\mathrm{L},\blacksquare}_{\mathrm{B}} \mathrm{M} \longrightarrow \underline{\mathrm{RHom}}_{\mathrm{B}}(\mathrm{P}, \mathrm{M})$$

is an equivalence. For a discrete analytic ring this condition is automatic; for example, $\mathrm{D}_\blacksquare(\mathbf{Z}_{\mathrm{disc}})^{\mathrm{nuc}} = \mathrm{D}(\mathbf{Z})$.

Lemma 3.9. *Let* B *be a derived p-complete solid analytic ring with a continuous action of a compact p-adic Lie group* G*. Give* $\mathrm{C}(\mathrm{G}^s, \mathrm{B})$ *its natural analytic ring structure. The natural functor*

$$\lim_{[s]\in\Delta} \mathrm{D}_\blacksquare(\mathrm{C}(\mathrm{G}^s, \mathrm{B})) \longrightarrow \mathsf{Rep}_{\mathrm{B}^t_\blacksquare}(\mathrm{G})$$

restricts to an equivalence from objects whose evaluation at $[0]$ *is p-completely nuclear*[4] *to representations whose underlying* B*-module is p-completely nuclear. If* $\mathrm{M} \in \mathrm{D}_{\blacksquare}(\mathrm{B})$ *is the underlying solid* B*-module under this equivalence,*

$$\varprojlim_{[s]\in\Delta} \mathrm{C}(\mathrm{G}^s, \mathrm{B}) \otimes_{\mathrm{B}}^{\mathrm{L},\blacksquare} \mathrm{M} \simeq \varprojlim_{[s]\in\Delta} \mathrm{C}(\mathrm{G}^s, \mathrm{M}) \simeq \mathrm{R}\Gamma_{\blacksquare}(\mathrm{G}, \mathrm{M}).$$

Proof. The Barr-Beck-Lurie theorem, as in [Mik26, Remark 1.49, Corollary 1.50] (which is a similar argument), identifies the source and target with coalgebras in $\mathrm{D}_{\blacksquare}(\mathrm{B})$ for the following comonads. The first is induced by the endofunctor

$$\mathbb{T} : \mathrm{D}_{\blacksquare}(\mathrm{B}) \xrightarrow{\mathrm{C}(\mathrm{G},\mathrm{B})\otimes_{\mathrm{B}}^{\mathrm{L},\blacksquare}-} \mathrm{D}_{\blacksquare}(\mathrm{C}(\mathrm{G}, \mathrm{B})) \longrightarrow \mathrm{D}_{\blacksquare}(\mathrm{B}).$$

Here the first morphism uses the map $\mathrm{B} \to \mathrm{C}(\mathrm{G}, \mathrm{B})$ sending $b \in \mathrm{B}$ to the constant function (valued at b). The second morphism is pushforward along the action map $\mathrm{act} : \mathrm{B} \to \mathrm{C}(\mathrm{G}, \mathrm{B})$ (i.e. b is sent to the function sending g to $g(b)$). The comonad structure is the obvious one (the comultiplication is induced by the multiplication on G)

The second is induced by the endofunctor $\mathbb{S} : \mathrm{D}_{\blacksquare}(\mathrm{B}) \to \mathrm{D}_{\blacksquare}(\mathrm{B})$ given by

$$\mathbb{S}(\mathrm{M}) = \mathrm{C}(\mathrm{G}, \mathrm{M}),$$

where $(bf)(g) = g(b)f(g)$ on $\mathbb{S}(\mathrm{M})$. Thus

$$\varprojlim_{[s]\in\Delta} \mathrm{D}_{\blacksquare}(\mathrm{C}(\mathrm{G}^s, \mathrm{B})) \simeq \mathrm{CoAlg}_{\mathbb{T}}(\mathrm{D}_{\blacksquare}(\mathrm{B})).$$

Similarly, $\mathsf{Rep}_{\mathrm{B}^t_{\blacksquare}}(\mathrm{G}) \simeq \mathrm{CoAlg}_{\mathbb{S}}(\mathrm{D}_{\blacksquare}(\mathrm{B}))$.

The map $f \otimes m \mapsto (g \mapsto f(g)m)$ defines a natural transformation $\mathbb{T} \to \mathbb{S}$ compatible with the counits and comultiplications, hence a map of comonads.

Suppose now that M is p-completely nuclear. For the compact object $\mathrm{P} = \mathrm{B}[\underline{\mathrm{G}}]$, nuclearity and the identifications $\mathrm{P}^\vee \simeq \mathrm{C}(\mathrm{G}, \mathrm{B})$ and $\underline{\mathrm{RHom}}_{\mathrm{B}}(\mathrm{P}, \mathrm{M}) \simeq \mathrm{C}(\mathrm{G}, \mathrm{M})$ give

$$\mathbb{T}(\mathrm{M}) = \mathrm{C}(\mathrm{G}, \mathrm{B}) \otimes_{\mathrm{B}}^{\mathrm{L},\blacksquare} \mathrm{M} \simeq \mathrm{P}^\vee \otimes_{\mathrm{B}}^{\mathrm{L},\blacksquare} \mathrm{M} \simeq \underline{\mathrm{RHom}}_{\mathrm{B}}(\mathrm{P}, \mathrm{M}) \simeq \mathrm{C}(\mathrm{G}, \mathrm{M}) = \mathbb{S}(\mathrm{M}).$$

This calculation shows both comonads restrict to endofunctors of the full subcategory of p-completely nuclear modules in $\mathrm{D}_{\blacksquare}(\mathrm{B})$ (since $\mathrm{C}(\mathrm{G}, \mathrm{B})$ is nuclear and the subcategory of nuclear objects is closed under tensor products); moreover it also shows $\mathbb{S}$ and $\mathbb{T}$ become isomorphic comonads. It follows that on p-completely nuclear objects we can describe the source and target as coalgebras over isomorphic comonads, thus the claim follows.

The cohomological comparison follows by taking RHom with the unit. □

[4] By this we mean it is derived p-complete and satisfies the nuclearity property for derived p-complete objects; equivalently, it is derived p-complete and nuclear mod p.

To compute solid group cohomology, we use the resolution of [Bos23, Proposition B.3 and Lemma B.4]. If $\mathrm{G} \simeq \mathbf{Z}_p^r$, let Λ_{G} be the solid algebra associated with $\mathbf{Z}_p[\![\mathrm{G}]\!]$. For topological generators $\gamma_1, \dots, \gamma_r$, there is a finite projective resolution in solid Λ_{G}-modules

$$0 \longrightarrow \Lambda_{\mathrm{G}}^{\binom{r}{r}} \longrightarrow \cdots \longrightarrow \Lambda_{\mathrm{G}}^{\binom{r}{1}} \xrightarrow{(\gamma_1-1,\dots,\gamma_r-1)} \Lambda_{\mathrm{G}} \longrightarrow (\mathbf{Z}_p)_{\blacksquare} \longrightarrow 0. \tag{1}$$

The omitted arrows are the Koszul differentials. Thus solid group cohomology of a complex of solid $\mathbf{Z}_p$-modules with continuous G-action is computed by the Koszul complex of the operators $\gamma_i - 1$. The topology of Definition 3.7 ensures that the translations and homotopies below are morphisms of solid modules.

One can carry out the needed cohomology calculation in the $\eta_{i,m}$ basis of Definition 3.7, but it is actually slightly easier to work modulo p where a more natural characterization is available. Via the coordinates $x = (x_0, \dots, x_n) : \Gamma \to \mathbf{Z}_p^{n+1}$ of Definition 3.7[5], a continuous function $f : \Gamma \to \mathrm{M}$ with values in a complete separated module has a unique *Mahler expansion* $f(x) = \sum_\alpha a_\alpha \binom{x}{\alpha}$. Its Mahler coefficient of degree α is $a_\alpha = (\Delta_0^{\alpha_0} \cdots \Delta_n^{\alpha_n} f)(0)$, where $\Delta_i f(x) = f(x+e_i) - f(x)$ and e_i is the ith standard basis vector of $\mathbf{Z}_p^{n+1}$. We will use this in the following to give an alternative characterization of C_q using Mahler expansions, at least modulo p.

Lemma 3.10. *Let $\bar{\mathrm{A}} = \mathrm{A}/p$ (similarly for $\bar{\mathrm{A}}^+$) and consider $f \in \mathrm{C}(\Gamma, \bar{\mathrm{A}})$. We take its Mahler expansion*

$$f(x) = \sum_{\alpha \in \mathbf{N}^{n+1}} a_\alpha \binom{x}{\alpha},$$

where $\binom{x}{\alpha} = \prod_i \binom{x_i}{\alpha_i}$. Suppose there are constants $\sigma > 1$ and C for which

$$a_\alpha \in (q-1)^{\lceil \sigma|\alpha| - \mathrm{C} \rceil} \bar{\mathrm{A}}^+$$

for every multi-index α. Then $f \in \mathrm{C}_q(\Gamma, \overline{\mathrm{A}})$.

Proof. We will prove the stronger assertion

$$\left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} \overline{\mathrm{A}}^+ \eta_\alpha \right)^\wedge_{(q-1)} = \left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} \overline{\mathrm{A}}^+ (q-1)^{|\alpha|} \binom{x}{\alpha} \right)^\wedge_{(q-1)}.$$

In one variable x, set $\tilde{\eta}_m = q^{p\binom{m}{2}} \binom{x}{m}_{q^p}$, a renormalized version of $\eta_{i,m}$ from Definition 3.7 (by a factor of $(q-1)^m$). For $m \geq 1$, the q-Pascal recurrence

$$\tilde{\eta}_m(x+1) - \tilde{\eta}_m(x) = (q^{pm} - 1)\tilde{\eta}_m(x) + q^{p(m-1)} \tilde{\eta}_{m-1}(x)$$

shows that the coefficient of $\binom{x}{j}$ in $\tilde{\eta}_m$ vanishes for $j < m$, is a unit for $j = m$, and belongs to $(q^p - 1)^{j-m} \overline{\mathrm{A}}^+$ for $j > m$. Here we use that the finite difference operator $\Delta(f) = f(x+1) - f(x)$ extracts the coefficient of $\binom{x}{j}$ in its Mahler expansion as $\Delta^j f(0)$.

[5] The map x is only a homeomorphism, and does not respect the group structures.

The first claim is simply because $\tilde{\eta}_m(j) = 0$ when $j < m$, the second because we obtain $q^{p\binom{m}{2}}$ at $j = m$, and the third follows by induction using the q-Pascal recurrence.

In $\overline{\mathrm{A}}^+$ one has $q^p - 1 = (q-1)^p$, and hence for all $0 \le i \le n$ (rescaling by the unit q^m at $i = 0$) one has

$$\eta_{i,m} = u_{i,m}(q-1)^m \binom{x_i}{m} + \sum_{j>m} c_{i,m,j}(q-1)^j \binom{x_i}{j}.$$

Here $u_{i,m} \in (\overline{\mathrm{A}}^+)^\times$, and $c_{i,m,j} \in (q-1)^{(p-1)(j-m)}\overline{\mathrm{A}}^+$. This shows the $(q-1)^{|\alpha|}\binom{x}{\alpha}$ and the η_α bases are related by an operator $\mathrm{D}(1+\mathrm{N})$ where D is diagonal and invertible and

$$\mathrm{N}^r \left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} \overline{\mathrm{A}}^+ (q-1)^{|\alpha|} \binom{x}{\alpha} \right)^\wedge_{(q-1)} \subset (q-1)^{(p-1)r} \left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} \overline{\mathrm{A}}^+ (q-1)^{|\alpha|} \binom{x}{\alpha} \right)^\wedge_{(q-1)},$$

hence N is topologically nilpotent. Thus we deduce the claimed isomorphism. □

We now carry out the needed cohomology calculation.

Lemma 3.11. *Right translation defines a continuous Γ-action on $\mathrm{C}_q(\Gamma, \mathrm{A})$ by $(\gamma f)(g) = f(g\gamma)$. Then there is a natural equivalence*

$$\mathrm{A} \xrightarrow{\sim} \mathrm{R}\Gamma_{\blacksquare}(\Gamma, \mathrm{C}_q(\Gamma, \mathrm{A})).$$

Proof. By derived Nakayama it is enough to work modulo p. The proof of Lemma 3.10 shows there is a lattice

$$\left(\bigoplus_{\alpha \in \mathbf{N}^{n+1}} \overline{\mathrm{A}}^+ (q-1)^{|\alpha|} \binom{x}{\alpha} \right)^\wedge_{q-1} \subset \mathrm{C}_q(\Gamma, \overline{\mathrm{A}})$$

Let $\mathrm{Fil}^\bullet$ be the $q-1$-adic filtration on this lattice; this filtration is complete and separated.

Let $\gamma_0, \ldots, \gamma_n$ be the usual generators of Γ, in the coordinates chosen in Lemma 3.10. Right translation by γ_i sends x_i to $x_i + 1 + px_0$, while γ_0 sends x_0 to $x_0 + 1 + px_0$. In characteristic p, one has the standard binomial identity $\binom{x+pz}{m} = \sum_{\ell \ge 0} \binom{x}{m-p\ell}\binom{z}{\ell}$. We claim that, for every $0 \le i \le n$, the two-term complex

$$\left[\mathrm{C}_q(\Gamma, \overline{\mathrm{A}}) \xrightarrow{\gamma_i - 1} \mathrm{C}_q(\Gamma, \overline{\mathrm{A}}) \right]$$

retracts onto the functions independent of x_i in degree 0. Indeed, let π_i be the projection onto the Mahler terms with $\alpha_i = 0$. We define

$$h_i \left((q-1)^{|\alpha|} \binom{x}{\alpha} \right) = (q-1)^{|\alpha|} \binom{x_i}{\alpha_i + 1} \prod_{j \ne i} \binom{x_j}{\alpha_j}.$$

This gives a continuous endomorphism of $\mathrm{C}_q(\Gamma, \overline{\mathrm{A}})$ such that $\Delta_i h_i = 1$, $h_i \Delta_i = 1 - \pi_i$, and $h_i \,\mathrm{Fil}^r \subset \mathrm{Fil}^{r-1}$. Here Δ_i is the finite difference operator used earlier; it simply lowers the ith component of α by one. The terms with $\ell > 0$ in $\binom{x+pz}{m} = \sum_{\ell\geq 0}\binom{x}{m-p\ell}\binom{z}{\ell}$ lower Mahler degree by at least $p-1$. It follows that $1-(\gamma_i-1)h_i$ carries Fil^r into Fil^{r+p-2}. Since $p>2$, the series in

$$h_i' = h_i \sum_{k\geq 0}(1-(\gamma_i-1)h_i)^k$$

converges, and $(\gamma_i-1)h_i' = 1$. Moreover, $\gamma_i - 1$ vanishes on the image of π_i. The identity $1 = h_i\Delta_i + \pi_i$ therefore gives $\gamma_i - 1 = (\gamma_i-1)h_i\Delta_i$, and hence $h_i'(\gamma_i-1) = h_i\Delta_i = 1-\pi_i$. This proves the claim.

Applying this successively to the factors of $\mathbf{Z}_p(1)^n$ in the resolution (1) shows that the natural $1+p\mathbf{Z}_p$-equivariant map

$$\mathrm{C}_q(1+p\mathbf{Z}_p, \overline{\mathrm{A}}) \longrightarrow \mathrm{R}\Gamma_{\blacksquare}\big(\mathbf{Z}_p(1)^n, \mathrm{C}_q(\Gamma, \overline{\mathrm{A}})\big)$$

is an equivalence after forgetting the action, hence is an equivalence. Applying the claim with $i=0$ and the Hochschild-Serre equivalence gives

$$\overline{\mathrm{A}} \xrightarrow{\sim} \mathrm{R}\Gamma_{\blacksquare}\big(\Gamma, \mathrm{C}_q(\Gamma, \overline{\mathrm{A}})\big).$$

The finite resolutions above show that solid group cohomology preserves derived p-completeness and commutes with reduction modulo p. Derived Nakayama proves the claim. ☐

Proposition 3.12. *Evaluation on the log q-de Rham prism attached to $\mathrm{A}^+ = \mathrm{R}_0^{\square}[\![q-1]\!]$ induces a fully faithful functor*

$$\rho_{\mathrm{A}}^{\Gamma,*} : \mathrm{D}_{\blacksquare}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}) \longrightarrow \mathrm{D}_{\blacksquare}(\mathrm{A})^{h\Gamma}.$$

When the evaluation on the log q-de Rham prism is p-completely nuclear, the target can be thought of as semilinear Γ-modules by the equivalence of Lemma 3.9.

Proof. We first note that ρ_{A}^* can be constructed as the map induced on $\mathrm{D}_{\blacksquare}(-)$ by the pullback functor along

$$\rho_{\mathrm{A}}^{\Gamma} : [\mathrm{AnSpec}\,\mathrm{A}/\Gamma] \to (\mathrm{Spf}\,\mathrm{R}, \mathcal{M})^{\mathcal{L},\blacksquare}.$$

Indeed, taking the presentation $\mathrm{colim}_{[n]\in\Delta^{\mathrm{op}}}\,\mathrm{AnSpec}\,\mathrm{A}^{(n)}$ of the target given by Proposition A.22 after base change to $(\mathrm{X}, \mathcal{M})^{\mathcal{L},\blacksquare}$ and the natural presentation $\mathrm{colim}_{[n]\in\Delta^{\mathrm{op}}}\,\mathrm{AnSpec}\,\mathrm{C}(\Gamma^n, \mathrm{A})$ of the source one obtains the description of the functor appearing before Lemma 3.8. Here, both presentations are obtained by taking the Čech nerve of the cover by $\mathrm{AnSpec}\,\mathrm{A}$. We note that one should not construct this map by claiming only that the map $\mathrm{AnSpec}\,\mathrm{A} \to (\mathrm{Spf}\,\mathrm{R}, \mathcal{M})^{\mathcal{L},\blacksquare}$ obtained from the fpqc epimorphism of formal stacks obtained in Lemma A.4 is Γ-equivariant, because a priori the action might not be continuous. But once one uses presentations of both stacks and Lemma 3.8 to check the action is continuous, as in the site-theoretic construction it becomes well-defined.

Now we explain why the assertion reduces to properness of $\rho_{\mathrm{A}}^{\Gamma}$ in the sense of [Kip26b, Definition 2.1.3.1]. We use the same conventions as [Kip26b], and work with the six functor formalism on analytic stacks. In our case the analytic stacks all lie over $\mathrm{AnSpec}\,\mathbf{Z}_{p,\blacksquare}$ with its solid analytic ring structure. We have a diagram

$$\begin{tikzcd} {[\mathrm{AnSpec}\,\mathrm{A}^{(1)}/\Gamma]} \arrow[r] \arrow[d, "f"] & {[\mathrm{AnSpec}\,\mathrm{A}/\Gamma]} \arrow[d, "\rho_{\mathrm{A}}^{\Gamma}"] \\ \mathrm{AnSpec}\,\mathrm{A} \arrow[r, "\rho_{\mathrm{A}}"] & (\mathrm{Spf}\,\mathrm{R}, \mathcal{M})^{\mathcal{L},\blacksquare}. \end{tikzcd}$$

This square is cartesian, using that $\mathrm{AnSpec}\,\mathrm{A} \times_{(\mathrm{Spf}\,\mathrm{R},\mathcal{M})^{\mathcal{L},\blacksquare}} \mathrm{AnSpec}\,\mathrm{A} = \mathrm{AnSpec}\,\mathrm{A}^{(1)}$ from the Čech nerve presentation of $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})^{\mathcal{L},\blacksquare}$. In particular, note that the Γ action is nontrivial in $[\mathrm{AnSpec}\,\mathrm{A}^{(1)}/\Gamma]$ as it is acting by right translation. Assuming $\rho_{\mathrm{A}}^{\Gamma}$ is cohomologically proper for $\mathrm{D}_\blacksquare$ we obtain base change for this diagram: in particular for the unit $\mathcal{O} \in \mathrm{D}_\blacksquare([\mathrm{AnSpec}\,\mathrm{A}/\Gamma])$ there is a base change isomorphism

$$\rho_{\mathrm{A}}^{*}\mathrm{R}\rho_{\mathrm{A},*}^{\Gamma}\mathcal{O} \xrightarrow{\sim} \mathrm{R}f_*\mathcal{O} \simeq \mathrm{R}\Gamma_{\mathrm{solid}}(\Gamma, \mathrm{C}_q(\Gamma, \mathrm{A})) \tag{2}$$

by p-complete nuclearity, Lemma 3.9 and Lemma 3.8 to identify $\mathrm{A}^{(1)}$ with $\mathrm{C}_q(\Gamma, \mathrm{A})$. We use the presentation in Lemma 3.6 and [AM24, Lemma 2.18(i)] to show that $\mathrm{A}^{(1)}$ is p-completely nuclear over A (though either map $\mathrm{A} \to \mathrm{A}^{(1)}$ in the Čech nerve).

Now using Lemma 3.11, on the right hand side of (2) the canonical map $\mathrm{A} \to \mathrm{R}f_*\mathcal{O} \simeq \mathrm{R}\Gamma_{\mathrm{solid}}(\Gamma, \mathrm{C}_q(\Gamma, \mathrm{A}))$ is an isomorphism. This shows that the canonical map

$$\mathcal{O}_{(\mathrm{Spf}\,\mathrm{R},\mathcal{M})^{\mathcal{L},\blacksquare}} \longrightarrow \mathrm{R}\rho_{\mathrm{A},*}^{\Gamma}\mathcal{O}$$

is an isomorphism, because applying the conservative evaluation ρ_{A}^{*} identifies it with the canonical isomorphism $\mathrm{A} \to \mathrm{R}\Gamma_{\mathrm{solid}}(\Gamma, \mathrm{C}_q(\Gamma, \mathrm{A})) \simeq \mathrm{A}$. Again using cohomological properness, full faithfulness of the pullback $\rho_{\mathrm{A}}^{\Gamma,*}$ is formal by the projection formula. We can compute

$$\mathrm{R}\rho_{\mathrm{A},*}^{\Gamma}(\rho_{\mathrm{A}}^{\Gamma,*}\mathcal{E}) \simeq \mathrm{R}\rho_{\mathrm{A},!}^{\Gamma}(\mathcal{O} \otimes^{\mathrm{L},\blacksquare} \rho_{\mathrm{A}}^{\Gamma,*}\mathcal{E}) \simeq \mathrm{R}\rho_{\mathrm{A},!}^{\Gamma}\mathcal{O} \otimes^{\mathrm{L},\blacksquare} \mathcal{E} \simeq \mathcal{E}$$

for $\mathcal{E} \in \mathrm{D}_\blacksquare((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})^{\mathcal{L},\blacksquare}) \simeq \mathrm{D}_\blacksquare((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})$, proving a left inverse witnessing full faithfulness.

Thus, all that remains is properness. Proposition A.22 and base change to the Laurent stack $(\mathrm{X}, \mathcal{M})^{\mathcal{L},\blacksquare}$ show that ρ_{A} is a representable proper effective epimorphism. Since properness is local on the target for such covers by [Kip26b, Remark 1.1.3.7], it suffices to check the claim for

$$[\mathrm{AnSpec}\,\mathrm{A}^{(1)}/\Gamma] \to \mathrm{AnSpec}\,\mathrm{A}$$

The property is stable under base change, so this can be checked using A^+ and $\mathrm{A}^{+,(1)}$. The right translation action fixes the image of A^+, giving a factorization of the map as

$$[\mathrm{AnSpec}\,\mathrm{A}^{+,(1)}/\Gamma] \longrightarrow \mathrm{AnSpec}\,\mathrm{A}^+ \times \mathrm{B}\Gamma \longrightarrow \mathrm{AnSpec}\,\mathrm{A}^+.$$

After base change, pulling the first map back along $\mathrm{AnSpec}\,\mathrm{A}^+ \to \mathrm{AnSpec}\,\mathrm{A}^+ \times B\Gamma$ gives $\mathrm{AnSpec}\,\mathrm{A}^{+,(1)} \to \mathrm{AnSpec}\,\mathrm{A}^+$, which is proper by Lemma A.13. It remains to show that

$$\mathrm{B}\Gamma \to \mathrm{AnSpec}\,\mathbf{Z}_{p,\blacksquare}$$

is proper for the six functor formalism on analytic stacks. One may use an argument similar to [HM24, Proposition 5.3.2]. The map $g : \mathrm{AnSpec}\,\mathbf{Z}_{p,\blacksquare} \to \mathrm{B}\Gamma$ is representable proper, while $\mathrm{B}\Gamma \to \mathrm{AnSpec}\,\mathbf{Z}_{p,\blacksquare}$ is 1-truncated and its composite with g is the identity. By [HM24, Corollary 4.7.5(iii)], it remains only to verify that $g_*\mathcal{O} = \mathrm{C}(\Gamma, \mathbf{Z}_p)$ is descendable. The finite resolutions in (1), applied to the normal subgroup $\mathbf{Z}_p(1)^n$ and quotient $1 + p\mathbf{Z}_p \simeq \mathbf{Z}_p$, show that the trivial representation $\mathbf{Z}_p$ is perfect over $\mathbf{Z}_p[\![\Gamma]\!]$. Taking derived solid $\mathbf{Z}_p$-duals and using $\mathbf{Z}_p[\![\Gamma]\!]^\vee \simeq \mathrm{C}(\Gamma, \mathbf{Z}_p)$ gives

$$\mathbf{Z}_p \in \mathrm{Thick}\big(\mathrm{C}(\Gamma, \mathbf{Z}_p)\big) \subset \mathrm{D}_\blacksquare(\mathrm{B}\Gamma),$$

as desired. □

3.2. **Roots in Laurent crystals.** We now turn our attention to proving Theorem 3.5. As we now impose a perfectness condition, condensed mathematics is not relevant in this subsection. In Appendix A we show in Corollary A.19 that perfect Laurent crystals have a canonical t-structure in the normal crossings setting, allowing us to define $\mathsf{Coh}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/\mathrm{I}_\Delta]^\wedge_p)$.

For a framed log affine, put $\mathrm{A} = \mathrm{R}_0^\square(\!(q-1)\!)^\wedge_p$ as before. We use A to mean this throughout this subsection, and use $\overline{\mathrm{A}} := \mathrm{A}/p$. One can understand the category of perfect A-complexes with continuous Γ action as the totalization

$$\lim\left(\mathsf{Perf}(\mathrm{A}) \underset{p_1}{\overset{p_0}{\rightrightarrows}} \mathsf{Perf}(\mathrm{C}(\Gamma, \mathrm{A})) \Rrightarrow \mathsf{Perf}(\mathrm{C}(\Gamma^2, \mathrm{A})) \ \substack{\rightarrow\\\rightarrow\\\rightarrow\\\rightarrow}\ \cdots \right) \tag{3}$$

where A has the topology specified after Lemma 3.6.

- This is compatible with the previous very general description $\lim_{[s]\in\Delta} \mathrm{D}_\blacksquare(\mathrm{C}(\Gamma^s, \mathrm{A}))$ by taking dualizable objects, which will only depend on the underlying ring (i.e. evaluation on $*$ is an equivalence with ordinary perfect complexes). These are automatically p-completely nuclear, so by Lemma 3.9 there is no subtlety about the relation to modules with semilinear Γ-action.

- Upon restricting to Coh the description is also compatible with classical theory describing Γ-modules, where one equips finite modules with their weak topology as in [SV11, Lemmas 8.2 and 8.22] (which discusses the case of a point). In particular, one may check that there is an equivalence with $\mathsf{Coh}(\mathrm{A})^{h\Gamma}$ (the heart of the cobar description described above) and finite A-modules with a continuous Γ-action, with the A-module given its weak topology. We omit the proof here as it is not strictly needed for this article (indeed one may use the cobar description everywhere); the key idea is to use that $\mathrm{C}(\Gamma, -)$ takes exact sequences of modules with their weak topologies to an exact sequence of topological modules (it is sufficient that the topology on A is an inverse limit of Banach rings), allowing one to reduce the compatibility to the case where the module is finite free.

For coherent objects, to write down the data it is enough to keep the terms through $\mathrm{C}(\Gamma^2, \overline{\mathrm{A}})$ in (3). The descent data can naturally be encoded as a comodule over the Hopf algebroid

$(\overline{\mathrm{A}}, \mathrm{C}(\Gamma, \overline{\mathrm{A}}))$. Explicitly, an object of the three-term totalization is a finite $\overline{\mathrm{A}}$-module M together with an isomorphism

$$\alpha : \mathrm{C}(\Gamma, \overline{\mathrm{A}}) \otimes^{\blacksquare}_{p_1,\overline{\mathrm{A}}} \mathrm{M} \xrightarrow{\sim} \mathrm{C}(\Gamma, \overline{\mathrm{A}}) \otimes^{\blacksquare}_{p_0,\overline{\mathrm{A}}} \mathrm{M}$$

satisfying the identity and cocycle conditions. Writing $p_0(a)(g) = a$ and $p_1(a)(g) = g(a)$, the rule $\delta(m) = \alpha(1 \otimes m)$ identifies α with a p_1-semilinear coaction $\delta : \mathrm{M} \to \mathrm{C}(\Gamma, \overline{\mathrm{A}}) \otimes^{\blacksquare}_{p_0,\overline{\mathrm{A}}} \mathrm{M}$. The identity and cocycle conditions become the counit and coassociativity axioms, respectively, giving the equivalence with finite comodules over the Hopf algebroid.

Lemmas A.17 and A.16 give the same dictionary for Laurent crystals. Put $\overline{\mathrm{A}}^{(m)} = \mathrm{A}^{(m)}/p$. In the coaction formulation, coherent Laurent crystals are described by replacing $\mathrm{C}(\Gamma, \overline{\mathrm{A}})$ and $\mathrm{C}(\Gamma^2, \overline{\mathrm{A}})$ with $\overline{\mathrm{A}}^{(1)}$ and $\overline{\mathrm{A}}^{(2)}$. For the following lemma, we note there is a natural identification $\mathrm{C}(\Gamma^s, \mathrm{N}) \simeq \mathrm{C}(\Gamma^s, \overline{\mathrm{A}}) \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{N}$ for a finite $\overline{\mathrm{A}}$-module N equipped with its weak topology.

Lemma 3.13. *Let $\overline{\mathrm{A}} = \mathrm{A}/p$, let $s = 1, 2$, and let $p_0 : \overline{\mathrm{A}} \to \overline{\mathrm{A}}^{(s)}$ be the zeroth Čech projection. For every finite $\overline{\mathrm{A}}$-module* N, *evaluation induces an injection*

$$\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{N} \longrightarrow \mathrm{C}(\Gamma^s, \mathrm{N}).$$

If $\mathrm{M} \subset \mathrm{N}$ *is finite, then inside* $\mathrm{C}(\Gamma^s, \mathrm{N})$ *one has*

$$\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{M} = \big(\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{N}\big) \cap \mathrm{C}(\Gamma^s, \mathrm{M}).$$

Proof. By Definition 3.7 and Lemma 3.8, an element of $\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{N}$ has a completed expansion

$$f = \sum_{\alpha \in \mathbf{N}^{s(n+1)}} m_\alpha \eta_\alpha.$$

One may verify that for $\alpha, \ell \in \mathbf{N}^{s(n+1)}$, one has $\eta_\alpha(\ell) = 0$ unless $\alpha \leq \ell$ coordinatewise, while $\eta_\ell(\ell) \in \overline{\mathrm{A}}^{\times}$. If f evaluates to zero, then

$$0 = f(\ell) = \sum_{\alpha \leq \ell} m_\alpha \eta_\alpha(\ell).$$

The sum is finite. Induction on $|\ell|$ therefore gives $m_\ell = 0$ for every ℓ, proving injectivity.

The map $p_0 : \mathrm{A} \to \mathrm{A}^{(s)}$ is p-completely flat: before localizing p-completely at $[p]_q$ this follows from [Kos22, Proposition 4.13], and the property is preserved by inverting $[p]_q$ and applying p-adic completion. Since A is Noetherian, [Bha20, Lemma 5.15] makes $p_0 : \mathrm{A} \to \mathrm{A}^{(s)}$ flat. Its reduction $p_0 : \overline{\mathrm{A}} \to \overline{\mathrm{A}}^{(s)}$ is then flat. Consider the commutative square

$$\begin{tikzcd} \overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{N} \arrow[r] \arrow[d] & \mathrm{C}(\Gamma^s, \mathrm{N}) \arrow[d] \\ \overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} (\mathrm{N}/\mathrm{M}) \arrow[r] & \mathrm{C}(\Gamma^s, \mathrm{N}/\mathrm{M}). \end{tikzcd}$$

The lower horizontal map is injective by the first part. Flatness identifies the kernel of the left vertical map with $\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{M}$. If an element of the upper left term maps into $\mathrm{C}(\Gamma^s, \mathrm{M})$, its image in the lower right term vanishes. It therefore vanishes in the lower left term and belongs to $\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{M}$ (using injectivity of the lower horizontal map). The reverse inclusion follows from naturality, proving

$$\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{M} = \left(\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{N}\right) \cap \mathrm{C}(\Gamma^s, \mathrm{M}).$$

□

Lemma 3.14. *Let* M *be a finite* $\overline{\mathrm{A}}$*-module with continuous semilinear* Γ*-action. For* $k \gg 0$*, the coaction of* $\varphi^{*k}\mathrm{M}$ *factors through*

$$\mathrm{C}_q(\Gamma, \overline{\mathrm{A}}) \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \varphi^{*k}\mathrm{M} \subset \mathrm{C}(\Gamma, \varphi^{*k}\mathrm{M}).$$

The same holds with Γ^2 *as needed for the cocycle condition.*

Proof. Put $\overline{\mathrm{A}}^+ = \mathrm{A}^+/p$, and choose a finite $\overline{\mathrm{A}}^+$-submodule $\mathrm{M}^+ \subset \mathrm{M}$ such that $\mathrm{M}^+[1/(q-1)] = \mathrm{M}$. Picking a finite set of generators S for M^+, its orbit $\Gamma\mathrm{S}$ must be contained $(q-1)^{-N}\mathrm{M}^+$ by compactness of Γ and continuity of the action. Putting $\mathrm{M}^{+'} = \overline{\mathrm{A}}^+ \cdot \Gamma\mathrm{S} \subset (q-1)^{-N}\mathrm{M}^+$, since $\overline{\mathrm{A}}^+$ is Noetherian this is a finite Γ-stable module so we may assume without loss of generality that M^+ is Γ-stable. We equip M with a complete separated valuation $\nu_{q-1} : m \mapsto \sup\{r \in \mathbf{Z} : m \in (q-1)^r\mathrm{M}^+\}$. The action is isometric since Γ preserves M^+ and sends $q-1$ to a unit multiple of itself; we will need this structure to apply results in [BR25] later in the proof.

We first note that Lemma 3.10 applies with coefficients, giving a practical criterion for when $f \in \mathrm{C}(\Gamma, \overline{\mathrm{A}}) \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{M} \simeq \mathrm{C}(\Gamma, \mathrm{M})$ belongs to $\mathrm{C}_q(\Gamma, \overline{\mathrm{A}}) \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{M}$. Indeed, taking the existing criterion for $\bar{\mathrm{A}}$ and choosing a finite free surjection $\overline{\mathrm{A}}^{+\oplus n} \twoheadrightarrow \mathrm{M}^+$ and inverting $q-1$ proves the criterion for finite coefficients. Equipped with this, our goal is now to show that a high Frobenius pullback automatically satisfies the criterion. For $j \geq 0$, put $\Gamma_j = \{g^{p^j} : g \in \Gamma\}$. Then Γ is uniform and

$$\Gamma_j = (1 + p^{j+1}\mathbf{Z}_p) \ltimes p^j\mathbf{Z}_p(1)^n.$$

Moreover, $gg'^{-1} \in \Gamma_j$ if and only if $x_i(g) - x_i(g') \in p^j\mathbf{Z}_p$ for every i. Thus $x = (x_0, \ldots, x_n) : \Gamma \to \mathbf{Z}_p^{n+1}$ is a coordinate in the sense of [BR25, Proposition 1.1.1].

Continuity on a finite set of generators, together with the formulas for the action of Γ on A, gives $\ell \geq 0$ and $d > 0$ such that

$$(g-1)(q-1)^r \mathrm{M}^+ \subset (q-1)^{r+d}\mathrm{M}^+$$

for $g \in \Gamma_\ell$ and $r \in \mathbf{Z}$. If $g \in \Gamma_{\ell+j}$, write $g = g_0^{p^j}$ with $g_0 \in \Gamma_\ell$. Since $g - 1 = (g_0 - 1)^{p^j}$ in characteristic p, we obtain

$$(g-1)(q-1)^r \mathrm{M}^+ \subset (q-1)^{r+dp^j}\mathrm{M}^+.$$

It follows from [BR25, Lemmas 1.2.8 and 1.2.10] that every orbit map is super-Hölder with $e = 1$ and $\lambda = \log_p(d) - \ell$ in the sense of [BR25, Definition 1.2.1]. Applying [BR25, Theorem 4.3.4] in the coordinate x, its Mahler coefficients satisfy

$$a_\alpha \in (q-1)^{\lceil c \max_i \alpha_i - \mathrm{C}\rceil}\mathrm{M}^+ \subset (q-1)^{\lceil c|\alpha|/(n+1) - \mathrm{C}\rceil}\mathrm{M}^+.$$

Here $c > 0$ and C are independent of α.

Since $\varphi^k(q-1) = (q-1)^{p^k}$ modulo p, the orbit map of $1 \otimes m$, for $m \in \mathrm{M}^+$, has Mahler coefficients in

$$(q-1)^{\lceil p^k c|\alpha|/(n+1) - p^k\mathrm{C}\rceil}(\overline{\mathrm{A}}^+ \otimes_{\varphi^k, \overline{\mathrm{A}}^+} \mathrm{M}^+).$$

Choose k so that $p^k c/(2(n+1)) > 1$. The criterion in Lemma 3.10 now shows that the coaction on $1 \otimes m$ belongs to $\mathrm{C}_q(\Gamma, \overline{\mathrm{A}}) \otimes^{\blacksquare}_{\overline{\mathrm{A}}, p_0} \varphi^{*k}\mathrm{M}$. Since $\mathrm{M}^+[1/(q-1)] = \mathrm{M}$, the elements $1 \otimes m$ with $m \in \mathrm{M}^+$ generate $\varphi^{*k}\mathrm{M}$ over $\overline{\mathrm{A}}$. The coaction is p_1-linear, and $p_1(\overline{\mathrm{A}}) \subset \mathrm{C}_q(\Gamma, \overline{\mathrm{A}})$ by Lemma 3.8, so this proves the assertion over Γ.

The same argument in the $2(n+1)$ product coordinates proves the assertion over Γ^2. □

Proposition 3.15. *Let* M *be a finite* $\overline{\mathrm{A}}$*-module with continuous semilinear* Γ*-action. For* $k \gg 0$*, the module* $\varphi^{*k}\mathrm{M}$ *arises from a coherent Laurent crystal killed by* p*.*

Proof. Choose k large enough for Lemma 3.14 in degrees one and two, and put $\mathrm{N} = \varphi^{*k}\mathrm{M}$. Lemmas 3.8 and 3.13 give injective vertical maps in the commutative diagram

$$\begin{tikzcd}
\mathrm{N} \arrow[r, shift left] \arrow[r, shift right] \arrow[d, equal] & \mathrm{C}(\Gamma, \mathrm{N}) \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] & \mathrm{C}(\Gamma^2, \mathrm{N}) \\
\mathrm{N} \arrow[r, shift left] \arrow[r, shift right] & \overline{\mathrm{A}}^{(1)} \otimes^{\blacksquare}_{\overline{\mathrm{A}}, p_0} \mathrm{N} \arrow[u, hook] \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] & \overline{\mathrm{A}}^{(2)} \otimes^{\blacksquare}_{\overline{\mathrm{A}}, p_0} \mathrm{N} \arrow[u, hook]
\end{tikzcd}$$

The coaction $m \mapsto (g \mapsto g(m))$ factors uniquely through the lower degree one term by Lemma 3.14; use this factorization for the corresponding lower coface. Compatibility of the evaluation maps with the Čech maps makes the diagram commute. The counit and coassociativity identities hold in the upper row, hence in the lower row by injectivity in degrees one and two. Thus the lower row defines a descent datum on N as a coherent Laurent crystal. □

Lemma 3.16. *Let* $\mathrm{M} \subset \mathrm{N}$ *be finite* $\overline{\mathrm{A}}$*-modules with continuous semilinear* Γ*-action. If* N *is the evaluation of a coherent Laurent crystal killed by* p *and* M *is* Γ*-stable, then the crystal coaction restricts to* M.

Proof. One can identify the crystal coaction with

$$\mathrm{N} \longrightarrow \overline{\mathrm{A}}^{(1)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{N}.$$

For $x \in \mathrm{M}$, its image in $\overline{\mathrm{A}}^{(1)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} (\mathrm{N}/\mathrm{M})$ evaluates to zero because M is Γ-stable. Lemma 3.13 makes this image zero, so the coaction restricts to M. The same argument after exchanging the two factors restricts the inverse coaction. For $s = 1, 2$, flatness of $p_0 : \overline{\mathrm{A}} \to \overline{\mathrm{A}}^{(s)}$ makes

$$\overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{M} \longrightarrow \overline{\mathrm{A}}^{(s)} \otimes^{\blacksquare}_{\overline{\mathrm{A}},p_0} \mathrm{N}$$

injective. The restricted coactions are therefore inverse, and the cocycle identity follows from that for N. The counit identity follows after composing with $\mathrm{M} \hookrightarrow \mathrm{N}$. Since $\overline{\mathrm{A}}$ is Noetherian regular and M is finite, the resulting subcrystal is coherent. □

Proof of Theorem 3.5. Let $\beta : \mathrm{M} \hookrightarrow \varphi^*\mathrm{M}$ be the given map. Since $\overline{\mathrm{A}}$ is regular, Frobenius is flat by Kunz's theorem. Iterating β therefore embeds both M and $\varphi^*\mathrm{M}$ as Γ-stable submodules of $\mathrm{N} = \varphi^{*k}\mathrm{M}$ for every $k \geq 1$.

Choose k large enough for Proposition 3.15, and let $\mathcal{F}$ be the resulting coherent crystal with evaluation N. Lemma 3.16 gives coherent subcrystals $\mathcal{E}, \varphi^*\mathcal{E} \subset \mathcal{F}$ with evaluations M and $\varphi^*\mathrm{M}$ (full faithfulness in Proposition 3.12 shows the second crystal is really the Frobenius pullback; the required nuclearity is automatic for coherent crystals). Proposition 3.12 also lifts β uniquely to a morphism

$$\widetilde{\beta} : \mathcal{E} \longrightarrow \varphi^*\mathcal{E}.$$

Its kernel has evaluation 0 on the log q de Rham prism, so t-exactness and conservativity from Lemma A.17 and Corollary A.19 make $\widetilde{\beta}$ injective. □

As promised we now investigate other bases, which is a straightforward modification.

Corollary 3.17. *Both Proposition 3.12 and Theorem 3.5 hold over* $\mathrm{W}(k)[\zeta_{p^\ell}]$ *for* $\ell > 0$*, by changing to* $\Gamma = (1 + p^\ell \mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)^n$ *and keeping the same base* $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)^{\wedge}_p$.

For $\ell = 0$*, again keeping the same base* $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)^{\wedge}_p$ *but using* $\Gamma = \mathbf{Z}_p^{\times} \ltimes \mathbf{Z}_p(1)^n$ *Proposition 3.12 holds and Theorem 3.5 also holds.*

Proof. For $\ell > 0$ the modification is very easy, and can be done by simply setting the prismatic ideal in the q-de Rham prism to be $\varphi^{\ell-1}([p]_q)$. The toric factor acts by $\mathrm{T}_i \mapsto q^{p^\ell b_i}\mathrm{T}_i$, so this action is trivial modulo $\varphi^{\ell-1}([p]_q)$.

For $\ell = 0$ and p odd, the needed argument is more substantial. A tempting argument is to try to use either the $\tilde{p}$-de Rham prism, or try to prove some μ_{p-1}-torsor statement on Laurent stacks – the first fails because there appears to be no natural variant incorporating the toric directions, and the second appears to be false. However what is needed to deduce the full faithfulness claim in Proposition 3.12 from the one over $\mathrm{W}(k)[\zeta_p]$ is a weaker assertion, which is that the pullback along

$$[\mathrm{W}(k)[\zeta_p]^{\mathscr{L},\blacksquare}/\mu_{p-1}] \to \mathrm{W}(k)^{\mathscr{L},\blacksquare}$$

is fully faithful on $\mathrm{D}_\blacksquare(-)$. Here the μ_{p-1} action comes from functoriality and the Galois action on $\mathrm{W}(k)[\zeta_p]/\mathrm{W}(k)$.

One can use the same argument as in Proposition 3.12 to reduce to checking if the pushforward of the structure sheaf on the source is $\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^\wedge_p$. Using Proposition 6.5, we may work site-theoretically, so one may reduce to checking the stronger claim

$$\mathrm{W}(k)[\![\tilde{p}]\!] \simeq \Delta^{h\mu_{p-1}}_{\mathrm{W}(k)[\zeta_p]/\mathrm{W}(k)[\![\tilde{p}]\!]}.$$

Here we use conservativity of evaluation on $\mathrm{W}(k)[\![\tilde{p}]\!]$ to make this reduction. This can be checked directly, using the Hodge-Tate comparison. Indeed, $\mathrm{L}_{\mathrm{W}(k)[\zeta_p]/\mathrm{W}(k)} \simeq \Omega^1_{\mathrm{W}(k)[\zeta_p]/\mathrm{W}(k)}$ compatibly with the Galois action on $\mathrm{W}(k)[\zeta_p]/\mathrm{W}(k)$, but the target has no ordinary μ_{p-1} invariants; higher exterior powers also follow, as in this case they are just shifts of $\Omega^1_{\mathrm{W}(k)[\zeta_p]/\mathrm{W}(k)}$. The μ_{p-1} invariants are exact so the desired result follows from

$$\mathrm{W}(k)[\![\tilde{p}]\!] = \mathrm{W}(k)[\![q-1]\!]^{\mu_{p-1}}.$$

For Theorem 3.5, the only additional check is that iterating Frobenius pullbacks to an object in $\mathsf{Coh}([\mathrm{W}(k)[\zeta_p]^{\mathscr{L},\blacksquare}/\mu_{p-1}])$ lands in the essential image of $\mathsf{Coh}(\mathrm{W}(k)^{\mathscr{L},\blacksquare})$ under the previous fully faithful functor. □

4. Finitely generated unit Laurent crystals

4.1. **Basic constructions.** Following [EK04b] and the more general version in [BL19] (which explains how to remove finiteness hypotheses), we first generalize the notion of finitely generated unit modules.

Definition 4.1. Let $(\mathrm{A}^+, \mathrm{I})$ be a prism with $\mathrm{I} = (d)$. Equip $\mathrm{A} = \mathrm{A}^+[1/d]^\wedge_p$ with the analytic ring structure in Definition A.11, and we use $\mathrm{D}_\blacksquare(\mathrm{A})$ as defined there. Note A has a natural Frobenius φ induced by the Frobenius on A^+ (since $\varphi(d) \equiv d^p \pmod p$).

Write $\mathrm{A}\langle \mathrm{F}\rangle$ for

$$\widehat{\bigoplus}_{i\geq 0} \mathrm{AF}^i,$$

where $\mathrm{F}x = \varphi(x)\mathrm{F}$. We put $\mathrm{D}_\blacksquare(\mathrm{A}\langle \mathrm{F}\rangle)$ for solid modules over this noncommutative analytic ring, with the analytic ring structure induced from A.

For $\mathrm{M} \in \mathrm{D}_\blacksquare(\mathrm{A}\langle \mathrm{F}\rangle)$, we say M is *unit* if

$$\varphi^*_{\mathrm{A}}\mathrm{M} := \mathrm{A} \otimes^{\mathrm{L},\blacksquare}_{\mathrm{A},\varphi_{\mathrm{A}}} \mathrm{M} \longrightarrow \mathrm{M}$$

given by $a \otimes m \mapsto a\mathrm{F}_{\mathrm{M}}(m)$ is an equivalence. A unit object is *finitely generated unit* if it is perfect over $\mathrm{A}\langle \mathrm{F}\rangle$; these objects form the full subcategory $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A})$. Here, we mean perfect in the sense that it lies in the thick subcategory generated by $\mathrm{A}\langle \mathrm{F}\rangle$ as we work over a noncommutative base.

In fact one can show the category on a fixed prism is insensitive to the solid structure, but we will still remember it so that our crystals live in an ambient category which is a stack.

Lemma 4.2. *Let* $(\mathrm{A}^+, (d))$ *be an orientable bounded prism, and let* $\mathrm{A} = \mathrm{A}^+[1/d]^\wedge_p$ *with the analytic ring structure from Definition A.11. Then*

$$\mathrm{Thick}_{\mathrm{A}\langle \mathrm{F}\rangle} \mathrm{D}(\mathrm{A}\langle \mathrm{F}\rangle) \simeq \mathrm{Thick}_{\mathrm{A}\langle \mathrm{F}\rangle} \mathrm{D}_{\blacksquare}(\mathrm{A}\langle \mathrm{F}\rangle)$$

via $\mathrm{M} \mapsto \mathrm{M}(*)$. *The solid modules in the target thick subcategory are all* p*-completely nuclear.*

Proof. The comparison is exact and carries $\mathrm{A}\langle \mathrm{F}\rangle$ to the corresponding solid module. We will show the natural functor induces an equivalence on their mapping spectra. It is then fully faithful on the thick closure of the generator, and its essential image is a thick subcategory containing the target generator.

We can compute

$$\mathrm{Map}_{\mathrm{D}_{\blacksquare}(\mathrm{A}\langle \mathrm{F}\rangle)}\big(\mathrm{A}\langle \mathrm{F}\rangle, \mathrm{A}\langle \mathrm{F}\rangle\big) \simeq \mathrm{Map}_{\mathrm{D}_{\blacksquare}(\mathrm{A})}\big(\mathrm{A}^+, \mathrm{A}\langle \mathrm{F}\rangle\big) \simeq \mathrm{A}\langle \mathrm{F}\rangle.$$

The first equivalence uses the free-forgetful adjunction. The second equivalence can be checked modulo p by derived Nakayama. It then follows from compactness of A^+/p: modulo p localization at d is a filtered colimit and $(\mathrm{A}\langle \mathrm{F}\rangle)/p = \bigoplus_{i\geq 0} (\mathrm{A}/p)\mathrm{F}^i$.

The unit A^+ is nuclear. Since A is the filtered colimit of multiplication by d in derived p-complete solid modules, it is p-completely nuclear. The p-completely nuclear objects are closed under colimits, so the underlying module of $\mathrm{A}\langle \mathrm{F}\rangle = \widehat{\bigoplus}_{i\geq 0} \mathrm{A}\mathrm{F}^i$ is p-completely nuclear. The forgetful functor from Frobenius modules preserves finite colimits and retracts, so the same holds throughout the thick subcategory generated by the free $\mathrm{A}\langle F\rangle$-module. ☐

Definition 4.3. Let $(\mathrm{X}, \mathcal{M})$ be a bounded fs log p-adic formal scheme. A crystal

$$\mathcal{E} \in \mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^\wedge_p)^{\varphi=1}$$

is finitely generated unit if it is t-bounded and, for every object $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf}\,\mathrm{A}})^a$ of $(\mathrm{X}, \mathcal{M})_{\Delta}$, its evaluation is perfect in

$$\mathrm{Mod}_{\mathrm{A}[1/\mathrm{I}]^\wedge_p\langle \mathrm{F}\rangle}\big(\mathrm{D}_{\blacksquare}(\mathrm{A}[1/\mathrm{I}]^\wedge_p)\big).$$

We write $\mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^\wedge_p)$ for the full subcategory of finitely generated unit crystals, and $\mathsf{Mod}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^\wedge_p)$ for its intersection with the heart of $\mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^\wedge_p)$.

Remark 4.4. Let B be a p-adically complete associative ring in which p is central, and let M be a derived p-complete left B-module. Then

$$\mathrm{M} \in \mathsf{Perf}(\mathrm{B}) \Longleftrightarrow (\mathrm{B}/p) \otimes^{\mathrm{L}}_{\mathrm{B}} \mathrm{M} \in \mathsf{Perf}(\mathrm{B}/p).$$

Indeed, apply [Sta26, Lemma 15.99.4, Tag 09AW] to the tower $\{\mathrm{M}/p^n\}_{n\geq 1}$; its proof applies unchanged when (p) is central. The same holds in the solid setting; the unit Frobenius condition can also be checked modulo p by derived Nakayama.

By Remark 4.12, it suffices to test Definition 4.3 on the log q-de Rham prisms associated to framed log affines.

Our first main goals will be to give an explicit local description of this category, as well as deduce strict-étale descent. We note that while $\mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)^{\varphi=1}$ is a strict-étale stack (by Corollary A.18), it is not immediate that the finitely generated unit subcategory is also a strict-étale stack. While ordinary dualizable objects can be detected termwise in a limit, because we work over $\mathrm{A}\langle\mathrm{F}\rangle$ we use the thick subcategory generated by the unit (which satisfies no such property). Instead, descent of the fgu condition will use its characterization by coherent roots, which we now study.

Definition 4.5. Let A be as in 4.1, and assume that A is Noetherian regular. If M is a unit Frobenius module over A, a *root* of M is a coherent A-submodule $\mathrm{M}_0 \subset \mathrm{M}$ together with an injective map $\beta : \mathrm{M}_0 \to \varphi_{\mathrm{A}}^*\mathrm{M}_0$ whose iterates identify

$$\mathrm{M} \simeq \mathrm{colim}_n\, \varphi_{\mathrm{A}}^{*n}\mathrm{M}_0.$$

It is *minimal* if it has no proper subroot. A distinguished minimal root is a minimal root contained in every other root of M as a submodule of the same ambient unit module.

One can make the analogous definitions for any unit object in the heart of $\mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)$. When referring to a root of a module we mean the former, and when referring to a root of a crystal we mean the latter.

The essential idea of the proof of strict-étale descent is now the following: Proposition 4.7 identifies the existence of distinguished minimal roots (as Laurent crystals) with the finitely generated unit condition. Distinguished minimal roots are unique as subcrystals and therefore glue on intersections, once we know strict-étale descent for the ambient category $\mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)^{\varphi=1}$. This reduces the descent problem to descent of coherent Laurent crystals, to which the main results of [Mat22] apply. There are of course many technical details to dispense of, but the core argument is very simple.

Let $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ be a framed log affine in the sense of Definition 3.2 and put $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)^{\wedge}_p$, the Laurent version of its logarithmic q-de Rham prism. The following theorem identifies finitely generated unit crystals with (φ, Γ)-modules and constructs distinguished minimal roots. A key aspect is that we may ignore solid modules locally and work with continuous semilinear Γ-actions on topological modules. By $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A})^{h\Gamma}$, we mean

$$\lim_{[s]\in\Delta} \mathrm{Thick}_{\mathrm{C}(\Gamma^s, \mathrm{A}\langle\mathrm{F}\rangle)}\mathrm{D}(\mathrm{C}(\Gamma^s, \mathrm{A}\langle\mathrm{F}\rangle))$$

with no solid structures involved. Here $\mathrm{A}\langle\mathrm{F}\rangle$ has its natural topology (induced from A and the completed direct sum). Note $\mathrm{C}(\Gamma^s, \mathrm{A})\widehat{\otimes}_{\mathrm{A}}\mathrm{A}\langle\mathrm{F}\rangle \simeq \mathrm{C}(\Gamma^s, \mathrm{A}\langle\mathrm{F}\rangle)$ as topological rings.

The condition of lying in these thick subcategories is stable under the pullbacks $\mathrm{C}(\Gamma^s, \mathrm{A})\otimes^{\mathrm{L}}_{\mathrm{A}} -$ by nuclearity of $\mathrm{A}\langle\mathrm{F}\rangle$ (passing to the associated solid condensed module recovers the solid tensor product). For a unit object in the heart, membership in the thick subcategory at $s = 0$ is equivalent to finite generation over $\mathrm{A}\langle\mathrm{F}\rangle$; the unit hypothesis is essential here. Indeed, the nontrivial implication may be checked modulo p by Remark 4.4: [Bli08, Corollary 2.26] supplies a coherent root, whose two-term induced resolution from [EK04b, Proposition 5.3.3] makes the module perfect. Thus on the unit heart one may take a more classical approach, endowing such a module with the weak topology induced by a surjection from $\mathrm{A}\langle\mathrm{F}\rangle^{\oplus i}$ (see the discussion at the beginning of §3.2; the argument is the same); this is essentially what we are doing here, just in a more derived way.

Theorem 4.6. *Let* $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ *be a framed log affine in the sense of Definition 3.2 and put* $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)^{\wedge}_p$. *Evaluation on the log q-de Rham prism* $(\mathrm{R}_0^{\square}[\![q-1]\!], [p]_q, \mathcal{M})^a$ *induces an equivalence*

$$\rho_{\mathrm{A}}^* : \mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^{\wedge}_p) \xrightarrow{\sim} \mathrm{D}_{\mathrm{fgu}}(\mathrm{A})^{h\Gamma}.$$

This is t-exact for the t-structure on $\mathrm{D}_{\blacksquare}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^{\wedge}_p)$.

Proof. Lemma 4.2 shows that the evaluation of every finitely generated unit crystal is p-completely nuclear. Proposition 3.12 therefore gives derived full faithfulness, where the target is solid finitely generated unit modules. Lemma 4.2 identifies the target with ordinary finitely generated unit complexes with a continuous Γ-action, as defined before the theorem as $\lim_{[s]\in\Delta} \mathrm{Thick}_{\mathrm{C}(\Gamma^s,\mathrm{A}\langle\mathrm{F}\rangle)}\mathrm{D}(\mathrm{C}(\Gamma^s, \mathrm{A}\langle\mathrm{F}\rangle))$. Strictly speaking it applies directly to only the first term $s = 0$, but an identical argument covers all terms.

We first prove essential surjectivity onto the heart of the target modulo p. Put $\overline{\mathrm{A}} = \mathrm{A}/p$, and let M be a finitely generated unit $\overline{\mathrm{A}}$-module with continuous semilinear Γ-action. Since $\overline{\mathrm{A}}$ is regular and F-finite, [Bli08, Corollary 2.26] gives a distinguished minimal root

$$\beta : \mathrm{M}_{\mathrm{min}} \hookrightarrow \varphi^*\mathrm{M}_{\mathrm{min}}.$$

Every Γ-translate is again a root. Moreover distinguished containment (i.e. its defining property) applied to a translate and its inverse gives $\gamma\mathrm{M}_{\mathrm{min}} = \mathrm{M}_{\mathrm{min}}$, so β is Γ-equivariant. Theorem 3.5 lifts it to a coherent Laurent crystal $\mathcal{E}_0$. Its unitalization $\mathcal{E}$ satisfies $\rho_{\mathrm{A}}^*\mathcal{E} \simeq \mathrm{M}$.

The existence of a root $\mathcal{E}_0$ so that $\mathcal{E} \simeq \mathrm{colim}_k\, \varphi^{*k}\mathcal{E}_0$ shows that $\mathcal{E}$ is finitely generated unit, as this presentation applied to evaluation on any log prism induces a two-term resolution showing it lies in the thick subcategory of Definition 4.1. Since evaluation is compatible with shifts and fibers, [Hau24, Lemma 5.4.3] upgrades essential surjectivity on the heart modulo p to essential surjectivity on the derived categories. Thus evaluation is an equivalence modulo p. Lemma 2.3 now gives the claim.

We note that in this proof of essential surjectivity implicitly the preimages constructed are all in the heart of $\mathrm{D}_{\blacksquare}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^{\wedge}_p)$. Indeed mod p the preimage is a unitalization of a coherent root, and coherent crystals are in the heart of the solid t-structure by Corollary A.19. The analogous procedure, instead run for exact sequences in the heart of $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A})^{h\Gamma}$ to iteratively write heart objects as extensions of p-torsion objects (and taking a limit), constructs preimages in the heart by full faithfulness.

For t-exactness, we note that if $\mathcal{E} \in \mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf\,R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$ is in

$$\mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf\,R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}) \cap \mathrm{D}_{\blacksquare}((\mathrm{Spf\,R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})^{\leq 0}$$

it is automatic that it lands in the ≤ 0 part on the target. For the ≥ 0 part, suppose it fails and pick the smallest r so $\mathrm{N} = \mathrm{H}^r(\rho_{\mathrm{A}}^{*}\mathcal{E}) \neq 0$. This is a finitely generated unit Γ-module, so by the previous paragraph's observation has a preimage $\mathcal{E}_{\mathrm{N}}$ which is in $\mathrm{D}_{\blacksquare}((\mathrm{Spf\,R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})^{\heartsuit}$. By full faithfulness we get a nonzero map $\mathcal{E}_{\mathrm{N}}[-r] \to \mathcal{E}$, but by the orthogonality defining the canonical t-structure on solid Laurent crystals in Corollary A.19 this must be the zero map. ☐

By applying Theorem 3.5 we can prove a root criterion for the finitely generated unit condition.

Proposition 4.7. *Let* $(\mathrm{Spf\,R}, \mathcal{M})$ *be a framed log affine. A crystal*

$$\mathcal{E} \in \mathrm{D}_{\blacksquare}((\mathrm{Spf\,R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})^{\varphi=1}$$

is finitely generated unit if and only if it is t-bounded and each crystal $\mathrm{H}^i(\mathcal{E}_{p=0})$ *admits a distinguished minimal root in* $\mathsf{Coh}((\mathrm{Spf\,R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})_{p=0}$.

Proof. Put $\mathrm{A} = \mathrm{R}_0^{\square}((q-1))^{\wedge}_{p}$ and $\overline{\mathrm{A}} = \mathrm{A}/p$. Suppose first that the condition on roots holds. By Remark 4.4, it is enough to work modulo p. A coherent root is a perfect Laurent crystal, so its evaluation on every log prism is perfect over the coefficient ring. On a log prism $(\mathrm{B}, \mathrm{J}, \mathcal{N})^{a}$, the two-term induced resolution of [EK04b, Proposition 5.3.3] makes its unitalization perfect over $(\mathrm{B}[1/\mathrm{J}]^{\wedge}_{p}/p)[\mathrm{F}]$. Thus every cohomology module of the evaluation of $\mathcal{E}_{p=0}$ on the log q-de Rham prism is perfect over $\overline{\mathrm{A}}[\mathrm{F}]$. The unit condition is detected on cohomology by [EK04b, 5.4.1]. Since $\mathcal{E}$ is t-bounded and perfect objects are closed under finite extensions, its evaluation is finitely generated unit. Theorem 4.6 and Proposition 3.12 show that $\mathcal{E}_{p=0}$ is finitely generated unit.

Conversely, Proposition 3.12 makes evaluation on the mod-p log q-de Rham prism t-exact and conservative. After forgetting the solid structure, Lemma 4.2 makes the evaluated complex perfect over $\overline{\mathrm{A}}[\mathrm{F}]$. Proposition 11.3.9 of [BL19] shows that the evaluation M of $\mathrm{H}^i(\mathcal{E}_{p=0})$ is an ordinary finitely generated unit module. As in the proof of Theorem 4.6, its distinguished minimal root $\mathrm{M}_{\min}$ is Γ-stable and its root map is Γ-equivariant. Theorem 3.5 lifts this map to a coherent Laurent crystal $\mathcal{E}_{\min}$. Proposition 3.12 identifies its unitalization with $\mathrm{H}^i(\mathcal{E}_{p=0})$, since both evaluate to M. Transporting the canonical inclusion into the unitalization gives

$$\mathcal{E}_{\min} \hookrightarrow \mathrm{H}^i(\mathcal{E}_{p=0}).$$

It is injective by the t-exactness and conservativity of Proposition 3.12, and its iterates unitalize to the ambient crystal. Thus it is a root.

Let $\mathcal{E}_0'$ be any other coherent root, with evaluation M_0'. Distinguished containment gives a Γ-equivariant inclusion $\mathrm{M}_{\min} \hookrightarrow \mathrm{M}_0'$. Proposition 3.12 lifts it uniquely to $\mathcal{E}_{\min} \to \mathcal{E}_0'$ and makes the lift injective by t-exact conservativity. The two composites to $\varphi^*\mathcal{E}_0'$ evaluate to the same map, so the same proposition makes the inclusion compatible with the root maps. Hence $\mathcal{E}_{\min}$ is contained in every root. Any subroot evaluates to a subroot of $\mathrm{M}_{\min}$, so minimality and conservativity in Proposition 3.12 make it equal to $\mathcal{E}_{\min}$. Thus $\mathcal{E}_{\min}$ is the distinguished minimal root as a crystal. □

4.2. **Strict-étale descent.** Proposition 4.7 reduces strict-étale descent to descent of coherent Laurent crystals on $(\mathrm{X}, \mathcal{M})_{\Delta}$. The following lemmas give the required compatibility of minimal roots with completion and étale base change.

Lemma 4.8. *Let* A *be a regular local* F*-finite* $\mathbf{F}_p$*-algebra, let* $\widehat{\mathrm{A}}$ *be its completion, and let* $\mathrm{N}_0 \subset \mathrm{N}$ *be a root of a finitely generated unit* $\mathrm{A}[\mathrm{F}]$*-module. Then* N_0 *is minimal if and only if* $\widehat{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{N}_0 \subset \widehat{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{N}$ *is minimal.*

Proof. The forward implication is [Bli04, Corollary 2.11]. Conversely, assume the completed root is minimal and let $\mathrm{N}_1 \subset \mathrm{N}_0$ be a subroot. Its completion is a subroot, hence $\widehat{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{N}_1 = \widehat{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{N}_0$.

Since $\mathrm{A} \to \widehat{\mathrm{A}}$ is faithfully flat, it follows that $\mathrm{N}_1 = \mathrm{N}_0$. Thus N_0 has no proper subroot, hence is minimal. □

Lemma 4.9. *Let* R *be a* $(q-1)$*-adically complete Noetherian* $k[\![q-1]\!]$*-algebra, topologically of finite type and formally smooth over* $k[\![q-1]\!]$*, and let* S *be another such algebra. Suppose* $\mathrm{R}(\!(q-1)\!) \to \mathrm{S}(\!(q-1)\!)$ *induces an étale morphism of* $k(\!(q-1)\!)$*-affinoid spaces. Then scalar extension sends minimal roots of finitely generated unit modules to minimal roots.*

Proof. Put $\mathrm{A} = \mathrm{R}(\!(q-1)\!)$ and $\mathrm{B} = \mathrm{S}(\!(q-1)\!)$, and let $\mathrm{N}_0 \subset \mathrm{N}$ be a minimal root over A, with root map $\beta : \mathrm{N}_0 \to \varphi_{\mathrm{A}}^*\mathrm{N}_0$. Both rings are regular F-finite, so the hypotheses in [Bli08] for the results we will use are satisfied. Flatness of $\mathrm{A} \to \mathrm{B}$ and naturality of Frobenius pullback show that $\mathrm{B} \otimes_{\mathrm{A}} \mathrm{N}_0$ is a root of $\mathrm{B} \otimes_{\mathrm{A}} \mathrm{N}$, so we only concern ourselves with minimality. More generally, flat pullbacks send roots to roots.

Following [Bli08, §2.4], let $\sigma_\beta^{-1}(\mathrm{N}_0) \subset \mathrm{N}_0$ be the smallest submodule N_0' such that $\beta(\mathrm{N}_0) \subset \varphi_{\mathrm{A}}^*\mathrm{N}_0'$. By [Bli08, Lemma 2.17 and Proposition 2.18] this construction commutes with localization, and a root is minimal precisely when $\sigma_\beta^{-1}(\mathrm{N}_0) = \mathrm{N}_0$. It follows that a root over A is minimal if and only if its localization at every maximal ideal is minimal, and similarly over B. Together with Lemma 4.8, this shows that minimality may be checked on the *completed* local rings at the classical points of the rigid spaces.

Fix a maximal ideal $\mathfrak{m} \subset \mathrm{B}$, let $x \in \mathrm{Spa}(\mathrm{B})$ be the corresponding rigid point, and put $\mathfrak{n} = \mathfrak{m} \cap \mathrm{A}$. By [KL15, Definition 8.2.16] and the affinoid basis following [KL15, Definition 8.2.19], the morphism factors near x into affinoid open immersions and a finite étale morphism.

Since affinoid open immersions induce isomorphisms on completed local rings by [BGR84, 7.3.2/3], the map

$$\widehat{\mathrm{A}_{\mathfrak{n}}} \longrightarrow \widehat{\mathrm{B}_{\mathfrak{m}}}$$

is finite étale. Both rings are again regular and F-finite. It therefore remains to show that finite étale scalar extension preserves minimal roots.

Let $\mathrm{A} \to \mathrm{B}$ now be a finite étale map between regular F-finite rings. Put $\mathrm{D}_{\mathrm{A}}^{(1)} = \mathrm{End}_{\mathrm{A}}(\varphi_{\mathrm{A},*}\mathrm{A})$, and define $\mathrm{D}_{\mathrm{B}}^{(1)}$ similarly. The relative Frobenius is an isomorphism by [Sta26, Lemma 41.14.3, Tag 0EBS], and F-finiteness together with Kunz's theorem gives

$$\mathrm{B} \otimes_{\mathrm{A}} \mathrm{D}_{\mathrm{A}}^{(1)} \simeq \mathrm{D}_{\mathrm{B}}^{(1)}.$$

Blickle's description in [Bli08, §2.4] then gives, for the base change β_{B} of a root map β,

$$\sigma_{\beta_{\mathrm{B}}}^{-1}(\mathrm{B} \otimes_{\mathrm{A}} \mathrm{N}_0) = \mathrm{B} \otimes_{\mathrm{A}} \sigma_{\beta}^{-1}(\mathrm{N}_0),$$

where faithful flatness of φ_{B} descends the equality after Frobenius pullback. Thus finite étale scalar extension preserves minimal roots. Applying this to the completed local rings above and using the completed-local criterion proves the claim. □

Corollary 4.10. *Let $f : (\mathrm{Spf}\,\mathrm{S}, \mathcal{M}_{\mathrm{Spf}\,\mathrm{S}}) \to (\mathrm{Spf}\,\mathrm{R}, \mathcal{M}_{\mathrm{Spf}\,\mathrm{R}})$ be a strict-étale map of framed log affines compatible with the framings. If $\mathcal{E}$ is a finitely generated unit crystal killed by p with distinguished minimal root $\mathcal{E}_{\min}$, then $f^*\mathcal{E}$ is finitely generated unit and $f^*\mathcal{E}_{\min}$ is its distinguished minimal root.*

Proof. The pullback is t-exact because f is strict-étale. Applying Lemma 4.9 to the evaluations on the logarithmic q-de Rham prisms for the compatible framings shows that $f^*\mathcal{E}_{\min}$ is minimal. By [Bli08, Corollary 2.26], the minimal root is unique and contained in every root, so it is distinguished. The claim for crystals now follows from Theorem 4.6 and the argument of Proposition 4.7. □

We can now deduce strict-étale descent.

Theorem 4.11. *Let $(\mathrm{X}, \mathcal{M})$ satisfy Assumption 3.1 over $\mathrm{W}(k)[\zeta_p]$. Then the assignment*

$$(\mathrm{X}, \mathcal{M}) \mapsto \mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})$$

is a stack for the strict-étale topology on such $(\mathrm{X}, \mathcal{M})$.

Proof. By Corollary A.18, $\mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})^{\varphi=1}$ satisfies strict-étale descent. The pullback is t-exact, and the cover may be taken finite, so boundedness is local. Let $\{\rho_i : \mathrm{U}_i \to (\mathrm{X}, \mathcal{M})\}_{i \in \mathrm{I}}$ be a strict-étale cover such that every $\rho_i^*\mathcal{E}$ is finitely generated unit.

Using Remark 4.4 and Proposition 4.7 we may reduce the claim to showing

$$\left\{\rho_i^*\mathcal{E} \in \mathsf{Mod}_{\mathrm{fgu}}((\mathrm{U}_i)_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})_{p=0}\right\} \implies \left\{\mathcal{E} \in \mathsf{Mod}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})_{p=0}\right\}$$

for $\mathcal{E} \in \mathsf{QCoh}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})^{\varphi=1}$ killed by p, or that the condition can be detected locally on a strict-étale cover. Note that the converse is obvious, because any test log prism for the U_i is also one for $(\mathrm{X}, \mathcal{M})$, so the finitely generated unit condition on the right side already includes the conditions imposed on the left hand side.

By Lemma 3.4, after a finite strict-étale refinement we may assume without loss of generality that every $\mathrm{U}_i = \operatorname{Spf} \mathrm{R}_i$ is a framed log affine. Put $\mathrm{U}_{ij} = \mathrm{U}_i \times_{(\mathrm{X},\mathcal{M})} \mathrm{U}_j$ with projections p_i, p_j; it admits the structure of a framing through either U_i or U_j.

Proposition 4.7 gives distinguished minimal roots $\mathcal{R}_i \subset \rho_i^* \mathcal{E}$. Corollary 4.10, applied with each framing, makes $p_i^* \mathcal{R}_i$ and $p_j^* \mathcal{R}_j$ distinguished minimal roots on any open affine in $\mathcal{E}|_{\mathrm{U}_{ij}}$. Since U_{ij} is quasicompact by virtue of X being quasiseparated, we may cover U_{ij} by finitely many such framed log affines.

Their intrinsic uniqueness in Proposition 4.7 gives

$$p_i^* \mathcal{R}_i = p_j^* \mathcal{R}_j \subset \mathcal{E}|_{\mathrm{U}_{ij}}.$$

Indeed by the remark after Corollary A.19 one has descent for coherent Laurent crystals $\mathsf{Coh}^{\varphi}((\mathrm{X}, \mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^{\wedge}_p)$ with a lax Frobenius. Thus we can check agreement locally on the chosen finite open cover of U_{ij} by framed log affines, which also shows agreement of the root maps. These equalities also satisfy the cocycle condition on triple overlaps.

Strict-étale descent for $\mathsf{Coh}^{\varphi}((\mathrm{X}, \mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^{\wedge}_p)_{p=0}$ and $\mathrm{D}^{\varphi}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^{\wedge}_p)^{\heartsuit}_{p=0}$ produces a global coherent root. Restricting any subroot or any other root to the cover shows that the descended root is again distinguished minimal. Injectivity and unitalization are local, so the unitalization of the descended coherent root is $\mathcal{E}$. On a log prism $(\mathrm{A}, \mathrm{I}, \mathrm{M})^a$, its two-term induced resolution is perfect over $(\mathrm{A}[1/\mathrm{I}]^{\wedge}_p/p)[\mathrm{F}]$. Hence $\mathcal{E}$ is finitely generated unit. □

Remark 4.12. Combined with Theorem 4.6, this shows that the finitely generated unit condition can equivalently be tested upon evaluation by a covering family of log q-de Rham prisms as claimed in the introduction.

Let us record an important fact we showed in the proof of the theorem.

Corollary 4.13. *Let $(\mathrm{X}, \mathcal{M})$ satisfy Assumption 3.1 over $\mathrm{W}(k)[\zeta_p]$. A crystal*

$$\mathcal{E} \in \mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^{\wedge}_p)^{\varphi=1}$$

is finitely generated unit if and only if it is t-bounded and, for every ℓ, the unit Laurent crystal $\mathrm{H}^{\ell}(\mathcal{E} \otimes^{\mathrm{L},\blacksquare} \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]/p)$ admits a unique global minimal coherent root.

Proof. The forward implication is the construction in the proof of Theorem 4.11, applied to the cohomology sheaves of $\mathcal{E} \otimes^{\mathrm{L},\blacksquare} \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]/p$ using Proposition 4.7 locally. The converse follows from the first paragraph of the proof of Proposition 4.7 and Theorem 4.11. □

Remark 4.14. This is the Laurent analogue of the root criterion for finitely generated unit modules on a k-scheme X. By [EK04b, Theorem 6.1.3], a finitely generated unit $\mathcal{O}_{\mathrm{F},\mathrm{X}}$-module admits a coherent root on every quasicompact open; as X is quasicompact, the root is global.

Such a root need not carry a crystal structure like in the previous corollary, but it turns out there always exists a root carrying such a structure. Cartier descent gives a fully faithful functor

$$\mathsf{QCoh}(\mathrm{X}^{(1)}) \to \mathsf{QCoh}((\mathrm{X}/k)_{\mathrm{cris}})$$

with essential image the crystals with vanishing p-curvature. If $\mathrm{Fr} : \mathrm{X} \to \mathrm{X}^{(1)}$ is relative Frobenius, it sends $\mathscr{F}$ to $(\mathrm{Fr}^*\mathscr{F}, \nabla_{\mathrm{can}})$, where $(\mathrm{Fr}^*\mathscr{F})^{\nabla_{\mathrm{can}}=0} \simeq \mathscr{F}$.

Write the absolute Frobenius as $\mathrm{F}_\mathrm{X} = \pi \circ \mathrm{Fr}$, with $\pi : \mathrm{X}^{(1)} \to \mathrm{X}$. Given a coherent root $\beta : \mathrm{M}_0 \to \mathrm{F}_\mathrm{X}^*\mathrm{M}_0$, replace it by

$$\mathrm{F}_\mathrm{X}^*\beta : \mathrm{F}_\mathrm{X}^*\mathrm{M}_0 \longrightarrow \mathrm{F}_\mathrm{X}^{*2}\mathrm{M}_0.$$

Since F_X is finite flat, this map is injective and has the same unitalization. Moreover, $\mathrm{F}_\mathrm{X}^*\mathrm{M}_0 = \mathrm{Fr}^*\pi^*\mathrm{M}_0$ and $\mathrm{F}_\mathrm{X}^*\beta = \mathrm{Fr}^*(\pi^*\beta)$, so the map is horizontal for the canonical Cartier connections. It is therefore a coherent crystalline root. Uniqueness in Cartier descent identifies its unitalization with the crystal structure on M (similarly) induced by being an $\mathcal{O}_\mathrm{X}$-module with unit Frobenius.

Corollary 4.15. *Let $f : (\mathrm{Y}, \mathcal{N}) \to (\mathrm{X}, \mathcal{M})$ be a morphism between pairs satisfying Assumption 3.1 over $\mathrm{W}(k)[\zeta_p]$. Then $\mathrm{L}f^*$ preserves finitely generated unit crystals.*

Proof. Let $\mathcal{E}$ be a finitely generated unit crystal on $(\mathrm{X}, \mathcal{M})_{\Delta}$, and let $(\mathrm{B}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf\,B}})^a$ be a log prism over $(\mathrm{Y}, \mathcal{N})$. Composition with f makes it a log prism over $(\mathrm{X}, \mathcal{M})$, and pullback of crystals gives a Frobenius-equivariant equivalence

$$(\mathrm{L}f^*\mathcal{E})(\mathrm{B}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf\,B}}) \simeq \mathcal{E}(\mathrm{B}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf\,B}}).$$

Thus every evaluation of $\mathrm{L}f^*\mathcal{E}$ is perfect over $\mathrm{B}[1/\mathrm{I}]_p^\wedge\langle \mathrm{F}\rangle$ by Definition 4.3. ☐

Theorem 4.11 and Lemma 3.4 permit strict-étale localization to framed log affines, where Theorem 4.6 describes finitely generated unit crystals explicitly.

4.3. The Kummer-étale realization and constructibility.

We construct a functor

$$\mathrm{T}_{\text{ét}} : \mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge)^{\mathrm{op}} \to \mathrm{D}((\mathrm{X}, \mathcal{M})_{\eta,\text{qpkét}}, \mathbf{Z}_p).$$

Its essential image consists of p-Kummer complexes constructible for the logarithmic stratification induced by D. We first note that we can use several equivalent formalisms to access the target category under our assumptions.

Lemma 4.16. *Let $(\mathrm{X}, \mathcal{M})$ be a locally Noetherian fs log p-adic formal scheme. Then the natural comparison functors induce equivalences of topoi*

$$(\mathrm{X}, \mathcal{M})_{\eta,\text{prokét}} \simeq (\mathrm{X}, \mathcal{M})_{\eta,\text{qpkét}} \simeq (\mathrm{X}, \mathcal{M})^\diamond_{\eta,\text{qpkét}}.$$

Proof. Put $(\mathrm{Y}, \mathcal{M}_\mathrm{Y}) = (\mathrm{X}, \mathcal{M})_\eta$. By [DLLZ23, Propositions 5.3.11 and 5.3.12], the log affinoid perfectoid objects form a basis B of $\mathrm{Y}_{\text{prokét}}$ stable under fiber products. For $\mathrm{U} \in \mathrm{B}$, let $\underline{\mathrm{U}}$ be its associated affinoid perfectoid space. Its characteristic monoid is divisible and saturated by [DLLZ23, Definition 5.3.1 and Remark 5.3.3].

Lemma 5.3.8 of [DLLZ23], together with [Sch26a, Lemma 15.6 and the proof of Proposition 14.8], gives functorial equivalences

$$(\mathrm{Y}_{\text{prokét}})_{/\mathrm{U}} \simeq \underline{\mathrm{U}}_{\text{proét}} \simeq \underline{\mathrm{U}}^{\diamond}_{\text{qproét}}.$$

The last term is the corresponding localization of the diamond quasi-pro-Kummer-étale topos by [KY25, Corollary 7.23]; the same basis computes the adic quasi-pro-Kummer-étale localization.

Saturated fiber products commute with passage to log diamonds by the proof of [KY25, Proposition 7.22], and quasi-pro-Kummer-étale morphisms are stable under saturated base change by [KY25, Proposition 7.16(2)]. Thus these equivalences identify fiber products and covering families on the three bases. The comparison lemma for sites proves the claim. □

Thus it suffices to define $\mathrm{T}_{\text{ét}}$ on log perfectoid test objects.

Definition 4.17. Let $(\mathrm{X}, \mathcal{M})$ be a bounded fs log formal scheme, and let $\mathcal{E}$ be a finitely generated unit Laurent crystal. For a saturated log perfectoid space $(\mathrm{Spa}(\mathrm{S}, \mathrm{S}^+), \mathcal{M}_{\mathrm{S}}) \to (\mathrm{X}, \mathcal{M})^{\diamond}_{\eta}$, let $\mathbb{\Delta}_{\mathrm{S}^+}$ be the perfect saturated log prism with underlying prism $(\mathrm{W}(\mathrm{S}^{+,\flat}), \ker\theta)$ and log structure induced by the tilted log structure on $\mathrm{S}^\flat$. Set

$$\mathrm{T}_{\text{ét}}(\mathcal{E})(\mathrm{Spa}(\mathrm{S}, \mathrm{S}^+), \mathcal{M}_{\mathrm{S}}) := \mathrm{RHom}_{\mathrm{D}^{\varphi}_{\blacksquare}(\mathrm{W}(\mathrm{S}^\flat))}(\mathcal{E}(\mathbb{\Delta}_{\mathrm{S}^+}), \mathrm{W}(\mathrm{S}^\flat)) \in \widehat{\mathrm{D}}(\mathbf{Z}_p).$$

These assignments define a presheaf on the saturated log affinoid perfectoid basis. We denote its right Kan extension to the qpkét site by $\mathrm{T}_{\text{ét}}(\mathcal{E})$.

Here $\mathrm{D}_{\blacksquare}(\mathrm{W}(\mathrm{S}^\flat))$ is the inverse limit over n of the categories of $\mathrm{W}(\mathrm{S}^\flat)/p^n$-modules in the ordinary solid category of the analytic ring $\mathrm{Spa}(\mathbb{A}_{\inf}(\mathrm{S}^+)/p^n, \mathbb{A}_{\inf}(\mathrm{S}^+)/p^n)$. We remark that the perfect (log) prism we use has underlying ring

$$\mathrm{W}(\mathrm{S}^{+,\flat}) = \mathbb{A}_{\inf}(\mathrm{S}^+),$$

whereas its Laurent variant is

$$\mathrm{W}(\mathrm{S}^\flat) = \mathbb{A}_{\inf}(\mathrm{S}^+)[1/\xi]^{\wedge}_p,$$

with $(\xi) = \ker(\theta)$ the prismatic ideal in $\mathbb{A}_{\inf}(\mathrm{S}^+)$ (one can see these are equal using that they are both strict p-rings). Thus $\mathcal{E}(\mathbb{\Delta}_{\mathrm{S}^+})$ in Definition 4.17 denotes the Laurent evaluation, which is naturally a module over $\mathrm{W}(\mathrm{S}^\flat)$.

Lemma 4.18. *Let $(\mathrm{X}, \mathcal{M})$ be a bounded fs log formal scheme. Definition 4.17 yields a well-defined functor*

$$\mathrm{T}_{\text{ét}} : \mathrm{D}_{\text{fgu}}((\mathrm{X}, \mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^{\wedge}_p)^{\mathrm{op}} \to \mathrm{D}((\mathrm{X}, \mathcal{M})^{\diamond}_{\eta,\text{qpkét}}, \mathbf{Z}_p).$$

Proof. It is enough to check hyperdescent on the saturated log affinoid perfectoid basis of $(\mathrm{X},\mathcal{M})^{\diamond}_{\eta,\mathrm{qpk\acute{e}t}}$. After refining by saturated log affinoid perfectoids with divisible characteristic monoid, the quasi-pro-Kummer-étale and quasi-pro-étale sites agree by [KY25, Corollary 7.23]. Thus let $(\mathrm{Spa}(\mathrm{S},\mathrm{S}^+),\mathcal{M}_\mathrm{S})$ be such a test object and let $(\mathrm{Spa}(\mathrm{S}_\bullet,\mathrm{S}^+_\bullet),\mathcal{M}_{\mathrm{S}_\bullet}) \to (\mathrm{Spa}(\mathrm{S},\mathrm{S}^+),\mathcal{M}_\mathrm{S})$ be a quasi-pro-étale hypercover by objects of the same kind.

Put $\mathrm{A} = \mathrm{W}(\mathrm{S}^\flat)$ and $\mathrm{A}_m = \mathrm{W}(\mathrm{S}^\flat_m)$. The v-acyclicity of the tilted structure sheaf on affinoid perfectoid spaces [Sch26a, Theorem 1.2] identifies the derived reductions modulo p of A and $\mathrm{R}\varprojlim_m \mathrm{A}_m$ (which agree with their ordinary mod p reductions). Both are derived p-complete, so derived Nakayama gives

$$\mathrm{A} \simeq \mathrm{R}\varprojlim_{[m]\in\Delta} \mathrm{A}_m$$

in $\mathrm{D}^{\varphi}_{\blacksquare}(\mathrm{A})$, where Frobenius compatibility follows from functoriality of Witt vectors.

Put $\mathrm{M} = \mathcal{E}(\mathbb{\Delta}_{\mathrm{S}^+})$ and $\mathrm{M}_m = \mathcal{E}(\mathbb{\Delta}_{\mathrm{S}^+_m})$. The crystal property shows M_m is simply a base change of M. Using $\mathrm{A} \simeq \mathrm{R}\varprojlim_{[m]\in\Delta} \mathrm{A}_m$, we obtain

$$\mathrm{R}\varprojlim_{[m]\in\Delta} \mathrm{RHom}_{\mathrm{D}^{\varphi}_{\blacksquare}(\mathrm{A}_m)}(\mathrm{M}_m,\mathrm{A}_m) \simeq \mathrm{RHom}_{\mathrm{D}^{\varphi}_{\blacksquare}(\mathrm{A})}(\mathrm{M},\mathrm{A}).$$

Thus $\mathrm{T}_{\mathrm{\acute{e}t}}(\mathcal{E})$ is a hypercomplete sheaf. □

It is easy to see that for every morphism $f : (\mathrm{Y},\mathcal{N}) \to (\mathrm{X},\mathcal{M})$ between pairs satisfying Assumption 3.1, the construction is compatible with pullback:

$$\mathrm{T}_{\mathrm{\acute{e}t},\mathrm{Y}}(\mathrm{L}f^*\mathcal{E}) \simeq \mathrm{L}f^*_\eta \mathrm{T}_{\mathrm{\acute{e}t},\mathrm{X}}(\mathcal{E}).$$

The functor $\mathrm{T}_{\mathrm{\acute{e}t}}$ actually lands in the full subcategory of p-Kummer sheaves, which we now define.

Definition 4.19. Let $(\mathrm{X},\mathcal{M})^\diamond$ be an fs log diamond. A complex $\mathrm{K} \in \mathrm{D}((\mathrm{X},\mathcal{M})^\diamond_{\mathrm{qpk\acute{e}t}},\mathbf{Z}_p)$ is *p-Kummer* if, for every saturated log perfectoid $(\mathrm{Y},\mathcal{M}_\mathrm{Y})^\diamond \to (\mathrm{X},\mathcal{M})^\diamond$ admitting charts quasi-pro-étale locally and such that $\mathcal{M}_\mathrm{Y}/\mathcal{M}^\times_\mathrm{Y}$ is p-divisible, its pullback lies in the essential image of

$$\epsilon^*_\mathrm{Y} : \mathrm{D}(\mathrm{Y}_{\mathrm{pro\acute{e}t}},\mathbf{Z}_p) \longrightarrow \mathrm{D}((\mathrm{Y},\mathcal{M}_\mathrm{Y})^\diamond_{\mathrm{qpk\acute{e}t}},\mathbf{Z}_p),$$

where ϵ_Y is the natural morphism of sites. For a bounded derived lisse complex, this is equivalent to requiring each cohomology sheaf to be p-Kummer in the sense of [IKY26, Lemma 1 and Definition 2]; equivalently, prime-to-p logarithmic inertia acts trivially on every cohomology stalk by [IKY26, Proposition 5].

We next prove that $\mathrm{T}_{\mathrm{\acute{e}t}}(\mathcal{E})$ is constructible for the stratification induced by D. We first need the following descent result.

Lemma 4.20. *On pairs satisfying Assumption 3.1 over $\mathcal{O}_K$, the full subcategories of p-Kummer sheaves in*

$$\mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}}((\mathrm{X},\mathcal{M})_\eta,\mathbf{Z}_p)$$

form a stack for the strict-étale topology.

Proof. Let $g : (\mathrm{U}, \mathcal{N}) \to (\mathrm{X}, \mathcal{M})$ be a strict-étale cover, and let $\mathrm{U}^{(\bullet)}$ be its Čech nerve. Passage to the log adic generic fiber preserves fiber products and carries g to a strict-étale cover. Descent for quasi-pro-Kummer-étale sheaves gives

$$\mathrm{D}^{(b)}((\mathrm{X}, \mathcal{M})_{\eta,\mathrm{qpk\acute{e}t}}, \mathbf{Z}_p) \simeq \varprojlim_{[m]\in\Delta} \mathrm{D}^{(b)}((\mathrm{U}^{(m)}, \mathcal{N}^{(m)})_{\eta,\mathrm{qpk\acute{e}t}}, \mathbf{Z}_p).$$

Here local boundedness descends because it is strict-étale local. By strictness, if $\mathrm{X}_\eta^{(a)}$ and $\mathrm{U}_\eta^{(a)}$ denote the loci where $\mathrm{rk}_{\mathbf{Z}}\, \overline{\mathcal{M}}_{\bar{x}}^{\mathrm{gp}} = a$ (respectively for $\mathcal{N}$), then

$$\mathrm{U}_\eta^{(a)} = \mathrm{U}_\eta \times_{\mathrm{X}_\eta} \mathrm{X}_\eta^{(a)}.$$

The logarithmic strata are therefore compatible, reducing descent for $\mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$ to descent for lisse sheaves.

It remains to check the claim still holds when we restrict to p-Kummer sheaves. By [IKY26, Proposition 5], a derived local system is p-Kummer if and only if prime-to-p logarithmic inertia acts trivially on its cohomology at every geometric log point. However strictness canonically identifies the logarithmic inertia groups, so the same reduction to lisse sheaves shows the claim. ☐

By Lemmas 3.4 and 4.20, constructibility for the logarithmic stratification induced by D may be checked on a framed log affine Spf R. Let $\mathrm{R}_0^\square[\![q-1]\!]_{\mathrm{perf}}$ be the completed perfection of the framed log q-de Rham prism. If $\mathrm{Spf}\,\mathrm{R}_m$ is the p-completed pullback of the framing along the Kummer map $\mathrm{T}_i \mapsto \mathrm{T}_i^{p^m}$ over $\mathrm{W}(k)[\zeta_{p^m}]$, then

$$(\mathrm{R}_0^\square)_\infty := \mathrm{R}_0^\square[\![q-1]\!]_{\mathrm{perf}}/([p]_q) = \left(\varinjlim_m \mathrm{R}_m\right)^\wedge_p.$$

Its log structure $\mathcal{M}_\infty$ is induced by the saturated chart $\mathbf{N}[1/p]^r \to (\mathrm{R}_0^\square)_\infty$ sending e_i/p^m to T_i^{1/p^m} for $1 \le i \le r$. Write $\mathrm{X}_{\infty,\eta} = \mathrm{Spa}((\mathrm{R}_0^\square)_\infty[1/p], (\mathrm{R}_0^\square)_\infty)$. We use ϖ for the tilted pseudouniformizer, and write $(\mathrm{R}_0^\square)^\flat_{\infty,\eta} = ((\mathrm{R}_0^\square)_\infty)^\flat[1/\varpi]$.

For $m \ge 1$, the Γ-action extends to this tower, for $1 \le i \le n$, by

$$\gamma(\mathrm{T}_i^{1/p^m}) = q^{pb_i/p^m}\mathrm{T}_i^{1/p^m} = \zeta_{p^m}^{b_i}\mathrm{T}_i^{1/p^m}.$$

Lemma 4.21. *Let* $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ *be a framed log affine and let* $\mathrm{K} \in \mathrm{D}^{(b)}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\eta,\mathrm{qpk\acute{e}t}}, \mathbf{Z}_p)$. *For* $\Lambda \subseteq \{1, \dots, r\}$, *put*

$$\mathrm{X}_{\infty,\eta,\Lambda} = \mathrm{X}_{\infty,\eta} \times_{\mathrm{X}_\eta} \mathrm{X}_{\eta,\Lambda},$$

and let $\epsilon_\infty : (\mathrm{X}_{\infty,\eta}, \mathcal{M}_\infty)_{\mathrm{qpk\acute{e}t}} \to (\mathrm{X}_{\infty,\eta})_{\mathrm{pro\acute{e}t}}$ *be the natural morphism of sites. The following are equivalent:*

(1) K *is constructible for the stratification induced by* D *and is p-Kummer;*

(2) *its pullback to* $X_{\infty,\eta}$ *is* $\epsilon_\infty^* \mathcal{L}_\infty$ *for some* $\mathcal{L}_\infty \in D^{(b)}((X_{\infty,\eta})_{\text{proét}}, \mathbf{Z}_p)$ *which is constructible for the finite stratification* $X_{\infty,\eta} = \bigsqcup_\Lambda X_{\infty,\eta,\Lambda}$.

Proof. Put $Y = X_{\infty,\eta}$ and $\epsilon = \epsilon_\infty$. We first work with $\mathbf{Z}/p^n$-coefficients. For a log geometric point $\bar{y}^{\log}$ over an ordinary geometric point $\bar{y}$ of Y, put

$$I_{\bar{y}} = \operatorname{Hom}\left(\overline{\mathcal{M}}^{\mathrm{gp}}_{\infty,\bar{y}}, \widehat{\mathbf{Z}}(1)\right).$$

The characteristic monoid is p-divisible, so $I_{\bar{y}}$ is pro-prime-to-p. The Kummer strict-localization calculation of [DLLZ23, Lemma 4.4.27] gives

$$(R\epsilon_* F)_{\bar{y}} \simeq R\Gamma_{\mathrm{cont}}(I_{\bar{y}}, F_{\bar{y}^{\log}})$$

for a bounded constructible complex F. To pass to the quasi-pro sites, work over an ordinary geometric strict localization and adjoin all roots of the saturated chart. The resulting divisible log structure has ordinary quasi-pro-étale site by [KY25, Corollary 7.23]. The strict-local components are acyclic for finite coefficients, and Čech descent computes the displayed continuous inertia cohomology.

Taking $I_{\bar{y}}$-invariants is exact on $\mathbf{Z}/p^n$-modules, by averaging over finite quotients of order prime to p. Hence the adjunction unit $L \to R\epsilon_*\epsilon^* L$ is an equivalence for every bounded constructible L. Moreover, the counit $\epsilon^* R\epsilon_* F \to F$ is an equivalence if and only if $I_{\bar{y}}$ acts trivially on every $H^a(F_{\bar{y}^{\log}})$. This calculation is compatible with restriction to strata. In the latter case, descent for local systems on each stratum shows that $R\epsilon_* F$ is constructible. Thus ϵ^* is fully faithful on constructible complexes, including their mixed extension groups, and its essential image is characterized by trivial logarithmic inertia on cohomology stalks. The assertions with $\mathbf{Z}_p$ coefficients follow by applying the unit and counit criteria modulo p^n and using derived p-completeness. The argument applies to any saturated log perfectoid with p-divisible characteristic monoid and charts quasi-pro-étale locally.

Write $\pi : (Y, \mathcal{M}_\infty) \to (X, \mathcal{M})_\eta$ for the tower map. If (1) holds, Definition 4.19 makes logarithmic inertia act trivially on the cohomology stalks of $\pi^* K$. The preceding calculation gives (2), with $\mathcal{L}_\infty = R\epsilon_* \pi^* K$.

Conversely, suppose (2) holds. Derived lissity descends along π on each stratum, so K is constructible. At a point of $X_{\eta,\Lambda}$, the map on characteristic groups is $\mathbf{Z}^{|\Lambda|} \to \mathbf{Z}[1/p]^{|\Lambda|}$, and hence

$$I^{\log}_{\bar{y}} \simeq \operatorname{Hom}\left(\mathbf{Z}[1/p]^{|\Lambda|}, \widehat{\mathbf{Z}}(1)\right) \simeq I^{\log,(p')}_{\pi(\bar{y})}.$$

Thus prime-to-p logarithmic inertia acts trivially on the cohomology stalks of K. For any test object in Definition 4.19, its logarithmic inertia is pro-prime-to-p and maps into this subgroup. The same counit calculation therefore places the pullback of K in the image of ordinary pro-étale pullback. This proves that K is p-Kummer. □

Let $\mathrm{T}_{\text{ét}}^{\text{perf}}$ denote the scheme-theoretic solution functor on $\mathrm{Spec}(\mathrm{R}_0^\square)^\flat_{\infty,\eta}$ of Theorem 2.6. For a finitely generated unit module M over $\mathrm{R}_0^\square((q-1))_p^\wedge$, we use the convention

$$\mathrm{T}_{\text{ét}}^{\text{perf}}(\mathrm{M}) := \mathrm{T}_{\text{ét}}^{\text{perf}}\left(\mathrm{M} \otimes^{\mathrm{L},\blacksquare}_{\mathrm{R}_0^\square((q-1))_p^\wedge} \mathrm{W}((\mathrm{R}_0^\square)^\flat_{\infty,\eta})\right),$$

For $\mathrm{M} \in \mathrm{D}_{\text{fgu}}(\mathrm{W}((\mathrm{R}_0^\square)^\flat_{\infty,\eta}))$, its value after base change to $\mathrm{Spec}\,\mathrm{S}^\flat$ is

$$\mathrm{RHom}_{\widehat{\mathrm{D}}^\varphi(\mathrm{W}(\mathrm{S}^\flat))}\left(\left(\mathrm{M} \otimes^{\mathrm{L}}_{\mathrm{W}((\mathrm{R}_0^\square)^\flat_{\infty,\eta})} \mathrm{W}(\mathrm{S}^\flat)\right)^\wedge_p, \mathrm{W}(\mathrm{S}^\flat)\right) \in \mathrm{D}(\mathbf{Z}_p).$$

The following lemma controls the Fitting ideals on each stratum. Using the minimal root characterization of being a finitely generated unit module, this will be crucial in sufficiently constraining $\mathsf{Mod}_{\text{fgu}}(\mathrm{R}_0^\square((q-1))_p^\wedge)^{h\Gamma}$ enough to show the essential image of $\mathrm{T}_{\text{ét}}$ is constructible.

Lemma 4.22. *Let* $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ *be a framed log affine with* $\overline{\mathrm{R}} = \mathrm{R}_0/p$. *If* $\mathfrak{a} \subset \overline{\mathrm{R}}^\square((q-1))$ *is stable under* $\mathbf{Z}_p(1)^n \subset \Gamma$, *then, on each connected component of*

$$\mathrm{Spa}\,\overline{\mathrm{R}}^\square((q-1)) \setminus \mathrm{V}(\mathrm{T}_1 \cdots \mathrm{T}_r)$$

the image of $\mathfrak{a}[1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r]$ *is either zero or the unit ideal.*

Proof. Put $\mathrm{X} = \mathrm{Spa}\,\overline{\mathrm{R}}^\square((q-1))$ and $\mathrm{X}^\circ = \mathrm{X} \setminus \mathrm{V}(\mathrm{T}_1 \cdots \mathrm{T}_r)$. The framing gives an étale map

$$\mathrm{X} \longrightarrow \mathrm{Spa}\,k((q-1))\langle \mathrm{T}_1, \ldots, \mathrm{T}_r, \mathrm{T}_{r+1}^{\pm 1}, \ldots, \mathrm{T}_n^{\pm 1}\rangle.$$

The target is reduced, hence so is X, and every T_i is a unit on X°. The ith copy of $\mathbf{Z}_p(1)$ acts by $\mathrm{T}_i \mapsto q^{pb_i}\mathrm{T}_i$ and fixes the other coordinates.

Put $\mathcal{I} = \mathfrak{a}\mathcal{O}_{\mathrm{X}^\circ}$. Let x be a classical point at which $\mathcal{I}_x$ is proper. After replacing $k((q-1))$ by a finite extension containing the residue field of x, choose a rational lift $\widetilde{x}$ of x. The rigid analytic inverse function theorem identifies a neighborhood of $\widetilde{x}$ with a polydisk. We note vanishing may be checked after passage to this extension. Taking the elements of $\mathbf{Z}_p(1)^n \leq \Gamma$ with coordinates $b_i = p^{m_i - 1}$ for integers m_i, stability gives zeros of every $f \in \mathfrak{a}$ at the points with coordinates $q^{p^{m_i}}\mathrm{T}_i(\widetilde{x})$ for large $m_i \geq 1$ lying in this polydisk. Since

$$q^{p^m} - 1 = (q-1)^{p^m} \longrightarrow 0.$$

these points form a product of convergent sequences.

Successive applications of the single variable rigid analytic identity theorem, as in [Pen25, Lemma 3.8], shows that a finite set of generators of $\mathfrak{a}$ vanishes on a neighborhood of $\widetilde{x}$. Faithful flatness gives $\mathcal{I}_x = 0$; if $\mathcal{I}_x$ is not proper, then $\mathcal{I}_x = \mathcal{O}_{\mathrm{X}^\circ, x}$.

The closed analytic locus

$$\mathrm{Supp}(\mathcal{I}) \cap \mathrm{Supp}(\mathcal{O}_{\mathrm{X}^\circ}/\mathcal{I})$$

has no classical points. If it were nonempty, a point in it would admit an affinoid neighborhood in X°. Its intersection with this neighborhood would be a nonempty closed analytic locus and hence would contain a classical point by the affinoid Nullstellensatz [BGR84, 5.2.6/3 and

7.1.3/1], a contradiction. The zero and unit loci of $\mathcal{I}$ are therefore complementary open and closed subsets of X°. On each connected component, $\mathcal{I}$ is zero or the unit ideal. $\square$

Lemma 4.23. *Let* $A = \overline{R}^{\square}((q-1))$ *for a framed log affine, and let* $J = (T_i \mid i \in \Lambda)$ *for some* $\Lambda \subset \{1, \ldots, r\}$. *If* $M \in D_{\mathrm{fgu}}(A)^{h\Gamma}$, *then*

$$(A/J) \otimes_A^L M$$

has bounded finitely generated unit cohomology over A/J *and carries the induced* Γ*-action.*

Proof. Frobenius preserves J, and extension of scalars gives

$$(A/J)[F] \otimes_{A[F]}^L M \simeq (A/J) \otimes_A^L M.$$

The resulting complex is unit because Frobenius pullback commutes with extension of scalars. It is perfect over $(A/J)[F]$, so its underlying A/J-complex is bounded. Since A/J is regular, Noetherian, and F-finite, [BL19, Proposition 11.3.9] makes its cohomology modules finitely generated unit. Finally, J is Γ-stable and Γ commutes with Frobenius, so the quotient map is Γ-equivariant. $\square$

Proposition 4.24. *Let* $(\operatorname{Spf} R, \mathcal{M}_{\operatorname{Spf} R})$ *be a framed log affine with framing* $\square$, *and let* $\mathcal{E}$ *be an object of* $D_{\mathrm{fgu}}((\operatorname{Spf} R, \mathcal{M}_{\operatorname{Spf} R})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^\wedge)$. *Then* $T_{\text{ét}}^{\mathrm{perf}}(\mathcal{E}(R_0^\square((q-1))_p^\wedge))$ *is constructible for the stratification of* $\operatorname{Spec}(R_0^\square)_{\infty,\eta}^\flat$ *cut out by the divisor* $D_\infty = \bigcup_i V(T_i^\flat)$.

Proof. The evaluated crystal is perfect over $W((R_0^\square)_{\infty,\eta}^\flat)\langle F\rangle$. Consequently,

$$T_{\text{ét}}^{\mathrm{perf}}(\mathcal{E}(R_0^\square((q-1))_p^\wedge)) \otimes_{\mathbf{Z}_p}^L \mathbf{F}_p \simeq T_{\text{ét}}^{\mathrm{perf}}((\mathcal{E}/p)(R_0^\square((q-1))_p^\wedge)).$$

Constructibility is detected modulo p, so assume that $\mathcal{E}$ is p-torsion and write $A = \overline{R}_0^\square((q-1))$. Evaluation identifies each cohomology module with a finitely generated unit Γ-module M. Let $\beta : M_0 \to \varphi^* M_0$ be its distinguished minimal coherent root. It is Γ-equivariant by the distinguished property.

Each $\mathrm{Fitt}_i(M_0)$ is $\mathbf{Z}_p(1)^n$-stable. On a connected component of the open stratum $\operatorname{Spec} \overline{R}_0^\square((q-1))[1/T_1, \ldots, 1/T_r]$, let ρ be the least index for which the restriction of $\mathrm{Fitt}_\rho(M_0)$ is the unit ideal (we see $\mathrm{Fitt}_{-1} = 0$). After restriction to this component, Lemma 4.22 gives $\mathrm{Fitt}_{\rho-1}(M_0) = 0$ and $\mathrm{Fitt}_\rho(M_0) = \mathcal{O}$. The Fitting ideal criterion [Sta26, Lemma 15.8.8, Tag 07ZD] makes M_0 locally free of rank ρ there. We note $\varphi^* M_0$ has the same rank.

Thus β is an injection between vector bundles of the same rank. Moreover β is Γ-equivariant, so by the same argument $\operatorname{coker} \beta$ must again be a vector bundle. This is forced to be rank zero, and we conclude that β is an isomorphism: in particular restricted to the open stratum $M = M_0$. It follows $T_{\text{ét}}^{\mathrm{perf}}(\mathcal{E}(R_0^\square((q-1))_p^\wedge))$ is lisse on the open stratum by (the dualized version of) Theorem 2.1. One can make a similar argument for any locally closed stratum, by first taking an appropriate quotient of $\overline{R}_0^\square((q-1))$ before inverting any of the T_i (using Lemma 4.23). $\square$

Remark 4.25. For the trivial log structure there is only the open stratum, so the same argument gives a derived lisse complex.

Let $\nu : (\mathrm{X}^\flat_{\infty,\eta})_{\text{ét}} \to \mathrm{Spec}((\mathrm{R}^\square_0)^\flat_{\infty,\eta})_{\text{ét}}$ be the support map of [Bha25, Notation 6.2.1], and let $\lambda : (\mathrm{X}^\flat_{\infty,\eta})_{\text{proét}} \to (\mathrm{X}^\flat_{\infty,\eta})_{\text{ét}}$ be the natural morphism. For a constructible $\mathbf{Z}_p$-complex K, set

$$\mathrm{an}^*\mathrm{K} := \mathrm{R}\varprojlim_m \lambda^*\nu^*(\mathrm{K}/p^m) \in \mathrm{D}((\mathrm{X}^\flat_{\infty,\eta})_{\text{proét}}, \mathbf{Z}_p).$$

For a Γ-equivariant constructible complex K, set

$$\mathrm{an}^*_\Gamma\mathrm{K} := \left(\epsilon^*_\infty((\mathrm{an}^*\mathrm{K})^\flat)\right)^{h\Gamma} \in \mathrm{D}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\eta,\text{qpkét}}, \mathbf{Z}_p),$$

where $(-)^\flat$ denotes transport across the tilting equivalence and $h\Gamma$ denotes descent along $\mathrm{X}_{\infty,\eta} \to \mathrm{X}_\eta$.

Lemma 4.26. *Let* $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ *be a framed log affine with framing* $\square : (\mathrm{Spf}\,\mathrm{R}, \mathcal{M}) \to \mathbb{T}^n_r$. *Evaluation on* $\Delta_{(\mathrm{R}^\square_0)_\infty}$ *gives a natural commutative diagram*

$$\begin{tikzcd}
\mathrm{D}_{\text{fgu}}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_\Delta, \mathcal{O}_\Delta[1/\mathrm{I}_\Delta]^\wedge_p)^{\text{op}} \arrow[r, "\mathrm{T}_{\text{ét}}"] \arrow[d, "\mathcal{E}\mapsto\mathcal{E}(\Delta_{(\mathrm{R}^\square_0)_\infty})"'] & \mathrm{D}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\eta,\text{qpkét}}, \mathbf{Z}_p) \\
\left(\mathrm{D}_{\text{fgu}}(\mathrm{W}((\mathrm{R}^\square_0)^\flat_{\infty,\eta}))^{h\Gamma}\right)^{\text{op}} \arrow[r, "\mathrm{T}^{\text{perf}}_{\text{ét}}"] & \mathrm{D}^{(b)}_{\text{cons}}(\mathrm{Spec}(\mathrm{R}^\square_0)^\flat_{\infty,\eta}, \mathbf{Z}_p)^{h\Gamma} \arrow[u, "\mathrm{an}^*_\Gamma"']
\end{tikzcd}$$

Proof. Put $\mathrm{A}_\infty = \mathrm{W}((\mathrm{R}^\square_0)^\flat_{\infty,\eta})$. Let $\mathrm{U} = (\mathrm{Spa}(\mathrm{S}, \mathrm{S}^+), \mathcal{M}_\mathrm{S})$ belong to the saturated log affinoid perfectoid basis of $(\mathrm{X}_{\infty,\eta}, \mathcal{M}_\infty)_{\text{qpkét}}$ and have divisible characteristic monoid, and put $\mathrm{A}_\mathrm{S} = \mathrm{W}(\mathrm{S}^\flat)$. Applying Lemma 4.2 termwise to the continuous Γ-cobar diagram, $\mathcal{E}(\Delta_{(\mathrm{R}^\square_0)_\infty})$ corresponds functorially to an object $\mathrm{M}_\infty \in \mathrm{D}_{\text{fgu}}(\mathrm{A}_\infty)^{h\Gamma}$. Since $(\mathrm{X}_{\infty,\eta}, \mathcal{M}_\infty) \to (\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_\eta$ is the Γ-cover defining $h\Gamma$-descent, it suffices to compare the two paths on U before descent. The definition of an^*, tilting, $((\mathrm{X}_{\infty,\eta}, \mathcal{M}_\infty)_{\text{qpkét}})_{/\mathrm{U}} \simeq \mathrm{Spa}(\mathrm{S}, \mathrm{S}^+)_{\text{proét}}$, and base change compatibility give

$$\epsilon^*_\infty((\mathrm{an}^*\mathrm{T}^{\text{perf}}_{\text{ét}}(\mathrm{M}_\infty))^\flat)(\mathrm{U}) \simeq \mathrm{RHom}_{\widehat{\mathrm{D}}^\varphi(\mathrm{A}_\mathrm{S})}\left(\left(\mathrm{M}_\infty \otimes^{\mathrm{L}}_{\mathrm{A}_\infty} \mathrm{A}_\mathrm{S}\right)^\wedge_p, \mathrm{A}_\mathrm{S}\right).$$

On the other hand, Definition 4.17 gives

$$\mathrm{T}_{\text{ét}}(\mathcal{E})(\mathrm{U}) \simeq \mathrm{RHom}_{\mathrm{D}^\varphi_\blacksquare(\mathrm{A}_\mathrm{S})}(\mathcal{E}(\Delta_{\mathrm{S}^+}), \mathrm{A}_\mathrm{S}).$$

The comparison functor of Lemma 4.2 and the crystal property give

$$\left(\left(\mathrm{M}_\infty \otimes^{\mathrm{L}}_{\mathrm{A}_\infty} \mathrm{A}_\mathrm{S}\right)^\wedge_p\right)(*) \simeq \mathrm{M}_\infty(*) \otimes^{\mathrm{L},\blacksquare}_{\mathrm{A}_\infty(*)} \mathrm{A}_\mathrm{S}(*) \simeq \mathcal{E}(\Delta_{\mathrm{S}^+}).$$

The first equivalence holds on the generator $\mathrm{A}_\infty\langle\mathrm{F}\rangle$, where both sides are $\mathrm{A}_\mathrm{S}(*)\langle\mathrm{F}\rangle$, and hence on its thick closure. Moreover $0 \to \mathrm{A}_\mathrm{S}\langle\mathrm{F}\rangle \xrightarrow{\cdot(\mathrm{F}-1)} \mathrm{A}_\mathrm{S}\langle\mathrm{F}\rangle \to \mathrm{A}_\mathrm{S} \to 0$ places A_S in this thick closure. Lemma 4.2 now identifies the two mapping complexes.

These equivalences are natural in U and equivariant for the log prism automorphisms defining the Γ-action. It follows on the saturated log affinoid perfectoid basis that the two paths agree; taking $h\Gamma$ proves the claim. □

Remark 4.27. The factorization in Lemma 4.26 shows that Definition 4.17 is symmetric monoidal. For $\mathrm{T}_{\text{ét}}^{\text{perf}}$, the mod-$p$ tensor comparison follows from [BL19, Theorem 8.4.1] and its identification with the Emerton-Kisin solution functor in [BL19, Theorems 12.5.4 and 12.6.1]; the integral statement follows by derived p-completeness. Analytification, tilting, and descent are symmetric monoidal.

The same factorization controls the essential image of $\mathrm{T}_{\text{ét}}$.

Proposition 4.28. *Every object in the essential image of* $\mathrm{T}_{\text{ét}}$ *on* $\mathrm{D}_{\text{fgu}}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})^{\text{op}}$ *is a* p*-Kummer sheaf in* $\mathrm{D}^{(b)}_{\text{cons},\mathrm{D}}((\mathrm{X},\mathcal{M})_{\eta},\mathbf{Z}_p)$.

Proof. The source satisfies strict-étale descent by Theorem 4.11, while Lemma 4.20 shows that being p-Kummer and constructible is strict-étale local. Lemma 3.4 therefore reduces the claim to a framed log affine.

Put $\mathscr{F} = \mathrm{T}_{\text{ét}}^{\text{perf}}(\mathcal{E}(\Delta_{(\mathrm{R}_0^{\square})_\infty}))$. By Proposition 4.24, it is constructible for the stratification cut out by $\bigcup_i \mathrm{V}(\mathrm{T}_i^{\flat})$. Analytification and tilting give a pro-étale complex whose restriction to every stratum cut out by $\bigcup_i \mathrm{V}(\mathrm{T}_i)$ is derived lisse. Lemma 4.26 gives

$$\pi^*\mathrm{T}_{\text{ét}}(\mathcal{E}) \simeq \epsilon_\infty^*((\mathrm{an}^*\mathscr{F})^{\flat}).$$

Finally, Lemma 4.21 proves the claim. □

5. The main equivalence

We now prove that

$$\mathrm{T}_{\text{ét}} : \mathrm{D}_{\text{fgu}}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})^{\text{op}} \longrightarrow \mathrm{D}^{(b)}_{\text{cons},\mathrm{D}}((\mathrm{X},\mathcal{M})_{\eta},\mathbf{Z}_p)$$

is fully faithful, with essential image the p-Kummer complexes. Theorem 4.11 and Lemmas 3.4 and 4.20 reduce the assertion to framed affines, where Theorem 4.6 describes the source by (φ,Γ)-modules.

The first step constructs crystals realizing $j_!\mathbb{L}$ for pro-étale $\mathbf{Z}_p$-local systems on $\mathrm{U}_\eta = \mathrm{X}_\eta \setminus \mathrm{D}_\eta$. It begins with $\mathcal{O}_{\Delta}(*\mathrm{D})$, whose realization is $j_!\mathbf{Z}_p$, where

$$j_! : \mathsf{Loc}_{\mathbf{Z}_p}(\mathrm{U}_\eta) \to \mathrm{D}((\mathrm{X},\mathcal{M})_{\eta,\text{qpkét}},\mathbf{Z}_p).$$

Lemma 4.26 reduces full faithfulness to that of analytification, supplied by Bhatt-Lurie's prismatic Riemann-Hilbert functor [Bha25].

For essential surjectivity on the open stratum, the basic strategy is as follows. Lütkebohmert's theorem extends $\mathbb{L}$ to a Kummer-étale local system $\mathbb{L}'$ on $(\mathrm{X},\mathcal{M})_\eta$; after this a finite prime-to-p Kummer Galois cover $\pi : \mathrm{Y}_\eta \to \mathrm{X}_\eta$ kills its prime-to-p monodromy. This allows us to apply Theorem 7 of [IKY26], which produces a crystal $\mathcal{E}_{\mathrm{Y}}$ for $\pi^*\mathbb{L}'$; tensoring with $\mathcal{O}_{\mathbb{\Delta}}(*\mathrm{D}_{\mathrm{Y}})$ realizes $j_{\mathrm{Y}_\eta,!}\pi^*_{\mathrm{U}_\eta}\mathbb{L}$, which we prove descends to a crystal realizing $j_!\mathbb{L}$.

We next prove a Γ-equivariant variant of Kashiwara's lemma, reducing to [EK04b]. This gives the induction on dimension, apart from full faithfulness for maps between open and closed strata. For essential surjectivity, the target is thickly generated by $j_!\mathbb{L}$ for inclusions j of locally closed strata, and the preceding constructions realize these generators.

5.1. **The open stratum.** We construct $\mathcal{O}_{\mathbb{\Delta}}(*\mathrm{D})$ from the line bundle with section attached to the log structure, following the analogous log-crystalline construction of [Tsu99, §5]. Assume $(\mathrm{X}, \mathcal{M})$ satisfies Assumption 3.1 over $\mathrm{W}(k)[\zeta_p]$

For a sheaf of groups G, a right G-torsor P, and a left G-sheaf V, we write $\mathrm{P} \times^{\mathrm{G}} \mathrm{V}$ for the contracted product, with $(pg, v) \sim (p, gv)$.

Definition 5.1. Let $\mathcal{M}(-\mathrm{D})$ be the $\mathcal{O}_{\mathrm{X}}^\times$-torsor of local sections $m \in \mathcal{M}$ such that $\alpha_{\mathrm{X}}(m)\mathcal{O}_{\mathrm{X}} = \mathcal{O}_{\mathrm{X}}(-\mathrm{D})$. Let $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf\,A}})^a$ be a saturated log prism in $(\mathrm{X}, \mathcal{M})_{\mathbb{\Delta}}$, with structure map

$$f : (\mathrm{Spf\,A/I}, \mathrm{M}_{\mathrm{Spf\,A/I}}) \to (\mathrm{X}, \mathcal{M}).$$

Let $i : \mathrm{Spf\,A/I} \to \mathrm{Spf\,A}$ be the closed immersion. On $(\mathrm{Spf\,A/I})_{\text{ét}}$, set $f^*\mathcal{M}(-\mathrm{D}) := f^{-1}\mathcal{M}(-\mathrm{D}) \times^{f^{-1}\mathcal{O}_{\mathrm{X}}^\times} \mathcal{O}^\times_{\mathrm{Spf\,A/I}}$. The log structure map gives $f^*\mathcal{M}(-\mathrm{D}) \to \mathrm{M}_{\mathrm{Spf\,A/I}}$. On $(\mathrm{Spf\,A})_{\text{ét}}$, set $\widetilde{\mathcal{M}}_{\mathrm{A}}(-\mathrm{D}) := \mathrm{M}_{\mathrm{Spf\,A}} \times_{i_*\mathrm{M}_{\mathrm{Spf\,A/I}}} i_*f^*\mathcal{M}(-\mathrm{D})$. Define

$$\mathcal{O}_{\mathbb{\Delta}}(-\mathrm{D})(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf\,A}}) := \Gamma\left(\mathrm{Spf\,A}, \widetilde{\mathcal{M}}_{\mathrm{A}}(-\mathrm{D}) \times^{\mathcal{O}^\times_{\mathrm{Spf\,A}}} \mathcal{O}_{\mathrm{Spf\,A}}\right).$$

From this, we can construct an important finitely generated unit crystal corresponding to $j_!\mathbf{Z}_p$ under $\mathrm{T}_{\text{ét}}$.

Definition 5.2. For $n \geq 0$, set

$$\mathcal{O}_{\mathbb{\Delta}}(n\mathrm{D}) := \underline{\mathrm{Hom}}_{\mathcal{O}_{\mathbb{\Delta}}}\left(\mathcal{O}_{\mathbb{\Delta}}(-\mathrm{D})^{\otimes^{\blacksquare} n}, \mathcal{O}_{\mathbb{\Delta}}\right).$$

The map $\mathcal{O}_{\mathbb{\Delta}}(-\mathrm{D}) \to \mathcal{O}_{\mathbb{\Delta}}$ gives transition maps $\mathcal{O}_{\mathbb{\Delta}}(n\mathrm{D}) \to \mathcal{O}_{\mathbb{\Delta}}((n+1)\mathrm{D})$. We define

$$\mathcal{O}_{\mathbb{\Delta}}(*\mathrm{D}) := \left((\mathrm{colim}_{n\geq 0}\,\mathcal{O}_{\mathbb{\Delta}}(n\mathrm{D}))^\wedge_p\,[1/\mathrm{I}_{\mathbb{\Delta}}]\right)^\wedge_p.$$

Strict-étale locally on a framed log affine, if D is cut out by parameters $\mathrm{T}_1, \ldots, \mathrm{T}_r$, then $\mathcal{O}_{\mathbb{\Delta}}(-\mathrm{D})$ is trivial and its map to $\mathcal{O}_{\mathbb{\Delta}}$ is multiplication by a lift of $\mathrm{T}_1 \cdots \mathrm{T}_r$. Hence $\mathcal{O}_{\mathbb{\Delta}}(*\mathrm{D})$ evaluates on the corresponding log q-de Rham prism as $\mathrm{R}_0^\square(\!(q-1)\!)^\wedge_p[1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r]^\wedge_p$. Logarithmic Frobenius carries the canonical section to its pth power up to a unit, so Frobenius on $\mathcal{O}_{\mathbb{\Delta}}$ extends to $\mathcal{O}_{\mathbb{\Delta}}(*\mathrm{D})$.

Since $\mathcal{O}_{\mathbb{\Delta}}(*\mathrm{D})$ is a sheaf of rings, we can consider $\mathsf{Perf}((\mathrm{X}, \mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}(*\mathrm{D}))^{\varphi=1}$.

This category has the following local description. On a framed log affine, put $A = R_0^{\square}((q-1))_p^{\wedge}$ and $A^{\circ} = A[1/T_1, \ldots, 1/T_r]_p^{\wedge}$. If $A^{(m)}$ is the degree m term of the Laurentized log q-de Rham Čech nerve, set

$$A^{\circ,(m)} = A^{(m)} \otimes_A^{L,\blacksquare} A^{\circ} \simeq A^{(m)}[1/T_1, \ldots, 1/T_r]_p^{\wedge}.$$

Here the T_i denote their pullbacks to $A^{(m)}$; strictness makes the resulting localization independent of the projection to the degree zero term. These rings are the evaluations of $\mathcal{O}_{\Delta}(*D)$ on the Čech nerve. Lemma A.17 and the compatibility of module objects with limits give

$$D_{\blacksquare}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*D)) \simeq \varprojlim_{[m]\in\Delta} D_{\blacksquare}(A^{\circ,(m)}).$$

Here Perf denotes dualizable objects. Dualizability is detected termwise in a limit of symmetric monoidal categories, so this equivalence restricts to

$$\mathsf{Perf}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*D)) \simeq \varprojlim_{[m]\in\Delta} \mathsf{Perf}(A^{\circ,(m)}).$$

Applying $\varphi = 1$, since this is an equalizer of stable ∞-categories, gives the analogous description.

Lemma 5.3. *We have* $\mathcal{O}_{\Delta}(*D) \in D_{\mathrm{fgu}}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$. *Moreover any crystal in* $\mathsf{Perf}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*D))^{\varphi=1}$ *is also in* $D_{\mathrm{fgu}}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$.

Proof. This can be checked strict-étale locally by Theorem 4.11 using the log q-de Rham prism $(R_0^{\square}[\![q-1]\!], [p]_q, M)^a$; set $A = R_0^{\square}((q-1))_p^{\wedge}$. Then we can apply the criterion in Theorem 4.6 to check the finitely generated unit condition. On a framed chart where D is defined by $T_1 \cdots T_r = 0$, the mod p coherent root is generated by $(T_1 \cdots T_r)^{-1}$, and Frobenius carries it to the root generated by $(T_1 \cdots T_r)^{-p}$. Its unitalization is precisely $A[1/T_1, \ldots, 1/T_r]_p^{\wedge}$.

For the second assertion, Remark 4.4 and a dévissage using [Hau24, Lemma 5.4.3] reduce to a finitely generated unit Frobenius module P over $A[1/T_1, \ldots, 1/T_r]/p$ whose underlying coefficient module is perfect. It is therefore coherent. Put $T = T_1 \cdots T_r$, choose a coherent A/p-submodule $P_0 \subset P$ with $P_0[1/T] = P$, and let $\beta : P \to \varphi^* P$ be the inverse linearized Frobenius. For some $N \geq 0$,

$$\beta(P_0) \subseteq T^{-N} \varphi^* P_0.$$

If $(p-1)a > N$, then $T^{-a}P_0 \to \varphi^*(T^{-a}P_0)$ is a coherent root. Indeed, if $\Phi : \varphi^* P \to P$ is the linearized Frobenius and $d = (p-1)a - N > 0$, then

$$\Phi^n\big(\varphi^{*n}(T^{-a}P_0)\big) \supseteq T^{-d(1+p+\cdots+p^{n-1})} T^{-a} P_0.$$

Its unitalization is therefore P. ☐

Theorem 5.4. *Let* $(X, \mathcal{M})$ *be as in Assumption 3.1 over* $W(k)[\zeta_p]$. *Then* $T_{\text{ét}}$ *fits in a commutative diagram*

$$\begin{tikzcd}
\mathsf{Perf}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}(*\mathrm{D}))^{\varphi=1,\mathrm{op}} \arrow[r, "\sim"] \arrow[d, hook] & \mathrm{D}^{(b)}_{\mathrm{lisse}}((\mathrm{X}_\eta \setminus \mathrm{D}_\eta)_{\mathrm{pro\acute{e}t}}, \mathbf{Z}_p) \arrow[d, hook, "j_!"] \\
\mathrm{D}_{\mathrm{fgu}}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)^{\mathrm{op}} \arrow[r, "\mathrm{T}_{\mathrm{\acute{e}t}}"] & \mathrm{D}^{(b)}_{\mathrm{cons,D}}((\mathrm{X},\mathcal{M})_\eta, \mathbf{Z}_p).
\end{tikzcd}$$

The top horizontal arrow is an equivalence.

Remark 5.5. Although [BS23, Corollary 3.7] describes the target using ordinary Laurent F-crystals, it is not directly suited to the functor $j_!$. The complement $\mathrm{X}_\eta \setminus \mathrm{D}_\eta$ need not be quasicompact; for example, $\mathbf{P}^1_{\mathbf{Q}_p} \setminus \{\infty\} = \mathbf{A}^1_{\mathbf{Q}_p}$. Moreover, a formal model of this complement need not map to the fixed smooth proper model X with generic fiber j. Raynaud models require noncanonical admissible blowups and need not remain smooth. We fix this problem by using log formal schemes, as in Assumption 3.1.

Lemma 4.26 computes $\mathrm{T}_{\mathrm{\acute{e}t}}(\mathcal{O}_{\Delta}(*\mathrm{D})) \simeq j_!\mathbf{Z}_p$, and Remark 4.27 identifies the bottom horizontal functor in the diagram of Theorem 5.4 with applying $j_!$ to the lisse sheaf on the open stratum.

Lemma 5.6. *Let* $(\mathrm{X},\mathcal{M})$ *be as in Assumption 3.1 over* $\mathrm{W}(k)[\zeta_p]$. *The category*

$$\mathsf{Perf}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}(*\mathrm{D}))^{\varphi=1,\mathrm{op}}$$

is a stack for the strict-étale topology on such $(\mathrm{X},\mathcal{M})$.

Proof. Let $g : (\mathrm{U},\mathcal{N}) \to (\mathrm{X},\mathcal{M})$ be strict-étale and put $\mathrm{D}_{\mathrm{U}} = \mathrm{D} \times_{\mathrm{X}} \mathrm{U}$. There is a canonical Frobenius-equivariant equivalence

$$g^*\mathcal{O}_{\Delta,\mathrm{X}}(*\mathrm{D}) \simeq \mathcal{O}_{\Delta,\mathrm{U}}(*\mathrm{D}_{\mathrm{U}}).$$

Indeed, strictness identifies $g^*\mathcal{M}(-\mathrm{D})$ with $\mathcal{N}(-\mathrm{D}_{\mathrm{U}})$. In particular $\mathcal{O}_{\Delta}(*\mathrm{D})$ commutes with strict-étale pullback.

Let $\mathrm{U}^{(\bullet)}$ be the Čech nerve of a strict-étale cover of X. Corollary A.18 gives symmetric monoidal descent for solid logarithmic Laurent crystals, under which $\mathcal{O}_{\Delta,\mathrm{X}}(*\mathrm{D})$ corresponds to the algebras $\mathcal{O}_{\Delta,\mathrm{U}^{(m)}}(*\mathrm{D}_{\mathrm{U}^{(m)}})$. One then also obtains descent for their module categories. Dualizability is detected termwise in a limit, and imposing the unit Frobenius structure is a limit. Hence the same descent statement holds for the category in the lemma. Taking opposites preserves limits. □

Remark 5.7. Let $\pi = \zeta_p - 1$ be a uniformizer of $\mathrm{W}(k)[\zeta_p]$. If one does not invert I_{Δ} and instead takes the unitalization of $\mathcal{O}_{\Delta}(\mathrm{D}) \to \varphi^*\mathcal{O}_{\Delta}(\mathrm{D})$, one can calculate that the crystalline realization (i.e. the unitalization $\mathcal{E}$ of $\mathcal{O}_{\mathrm{cris}}(\mathrm{D}) \to \varphi^*\mathcal{O}_{\mathrm{cris}}(\mathrm{D})$) satisfies

$$\mathrm{RHom}_{\mathrm{D}^{\varphi}(((\mathrm{X},\mathcal{M})_{\pi=0}/\mathrm{W}(k))_{\mathrm{cris}},\mathcal{O}_{\mathrm{cris}})}(\mathcal{E},\mathcal{O}_{\mathrm{cris}}) \simeq (\mathrm{R}\Gamma_{\mathrm{cris},c}((\mathrm{X},\mathcal{M})_{\pi=0}/\mathrm{W}(k)))^{\varphi=1}.$$

Here compactly supported log crystalline cohomology is as defined in [Tsu99]. This shows the crystalline realization of the natural non-Laurentized variant is compatible with the expectation one would have from the open C_{st} conjecture.

Lemma 5.8. *Let* $(\operatorname{Spf} \mathrm{R}, \mathcal{M})$ *be a framed log affine. Then*

$$\mathsf{Perf}((\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*\mathrm{D}))^{\varphi=1} \simeq \mathsf{Perf}(\mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}[1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r]_p^{\wedge})^{\varphi=1,h\Gamma}.$$

Proof. Put $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}$ and $\mathrm{A}^{\circ} = \mathrm{A}[1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r]_p^{\wedge}$. Evaluation is symmetric monoidal and carries $\mathcal{O}_{\Delta}(*\mathrm{D})$ to A° with its Γ-action. Lemma 5.3 embeds the source into the fgu category, and Theorem 4.6 is fully faithful there. Modules over an idempotent localization form a full subcategory, so the displayed functor is fully faithful.

Conversely, let M be a perfect unit Frobenius A°-complex with Γ action. The algebraic argument in the proof of Lemma 5.3 makes its restriction to A finitely generated unit. Theorem 4.6 gives an fgu crystal $\mathcal{E}$ with evaluation M. The localization map

$$\mathcal{E} \longrightarrow \mathcal{E} \otimes \mathcal{O}_{\Delta}(*\mathrm{D})$$

is an equivalence after evaluation, hence is an equivalence by conservativity. Thus $\mathcal{E}$ is an $\mathcal{O}_{\Delta}(*\mathrm{D})$-module. The local description $\mathsf{Perf}((\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*\mathrm{D})) \simeq \varprojlim_{[m]\in\Delta} \mathsf{Perf}(\mathrm{A}^{\circ,(m)})$ and the crystal property identify its degree m evaluation with

$$\mathrm{A}^{\circ,(m)} \widehat{\otimes}^{\mathrm{L}}_{\mathrm{A}^{\circ}} \mathrm{M},$$

which is perfect over $\mathrm{A}^{\circ,(m)}$. Since dualizability is detected termwise, $\mathcal{E}$ is perfect over $\mathcal{O}_{\Delta}(*\mathrm{D})$. Hence the fgu equivalence restricts to the displayed equivalence. $\square$

We now tackle full faithfulness. Our strategy will be to show this in the perfectoid setting (using the formulation of $\mathrm{T}_{\text{ét}}$ in Lemma 4.26). However unlike Laurent F-crystals in perfect complexes, it is not possible to use only the unit Frobenius to reduce to the perfectoid setting. We explain this further in Remark 5.13.

Proposition 5.9. *Let* $(\operatorname{Spf} \mathrm{R}, \mathcal{M})$ *be a framed log affine. The base-change functor*

$$\mathsf{Perf}((\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*\mathrm{D}))^{\varphi=1} \to \mathsf{Perf}(\mathbb{A}_{\inf}((\mathrm{R}_0^{\square})_{\infty})[1/\xi, 1/[\mathrm{T}_1^{\flat}], \ldots, 1/[\mathrm{T}_r^{\flat}]]_p^{\wedge})^{\varphi=1,h\Gamma}$$

is fully faithful.

Proof. As usual, associated to the framed log affine we have a logarithmic q-de Rham prism, so we let $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}$. For this particular proof we only care about a localization, so put $\mathrm{A}^{\circ} := \mathrm{A}[1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r]_p^{\wedge}$ and $\mathrm{A}^{\circ}_{\text{perf}} = \mathbb{A}_{\inf}((\mathrm{R}_0^{\square})_{\infty})[1/\xi, 1/[\mathrm{T}_1^{\flat}], \ldots, 1/[\mathrm{T}_r^{\flat}]]_p^{\wedge}$. We can express the source as $\mathsf{Perf}(\mathrm{A}^{\circ})^{\varphi=1,h\Gamma}$ using Lemma 5.8; the target is already $\mathsf{Perf}(\mathrm{A}^{\circ}_{\text{perf}})^{\varphi=1,h\Gamma}$.

By the tensor-Hom adjunction, it suffices to show that

$$\mathrm{R}\Gamma_{\text{cont}}(\Gamma, \mathrm{M})^{\varphi=1} \longrightarrow \mathrm{R}\Gamma_{\text{cont}}(\Gamma, \mathrm{M} \otimes_{\mathrm{A}^{\circ}} \mathrm{A}^{\circ}_{\text{perf}})^{\varphi=1} \tag{4}$$

is an isomorphism for $\mathrm{M} \in \mathsf{Perf}(\mathrm{A}^{\circ})^{\varphi=1,h\Gamma}$. We can also use the ordinary tensor product $\mathrm{M} \otimes_{\mathrm{A}^{\circ}} \mathrm{A}^{\circ}_{\text{perf}}$ in the target because the relevant map $\mathrm{A}^{\circ} \to \mathrm{A}^{\circ}_{\text{perf}}$ is faithfully flat. Indeed, Kunz's theorem makes $A^+/p \to A^+_{\mathrm{perf}}/p$ faithfully flat, so $A^+ \to A^+_{\mathrm{perf}}$ is p-completely faithfully flat. The map $A^\circ \to A^\circ_{\mathrm{perf}}$ is its p-completed base change after inverting $[p]_q, T_1, \ldots, T_r$, so it is p-completely faithfully flat; since A° is Noetherian, it is faithfully flat.

The source and target also both have standard t-structures obtained by restricting the t-structure on $D_\blacksquare(A)$ and $D_\blacksquare(A_{\mathrm{perf}})$. To check these are well-defined one must check truncations remain perfect, which can be done modulo p by derived p-completeness. In this setting the t-structure exists even before applying $(-)^{h\Gamma}$, which is implicit in the proof [BS23, Proposition 3.4]: in particular they show that the top cohomology of a perfect unit Frobenius module over an $\mathbf{F}_p$-algebra is a vector bundle. By derived Nakayama we can work modulo p. Using a Postnikov dévissage ([Hau24, Lemma 5.4.3]) we reduce to testing the claim for p-torsion M in the heart, which must lie in $\mathsf{Vect}(A^\circ/p)^{\varphi=1,h\Gamma}$.

We now prove (4) for M in $\mathsf{Vect}(A^\circ/p)^{\varphi=1,h\Gamma}$. The faithfully flat map $A^\circ/p \to A^\circ_{\mathrm{perf}}/p$ is injective, with cokernel $(A^\circ_{\mathrm{perf}}/A^\circ)/p$. Since M is finite projective over A°/p, tensoring this sequence with M is exact. It is therefore enough to prove $R\Gamma_{\mathrm{cont}}(\Gamma, M \otimes_{A^\circ} (A^\circ_{\mathrm{perf}}/A^\circ))^{\varphi=1} = 0$.

Choose $m \gg 0$ and let $(R_0^\square)_m$ be obtained from the framing by adjoining p^mth roots of the coordinates T_i. Define $\varphi^{-m}A^\circ = (R_0^\square)_m(\!(q^{1/p^m}-1)\!)[1/T_1, \ldots, 1/T_r]^\wedge_p$ and $\varphi^{-m}A^{\circ,\mathrm{cyc}} = (R_0^\square)_m(\!(q^{1/p^\infty}-1)\!)^\wedge[1/T_1, \ldots, 1/T_r]^\wedge_p$, viewed as intermediate subrings

$$A^\circ \subset \varphi^{-m}A^\circ \subset \varphi^{-m}A^{\circ,\mathrm{cyc}} \subset A^\circ_{\mathrm{perf}}.$$

Since φ and Γ commute, the unit Frobenius of M gives a Γ-equivariant equivalence

$$R\Gamma_{\mathrm{cont}}(\Gamma, M)^{\varphi=1} \simeq R\Gamma_{\mathrm{cont}}\big(\Gamma, M \otimes_{A^\circ} \varphi^{-m}A^\circ\big)^{\varphi=1}.$$

For $m \gg 0$, the exact triangles associated to $\varphi^{-m}A^\circ/A^\circ \longrightarrow A^\circ_{\mathrm{perf}}/A^\circ \longrightarrow A^\circ_{\mathrm{perf}}/\varphi^{-m}A^\circ$ and to $\varphi^{-m}A^{\circ,\mathrm{cyc}}/\varphi^{-m}A^\circ \longrightarrow A^\circ_{\mathrm{perf}}/\varphi^{-m}A^\circ \longrightarrow A^\circ_{\mathrm{perf}}/\varphi^{-m}A^{\circ,\mathrm{cyc}}$ show

$$R\Gamma_{\mathrm{cont}}(\Gamma, M \otimes_{A^\circ} (A^\circ_{\mathrm{perf}}/A^\circ))^{\varphi=1} = 0$$

by applying Lemmas 5.11 and 5.12. The claim then follows. □

We now provide the needed technical lemmas. We retain the notation of Proposition 5.9, and use the t-structure constructed in the proof.

Put $T = T_1 \cdots T_r$. Through the identification in Theorem 4.6 (using a resolution in $D_\blacksquare(A)^{h\Gamma}$ by $A\langle F\rangle$ induced by a root), the mod-p coefficient ring

$$\overline{A}^\circ := A^\circ/p = \bigcup_{i \geq 0} T^{-i}\overline{R}^\square(\!(q-1)\!)$$

is equipped with the inductive limit topology, where $T^{-i}\overline{R}^\square[\![q-1]\!]$ is an open lattice in $T^{-i}\overline{R}^\square(\!(q-1)\!)$ with its $(q-1)$-adic topology. This is a strict inductive limit of Banach modules with closed transition maps, and every compact subset is contained in $T^{-i}\overline{R}^\square(\!(q-1)\!)$ for some i. It is worth noting that this topology agrees with the weak topology induced on a finitely generated unit module, by using the resolution by $\mathrm{A}\langle\mathrm{F}\rangle$ induced by the root and showing this weak topology, modulo p, agrees with the natural topology on the colimit presentation by the root (which is what we use here).

Choose a finite $\overline{\mathrm{R}}^{\square}[\![q-1]\!]$-submodule $\mathrm{M}_0 \subset \mathrm{M}$ with $\mathrm{M}_0[1/(q-1), 1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r] = \mathrm{M}$. The orbit under the compact group Γ of a finite generating set of M_0 is contained in $(q-1)^{-c}\mathrm{T}^{-i}\mathrm{M}_0$ for some $c, i \geq 0$. Its $\overline{\mathrm{R}}^{\square}[\![q-1]\!]$-span is therefore a finite Γ-stable lattice $\mathrm{M}^+ \subset \mathrm{M}$ with $\mathrm{M}^+[1/(q-1), 1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r] = \mathrm{M}$.

Let $\mathrm{H} = (1+p^m\mathbf{Z}_p) \ltimes (p^{m-1}\mathbf{Z}_p(1))^n \subset \Gamma$. Choose a topological generator γ_0 of $1+p^m\mathbf{Z}_p$ and generators $\gamma_1, \ldots, \gamma_n$ of $(p^{m-1}\mathbf{Z}_p(1))^n$ such that $\gamma_i(\mathrm{T}_i) = q^{p^m}\mathrm{T}_i$. For $N \geq 0$, put $\mathrm{M}^{(N)} = \mathrm{T}^{-N}\mathrm{M}^+[1/(q-1)]$ and $\mathrm{Fil}^a \mathrm{M}^{(N)} = \mathrm{T}^{-N}(q-1)^a\mathrm{M}^+$. Then $\mathrm{M} = \mathrm{colim}_N \mathrm{M}^{(N)}$ with its strict inductive limit topology. Increasing m if necessary, H is normal and uniform, and

$$(\gamma_i - 1)(\mathrm{Fil}^a \mathrm{M}^{(N)}) \subset \mathrm{Fil}^{a+1} \mathrm{M}^{(N)} \text{ for } 0 \leq i \leq n. \tag{5}$$

We use the following elementary lemma, which is similar to [KL16, Lemmas 5.6.3, 5.6.4]. For these mod-p calculations, we work with a more restrictive variant of Ind-Banach $\mathbf{F}_p$-modules. We will abbreviate this just to "LB" modules and rings, but our convention is the following. An LB module is a presentation $\mathrm{M} = \mathrm{colim}_{N\in\mathbf{N}} \mathrm{M}^{(N)}$ by Banach modules with closed contractive embeddings. LB rings are algebra objects with LB underlying modules, and group actions satisfy $h \cdot \mathrm{M}^{(N)} = \mathrm{M}^{(N)}$. For compatible presentations, write

$$\widehat{\bigoplus_j}^{\mathrm{LB}} \mathrm{M}_j := \mathrm{colim}_N \widehat{\bigoplus_j} \mathrm{M}_j^{(N)},$$

where the completed sum is taken for the supremum norm. In the applications, all norms are normalized by the same value of $|q-1|$.

For a closed subgroup $\mathrm{H} \leq \Gamma$, choose topological generators $h_1, \ldots, h_k$ of $\mathrm{H} \cap \mathbf{Z}_p(1)^n \simeq \mathbf{Z}_p^k$. If the image of H in $1+p\mathbf{Z}_p$ is nontrivial, choose $h_0 \in \mathrm{H}$ lifting a topological generator of this image. The resolution (1) and Hochschild-Serre compute $\mathrm{R}\Gamma_{\mathrm{cont}}(\mathrm{H}, \mathrm{M})$ as the fiber of $h_0 - 1$ on

$$\mathrm{Kos}^\bullet(\mathrm{M}; h_1 - 1, \ldots, h_k - 1),$$

where h_0 acts on the coefficients and by conjugation on the Koszul generators. If the cyclotomic image is trivial, the Koszul complex itself computes cohomology.

This cochain model of cohomology allows us to make the following definitions, subject to the choices of generators above. A complex is *strictly contractible* if it admits a continuous contracting homotopy. For a family of complexes $\mathrm{C}_j^\bullet$, such homotopies h_j^i are *uniformly bounded* if they preserve the Banach stages and

$$\sup_{j,i} \| h_j^i : \mathrm{C}_j^{i,(N)} \longrightarrow \mathrm{C}_j^{i-1,(N)} \| < \infty$$

for every N.

Lemma 5.10. *Let* $\mathrm{H} \leq \Gamma$ *be a closed subgroup and let* $\mathrm{B}_0 \subseteq \mathrm{B}_1 \subseteq \cdots \subseteq \mathrm{B}$ *be an increasing sequence of LB rings with continuous* H*-actions. Suppose there are* H*-equivariant* B_j*-linear retractions* $\psi_j : \mathrm{B}_{j+1} \longrightarrow \mathrm{B}_j$ *of LB modules, and that the natural map is an isomorphism of LB modules*

$$\widehat{\bigoplus}^{\mathrm{LB}}_{j\geq 0} \ker(\psi_j) \xrightarrow{\sim} \mathrm{B}/\mathrm{B}_0.$$

Let $\mathrm{M} \in \mathsf{Vect}(\mathrm{B}_0)$ *have continuous semilinear* H*-action, and write* $\mathrm{M}_j = \mathrm{B}_j \widehat{\otimes}_{\mathrm{B}_0} \mathrm{M}$ *and* $\mathrm{M}_{\mathrm{B}} = \mathrm{B}\widehat{\otimes}_{\mathrm{B}_0}\mathrm{M}$. *If the complexes computing continuous* H*-cohomology of* $\mathrm{M}_j \widehat{\otimes}_{\mathrm{B}_j} \ker(\psi_j)$ *are strictly contractible by compatible uniformly bounded homotopies, then*

$$\mathrm{R}\Gamma_{\mathrm{cont}}(\mathrm{H}, \mathrm{M}_{\mathrm{B}}/\mathrm{M}) = 0.$$

Proof. Finite projectivity of M, with the induced direct-summand topologies, carries the ring decomposition to an isomorphism of LB modules

$$\mathrm{M}_{\mathrm{B}}/\mathrm{M} \simeq \widehat{\bigoplus}^{\mathrm{LB}}_{j\geq 0} \left(\mathrm{M}_j \widehat{\otimes}_{\mathrm{B}_j} \ker(\psi_j)\right).$$

The finite Koszul complexes and their mapping fibers carry this to the corresponding completed sum of complexes. The uniform bounds extend the contracting homotopies to each Banach completed sum, and compatibility gives a contracting homotopy on the colimit. ☐

All LB structures below use the increasing localization filtration by T^{-N}, with the $(q-1)$-adic norm on each Banach stage. Put $\mathrm{M}_j = \mathrm{B}_j \widehat{\otimes}_{\overline{\mathrm{A}}^{\circ}} \mathrm{M}$ and write $\|-\|_N$ for operator norms on the Nth stage, using supremum norms on monomial decompositions and Koszul complexes.

Lemma 5.11. *Let* M *be* p*-torsion and in the heart of* $\mathsf{Perf}(\mathrm{A}^\circ)^{\varphi=1,h\Gamma}$. *For* $m \gg 0$,

$$\mathrm{R}\Gamma_{\mathrm{cont}}\big(\Gamma, \mathrm{M} \otimes_{\mathrm{A}^\circ} (\mathrm{A}^\circ_{\mathrm{perf}}/\varphi^{-m}\mathrm{A}^{\circ,\mathrm{cyc}})\big) = 0.$$

Proof. Apply Lemma 5.10 with $\mathrm{B}_j = \varphi^{-(m+j)}\mathrm{A}^{\circ,\mathrm{cyc}}/p$, $\mathrm{B} = \mathrm{A}^\circ_{\mathrm{perf}}/p$, $\mathrm{H} = (p^{m-1}\mathbf{Z}_p(1))^n$, and coefficient module $\mathrm{B}_0 \widehat{\otimes}_{\overline{\mathrm{A}}^\circ} \mathrm{M}$, which is finite projective by the t-structure reduction in the proof of Proposition 5.9. For $\alpha = (a_1, \ldots, a_n)$, the decomposition

$$\mathrm{B}_{j+1} = \bigoplus_{\alpha \in \{0,\ldots,p-1\}^n} \mathrm{B}_j \prod_{\nu=1}^{n} \mathrm{T}_\nu^{a_\nu/p^{m+j+1}}$$

defines Γ-equivariant B_j-linear retractions ψ_j by projection onto $\alpha = 0$. With the prescribed LB structures, these give

$$\widehat{\bigoplus}^{\mathrm{LB}}_{j\geq 0} \ker(\psi_j) \simeq \mathrm{B}/\mathrm{B}_0.$$

For $\alpha \neq 0$, choose the least i with $a_i \neq 0$. On $\mathrm{M}_j \prod_\nu \mathrm{T}_\nu^{a_\nu/p^{m+j+1}}$, the action on the coordinates and (5) give

$$\left\|(q^{a_i/p^{j+1}} - 1)^{-1}(\gamma_i - q^{a_i/p^{j+1}})\right\|_N \leq |q-1|^{(p-1)p^{-j-1}} < 1.$$

The geometric series gives $(\gamma_i - 1)^{-1}$ with norm at most $|q-1|^{-1}$. Thus $h_{j,\alpha} = (\gamma_i - 1)^{-1}\iota_i$ contracts the Koszul complex of this module, where ι_i is exterior contraction with $\iota_i(e_\nu) = \delta_{i\nu}$ for the basis $e_1, \dots, e_n$. These contractions preserve the stages, commute with their inclusions, and satisfy $\|h_{j,\alpha}\|_N \leq |q-1|^{-1}$. Lemma 5.10 gives the vanishing for H, and Hochschild-Serre gives it for Γ. ☐

Lemma 5.12. *Let* M *be* p*-torsion and in the heart of* $\mathsf{Perf}(\mathrm{A}^\circ)^{\varphi=1,h\Gamma}$. *For* $m \gg 0$,

$$\mathrm{R}\Gamma_{\mathrm{cont}}\big(\Gamma, \mathrm{M} \otimes_{\mathrm{A}^\circ} (\varphi^{-m}\mathrm{A}^{\circ,\mathrm{cyc}}/\varphi^{-m}\mathrm{A}^\circ)\big) = 0.$$

Proof. Apply Lemma 5.10 with $\mathrm{B} = \varphi^{-m}\mathrm{A}^{\circ,\mathrm{cyc}}/p$,

$$\mathrm{B}_j = (\overline{\mathrm{R}}_0^\square)_m(\!(q^{1/p^{m+j}} - 1)\!)[1/\mathrm{T}_1, \dots, 1/\mathrm{T}_r],$$

coefficient module $\mathrm{B}_0\widehat{\otimes}_{\overline{\mathrm{A}}^\circ}\mathrm{M}$, and $\mathrm{H} = (1 + p^m\mathbf{Z}_p) \ltimes \mathrm{N}$, where $\mathrm{N} = (p^{m-1}\mathbf{Z}_p(1))^n$. The decomposition

$$\mathrm{B}_{j+1} = \bigoplus_{a=0}^{p-1} \mathrm{B}_j q^{a/p^{m+j+1}}$$

defines Γ-equivariant B_j-linear retractions ψ_j by projection onto $a = 0$. As in the preceding proof, $\widehat{\bigoplus}_j^{\mathrm{LB}} \ker(\psi_j) \simeq \mathrm{B}/\mathrm{B}_0$.

Choose $\gamma_0 = 1 + p^m u$ generating $1 + p^m\mathbf{Z}_p$, with $u \in \mathbf{Z}_p^\times$. For $1 \leq a < p$, put $\mathrm{C}^\bullet_{j,a} = \mathrm{Kos}^\bullet_{\mathrm{N}}(\mathrm{M}_j q^{a/p^{m+j+1}})$. The relation $\gamma_0\gamma_i\gamma_0^{-1} = \gamma_i^{1+p^m u}$ gives the action

$$\gamma_0(ye_i) = \left(\frac{\gamma_i^{1+p^m u} - 1}{\gamma_i - 1}\right)^{-1} \gamma_0(y)e_i,$$

with the products of these factors in higher degrees. For m sufficiently large, this action satisfies

$$\left\|(q^{au/p^{j+1}} - 1)^{-1}(\gamma_0 - q^{au/p^{j+1}})\right\|_N \leq |q-1|^{p^{-j}(1-p^{-m}-p^{-1})} < 1$$

on $\mathrm{C}^\bullet_{j,a}$. The geometric series gives an inverse chain map $(\gamma_0 - 1)^{-1}$ of norm at most $|q-1|^{-1}$. It supplies compatible uniformly bounded contractions of $\mathrm{fib}(\gamma_0 - 1 : \mathrm{C}^\bullet_{j,a} \to \mathrm{C}^\bullet_{j,a})$. The sum of these fibers over $1 \leq a < p$ computes H-cohomology of $\mathrm{M}_j\widehat{\otimes}_{\mathrm{B}_j}\ker(\psi_j)$. Lemma 5.10 gives the vanishing for H, and Hochschild-Serre gives it for Γ. ☐

Remark 5.13. It is tempting to try to argue the previous result would only need to use φ, as this is the case for ordinary (φ, Γ) modules using invariance of étale φ-modules under completed perfections. Consider the map

$$\mathrm{A} = k(\!(q-1)\!)\langle\mathrm{T}\rangle[1/\mathrm{T}] \to \mathrm{B} = k(\!(q^{1/p^\infty} - 1)\!)^\wedge\langle(\mathrm{T}^\flat)^{1/p^\infty}\rangle[1/\mathrm{T}^\flat],$$

with $\mathrm{T} \mapsto \mathrm{T}^\flat$. If full faithfulness followed just from invariance under this completed perfection, then $\mathsf{Perf}(\mathrm{A})^{\varphi=1} \to \mathsf{Perf}(\mathrm{B})^{\varphi=1}$ would be fully faithful. By Katz's theorem this would force $\mathrm{H}^1_{\text{ét}}(\mathrm{A}, \mathbf{F}_p) \to \mathrm{H}^1_{\text{ét}}(\mathrm{B}, \mathbf{F}_p)$ to be an isomorphism. One can check that the Artin-Schreier class on B defined by

$$f = \sum_{n \geq 1} (q-1)^n (\mathrm{T}^\flat)^{-1+1/p^n}$$

is not pulled back from A, so the étale cohomology groups are not isomorphic via the natural base change map. The issue is passage to B mixes completed and decompleted perfections.

Proposition 5.14. *The top horizontal functor in Theorem 5.4 is fully faithful.*

Proof. Lemma 5.6 and strict-étale descent for $\mathrm{D}^{(b)}_{\text{lisse}}((\mathrm{X}_\eta \setminus \mathrm{D}_\eta)_{\text{proét}}, \mathbf{Z}_p)$, together with Lemma 3.4, reduce the claim to a framed log affine. Lemma 5.8 identifies the source with $\mathsf{Perf}(\mathrm{R}_0^\square((q-1))^\wedge_p[1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r]^\wedge_p)^{\varphi=1, h\Gamma}$, and Proposition 5.9 makes base change to $\mathbb{A}_{\text{inf}}((\mathrm{R}_0^\square)_\infty)[1/\xi, 1/[\mathrm{T}_1^\flat], \ldots, 1/[\mathrm{T}_r^\flat]]^\wedge_p$ fully faithful. Lemma 4.26 then identifies $\mathrm{T}_{\text{ét}}$ with tilting, analytification, and continuous descent along the perfectoid pro-Kummer Γ-torsor. Tilting and descent are equivalences.

Thus we must investigate full faithfulness of analytification. Put

$$\mathrm{U} = \operatorname{Spec}(\mathrm{R}_0^\square)^\flat_{\infty,\eta}[1/\mathrm{T}_1^\flat, \ldots, 1/\mathrm{T}_r^\flat].$$

For mod-p objects $\mathbb{L}, \mathbb{L}' \in \mathrm{D}^{(b)}_{\text{lisse}}(\mathrm{U}_{\text{ét}}, \mathbf{F}_p)$, the monodromy of each $\mathrm{H}^i\underline{\mathrm{RHom}}(\mathbb{L}, \mathbb{L}')$ is contained in a finite group $\mathrm{GL}_{m_i}(\mathbf{F}_p)$. Boundedness gives one finite étale cover $\mathrm{U}' \to \mathrm{U}$ on which $\mathrm{H}^i\underline{\mathrm{RHom}}(\mathbb{L}, \mathbb{L}') \simeq \mathbf{F}_p^{m_i}$ for every i. On U', [Bha25, Remark 6.2.7] gives

$$\mathrm{R}\Gamma(\mathrm{U}', \mathbf{F}_p) \xrightarrow{\sim} \mathrm{R}\Gamma(\mathrm{U}'^{\text{an}}, \mathbf{F}_p).$$

Postnikov dévissage and finite étale descent give full faithfulness modulo p; derived Nakayama gives the assertion. $\square$

We next prove essential surjectivity after passing, on a framed strict-étale chart, to a finite prime-to-p Kummer cover. Proposition 5.15 descends the object constructed on the cover, using the full faithfulness already proved.

Proposition 5.15. *Let* $(\operatorname{Spf} \mathrm{R}, \mathcal{M})$ *be a framed log affine with boundary coordinates* $\mathrm{T}_1, \ldots, \mathrm{T}_r$, *and let* N *be prime to* p. *Put*

$$\mathrm{R}_\mathrm{N} = (\mathrm{R}[\mathrm{S}_1, \ldots, \mathrm{S}_r]/(\mathrm{S}_1^\mathrm{N} - \mathrm{T}_1, \ldots, \mathrm{S}_r^\mathrm{N} - \mathrm{T}_r))^\wedge_p,$$

and let $(\operatorname{Spf} \mathrm{R}_\mathrm{N}, \mathcal{M}_\mathrm{N})$ *with its induced log structure. Then we have*

$$\left(\mathsf{Perf}((\operatorname{Spf} \mathrm{R}_\mathrm{N}, \mathcal{M}_\mathrm{N})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}(*\mathrm{D}_\mathrm{N}))^{\varphi=1,\text{op}}\right)^{h\mu_\mathrm{N}^r} \simeq \mathsf{Perf}((\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}(*\mathrm{D}))^{\varphi=1,\text{op}}.$$

Here $(-)^{h\mu_\mathrm{N}^r}$ *denotes the totalization of the* μ_N^r*-action groupoid; after localizing along* D_N, *this is the Čech nerve of a finite étale torsor. The equivalence is compatible with finite étale descent for the étale realization.*

Proof. Let $f_{\mathrm{N}} : (\operatorname{Spf} \mathrm{R}_{\mathrm{N}}, \mathcal{M}_{\mathrm{N}}) \to (\operatorname{Spf} \mathrm{R}, \mathcal{M})$ be the Kummer map. Since $f_{\mathrm{N}}^{*}\mathrm{D} = \mathrm{N}\mathrm{D}_{\mathrm{N}}$, there is a canonical equivalence

$$f_{\mathrm{N}}^{*}\mathcal{O}_{\mathbb{A}}(*\mathrm{D}) \simeq \mathcal{O}_{\mathbb{A}}(*(\mathrm{N}\mathrm{D}_{\mathrm{N}})) \simeq \mathcal{O}_{\mathbb{A}}(*\mathrm{D}_{\mathrm{N}}).$$

Thus pullback induces a functor on the corresponding module categories

$$\mathsf{Perf}((\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\mathbb{A}}, \mathcal{O}_{\mathbb{A}}(*\mathrm{D}))^{\varphi=1,\mathrm{op}} \to \mathsf{Perf}((\operatorname{Spf} \mathrm{R}_{\mathrm{N}}, \mathcal{M}_{\mathrm{N}})_{\mathbb{A}}, \mathcal{O}_{\mathbb{A}}(*\mathrm{D}_{\mathrm{N}}))^{\varphi=1,\mathrm{op}}$$

we will use to induce the desired equivalence.

Base change of the framing along $\mathrm{T}_i \mapsto \mathrm{S}_i^{\mathrm{N}}$ gives R_{N} a framing with boundary coordinates $\mathrm{S}_1, \ldots, \mathrm{S}_r$, so it also has an associated q-de Rham prism. On the degree zero log q-de Rham term, the pullback is base change along

$$\mathrm{A} := \mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}[1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r]_p^{\wedge} \longrightarrow \mathrm{B} := \mathrm{A}[\mathrm{S}_1, \ldots, \mathrm{S}_r]/(\mathrm{S}_1^{\mathrm{N}} - \mathrm{T}_1, \ldots, \mathrm{S}_r^{\mathrm{N}} - \mathrm{T}_r).$$

Because $\mathrm{N} \in \mathbf{Z}_p^{\times}$ and the T_i are units in A, this is a finite étale μ_{N}^{r}-torsor.

After rescaling the first r coordinates by N^{-1}, the two actions of Γ make $\mathrm{A} \to \mathrm{B}$ a Γ-equivariant μ_{N}^{r}-torsor. It is also Frobenius-equivariant by functoriality of the q-de Rham prisms. Lemma 5.8 identifies pullback of localized perfect crystals with extension of scalars from A to B. The proposition therefore reduces to finite étale descent in (φ, Γ)-equivariant perfect complexes:

$$\mathsf{Perf}(\mathrm{A})^{\varphi=1,h\Gamma} \simeq \left(\mathsf{Perf}(\mathrm{B})^{\varphi=1,h\Gamma}\right)^{h\mu_{\mathrm{N}}^{r}}.$$

Taking opposites proves the assertion.

Pullback compatibility is clear. □

Remark 5.16. On a framed log affine, $[(\mathrm{X}_{\mathrm{N}}, \mathcal{M}_{\mathrm{N}})/\mu_{\mathrm{N}}^{r}]$ is the Nth root stack of $(\mathrm{X}, \mathcal{M})$. Thus the proposition is the local form of finite étale descent from this root stack; the local form suffices because the equivalence can be checked strict-étale locally.

Corollary 5.17. *The top horizontal functor in Theorem 5.4 is an equivalence.*

Proof. By Proposition 5.14, it remains to prove essential surjectivity. Let $\mathbb{K} \in \mathrm{D}_{\mathrm{lisse}}^{(b)}((\mathrm{X}_\eta \setminus \mathrm{D}_\eta)_{\mathrm{pro\acute{e}t}}, \mathbf{Z}_p)$. Since X_η is quasicompact and smooth, it has finitely many connected components, each irreducible; their nonempty intersections with $\mathrm{X}_\eta \setminus \mathrm{D}_\eta$ remain irreducible. The amplitude of a derived-lisse complex is locally constant, so $\mathbb{K}$ is globally bounded. Bounded objects are generated under shifts and fibers by their hearts [Hau24, Lemma 5.4.3], so it suffices to treat $\mathbb{L} \in \mathrm{D}_{\mathrm{lisse}}^{(b)}((\mathrm{X}_\eta \setminus \mathrm{D}_\eta)_{\mathrm{pro\acute{e}t}}, \mathbf{Z}_p)^{\heartsuit}$.

This assertion is strict-étale local on X. Indeed, if a local system $\mathbb{L}$ is realized locally by objects $\mathcal{E}_i$, then full faithfulness gives unique identifications of the $\mathcal{E}_i$ on pairwise overlaps inducing the identity on $\mathbb{L}$. The cocycle condition holds after applying the fully faithful realization functor, and Lemma 5.6 glues the $\mathcal{E}_i$. Thus we may assume that $\mathrm{X} = \operatorname{Spf} \mathrm{R}$ is framed.

Let $\mathbb{L}$ be such a local system. By [DLLZ23, Corollary 6.3.4], whose proof uses the rigid Abhyankar lemma, $\mathbb{L}^{\mathrm{ext}} := j_{\mathrm{k\acute{e}t},*}\mathbb{L}$ is a Kummer-étale $\mathbf{Z}_p$-local system on the log generic fiber.

At every log geometric point, the congruence kernel of $\mathrm{Aut}_{\mathbf{Z}_p}(\mathbb{L}^{\mathrm{ext}}_{\bar{x}}) \to \mathrm{Aut}_{\mathbf{F}_p}(\mathbb{L}^{\mathrm{ext}}_{\bar{x}}/p)$ is pro-p, as follows from its filtration by the congruence kernels modulo p^n. Its intersection with the image of the prime-to-p Kummer inertia is therefore trivial, so this image is finite and injects into $\mathrm{Aut}_{\mathbf{F}_p}(\mathbb{L}^{\mathrm{ext}}_{\bar{x}}/p)$.

Working componentwise, these images have exponents dividing the prime-to-p exponent of some $\mathrm{GL}_m(\mathbf{F}_p)$. Choose N prime to p divisible by these exponents, and let $f_{\mathrm{N}} : (\mathrm{X}_{\mathrm{N}}, \mathcal{M}_{\mathrm{N}}) \to (\mathrm{X}, \mathcal{M})$ be the cover of Proposition 5.15. Then $f_{\mathrm{N}}^*\mathbb{L}^{\mathrm{ext}}$ is p-Kummer.

By the (dualized) Koshikawa-Yao correspondence [IKY26, Theorem 7]

$$\mathsf{Perf}((\mathrm{X}_{\mathrm{N}}, \mathcal{M}_{\mathrm{N}})_{\mathbb{A}}, \mathcal{O}_{\mathbb{A}}[1/\mathrm{I}_{\mathbb{A}}]^{\wedge}_p)^{\varphi=1,\mathrm{op}} \simeq \mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{lisse}}((\mathrm{X}_{\mathrm{N}}, \mathcal{M}_{\mathrm{N}})_\eta, \mathbf{Z}_p),$$

this p-Kummer local system is realized by a Laurent prismatic F-crystal $\mathcal{E}$ on $(\mathrm{X}_{\mathrm{N}}, \mathcal{M}_{\mathrm{N}})_{\mathbb{A}}$, so $\mathrm{T}_{\text{ét}}(\mathcal{E}) \simeq f_{\mathrm{N}}^*\mathbb{L}^{\mathrm{ext}}$. Let $j_{\mathrm{N}} : \mathrm{X}_{\mathrm{N},\eta} \setminus \mathrm{D}_{\mathrm{N},\eta} \hookrightarrow (\mathrm{X}_{\mathrm{N}}, \mathcal{M}_{\mathrm{N}})_\eta$ denote the open-stratum inclusion. Since $\mathrm{T}_{\text{ét}}(\mathcal{O}_{\mathbb{A}}(*\mathrm{D}_{\mathrm{N}})) \simeq j_{\mathrm{N},!}\mathbf{Z}_p$ by the compatibility of the vertical arrows before Theorem 5.4, the symmetric monoidality of $\mathrm{T}_{\text{ét}}$ gives

$$\mathrm{T}_{\text{ét}}(\mathcal{E} \otimes \mathcal{O}_{\mathbb{A}}(*\mathrm{D}_{\mathrm{N}})) \simeq j_{\mathrm{N},!}f_{\mathrm{N}}^*\mathbb{L}.$$

The sheaf $j_{\mathrm{N},!}f_{\mathrm{N}}^*\mathbb{L}$ has its canonical descent datum along the finite étale torsor on the open stratum. The Čech nerve of the localized map in Proposition 5.15 is finite étale in every degree. Proposition 5.14 therefore lifts the descent datum uniquely to $\mathcal{E} \otimes \mathcal{O}_{\mathbb{A}}(*\mathrm{D}_{\mathrm{N}})$, and full faithfulness gives the cocycle condition. Proposition 5.15 now descends this object to $\mathsf{Perf}((\mathrm{X}, \mathcal{M})_{\mathbb{A}}, \mathcal{O}_{\mathbb{A}}(*\mathrm{D}))^{\varphi=1,\mathrm{op}}$. Compatibility with realization shows that it realizes $j_!\mathbb{L}$. $\square$

5.2. **Proof of the main theorem.** We will show that

$$\mathrm{T}_{\text{ét}} : \mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\mathbb{A}}, \mathcal{O}_{\mathbb{A}}[1/\mathrm{I}_{\mathbb{A}}]^{\wedge}_p)^{\mathrm{op}} \longrightarrow \mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$$

is an equivalence onto the full subcategory of p-Kummer sheaves $\mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons},\mathrm{D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$. Strict-étale descent on the source and target, established in Theorem 4.11 and Lemma 4.20, permits us to work on framed log affines for the proof.

The remainder of the proof beyond the already established equivalence for the open stratum is an induction on dimension using a variant of Kashiwara's lemma in Lemma 5.18 for closed strata, together with full faithfulness for maps between open and closed strata. The following toy model summarizes the argument.

A toy model. Consider the disk $\mathbb{D}_{\mathbf{Q}_p(\zeta_p)}$ punctured at the origin; we use the coordinate T so $\mathbb{D} = \mathrm{Spa}\,\mathbf{Q}_p(\zeta_p)\langle \mathrm{T} \rangle$, equipped with the log structure $\mathcal{M}_{\mathrm{V(T)}}$. We use the usual formal model $(\mathrm{Spf}\,\mathbf{Z}_p[\zeta_p]\langle \mathrm{T} \rangle, \mathcal{M}_{\mathrm{V(T)}})$.

Let $j : \mathbb{D}^\times \hookrightarrow \mathbb{D}$ and $i : \mathrm{V(T)} \hookrightarrow \mathbb{D}$ denote the two strata. If $\mathrm{A} = \mathbf{Z}_p\langle \mathrm{T} \rangle(\!(q-1)\!)^{\wedge}_p$ and $\Gamma = (1 + p\mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)$, the setup of the argument is summarized by the diagram

$$\begin{tikzcd}
\left\{\begin{matrix}\text{open stratum unit}\\ \text{Laurent crystals}\end{matrix}\right\}^{\mathrm{op}} \arrow[r, hook] \arrow[d, "\mathrm{T}_{\acute{e}t}"] & \left\{\begin{matrix}\text{fgu Laurent crystals}\\ \text{on } (\mathrm{Spf}\, \mathbf{Z}_p[\zeta_p]\langle \mathrm{T}\rangle, \mathcal{M}_{\mathrm{V(T)}})_{\mathbb{\Delta}}\end{matrix}\right\}^{\mathrm{op}} \arrow[d, "\mathrm{T}_{\acute{e}t}"] & \left(\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/(\mathrm{T}))^{h\Gamma}\right)^{\mathrm{op}} \arrow[l, dashed, "(i_+(-))[-1]"'] \arrow[d, "\mathrm{T}_{\acute{e}t}"] \\
\left\{\begin{matrix}p\text{-Kummer local systems}\\ \text{on } \mathbb{D}^{\times}\end{matrix}\right\} \arrow[r, hook, "j_!"] & \left\{\begin{matrix}p\text{-Kummer constructible sheaves}\\ \text{on } (\mathbb{D}, \mathcal{M}_{\mathrm{V(T)}})\end{matrix}\right\} & \left\{\begin{matrix}p\text{-Kummer local systems}\\ \text{on } (\mathrm{V(T)}, i^*\mathcal{M}_{\mathrm{V(T)}})\end{matrix}\right\}. \arrow[l, hook', "i_*"']
\end{tikzcd}$$

The left vertical arrow is an equivalence by Theorem 5.4, and tensoring with $\mathcal{O}_{\mathbb{\Delta}}(*\mathrm{V(T)})$ identifies with $j_!$ under $\mathrm{T}_{\acute{e}t}$. The factor of Γ corresponding to T acts trivially on $\mathrm{A}/(\mathrm{T})$, but its action on objects records the normal Kummer inertia on the logarithmic point. The dashed arrow is supplied by the local Kashiwara equivalence below: since $i^! = \mathrm{L}i^*[-1]$ and $\mathrm{T}_{\acute{e}t}$ is contravariant, $(i_+(-))[-1]$ identifies with i_* under realization.

With Kashiwara's lemma what we have so far (generalized to a hollow log point to deal with the origin) is enough for essential surjectivity, because étale sheaves of the form $j_!\mathbb{L}$ and $i_*\mathbb{L}'$ generate $\mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{V(T)}}((\mathbb{D}, \mathcal{M}_{\mathrm{V(T)}}), \mathbf{Z}_p)$ under shifts and fibers; the same holds after imposing the p-Kummer condition. The functor $\mathrm{T}_{\acute{e}t}$ lands in the full subcategory of p-Kummer constructible sheaves along the stratification $(\mathbb{D}^{\times}, \mathrm{V(T)})$ of the log adic generic fiber and is compatible with shifts and fibers. One need only note that $\mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf}\, \mathbf{Z}_p[\zeta_p]\langle \mathrm{T}\rangle, \mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^{\wedge}_p)$ is closed under shifts and fibers to conclude the argument.

For full faithfulness, it is not quite enough. The strategy will be to place any $\mathcal{E} \in \mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf}\, \mathbf{Z}_p[\zeta_p]\langle \mathrm{T}\rangle, \mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^{\wedge}_p)$ into a fiber sequence with an "open" and "closed" part. We will use the corresponding stratum on the adic generic fiber as a subscript to indicate the support of a crystal. When comparing $\mathrm{RHom}(\mathcal{E}, \mathcal{E}')$ with $\mathrm{RHom}(\mathrm{T}_{\acute{e}t}(\mathcal{E}'), \mathrm{T}_{\acute{e}t}(\mathcal{E}))$ to deduce the comparison one must ultimately show full faithfulness in the form

$$\mathrm{RHom}(\mathcal{E}_{\mathbb{D}^{\times}}, \mathcal{E}'_{\mathrm{V(T)}}) \simeq \mathrm{RHom}(\mathrm{T}_{\acute{e}t}(\mathcal{E}'_{\mathrm{V(T)}}), \mathrm{T}_{\acute{e}t}(\mathcal{E}_{\mathbb{D}^{\times}})).$$

Here the corresponding étale realizations are supported on the logarithmic point at the origin and on the punctured disk $\mathbb{D}^{\times}$. The trick is similar to what was already done in the proof of essential surjectivity of Theorem 5.4, where we reduce the statement to a prime-to-p Kummer cover $\mathbb{D}_{\mathrm{N}}$. Unlike that case we do not work on the open stratum, so this does not become an étale cover. However as we are only interested in mapping complexes, we can show that one can still get away with taking μ_N^r-invariants to obtain the claim for $\mathbb{D}$.

Put $\mathbb{L} = \mathrm{T}_{\acute{e}t}(\mathcal{E}_{\mathbb{D}_{\mathrm{N}}^{\times}})$ and $i_*\mathbb{L}' = \mathrm{T}_{\acute{e}t}(\mathcal{E}'_{\mathrm{V(T)}})$. We may assume that $\mathbb{L}$ extends across the origin to the p-Kummer derived local system $\mathrm{R}j_*\mathbb{L}$. Thus there is

$$\mathcal{E}_{\mathbb{D}_{\mathrm{N}}} \in \mathsf{Perf}((\mathrm{Spf}\, \mathbf{Z}_p[\zeta_p]\langle \mathrm{T}^{1/\mathrm{N}}\rangle, \mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]^{\wedge}_p)^{\varphi=1}$$

such that $\mathrm{T}_{\acute{e}t}(\mathcal{E}_{\mathbb{D}_{\mathrm{N}}}) \simeq \mathrm{R}j_*\mathbb{L}$ and $\mathcal{E}_{\mathbb{D}_{\mathrm{N}}^{\times}} \simeq \mathcal{E}_{\mathbb{D}_{\mathrm{N}}}(*\mathrm{V(T)})$. On the Kummer-étale side, one has

$$\mathrm{RHom}(i_*\mathbb{L}', \mathrm{R}j_*\mathbb{L}) = 0.$$

Indeed, the adjunction between j^* and $\mathrm{R}j_*$ identifies this with $\mathrm{RHom}(j^*i_*\mathbb{L}', \mathbb{L}) = 0$. Applying $\mathrm{RHom}(i_*\mathbb{L}', -)$ to the triangle

$$j_!\mathbb{L} \longrightarrow \mathrm{R}j_*\mathbb{L} \longrightarrow i_*i^*\mathrm{R}j_*\mathbb{L}$$

and using the adjunction between i^* and i_* gives $\mathrm{RHom}(i_*\mathbb{L}', j_!\mathbb{L}) \simeq \mathrm{RHom}(\mathbb{L}', i^*\mathrm{R}j_*\mathbb{L})[-1]$. This now consists of sheaves supported at the origin $\mathrm{V(T)} \subset \mathbb{D}_{\mathrm{N}}$, so producing an analogous isomorphism on the crystal side would reduce the needed result down one dimension to a point where the claim is known (we will use an induction on dimension). For this to be plausible one needs $\mathrm{R}j_*\mathbb{L}$ to be p-Kummer, so there is a corresponding crystal; this is the point of passing to the cover. Using exactly this strategy, one can show

$$\mathrm{RHom}(\mathcal{E}_{\mathbb{D}_{\mathrm{N}}^{\times}}, \mathcal{E}'_{\mathrm{V(T)}}) \simeq \mathrm{RHom}(\mathrm{fib}(\mathcal{E}_{\mathbb{D}_{\mathrm{N}}} \to \mathcal{E}_{\mathbb{D}_{\mathrm{N}}}(*\mathrm{V(T)})), \mathcal{E}'_{\mathrm{V(T)}})[-1].$$

The remaining closed term is computed over the quotient by T, so it is indeed a dimension lower. Thus one can reduce to a hollow point where the claim can be checked directly.

Notation for this section. Consider a framed log affine $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ with framing $\square$ as in Definition 3.2, and put $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}$. For $\Lambda \subseteq \{1, \dots, r\}$, put

$$\mathrm{J}_\Lambda = (\mathrm{T}_i \mid i \in \Lambda).$$

For $\mathrm{Z}_\Lambda = \mathrm{V}(\mathrm{J}_\Lambda)$, put $\mathrm{D}|_{\mathrm{Z}_\Lambda} = \bigcup_{i \notin \Lambda}(\mathrm{D}_i \cap \mathrm{Z}_\Lambda)$ and consider $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_\Lambda)^{h\Gamma}$. The group $\Gamma_\Lambda = \prod_{i\in\Lambda} \mathbf{Z}_p(1)_i$ fixes $\mathrm{A}/\mathrm{J}_\Lambda$. Give Z_Λ the pullback log structure, which now has a hollow component. Since we continue using the full group Γ, we see $\mathrm{T}_{\text{ét}}$ factors as

$$\mathrm{T}_{\text{ét}} : \left(\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_\Lambda)^{h\Gamma}\right)^{\mathrm{op}} \longrightarrow \mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons},\mathrm{D}|_{\mathrm{Z}_\Lambda}}((\mathrm{Z}_\Lambda, \mathcal{M}|_{\mathrm{Z}_\Lambda})_\eta, \mathbf{Z}_p)$$

followed by extension by zero. One can see this using Lemma 4.26. The target is constructible for the remaining coordinate stratification.

We now prove a local version of Kashiwara's lemma. We write $\mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})_{\mathrm{Z}}$ for the full subcategory of finitely generated unit Laurent crystals supported on Z. Let us explain the general globalized notion of this when $\mathrm{D} = \bigcup_i \mathrm{D}_i$ is a simple normal crossings divisor with irreducible components D_i. For any nonempty intersection $\mathrm{Z} = \mathrm{D}_1 \cap \cdots \cap \mathrm{D}_c$ of distinct irreducible components, define

$$\mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})_{\mathrm{Z}} := \bigcap_{a=1}^{c} \ker\left(- \otimes^{\mathrm{L},\blacksquare} \mathcal{O}_{\Delta}(*\mathrm{D}_a)\right).$$

Similarly one may make a definition for connected components. This covers all Z which are smooth closed log strata of the stratification induced by D; connected components are necessarily of pure codimension. On a compatible framing, Theorem 4.6 identifies this supported ambient category with the source of the following lemma.

Lemma 5.18 (Kashiwara's lemma). *Let* $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ *be a framed log affine, put* $\mathrm{A} = \mathrm{R}_0^{\square}(\!(q-1)\!)_p^{\wedge}$ *and* $\mathrm{J} = \mathrm{J}_\Lambda = (\mathrm{T}_\lambda \mid \lambda \in \Lambda)$*, and let* $i : \mathrm{Z} = \mathrm{V}(\mathrm{J}) \hookrightarrow \mathrm{Spf}\,\mathrm{R}$ *so that under Theorem 4.6 one has*

$$\mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})_{\mathrm{Z}} \simeq \mathrm{D}_{\mathrm{fgu}}(\mathrm{A})_{\mathrm{J}}^{h\Gamma},$$

the analogous category of finitely generated unit modules supported on Z.

Then one has an equivalence

$$i^! := \left((\mathrm{A}/\mathrm{J}) \otimes_{\mathrm{A}}^{\mathrm{L},\blacksquare} -\right)[-|\Lambda|] : \mathrm{D}_{\mathrm{fgu}}(\mathrm{A})_{\mathrm{J}}^{h\Gamma} \xrightarrow{\sim} \mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J})^{h\Gamma}$$

We write i_+ for the inverse equivalence.

Proof. Put $\overline{\mathrm{A}} = \mathrm{A}/p$ and $\overline{\mathrm{J}} = \mathrm{J}\overline{\mathrm{A}}$. The categories and functor above satisfy the hypotheses of Lemma 2.3; the support condition is detected modulo p by derived Nakayama. It is therefore enough to prove that

$$i^! := \left((\overline{\mathrm{A}}/\overline{\mathrm{J}}) \otimes_{\overline{\mathrm{A}}}^{\mathrm{L},\blacksquare} -\right)[-|\Lambda|] : \mathrm{D}_{\mathrm{fgu}}(\overline{\mathrm{A}})_{\overline{\mathrm{J}}}^{h\Gamma} \longrightarrow \mathrm{D}_{\mathrm{fgu}}(\overline{\mathrm{A}}/\overline{\mathrm{J}})^{h\Gamma}$$

is an equivalence. The functor is well-defined: one can use Lemma 4.23.

In this case, the functor is an equivalence before we apply $(-)^{h\Gamma}$. One may use the proof of [EK04b, Corollary 5.11.3] to the present regular F-finite situation (noting we have access to important ingredients like roots by [Bli08]). Thus one obtains an adjoint equivalence

$$i_+ : \mathrm{D}_{\mathrm{fgu}}(\overline{\mathrm{A}}/\overline{\mathrm{J}}) \rightleftarrows \mathrm{D}_{\mathrm{fgu}}(\overline{\mathrm{A}})_{\overline{\mathrm{J}}} : i^!.$$

This survives after applying a bar construction $(-)^{h\Gamma}$ to both sides, because evaluation on the degree zero term is conservative. Thus one may test $i^! i_+ \simeq \mathrm{id}$ and $i_+ i^! \simeq \mathrm{id}$ after evaluation on the degree zero term, at which point it reduces to the ordinary Kashiwara lemma for finitely generated unit modules. ☐

For $i : \mathrm{Z}_\Lambda \hookrightarrow \operatorname{Spf} \mathrm{R}$ and $\mathrm{M} \in \mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_\Lambda)^{h\Gamma}$, the equivalence above in Lemma 5.18 gives

$$\mathrm{T}_{\text{ét}}(i_+\mathrm{M}) \simeq i_* \mathrm{T}_{\text{ét}}(\mathrm{M})[-|\Lambda|].$$

It is also easy to give a formula for $i_+ i^!(-)$ even when the input is not supported on Z. On the same chart, we also remark that local cohomology along Z is computed by

$$\mathrm{R}\Gamma_{\mathrm{Z}}(\mathrm{M}) \simeq \mathrm{M} \otimes_{\mathrm{A}}^{\mathrm{L},\blacksquare} \bigotimes_{i\in\Lambda}^{\mathrm{L},\blacksquare} \left[\mathrm{A} \longrightarrow (\mathrm{A}[1/\mathrm{T}_i])_p^\wedge\right].$$

Completed base change to A/J kills every localized term, so $i^! \mathrm{R}\Gamma_{\mathrm{Z}}(\mathrm{M}) \simeq i^! \mathrm{M}$. The complex $\mathrm{R}\Gamma_{\mathrm{Z}}(\mathrm{M})$ is canonically Γ-equivariant, since each $\gamma \in \Gamma$ carries every T_i to a unit multiple of itself. Checking $\mathrm{R}\Gamma_{\mathrm{Z}}(\mathrm{M})$ is finitely generated unit is straightforward, so Kashiwara's lemma therefore gives

$$\mathrm{R}\Gamma_{\mathrm{Z}}(\mathrm{M}) \simeq i_+ i^! \mathrm{M}.$$

In the toy model at the start of the subsection, the following is the Laurent crystal analogue of $\mathrm{RHom}(i_*\mathbb{L}', \mathrm{R}j_*\mathbb{L}) = 0$ (any Kummer-étale local system in the second argument can be written this way using [DLLZ23]).

Lemma 5.19. *Let $(\operatorname{Spf} \mathrm{R}, \mathcal{M})$ be a framed log affine, put $\mathrm{A} = \mathrm{R}_0^\square((q-1))_p^\wedge$, and fix one boundary component $\mathrm{Z} = \mathrm{D}_i = \mathrm{V}(\mathrm{T}_i)$. If*

$$\mathrm{E} \in \mathsf{Perf}(\mathrm{A})^{\varphi=1,h\Gamma}$$

and $\mathrm{Q} \in \mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/(\mathrm{T}_i))^{h\Gamma}$, *then*

$$\mathrm{RHom}(\mathrm{E}, i_+\mathrm{Q}) = 0.$$

Proof. Put $\mathrm{T} = \mathrm{T}_i$ and $\overline{\mathrm{A}} = \mathrm{A}/p$. Since Kashiwara's equivalence commutes with reduction modulo p, derived Nakayama reduces the assertion modulo p. A Postnikov dévissage (see [Hau24]) then reduces to Q concentrated in degree zero.

We use the distinguished minimal root of Q to give an explicit formula for $i_+\mathrm{Q}$. Consider

$$\mathcal{L} = \mathrm{T}^{-1}\overline{\mathrm{A}}/\overline{\mathrm{A}} \simeq (\mathrm{T}/\mathrm{T}^2)^\vee.$$

The inclusion $\mathrm{T}^{-1}\overline{\mathrm{A}}/\overline{\mathrm{A}} \to \mathrm{T}^{-p}\overline{\mathrm{A}}/\overline{\mathrm{A}}$ defines an injective map $\mathcal{L} \to \varphi^*\mathcal{L}$; this is the root for $i_+(\overline{\mathrm{A}}/(\mathrm{T}))$. It is a distinguished minimal root, so it is Γ-equivariant.

Let Q_0 be the distinguished minimal root of Q, which is Γ-stable by its distinguished property, and put

$$\mathrm{K} = \mathrm{colim}_{n \geq 0}\, \varphi^{*n}\big(\mathcal{L} \otimes^{\blacksquare}_{\overline{\mathrm{A}}/(\mathrm{T})} \mathrm{Q}_0\big).$$

This is a finitely generated unit Γ-module supported on $\mathrm{V}(\mathrm{T})$, and

$$i^!\mathrm{K} \simeq \Big((\overline{\mathrm{A}}/(\mathrm{T})) \otimes^{\mathrm{L},\blacksquare}_{\overline{\mathrm{A}}} \mathrm{K}\Big)[-1] \simeq \mathrm{Q}.$$

Lemma 5.18 therefore identifies K with $i_+\mathrm{Q}$.

There is a natural exhaustive "pole order" filtration on $i_+\mathrm{Q}$, with $\mathrm{Fil}_0 = 0$ and

$$\mathrm{Fil}_m = \mathrm{colim}_{p^n \geq m}\, \mathrm{T}^{p^n - m}\varphi^{*n}(\mathcal{L} \otimes^{\blacksquare} \mathrm{Q}_0).$$

Put $\mathrm{C} = \underline{\mathrm{RHom}}_{\overline{\mathrm{A}}/(\mathrm{T})}((\overline{\mathrm{A}}/(\mathrm{T})) \otimes^{\mathrm{L},\blacksquare}_{\overline{\mathrm{A}}} \mathrm{E}, \mathrm{Q})$. Using the unit Frobenius isomorphisms of Q, its graded pieces give isomorphisms of underlying complexes

$$\underline{\mathrm{RHom}}_{\overline{\mathrm{A}}}(\mathrm{E}, \mathrm{gr}_m(i_+\mathrm{Q})) \simeq \mathrm{C} =: \mathrm{C}_m.$$

We equip C_m with the Γ-action induced by the left side; relative to the natural action on the right side, it is twisted by the character through which Γ acts on T^{-m}.

The complex C is a finitely generated unit (φ, Γ)-module, and each C_m has the same underlying module with a twisted Γ-action. Every $\mathrm{H}^i(\mathrm{C})$ is $\mathrm{F}_{\overline{\mathrm{A}}/(\mathrm{T})}$-finite in the convention of [Hoc07], using the inverse of its unit isomorphism. By [Hoc07, Theorem 5.1], $\mathrm{End}_{\mathrm{F}_{\overline{\mathrm{A}}/(\mathrm{T})}}(\mathrm{H}^j(\mathrm{C}))$ is finite. The subgroup $\mathbf{Z}_p(1)_i \subset \Gamma$ corresponding to T fixes $\overline{\mathrm{A}}/(\mathrm{T})$ and commutes with Frobenius, so its action on each $\mathrm{H}^j(\mathrm{C})$ factors through the finite group of invertible Frobenius-linear endomorphisms. This subgroup is normal: every $\mathbf{Z}_p(1)_j$ commutes with it, while conjugation by $a \in 1 + p\mathbf{Z}_p$ sends $b \in \mathbf{Z}_p(1)_i$ to ab. Since C is bounded, $\mathrm{H} = p^\ell\mathbf{Z}_p(1)_i$ acts trivially on all its cohomology modules for some ℓ. Let γ generate H, with $\gamma(\mathrm{T}) = q^{p^{\ell+1}}\mathrm{T}$.

Once we restrict to this $\mathrm{H} \subset \mathbf{Z}_p(1)_i$, only the twisted part of the action remains. On each C_m, the action of γ on $\mathrm{H}^j(\mathrm{C}_m)$ is therefore multiplication by $q^{-mp^{\ell+1}}$. Thus $\gamma - 1$ acts by the unit $q^{-mp^{\ell+1}}-1$ modulo p, since $q-1$ is invertible. For fixed m the filtration on $\underline{\mathrm{RHom}}_{\overline{\mathrm{A}}}(\mathrm{E},\mathrm{Fil}_m(i_+\mathrm{Q}))$ is finite, so $\gamma-1$ is an automorphism on this complex.

The filtration on $i_+\mathrm{Q}$ is exhaustive. Since E is perfect, $\underline{\mathrm{RHom}}_{\overline{\mathrm{A}}}(\mathrm{E},-)\simeq \mathrm{E}^\vee\otimes^{\mathrm{L},\blacksquare}_{\overline{\mathrm{A}}}-$ preserves filtered colimits. Hence $\underline{\mathrm{RHom}}_{\overline{\mathrm{A}}}(\mathrm{E},i_+\mathrm{Q})\simeq \mathrm{colim}_m\,\underline{\mathrm{RHom}}_{\overline{\mathrm{A}}}(\mathrm{E},\mathrm{Fil}_m(i_+\mathrm{Q}))$, and $\gamma-1$ is an equivalence on this complex. Since $\mathrm{H}\simeq\mathbf{Z}_p$, taking the fiber of $\gamma-1$ gives

$$\mathrm{R}\Gamma_{\mathrm{cont}}(\mathrm{H},\underline{\mathrm{RHom}}_{\overline{\mathrm{A}}}(\mathrm{E},i_+\mathrm{Q}))\simeq[\underline{\mathrm{RHom}}_{\overline{\mathrm{A}}}(\mathrm{E},i_+\mathrm{Q})\xrightarrow{\gamma-1}\underline{\mathrm{RHom}}_{\overline{\mathrm{A}}}(\mathrm{E},i_+\mathrm{Q})]=0.$$

Applying Hochschild-Serre gives

$$\mathrm{RHom}(\mathrm{E},i_+\mathrm{Q})\simeq\mathrm{R}\Gamma_{\mathrm{cont}}\big(\Gamma/\mathrm{H},\mathrm{R}\Gamma_{\mathrm{cont}}(\mathrm{H},\underline{\mathrm{RHom}}_{\overline{\mathrm{A}}}(\mathrm{E},i_+\mathrm{Q}))\big)^{\varphi=1}=0.$$

Thus the claim follows. □

We now introduce an assumption which will be our induction step for the main theorem.

Assumption 5.20. Let $(\mathrm{Spf}\,\mathrm{R},\mathcal{M})$ be a framed log affine. Assume that the restriction of $\mathrm{T}_{\text{ét}}$ to $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_{\{i\}})^{h\Gamma}$ is an equivalence onto the full subcategory of $\mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons},\mathrm{D}}((\mathrm{Spf}\,\mathrm{R},\mathcal{M})_\eta,\mathbf{Z}_p)$ consisting of sheaves supported on $\mathrm{V}(\mathrm{T}_i)$ for all i.

As in the toy model, what we will try to do is reduce the general claim to a claim which is one dimension lower, which is where the assumption appears.

Lemma 5.21. *Suppose Assumption 5.20 holds.*

If $\mathcal{E},\mathcal{E}'\in\mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf}\,\mathrm{R},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^\wedge_p)$ satisfy

$$\mathcal{E}(*\mathrm{D})=\mathcal{E}'(*\mathrm{D})=0,$$

then the natural map

$$\mathrm{RHom}(\mathcal{E},\mathcal{E}')\longrightarrow\mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathcal{E}'),\mathrm{T}_{\text{ét}}(\mathcal{E}))$$

is an equivalence.

Moreover, if $\mathcal{E}_0\in\mathsf{Perf}((\mathrm{Spf}\,\mathrm{R},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^\wedge_p)^{\varphi=1}$ and $\mathrm{Q}\in\mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf}\,\mathrm{R},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^\wedge_p)$ satisfies $\mathrm{Q}(*\mathrm{D})=0$, then $\mathrm{RHom}(\mathcal{E}_0,\mathrm{Q})=0$.

Proof. Write $\mathrm{D}=\bigcup_{i=1}^r\mathrm{D}_i$, where $\mathrm{D}_i=\mathrm{V}(\mathrm{T}_i)$. Kashiwara's equivalence identifies the category supported on D_i with $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_{\{i\}})^{h\Gamma}$, and its compatibility with realization identifies the supported comparison with the restriction of $\mathrm{T}_{\text{ét}}$ to this category.

The first statement does not immediately follow from Lemma 5.18, because D need not be smooth. However it is not too far from the statement: we will use localization triangles to break D into smooth components by peeling off pieces supported on D_i one at a time.

Let $\mathcal{E}(*\mathrm{D})=0$. Put $\mathcal{E}_0=\mathcal{E}$ and $\mathcal{E}_i=\mathcal{E}_{i-1}(*\mathrm{D}_i)$. For each i,

$$\mathrm{R}\Gamma_{\mathrm{D}_i}(\mathcal{E}_{i-1})\longrightarrow\mathcal{E}_{i-1}\longrightarrow\mathcal{E}_i$$

is a localization triangle. Indeed if $\mathcal{E}$ is bounded finitely generated unit, then so are $\mathcal{E}(*\mathrm{D}_i)$ and $\mathrm{R}\Gamma_{\mathrm{D}_i}(\mathcal{E})$ by the calculation following Lemma 5.18 (both also inherit continuous Γ actions commuting with Frobenius). Symmetric monoidality and the identification of $\mathrm{T}_{\text{ét}}(\mathcal{O}_{\Delta}(*\mathrm{D}_i))$ with extension by zero from the complement of D_i show $\mathrm{T}_{\text{ét}}$ produces the corresponding Kummer-étale localization triangle. Moreover,

$$\mathcal{E}_r \simeq \mathcal{E} \otimes^{\mathrm{L},\blacksquare} \mathcal{O}_{\Delta}(*\mathrm{D}_1) \otimes^{\mathrm{L},\blacksquare} \cdots \otimes^{\mathrm{L},\blacksquare} \mathcal{O}_{\Delta}(*\mathrm{D}_r) \simeq \mathcal{E}(*\mathrm{D}) = 0.$$

Each $\mathrm{R}\Gamma_{\mathrm{D}_i}(\mathcal{E}_{i-1})$ is supported on D_i. For fixed $\mathcal{E}'$, the comparison is a natural transformation of exact functors

$$\mathrm{RHom}(-, \mathcal{E}') \longrightarrow \mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathcal{E}'), \mathrm{T}_{\text{ét}}(-)).$$

Applying it to the preceding triangles for $\mathcal{E}$ and working downward from $\mathcal{E}_r = 0$ reduces the first assertion to the case where $\mathcal{E}$ is supported on some D_i. We can run a similar reduction for $\mathcal{E}'$. The localization triangle for $\mathcal{E}'$ is

$$\mathrm{R}\Gamma_{\mathrm{D}_i}(\mathcal{E}') \longrightarrow \mathcal{E}' \longrightarrow \mathcal{E}'(*\mathrm{D}_i).$$

Localization gives $\mathrm{RHom}(\mathcal{E}, \mathcal{E}'(*\mathrm{D}_i)) = 0$. One sees $\mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathcal{E}'(*\mathrm{D}_i)), \mathrm{T}_{\text{ét}}(\mathcal{E})) = 0$, so the comparison is therefore reduced to

$$\mathrm{RHom}(\mathcal{E}, \mathrm{R}\Gamma_{\mathrm{D}_i}(\mathcal{E}')) \longrightarrow \mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathrm{R}\Gamma_{\mathrm{D}_i}(\mathcal{E}')), \mathrm{T}_{\text{ét}}(\mathcal{E})).$$

Both crystals are now supported on D_i. Kashiwara's equivalence and full faithfulness of $\mathrm{T}_{\text{ét}}$ on $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_{\{i\}})^{h\Gamma}$ show that this map is an equivalence.

The localization dévissage above shows that every Q with $\mathrm{Q}(*\mathrm{D}) = 0$ lies in the thick subcategory generated by bounded finitely generated unit crystals supported on the smooth divisors D_i. By Lemma 5.18, each such crystal is of the form $i_+\mathrm{Q}'$ for $\mathrm{Q}' \in \mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_{\{i\}})^{h\Gamma}$; hence Lemma 5.19 and exactness of $\mathrm{RHom}(\mathcal{E}_0, -)$ give the last assertion.

The same proof applies in $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_{\Lambda})^{h\Gamma}$, using the remaining boundary coordinates. Its induction hypothesis is full faithfulness of $\mathrm{T}_{\text{ét}}$ on $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_{\Lambda\cup\{i\}})^{h\Gamma}$ for $i \notin \Lambda$. □

We next treat maps between objects supported on the open stratum and its complement.

Lemma 5.22. *Under Assumption 5.20, let*

$$\mathrm{P} \in \mathsf{Perf}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*\mathrm{D}))^{\varphi=1}$$

and let $\mathrm{Q} \in \mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)$ *satisfy* $\mathrm{Q}(*\mathrm{D}) = 0$. *Then the natural map*

$$\mathrm{RHom}(\mathrm{P}, \mathrm{Q}) \longrightarrow \mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathrm{Q}), \mathrm{T}_{\text{ét}}(\mathrm{P}))$$

is an equivalence.

Proof. Let $j : \mathrm{U}_\eta \hookrightarrow (\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_\eta$ be the complement of D. Theorem 5.4 identifies $\mathrm{T}_{\text{ét}}(\mathrm{P})$ with $j_!\mathbb{L}$ for a p-Kummer derived local system $\mathbb{L}$ on the open stratum.

By [DLLZ23, Corollary 6.3.4], the cohomology sheaves of $\mathbb{L}$ extend across the boundary as Kummer-étale local systems. The vanishing in [DLLZ23, Theorem 4.6.1], applied modulo p^n and passed to the inverse limit, makes the spectral sequence

$$\mathrm{E}_2^{a,b} = \mathrm{R}^a j_* \mathrm{H}^b(\mathbb{L}) \Longrightarrow \mathrm{H}^{a+b}(\mathrm{R}j_*\mathbb{L})$$

degenerate. Thus $\mathrm{R}j_*\mathbb{L}$ is derived lisse, and, as in the essential surjectivity component of Theorem 5.4, for every log geometric point $\bar{x}$ and $b \in \mathbf{Z}$,

$$\mathrm{im}\left(\mathrm{I}_{\bar{x}}^{\log,(p')} \longrightarrow \mathrm{Aut}_{\mathbf{Z}_p}\left(\mathrm{H}^b(\mathrm{R}j_*\mathbb{L})_{\bar{x}}\right)\right)$$

is finite. Thus for each cohomology sheaf there is some prime-to-p Kummer cover making it p-Kummer by [IKY26, Proposition 5]. A Postnikov dévissage as in [Hau24, Lemma 5.4.3] and boundedness give a common exponent for these finite images, so a single prime-to-p root cover $\mathrm{Spf}\,\mathrm{R}_\mathrm{N}$ makes $\mathrm{R}j_*\mathbb{L}$ p-Kummer.

Let $f_\mathrm{N} : (\mathrm{Spf}\,\mathrm{R}_\mathrm{N}, \mathcal{M}_\mathrm{N}) \to (\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ be the corresponding map. We claim that there is a commutative diagram

$$\begin{tikzcd}
\mathrm{RHom}(f_\mathrm{N}^*\mathrm{P}, f_\mathrm{N}^*\mathrm{Q})^{h\mu_\mathrm{N}^r} \arrow[r, "\mathrm{T}_{\text{ét}}"] & \mathrm{RHom}(\mathrm{T}_{\text{ét}}(f_\mathrm{N}^*\mathrm{Q}), \mathrm{T}_{\text{ét}}(f_\mathrm{N}^*\mathrm{P}))^{h\mu_\mathrm{N}^r} \\
\mathrm{RHom}(\mathrm{P}, \mathrm{Q}) \arrow[u, "\sim"] \arrow[r, "\mathrm{T}_{\text{ét}}"] & \mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathrm{Q}), \mathrm{T}_{\text{ét}}(\mathrm{P})) \arrow[u, "\sim"']
\end{tikzcd}$$

which would then reduce the claim to showing $\mathrm{RHom}(f_\mathrm{N}^*\mathrm{P}, f_\mathrm{N}^*\mathrm{Q}) \simeq \mathrm{RHom}(\mathrm{T}_{\text{ét}}(f_\mathrm{N}^*\mathrm{Q}), \mathrm{T}_{\text{ét}}(f_\mathrm{N}^*\mathrm{P}))$ via the canonical map induced by $\mathrm{T}_{\text{ét}}$. We first must explain the natural actions of μ_N^r. For the Kummer-étale side, we note $\mathrm{T}_{\text{ét}}$ is compatible with pullbacks and $f_{\mathrm{N},\eta}$ is a μ_N^r-torsor for the Kummer-étale topology; this gives the desired action, and shows the vertical arrow on the right is an equivalence.

For the finitely generated unit (φ, Γ) module or Laurent crystal side it is slightly more subtle. As usual let $\mathrm{A} = \mathrm{R}_0^\square(\!(q-1)\!)_p^\wedge$ be the base so that $\mathrm{D}_{\mathrm{fgu}}((\mathrm{Spf}\,\mathrm{R}, \mathcal{M})_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[1/\mathrm{I}_{\mathbb{\Delta}}]_p^\wedge) \simeq \mathrm{D}_{\mathrm{fgu}}(\mathrm{A})^{h\Gamma}$ under Theorem 4.6, and let A_N be the corresponding base for the prime-to-p Kummer cover $(\mathrm{Spf}\,\mathrm{R}_\mathrm{N}, \mathcal{M}_\mathrm{N})$. Similarly to Proposition 5.15, $\mathrm{A}_\mathrm{N} = \bigoplus_{\alpha \in (\mathbf{Z}/\mathrm{N}\mathbf{Z})^r} \mathrm{A} \cdot \mathrm{S}^\alpha$ with a natural μ_N^r action, which gives $\mathrm{RHom}(f_\mathrm{N}^*\mathrm{P}, f_\mathrm{N}^*\mathrm{Q})$ a μ_N^r-action.

Since N is invertible, for any $\mathrm{M} \in \mathrm{D}_{\mathrm{fgu}}(\mathrm{A})^{h\Gamma}$ the homotopy fixed points of its pullback are computed by taking the degree zero summand in the decomposition $\mathrm{A}_\mathrm{N} = \bigoplus_{\alpha \in (\mathbf{Z}/\mathrm{N}\mathbf{Z})^r} \mathrm{A} \cdot \mathrm{S}^\alpha$. This projector is Γ-equivariant. It is also Frobenius equivariant, since Frobenius permutes the summands by $\alpha \mapsto p\alpha$ and $(p, \mathrm{N}) = 1$. The extension/restriction adjunction and preservation of limits by $\mathrm{RHom}(\mathrm{P}, -)$ (to move $h\mu_\mathrm{N}^r$ inside) therefore give

$$\begin{aligned}
\mathrm{RHom}(f_\mathrm{N}^*\mathrm{P}, f_\mathrm{N}^*\mathrm{Q})^{h\mu_\mathrm{N}^r} &\simeq \mathrm{RHom}\left(\mathrm{P}, (\mathrm{A}_\mathrm{N} \widehat{\otimes}_\mathrm{A}^\mathrm{L} \mathrm{Q})^{h\mu_\mathrm{N}^r}\right) \\
&\simeq \mathrm{RHom}(\mathrm{P}, \mathrm{Q}).
\end{aligned}$$

Pullback compatibility of $\mathrm{T}_{\text{ét}}$ and naturality makes the comparison on the cover μ_{N}^{r}-equivariant, proving commutativity of the diagram and reducing the claim to the prime-to-p Kummer cover.

On the cover $\operatorname{Spf} \mathrm{R}_{\mathrm{N}}$, $\mathrm{R}j_*\mathbb{L}$ is p-Kummer. The contravariant version of the Koshikawa-Yao correspondence [IKY26, Theorem 7] therefore gives a perfect unit Laurent crystal $\mathcal{E}_0$ realizing it. Since $\mathcal{E}_0(*\mathrm{D}_{\mathrm{N}})$ and $f_{\mathrm{N}}^*\mathrm{P}$ both realize $j_! f_{\mathrm{N}}^*\mathbb{L}$, open stratum full faithfulness in Theorem 5.4 gives

$$f_{\mathrm{N}}^*\mathrm{P} \simeq \mathcal{E}_0(*\mathrm{D}_{\mathrm{N}})$$

with $\mathcal{E}_0 \in \mathsf{Perf}((\mathrm{X}_{\mathrm{N}}, \mathcal{M}_{\mathrm{N}})_{\mathbb{A}}, \mathcal{O}_{\mathbb{A}}[1/\mathrm{I}_{\mathbb{A}}]_p^{\wedge})^{\varphi=1}$. We now work on the root cover and omit the pullbacks and subscripts, as we have already made the reduction. Put $\mathcal{E}_{\mathrm{Z}} = \mathrm{R}\Gamma_{\mathrm{D}}(\mathcal{E}_0)$. The localization triangle is

$$\mathcal{E}_{\mathrm{Z}} \longrightarrow \mathcal{E}_0 \longrightarrow \mathrm{P}.$$

The proofs of [EK04b, Propositions 5.11.5 and 6.8.1] show that $\mathcal{E}_{\mathrm{Z}}$ is bounded finitely generated unit and satisfies $\mathcal{E}_{\mathrm{Z}}(*\mathrm{D}) = 0$. The last assertion of Lemma 5.21 gives $\mathrm{RHom}(\mathcal{E}_0, \mathrm{Q}) = 0$.

By construction, $\mathrm{T}_{\text{ét}}(\mathcal{E}_0) \simeq \mathrm{R}j_*\mathbb{L}$. Since $\mathrm{Q}(*\mathrm{D}) = 0$, symmetric monoidality and the calculation preceding Theorem 5.4 give $j^*\mathrm{T}_{\text{ét}}(\mathrm{Q}) = 0$. The adjunction between j^* and $\mathrm{R}j_*$ therefore gives

$$\mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathrm{Q}), \mathrm{T}_{\text{ét}}(\mathcal{E}_0)) \simeq \mathrm{RHom}(j^*\mathrm{T}_{\text{ét}}(\mathrm{Q}), \mathbb{L}) = 0.$$

Applying $\mathrm{RHom}(-, \mathrm{Q})$ to the localization triangle and $\mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathrm{Q}), -)$ to its realization gives a map of fiber sequences

$$\begin{tikzcd}
\mathrm{RHom}(\mathrm{P}, \mathrm{Q}) \arrow[r] \arrow[d] & \mathrm{RHom}(\mathcal{E}_0, \mathrm{Q}) \arrow[r] \arrow[d] & \mathrm{RHom}(\mathcal{E}_{\mathrm{Z}}, \mathrm{Q}) \arrow[d] \\
\mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathrm{Q}), \mathrm{T}_{\text{ét}}(\mathrm{P})) \arrow[r] & \mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathrm{Q}), \mathrm{T}_{\text{ét}}(\mathcal{E}_0)) \arrow[r] & \mathrm{RHom}(\mathrm{T}_{\text{ét}}(\mathrm{Q}), \mathrm{T}_{\text{ét}}(\mathcal{E}_{\mathrm{Z}})).
\end{tikzcd}$$

The two middle terms vanish by the preceding two paragraphs, while the right vertical map is an equivalence by the first assertion of Lemma 5.21. The left vertical map is therefore an equivalence. This proves the comparison on the root cover; by the earlier reduction we are done. □

Proposition 5.23. *Let* $(\operatorname{Spf} \mathrm{R}, \mathcal{M})$ *be a framed log affine, and suppose Assumption 5.20 holds. Then the étale realization*

$$\mathrm{T}_{\text{ét}} : \mathrm{D}_{\text{fgu}}((\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\mathbb{A}}, \mathcal{O}_{\mathbb{A}}[1/\mathrm{I}_{\mathbb{A}}]_p^{\wedge})^{\mathrm{op}} \longrightarrow \mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons},\mathrm{D}}((\mathrm{X}, \mathcal{M})_{\eta}, \mathbf{Z}_p)$$

is an equivalence.

Proof. The hypothesis gives us full faithfulness on full subcategories of $\mathrm{D}_{\text{fgu}}((\mathrm{X}, \mathcal{M})_{\mathbb{A}}, \mathcal{O}_{\mathbb{A}}[1/\mathrm{I}_{\mathbb{A}}]_p^{\wedge})$ that have the form

$$\mathrm{D}_{\text{fgu}}(\mathrm{A}/\mathrm{J}_{\Lambda})^{h\Gamma}$$

with $|\Lambda| = 1$ under Theorem 4.6 (here as usual $\mathrm{A} = \mathrm{R}_0^{\square}((q-1))_p^{\wedge}$ with its Γ action).

The desired claim is for $\Lambda = \emptyset$, so we will show full faithfulness $|\Lambda| = 1$ implies the claim for $\Lambda = \emptyset$. We reduce full faithfulness to maps between two objects supported on D, between two objects supported on the open stratum, and from an object supported on the open stratum to an object supported on D. These cases are Lemma 5.21, Theorem 5.4, and Lemma 5.22, respectively. The middle case does not use the induction hypothesis, but the others do.

Let $\mathcal{E}, \mathcal{E}' \in \mathrm{D}_{\mathrm{fgu}}((\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})$. Put $\mathcal{E}_{\mathrm{Z}} = \mathrm{R}\Gamma_{\mathrm{D}}(\mathcal{E})$ and $\mathcal{E}_{\mathrm{U}} = \mathcal{E}(*\mathrm{D})$, and use primes for the variants applied to $\mathcal{E}'$. It is straightforward to check that $\mathcal{E}_{\mathrm{Z}}$ and $\mathcal{E}_{\mathrm{U}}$ remain finitely generated unit, and similarly for $\mathcal{E}'_{\mathrm{Z}}$ and $\mathcal{E}'_{\mathrm{U}}$. We will later check the open terms are actually in $\mathsf{Perf}((\operatorname{Spf} \mathrm{R}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*\mathrm{D}))^{\varphi=1}$.

The associated localization triangles give a square of fiber sequences

$$\begin{tikzcd}
\mathrm{RHom}(\mathcal{E}_{\mathrm{U}}, \mathcal{E}'_{\mathrm{Z}}) \arrow[r] \arrow[d] & \mathrm{RHom}(\mathcal{E}, \mathcal{E}'_{\mathrm{Z}}) \arrow[r] \arrow[d] & \mathrm{RHom}(\mathcal{E}_{\mathrm{Z}}, \mathcal{E}'_{\mathrm{Z}}) \arrow[d] \\
\mathrm{RHom}(\mathcal{E}_{\mathrm{U}}, \mathcal{E}') \arrow[r] \arrow[d] & \mathrm{RHom}(\mathcal{E}, \mathcal{E}') \arrow[r] \arrow[d] & \mathrm{RHom}(\mathcal{E}_{\mathrm{Z}}, \mathcal{E}') \arrow[d] \\
\mathrm{RHom}(\mathcal{E}_{\mathrm{U}}, \mathcal{E}'_{\mathrm{U}}) \arrow[r] & \mathrm{RHom}(\mathcal{E}, \mathcal{E}'_{\mathrm{U}}) \arrow[r] & \mathrm{RHom}(\mathcal{E}_{\mathrm{Z}}, \mathcal{E}'_{\mathrm{U}}).
\end{tikzcd}$$

In fact every row and column is a fiber sequence. Since $\mathcal{E}_{\mathrm{Z}}(*\mathrm{D}) = 0$ and $\mathcal{E}'_{\mathrm{U}} \simeq \mathcal{E}'_{\mathrm{U}}(*\mathrm{D})$, the lower right term $\mathrm{RHom}(\mathcal{E}_{\mathrm{Z}}, \mathcal{E}'_{\mathrm{U}})$ vanishes. After applying $\mathrm{T}_{\text{ét}}$, let $j : \mathrm{U}_\eta \hookrightarrow (\mathrm{X}, \mathcal{M})_\eta$ denote the complement of D. The adjunction between $j_!$ and j^* identifies the analogous term with

$$\mathrm{RHom}(j^*\mathrm{T}_{\text{ét}}(\mathcal{E}'_{\mathrm{U}}), j^*\mathrm{T}_{\text{ét}}(\mathcal{E}_{\mathrm{Z}})) = 0,$$

since $\mathrm{T}_{\text{ét}}(\mathcal{E}_{\mathrm{Z}})$ is supported on the complement of U. The same vanishing gives $\mathrm{RHom}(\mathcal{E}_{\mathrm{Z}}, \mathcal{E}') \simeq \mathrm{RHom}(\mathcal{E}_{\mathrm{Z}}, \mathcal{E}'_{\mathrm{Z}})$ and $\mathrm{RHom}(\mathcal{E}, \mathcal{E}'_{\mathrm{U}}) \simeq \mathrm{RHom}(\mathcal{E}_{\mathrm{U}}, \mathcal{E}'_{\mathrm{U}})$. Taking the total fiber of the bottom right square by columns gives $\mathrm{RHom}(\mathcal{E}_{\mathrm{U}}, \mathcal{E}'_{\mathrm{Z}})$; by rows, it is the fiber of

$$\mathrm{RHom}(\mathcal{E}, \mathcal{E}') \to \mathrm{RHom}(\mathcal{E}_{\mathrm{Z}}, \mathcal{E}'_{\mathrm{Z}}) \oplus \mathrm{RHom}(\mathcal{E}_{\mathrm{U}}, \mathcal{E}'_{\mathrm{U}})$$

and hence

$$\mathrm{RHom}(\mathcal{E}, \mathcal{E}') \longrightarrow \mathrm{RHom}(\mathcal{E}_{\mathrm{Z}}, \mathcal{E}'_{\mathrm{Z}}) \oplus \mathrm{RHom}(\mathcal{E}_{\mathrm{U}}, \mathcal{E}'_{\mathrm{U}}) \longrightarrow \mathrm{RHom}(\mathcal{E}_{\mathrm{U}}, \mathcal{E}'_{\mathrm{Z}}[1]).$$

after rotating the fiber sequence. Exactness of the étale realization gives the corresponding fiber sequence after applying $\mathrm{T}_{\text{ét}}$, and naturality gives a map between them. To deduce that the map induced by $\mathrm{T}_{\text{ét}}$ on the first term is an equivalence, since it is a fiber sequence we need only check the induced maps on the second and third terms are isomorphisms.

We verify that the open terms satisfy the hypotheses of Theorem 5.4 and Lemma 5.22. Put $\mathrm{A}^\circ = \mathrm{A}[1/\mathrm{T}_1, \ldots, 1/\mathrm{T}_r]_p^{\wedge}$. Theorem 4.6 identifies $\mathcal{E}_{\mathrm{U}}$ with an object $\mathrm{M} \in \mathrm{D}_{\mathrm{fgu}}(\mathrm{A})^{h\Gamma}$ satisfying $\mathrm{M} \simeq \mathrm{A}^\circ \otimes_{\mathrm{A}}^{\mathrm{L},\blacksquare} \mathrm{M}$. Modulo p, the Fitting ideal argument in the proof of Proposition

4.24 makes the minimal roots of the cohomology modules of M vector bundles over A°/p and their root maps isomorphisms. Hence M/p is perfect over A°/p, and therefore M is perfect over A°. Lemma 5.8 then gives

$$\mathcal{E}_U \in \mathsf{Perf}((X, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}(*D))^{\varphi=1}.$$

The same argument of course applies to $\mathcal{E}'_U$.

Lemma 5.21 treats $\mathrm{RHom}(\mathcal{E}_Z, \mathcal{E}'_Z)$ using the induction hypothesis. Theorem 5.4 treats $\mathrm{RHom}(\mathcal{E}_U, \mathcal{E}'_U)$. Lemma 5.22 treats $\mathrm{RHom}(\mathcal{E}_U, \mathcal{E}'_Z[1])$ again under the induction hypothesis. Hence the comparison for $\mathcal{E}$ and $\mathcal{E}'$ is an equivalence.

Finally, we address essential surjectivity. Since the source is idempotent complete and $T_{\text{ét}}$ is exact and fully faithful, its essential image is thick. The target is thickly generated by extensions by zero from the open stratum and by its full subcategories supported on the divisors $V(T_i)$. The former lie in the essential image by Theorem 5.4, and the latter do by Assumption 5.20 and Lemma 5.18. □

Corollary 5.24. *Let* $(\operatorname{Spf} R, \mathcal{M})$ *be a framed log affine. Then*

$$T_{\text{ét}} : D_{\text{fgu}}((\operatorname{Spf} R, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]^{\wedge}_p)^{\text{op}} \longrightarrow D^{(b),\text{pKum}}_{\text{cons,D}}((\operatorname{Spf} R, \mathcal{M})_\eta, \mathbf{Z}_p)$$

is an equivalence.

Proof. We run a descending induction on $|\Lambda|$ for the categories $\left(D_{\text{fgu}}(A/J_\Lambda)^{h\Gamma}\right)^{\text{op}}$, in the notation notation fixed at the start of this subsection. We claim these are equivalent to the full subcategories of $D^{(b),\text{pKum}}_{\text{cons,D}}((\operatorname{Spf} R, \mathcal{M})_\eta, \mathbf{Z}_p)$ supported on $(\bigcap_{i\in\Lambda} D_i)_\eta$ (with its pullback log structure). Equivalently, they should be equivalent to p-Kummer constructible sheaves on $(\bigcap_{i\in\Lambda} D_i)_\eta$ with its induced stratification by restricting the divisor.

In the base case $|\Lambda| = r$, the equivalence can be checked directly. Here this is because the corresponding stratum on the log adic generic fiber is an ordinary framed affine $Z = \bigcap_{i=1}^r D_i$ over a rank $n-r$ torus with a rank r hollow log structure. Equipping the target torus with the hollow chart $\mathbf{N}^r \to 0$, we get an induced strict-étale framing

$$\square : (Z, \mathcal{M}|_Z) \longrightarrow \left(\operatorname{Spf} W(k)[\zeta_p]\langle T^{\pm}_{r+1}, \ldots, T^{\pm}_n\rangle, (\mathbf{N}^r \to 0)^a\right).$$

The torus has a natural perfectoid cover $\operatorname{Spf} W(k)^{\text{cyc}}\langle T^{\pm 1/p^\infty}_{r+1}, \ldots, T^{\pm 1/p^\infty}_n\rangle$ with hollow log structure $\mathbf{N}[1/p]^r \to 0$, so using $\square$ we obtain an associated perfectoid cover $(R^{\square}_0)_{\infty,Z}[1/p]$ again with the chart $\mathbf{N}[1/p]^r \to 0$; we let $(A/J_\Lambda)_{\text{perf}}$ denote $\mathbb{A}_{\text{inf}}((R^{\square}_0)_{\infty,Z}[1/p]^\flat)$. Katz's theorem, tilting, and pro-Kummer-étale descent for the same full group Γ give

$$\left(\mathsf{Perf}((A/J_\Lambda)_{\text{perf}})^{\varphi=1,h\Gamma}\right)^{\text{op}} \xrightarrow{\sim} D^{(b),\text{pKum}}_{\text{lisse}}\left((Z, \mathbf{N}^r)_\eta, \mathbf{Z}_p\right) \xhookrightarrow{i_*} D^{(b)}_{\text{cons,D}}((\operatorname{Spf} R, \mathcal{M})_\eta, \mathbf{Z}_p).$$

In this case the finitely generated unit condition is equivalent to being perfect with unit Frobenius by the Fitting ideal argument of Proposition 4.24. Moreover,

$$D_{\text{fgu}}(A/J_\Lambda)^{h\Gamma} \simeq \mathsf{Perf}((A/J_\Lambda)_{\text{perf}})^{\varphi=1,h\Gamma}$$

by [BS23, Proposition 3.6] and Lemma 2.3; thus the desired claim follows. We note Lemma 4.26 identifies construction with $\mathrm{T}_{\text{ét}}$.

The proof of Proposition 5.23 applied to $\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_\Lambda)^{h\Gamma}$ supplies the induction step, deducing the claim for Λ from the claims for $\Lambda \cup \{i\}$ for all $i \notin \Lambda$. Strictly speaking, as stated the proposition would only apply with the group Γ/Γ_Λ, where $\Gamma_\Lambda = \prod_{i\in\Lambda} \mathbf{Z}_p(1)_i \simeq \mathbf{Z}_p(1)^{|\Lambda|}$. We actually need to deduce the equivalence

$$\mathrm{D}_{\mathrm{fgu}}(\mathrm{A}/\mathrm{J}_\Lambda)^{h\Gamma,\mathrm{op}} \simeq \mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons},\mathrm{D}|_{\mathrm{Z}_\Lambda}}((\mathrm{Z}_\Lambda, \mathcal{M}|_{\mathrm{Z}_\Lambda})_\eta, \mathbf{Z}_p)$$

for the induction step; here the target has a hollow component to its log structure, and is therefore not covered by Assumption 3.1. Fortunately, the proof of the open stratum comparison (Theorem 5.4) still applies with the full group Γ, or equivalently where a rank $|\Lambda|$ hollow log structure is allowed on the open stratum. The argument is identical once we can pass fully faithfully to the perfectoid level; thus since we only need a local comparison, the only modification needed is the decompletion result in Proposition 5.9. The reduction to cohomology lemmas holds verbatim, so one simply observes that the helper lemmas (applying Lemma 5.10) work with the larger group. □

The main theorem immediately follows.

Theorem 5.25. *Let* $(\mathrm{X}, \mathcal{M})$ *satisfy Assumption 3.1 over* $\mathrm{W}(k)[\zeta_p]$. *Let* $\mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons},\mathrm{D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$ *denote the full subcategory of* p*-Kummer constructible sheaves in* $\mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$. *Then*

$$\mathrm{T}_{\text{ét}} : \mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M})_{\mathbb{A}}, \mathcal{O}_{\mathbb{A}}[1/\mathrm{I}_{\mathbb{A}}]^\wedge_p)^{\mathrm{op}} \xrightarrow{\sim} \mathrm{D}^{(b),\mathrm{pKum}}_{\mathrm{cons},\mathrm{D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$$

is a symmetric monoidal equivalence.

Proof. By Lemma 3.4 and strict-étale descent (for the source Theorem 4.11 and for the target Lemma 4.20), we may first work on a framed log affine $(\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$. The claim is then Corollary 5.24.

The fact that the equivalence is symmetric monoidal can also be tested locally, where via Lemma 4.26 it reduces to symmetric monoidality in Theorem 2.6. □

We must also understand perverse t-exactness. We use a definition of the perverse t-structure on the Kummer-étale side of the correspondence which is similar to [BH22].

Definition 5.26. For each connected logarithmic stratum $j_{\mathrm{U}} : \mathrm{U}_\eta \hookrightarrow (\mathrm{X}, \mathcal{M})_\eta$, put $d_{\mathrm{U}} = \dim(\mathrm{U}_\eta)$. The p-perverse t-structure of [Gab04] on the full subcategory of p-Kummer sheaves in $\mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}}((\mathrm{X}, \mathcal{M})_\eta, \mathbf{Z}_p)$ is defined by

$${}^p\mathrm{D}^{\leq 0} = \{\mathrm{K} \mid j_{\mathrm{U}}^*\mathrm{K} \in \mathrm{D}^{\leq -d_{\mathrm{U}}}(\mathrm{U}_\eta, \mathbf{Z}_p) \text{ for every } \mathrm{U}_\eta\},$$

$${}^p\mathrm{D}^{\geq 0} = \{\mathrm{K} \mid j_{\mathrm{U}}^!\mathrm{K} \in \mathrm{D}^{\geq -d_{\mathrm{U}}}(\mathrm{U}_\eta, \mathbf{Z}_p) \text{ for every } \mathrm{U}_\eta\}.$$

Write ${}^{p}\mathrm{H}^{i}$ for its cohomology functors. The p^{+}-perverse t-structure is

$$ {}^{p^+}\mathrm{D}^{\leq 0} = \{\mathrm{K} \in {}^{p}\mathrm{D}^{\leq 1} \mid {}^{p}\mathrm{H}^{1}(\mathrm{K}) \text{ is locally bounded } p\text{-power torsion}\}, $$
$$ {}^{p^+}\mathrm{D}^{\geq 0} = \{\mathrm{K} \mid \mathrm{K}/p \in {}^{p}\mathrm{D}^{\geq 0}((\mathrm{X},\mathcal{M})_\eta, \mathbf{F}_p)\}. $$

One may check these t-structures restrict to the p-Kummer subcategory. The argument of [BH22, Construction 4.3 and Proposition 4.6], applied to the defining bounds after j_{U}^{*} and $j_{\mathrm{U}}^{!}$, gives the following reduction criteria. For every $n \geq 1$, derived reduction modulo p^n is right t-exact for p-perversity and left t-exact for p^{+}-perversity. Moreover, $\mathrm{K} \in {}^{p}\mathrm{D}^{\leq 0}$ if and only if $\mathrm{K}/p \in {}^{p}\mathrm{D}^{\leq 0}((\mathrm{X},\mathcal{M})_\eta, \mathbf{F}_p)$, while $\mathrm{K} \in {}^{p^+}\mathrm{D}^{\geq 0}$ if and only if $\mathrm{K}/p \in {}^{p}\mathrm{D}^{\geq 0}((\mathrm{X},\mathcal{M})_\eta, \mathbf{F}_p)$.

Using the equivalence with the essential image, the main content of the following proposition can be reduced to the analogous perverse t-exactness of the Emerton-Kisin correspondence.

Proposition 5.27. *Assume that $(\mathrm{X},\mathcal{M})_\eta$ is pure of dimension d. The equivalence of Theorem 5.25 with its essential image, shifted by $[d]$, is t-exact for the standard t-structure on the source and the p^{+}-perverse t-structure of Definition 5.26 on the target.*

Proof. The source t-structure is strict-étale local by Corollary A.19, and the same holds for the bounds in Definition 5.26. We may therefore work on a framed log affine.

We may work modulo p and therefore use the base $\overline{\mathrm{A}} = \overline{\mathrm{R}}^{\square}(\!(q-1)\!)$, as one may repeat the argument of Theorem 2.5 to obtain the general case. We consider $\mathrm{M} \in \mathrm{D}_{\mathrm{fgu}}(\overline{\mathrm{A}})^{h\Gamma}$, using Theorem 4.6.

For $\Lambda \subseteq \{1,\ldots,r\}$, put

$$ \overline{\mathrm{A}}_\Lambda = \big(\overline{\mathrm{A}}/(\mathrm{T}_i \mid i \in \Lambda)\big)\,[1/\mathrm{T}_j \mid j \notin \Lambda,\ j \leq r] $$

and write M_Λ for the derived base change $\mathrm{M} \otimes^{\mathrm{L}}_{\overline{\mathrm{A}}} \overline{\mathrm{A}}_\Lambda$.

Let $j_\Lambda : \mathrm{X}_{\eta,\Lambda} \hookrightarrow (\mathrm{X},\mathcal{M})_\eta$ be the corresponding stratum, of dimension $d-|\Lambda|$. Lemma 4.23 and the Fitting ideal argument of Proposition 4.24 show that each $\mathrm{H}^i(\mathrm{M}_\Lambda)$ is finite projective over $\overline{\mathrm{A}}_\Lambda$. Compatibility with restriction, Lemma 4.26, and the contravariant Katz correspondence therefore yield

$$ j_\Lambda^{*}\mathrm{T}_{\text{ét},\mathbf{F}_p}(\mathrm{M})[d] \in \mathrm{D}^{\leq -(d-|\Lambda|)} \iff \mathrm{M}_\Lambda \in \mathrm{D}^{\geq -c_\Lambda}. $$

We claim that $\mathrm{M} \in \mathrm{D}^{\geq 0}$ if and only if $\mathrm{M}_\Lambda \in \mathrm{D}^{\geq -|\Lambda|}$ for every Λ, where we use the ordinary t-structure on $\mathrm{D}_{\mathrm{fgu}}(\overline{\mathrm{A}})^{h\Gamma}$. Applying the coconnectivity criterion of [ATJLS10, Lemma 3.1] along the coordinate stratification, using $i_\Lambda^{!}\mathrm{M} \simeq \mathrm{M}_\Lambda[-|\Lambda|]$, gives the claim.

Definition 5.26 now gives

$$ \mathrm{T}_{\text{ét},\mathbf{F}_p}(\mathrm{M})[d] \in {}^{p}\mathrm{D}^{\leq 0} \iff \mathrm{M} \in \mathrm{D}^{\geq 0}. $$

Thus the contravariant equivalence exchanges coconnectivity on the source with perverse connectivity on the target. By orthogonality, it also exchanges connectivity with perverse coconnectivity, proving mod p perverse t-exactness. $\square$

It is also possible to extract ordinary perverse sheaves from the correspondence. We warn that extracting ordinary constructible sheaves in the derived sense cannot work without altering the crystal side of the correspondence, as these are not a full subcategory of the derived category of Kummer-étale sheaves (simply because Kummer-étale cohomology and étale cohomology differ).

Remark 5.28. Following [KL16, Theorem 7.5.6], call $\mathcal{E} \in \mathrm{D}_{\mathrm{fgu}}((\mathrm{X},\mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})$ corresponding to an object in the ordinary heart of constructible sheaves *effective* if, on every framed strict-étale chart, its associated (φ,Γ)-module M satisfies the following condition for every boundary coordinate T_i: the ith $\mathbf{Z}_p(1)$-factor acts trivially on

$$(\mathrm{A}/(\mathrm{T}_i)) \otimes_{\mathrm{A}}^{\mathrm{L},\blacksquare} \mathrm{M}$$

in the derived category. This condition is strict-étale local and independent of the framing. Under Theorem 5.25, it is equivalent to trivial Kummer monodromy along each local branch. Theorem 5.25 therefore gives an equivalence

$$\mathrm{T}_{\text{ét}} : \mathrm{D}_{\mathrm{fgu}}((\mathrm{X},\mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})^{\mathrm{eff,op}} \xrightarrow{\sim} \mathrm{D}^{\heartsuit}_{\mathrm{cons,D}}(\mathrm{X}_{\eta,\text{proét}}, \mathbf{Z}_p).$$

where on the target the ordinary t-structure is used.

In the derived setting, it seems plausible that one could take the "effective part" of an RHom to obtain the derived category of constructible sheaves along D.

We remark that trivial modifications of the previous arguments allow one to deduce the theorem over a slightly more general base. In particular our restriction to $\mathrm{W}(k)[\zeta_p]$ up until this point is mainly for expository reasons to avoid tracking additional variables and cases throughout.

Corollary 5.29. *Theorem 5.25 and all preceding results stated over* $\mathrm{W}(k)[\zeta_p]$ *admit corresponding variants over* $\mathrm{W}(k)[\zeta_{p^\ell}]$ *for every* $\ell \geq 0$.

Proof. Corollary 3.17 is the only nontrivial change needed in the argument. With this, one can deduce Theorem 4.11 verbatim (as it uses nothing about Γ). The following arguments are insensitive to the change in prismatic ideal or Γ, except for Proposition 5.9; however there the reduction is to vanishing of a cohomology group, so changing Γ from $(1+p\mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)^n$ to $\mu_{p-1} \times (1+p\mathbf{Z}_p) \ltimes \mathbf{Z}_p(1)^n$ does not change the result by using Hochschild-Serre. □

Remark 5.30. The entire local theory seems to have a variant which will work over $\mathbf{A}_K$, but unfortunately the globalization step does not appear to work. One would want to have a prism $(\mathbf{A}_K^+, \varphi^n([p]_q))$ which satisfies

$$\mathbf{A}_K^+[1/\varphi^n([p]_q)]_p^{\wedge} \simeq \mathbf{A}_K.$$

How it seems such a prism cannot exist in general without dropping some desirable properties. See [Wat26, Appendix 2, Proposition 23] for more details. The argument does use that the mod p reduction should be E_K^+, the ring of integers of the field of norms. It is still possible that $\mathbf{A}_K^+$ exists by dropping this condition, but this seems unnatural to do.

The following explicit (φ, Γ)-module gives a constructible sheaf with nontrivial gluing between the two strata.

Example 5.31. Let $(\mathrm{X}, \mathcal{M}) = (\mathrm{Spf}\, \mathbf{Z}_p[\zeta_p]\langle \mathrm{T}\rangle, \mathbf{N})$ with log structure induced by $\mathrm{D} = \{\mathrm{T} = 0\}$, and let $j : \mathrm{U}_\eta = \mathrm{X}_\eta \setminus \mathrm{D}_\eta \hookrightarrow (\mathrm{X}_\eta, \mathbf{N})$. Put $\overline{\mathrm{R}} = \mathbf{F}_p\langle \mathrm{T}\rangle$ and $\mathrm{A} = \overline{\mathrm{R}}^{\square}(\!(q-1)\!)$. For $\gamma \in \Gamma$, define

$$\mathrm{H}_\gamma = -\sum_{n \geq 0} \varphi^n(\gamma(\mathrm{T}) - \mathrm{T}).$$

If $\gamma = (a, b)$, then $\gamma(\mathrm{T}) - \mathrm{T} = (q^{pb} - 1)\mathrm{T}$, whose $(q-1)$-adic valuation is at least p. Frobenius multiplies this valuation by p, so the series converges. Since the actions of Γ and φ on A commute, one has $\mathrm{H}_{\gamma\delta} = \mathrm{H}_\gamma + \gamma(\mathrm{H}_\delta)$ and $\mathrm{H}_\gamma - \varphi(\mathrm{H}_\gamma) = \mathrm{T} - \gamma(\mathrm{T})$.

Set

$$\mathrm{M} = \mathrm{A}[1/\mathrm{T}]e_1 \oplus \mathrm{A}e_2.$$

Define $\varphi(e_1) = e_1$ and $\varphi(e_2) = e_2 + \mathrm{T}e_1$, and define $\gamma(e_1) = e_1$ and $\gamma(e_2) = e_2 + \mathrm{H}_\gamma e_1$. The preceding identities give a continuous semilinear Γ-action (for M equipped with its weak topology) commuting with Frobenius, so we obtain a finitely generated unit (φ, Γ)-module.

There is an exact sequence of (φ, Γ)-modules

$$0 \longrightarrow \mathrm{A}[1/\mathrm{T}]e_1 \longrightarrow \mathrm{M} \longrightarrow \mathrm{A}\bar{e}_2 \longrightarrow 0.$$

It remains nonsplit after inverting T: a Frobenius-equivariant splitting would give $g \in \mathrm{A}[1/\mathrm{T}]$ satisfying

$$g - \varphi(g) = \mathrm{T}.$$

The $(q-1)$-adic valuation forces $v_{q-1}(g) = 0$. Its constant coefficient would then satisfy $a(\mathrm{T}) - a(\mathrm{T}^p) = \mathrm{T}$ in $\mathbf{F}_p[\mathrm{T}^{\pm 1}]$, which recursively forces the coefficients of T^{p^n} to equal one for every n (which is not possible).

Since $p\mathrm{M} = 0$, set

$$\mathscr{F} := \mathrm{T}_{\text{ét}, \mathbf{F}_p}(\mathrm{M}) \simeq \mathrm{T}_{\text{ét}, \mathbf{Z}_p}(\mathrm{M})[1].$$

Contravariance and the previous exact sequence give a short exact sequence of $\mathbf{F}_p$-sheaves

$$0 \longrightarrow \mathbf{F}_{p,(\mathrm{X}_\eta, \mathbf{N})} \longrightarrow \mathscr{F} \longrightarrow j_!\mathbf{F}_{p,\mathrm{U}_\eta} \longrightarrow 0.$$

Its restriction to U_η is still nonsplit, so in particular this gives a nontrivial example (i.e. not a direct sum of trivial examples).

6. Pushforward compatibility

6.1. **Six functors and Laurent stacks.** The category $D_{\blacksquare}((X,\mathcal{M})_{\Delta},\mathcal{O}_{\Delta})$ is large enough that the six-functor results imply their Laurent analogues. For ordinary formal schemes, this is carried out in [Kip26b, Kip26a] by constructing an analytic stack $X^{\Delta,\blacksquare}$. We construct the logarithmic analogue and compare it with our site-theoretic category, and deduce several useful six functors inputs that we will use.

Let $(X,\mathcal{M})$ satisfy Assumption 3.1 over $\mathcal{O}_K$ for a finite extension $K/\mathbf{Q}_p$. As explained in more detail in Definition A.1, the ordinary log prismatic stack is defined by the fiber product

$$\begin{tikzcd} (X,\mathcal{M})^{\Delta} \arrow[r] \arrow[d] & X^{\Delta} \arrow[d] \\ \mathbf{Z}_p^{\Delta}\times \mathrm{Log} \arrow[r] & \mathrm{Log}^{\Delta} \end{tikzcd}$$

where Log is the formal p-completion of Olsson's log stack [Ols03]. This is a relative prismatic stack over Log, viewed as a "δ-stack", so the constructions of [Kip26a] apply by base change.

In [Kip26a, Construction 2.2.7], the éid topology is defined. Roughly, one takes the union of the étale topology and integral descendable maps; as explained in [Kip26a, 2.2.5], integrality for formal schemes is tested on ordinary schemes, so formal affines one checks it on rings modulo an ideal of definition. We will always use integrality in this sense.

Definition 6.1. For a formal prestack $\mathcal{Y}$ over $\mathbf{Z}_p^{\Delta}$, let $a_{\text{éid}}\mathcal{Y}$ denote the éid sheafification of the induced presheaf on $\mathrm{fSch}^{\mathrm{aff}}_{/\mathbf{Z}_p^{\Delta}}$ from [Kip26a, Construction 2.2.7]. We define its *analytic realization* by

$$\mathcal{Y}^{\blacksquare} := (-)^{\blacksquare}_{\mathrm{Kipp}}(a_{\text{éid}}\mathcal{Y}),$$

where

$$(-)^{\blacksquare}_{\mathrm{Kipp}} : (\mathbf{Z}_p^{\Delta})_{\text{éid}} \longrightarrow \mathrm{AnStack}_{/\mathbf{Z}_p^{\Delta,\blacksquare}}$$

is the colimit-preserving left exact functor of [Kip26a, Theorem 2.3.3(A) and §3.6.1]. We note $(-)^{\blacksquare}$ preserves finite limits, and $(\mathrm{Spf}\, B)^{\blacksquare} = \mathrm{Spa}(B,B)$ for a bounded prism B.

Strictly speaking, it is defined differently in [Kip26a]; however this definition will be more convenient for us, as the log prismatic stack is a type of relative stack so we need a version of analytification which also works there. We briefly explain why it agrees with Kipp's construction. In [Kip26a] he introduces $\mathsf{Sh}(\mathrm{Spf}(\mathbf{Z}_p)_{\infty,\text{ét}})$, or sheaves for the topology generated by infinite root covers and étale covers. The functor $\mathrm{Spf}\, R \mapsto \mathrm{Spa}(\Delta_R,\Delta_R)$ sends covers in this topology to effective epimorphisms of analytic stacks. It is shown that semiperfectoids form a basis of the topology $\mathrm{Spf}(\mathbf{Z}_p)_{\infty,\text{ét}}$, and so one obtains a colimit preserving functor

$$\mathsf{Sh}(\mathrm{Spf}(\mathbf{Z}_p)_{\infty,\text{ét}}) \to \mathrm{AnStack}_{/\mathbf{Z}_p^{\Delta,\blacksquare}}$$

induced by $\mathrm{Spf}\,\mathrm{R} \mapsto \mathrm{Spa}(\Delta_{\mathrm{R}}, \Delta_{\mathrm{R}})$ on the semiperfectoid basis. This is fine in the non-logarithmic setting, but unfortunately in the logarithmic setting semiperfectoids are problematic (they may not even have maps to the log prismatic stack, as they often lack fine log structures).

Fortunately the two constructions are the same, although this is not obvious. What is required is to show that for an infinite root/étale cover $\mathrm{Spf}\,\mathrm{R} \to \mathrm{X}$ that

$$\mathrm{Spf}\,\Delta_{\mathrm{R}} \to \mathrm{X}^{\Delta}$$

is an effective epimorphism in the éid topos. Then we may use the corresponding colimit presentation coming from the Čech nerve to present X^{Δ} in the éid topos, and then since both functors are colimit preserving we get the same colimit presentation in analytic stacks.

In fact we claim more generally that for an infinite root/étale cover $\mathrm{Spf}\,\mathrm{A} \to \mathrm{Spf}\,\mathrm{B}$ that after prismatization and éid sheafification we get an effective epimorphism in the éid topos. It suffices to check this generating covers; for étale covers it's clear, so one must show

$$a_{\text{éid}}\mathrm{V}^{\Delta}_{\infty} \to a_{\text{éid}}\mathrm{V}^{\Delta}$$

is an effective epimorphism where $\mathrm{V} = \mathrm{Spf}\,\mathbf{Z}_p[t_i]_{i\in\mathrm{I}}$ and $\mathrm{V}_\infty = \mathrm{Spf}\,\mathbf{Z}_p[t_i^{1/p^\infty}]_{i\in\mathrm{I}}$. In the affine case it is easy to see the éid sheafification is in fact redundant, because $\mathrm{W}(-)$ is an éid sheaf in spectra by [Kip26a, 3.2.1] (in particular one may apply the argument in the easy affine case of [BL22, Lemma 7.3]). The argument of [BL22, Lemma 6.3] applies to see we get an effective epimorphism. Let consider an R-point of V^{Δ}. One explicit cover $\mathrm{R} \to \mathrm{S}$ constructed there (for S-point lift of $\mathrm{V}^{\Delta}_{\infty}$) is in fact an éid cover. Indeed unwinding their proof one first constructs, after orienting the Cartier-Witt divisor as $(d) \to \mathrm{W}(\mathrm{R})$ by localizing,

$$\mathrm{B} = \mathrm{W}(\mathrm{R})\langle x_i^{1/p^\infty}\rangle_{i\in\mathrm{I}}\left\{\frac{x_i - \tilde{t}_i}{d}\right\}^{\wedge}_{(p,d)}$$

where $\tilde{t}_i$ are lifts of t_i to $\mathrm{W}(\mathrm{R})$ and $\delta(x_i^{1/p^n}) = 0$. Then $\mathrm{S} = \mathrm{B} \otimes_{\mathrm{W}(\mathrm{R})} \mathrm{R}$ (underived by their flatness result). Descendability follows from the proof of [Kip26b, Proposition 3.2.1.8]. For integrality, Lemma A.14 shows that $\pi_0(\mathrm{B}/(p,d))$ is integral over the algebra obtained from $\pi_0(\mathrm{W}(\mathrm{R})/(p,d))$ by adjoining compatible p-power roots of the images of the $\tilde{t}_i$. This algebra is itself integral over $\pi_0(\mathrm{W}(\mathrm{R})/(p,d))$. Since the image of (p,d) in R is nilpotent, base change and lifting across nilpotent ideals show that $\mathrm{R} \to \mathrm{S}$ is integral, hence an éid cover.

Thus, we use Definition 6.1 throughout. In particular, we may then safely define

$$\begin{tikzcd}
(\mathrm{X}, \mathcal{M})^{\Delta,\blacksquare} \arrow[r] \arrow[d] & \mathrm{X}^{\Delta,\blacksquare} \arrow[d] \\
(\mathbf{Z}_p^{\Delta} \times \mathrm{Log})^{\blacksquare} \arrow[r] & \mathrm{Log}^{\Delta,\blacksquare}
\end{tikzcd}$$

where the notation on the bottom is interpreted according to Definition 6.1, and thus includes éid sheafification. Finite limit preservation in that definition shows one may also construct $(\mathrm{X},\mathcal{M})^{\Delta,\blacksquare}$ by applying analytic realization to the already existing formal stack over $\mathbf{Z}_p^{\Delta}$.

The logarithmic Hodge-Tate substack is the fiber product $(\mathrm{X},\mathcal{M})^{\Delta,\blacksquare}\times_{\mathbf{Z}_p^{\Delta,\blacksquare}}\mathbf{Z}_p^{\mathrm{HT},\blacksquare}$, where $\mathbf{Z}_p^{\mathrm{HT},\blacksquare}$ is as in [Kip26a]. Following [Kip26a, Construction 4.3.7 and §4.3.9], we write $(\mathrm{X},\mathcal{M})^{\mathcal{L},\blacksquare}$ for the algebraic nonvanishing locus of the Hodge-Tate ideal, further base changed along $\mathrm{colim}_n \operatorname{Spa}(\mathbf{Z}/p^n)\to \operatorname{Spa}\mathbf{Z}_p$. Thus, after pullback along an orientable prism $(\mathrm{B},(d))$, it is given by

$$\mathrm{colim}_{n\geq 1}\operatorname{Spa}\left(((\mathrm{B}/^{\mathrm{L}}p^n)[1/d],\mathrm{B}/^{\mathrm{L}}p^n)\right).$$

Since this stack is obtained from $\mathrm{X}^{\Delta,\blacksquare}$ by base change, the relevant results of [Kip26a] apply to it. See [Kip26a, §1.1] for cohomological smoothness and weak cohomological properness in six-functor formalisms.

Lemma 6.2. *Let $f:\mathrm{X}\to\mathrm{Y}$ be a morphism of analytic stacks, and suppose it is cohomologically smooth and weakly cohomologically proper. Then given a square*

$$\begin{tikzcd} \mathrm{X}\times_{\mathrm{Y}}\mathrm{Z} \arrow[r, "\iota_{\mathrm{X}}"] \arrow[d, "f_{\mathrm{Z}}"] & \mathrm{X} \arrow[d, "f"] \\ \mathrm{Z} \arrow[r, "\iota_{\mathrm{Y}}"] & \mathrm{Y} \end{tikzcd}$$

the map $f_{\mathrm{Z}}:\mathrm{X}\times_{\mathrm{Y}}\mathrm{Z}\to\mathrm{Z}$ is cohomologically smooth and weakly cohomologically proper. Moreover, $f_{\mathrm{Z}}^!\mathcal{O}_{\mathrm{Z}}\simeq\iota_{\mathrm{X}}^*f^!\mathcal{O}_{\mathrm{Y}}$.

Proof. The property of being weakly cohomologically proper is stable under arbitrary base change by [Kip26a, Remark 1.1.8], and cohomological smoothness is defined after arbitrary base change, so f_{Z} is again cohomologically smooth and weakly cohomologically proper. Put $\omega_f=f^!\mathcal{O}_{\mathrm{Y}}$. For a cohomologically smooth map, ω_f is invertible, the natural map $\omega_f\otimes f^*(-)\to f^!(-)$ is an equivalence, and ω_f commutes with arbitrary base change; compare [Kip26b, Definition 1.2.1.1]. Hence the displayed square gives $\omega_{f_{\mathrm{Z}}}\simeq\iota_{\mathrm{X}}^*\omega_f$. Applying this to the unit object gives $f_{\mathrm{Z}}^!\mathcal{O}_{\mathrm{Z}}\simeq\omega_{f_{\mathrm{Z}}}\simeq\iota_{\mathrm{X}}^*f^!\mathcal{O}_{\mathrm{Y}}$. ☐

Lemma 6.3. *Let $f:\mathrm{X}\to\mathrm{Y}$ be a cohomologically smooth and weakly cohomologically proper morphism of analytic stacks, and put $\omega_f=f^!\mathcal{O}_{\mathrm{Y}}$. Suppose further that*

$$\mathrm{R}f_!\simeq\mathrm{R}f_*$$

via the canonical map.

Then $\mathrm{R}f_*$ preserves dualizable objects, and for every dualizable object $\mathcal{E}\in\mathrm{D}_\blacksquare(\mathrm{X})$ there is a natural equivalence

$$(\mathrm{R}f_*\mathcal{E})^\vee\simeq\mathrm{R}f_*(\mathcal{E}^\vee\otimes\omega_f).$$

Proof. For every $\mathcal{K} \in \mathrm{D}_{\blacksquare}(\mathrm{Y})$, one has

$$\underline{\mathrm{RHom}}_{\mathrm{Y}}(\mathrm{R}f_*\mathcal{E}, \mathcal{K}) \simeq \mathrm{R}f_*\underline{\mathrm{RHom}}_{\mathrm{X}}(\mathcal{E}, f^!\mathcal{K}) \simeq \mathrm{R}f_*(\mathcal{E}^\vee \otimes \omega_f) \otimes \mathcal{K}.$$

The first isomorphism follows from $\underline{\mathrm{RHom}}_{\mathrm{Y}}(\mathrm{R}f_!\mathcal{E}, \mathcal{K}) \simeq \mathrm{R}f_*\underline{\mathrm{RHom}}_{\mathrm{X}}(\mathcal{E}, f^!\mathcal{K})$ (which one can check by showing maps into each are the same, and using the tensor-Hom adjunction and the projection formula) and $\mathrm{R}f_! \simeq \mathrm{R}f_*$, while the second is just using the formula $f^!(-) \simeq \omega_f \otimes \mathrm{L}f^*(-)$.

Thus $\mathrm{R}f_*\mathcal{E}$ is dualizable with dual $\mathrm{R}f_*(\mathcal{E}^\vee \otimes \omega_f)$. □

Lemma 6.4. *Let $f : (\mathrm{X}, \mathcal{M}_{\mathrm{X}}) \to (\mathrm{Y}, \mathcal{M}_{\mathrm{Y}})$ be a strict log smooth proper morphism of bounded fs log p-adic formal schemes of pure relative dimension d. Then the induced maps on $(\mathrm{X}, \mathcal{M}_{\mathrm{X}})^{\mathbb{\Delta},\blacksquare}$ and $(\mathrm{X}, \mathcal{M}_{\mathrm{X}})^{\mathcal{L},\blacksquare}$ are cohomologically smooth and weakly cohomologically proper, and on both stacks one has*

$$f^!\mathcal{O} \simeq \mathcal{O}\{d\}[2d].$$

Moreover $\mathrm{R}f_! \simeq \mathrm{R}f_*$ via the canonical map.

Proof. The underlying map of formal schemes is smooth and proper. Theorem 4.1.3(C),(F) and Example 4.3.2 of [Kip26a] therefore show that the induced map

$$\mathrm{X}^{\mathbb{\Delta},\blacksquare} \to \mathrm{Y}^{\mathbb{\Delta},\blacksquare}$$

is cohomologically smooth and weakly cohomologically proper.[6] We show in addition that there is a canonical equivalence $\mathrm{R}f_! \simeq \mathrm{R}f_*$.

We use $\mathrm{D}_{\blacksquare}$ to denote the six functor formalism on analytic stacks.[7] Let π_2 denote the second projection from $\mathrm{X}^{\mathbb{\Delta},\blacksquare} \times_{\mathrm{Y}^{\mathbb{\Delta},\blacksquare}} \mathrm{X}^{\mathbb{\Delta},\blacksquare}$. Weak cohomological properness implies that f is $\mathrm{D}_{\blacksquare}$-prim, so [Mik25, Corollary 1.28] gives a codualizing object

$$\delta_f \simeq \mathrm{R}\pi_{2,*}\mathrm{R}\Delta_{f,!}\mathcal{O}$$

and a natural equivalence $\mathrm{R}f_!(\delta_f \otimes -) \xrightarrow{\sim} \mathrm{R}f_*$.

Thus it remains to trivialize δ_f canonically. Since $\pi_2 \circ \Delta_f = \mathrm{id}$, it is enough to construct a canonical equivalence $\mathrm{R}\Delta_{f,!} \simeq \mathrm{R}\Delta_{f,*}$. Since $f : \mathrm{X} \to \mathrm{Y}$ is smooth and separated, the diagonal $\mathrm{X} \to \mathrm{X} \times_{\mathrm{Y}} \mathrm{X}$ on the formal schemes is a regular closed immersion and hence locally of finite presentation. The induced map Δ_f on analytifications of prismatic stacks is therefore !-able by [Kip26a, Theorem 4.1.3(A) and Example 4.3.2]. Using [Mik25, Lemma 1.50] (see the footnote in [Mik25, Definition 1.10] for why it applies to analytic stacks), the desired equivalence holds provided that Δ_f is, universally $\mathrm{D}_{\blacksquare}(-)^*$-locally on its target, represented by morphisms in the class P of [Mik25, Definition 1.5(2)]. This class consists of affine morphisms $\mathrm{AnSpec}(\mathcal{B}) \to \mathrm{AnSpec}(\mathcal{A})$ for which the analytic structure on $\mathcal{B}$ is induced from that on $\mathcal{A}$, so it simply asks for the morphism to be affine proper (see [Kip26a, §2.1.10]) in our setting.

[6] We note [Kip26a, Lemma 3.5.6] is false for a general connective perfect complex. In its applications to smooth morphisms, the relevant cotangent complex is a vector bundle, in which case there is no issue.

[7] We abuse notation here somewhat, as of course in general $\mathrm{D}(\mathrm{X})$ for an analytic stack has nothing to do with solid quasicoherent sheaves. However in all of our use cases this will be the case.

We verify this remaining hypothesis. Since the functor $X \mapsto X^{\Delta,\blacksquare}$ preserves finite limits by [Kip26a, Construction 3.7.1], the target of Δ_f is $(X \times_Y X)^{\Delta,\blacksquare}$. Work on the semiperfectoid atlas $\mathrm{Spa}(\Delta_{R_0}, \Delta_{R_0})$ of this target from [Kip26a, Construction 3.7.1]. After a Zariski refinement, we can assume f is smooth of pure relative dimension d (as a map of formal schemes), so the diagonal is a regular immersion of codimension d. Thus, after this localization, the pullback of Δ_f on analytic stacks is induced by a derived quotient $S_0 = R_0/(g_1, \ldots, g_d)$. Choose a perfectoid ring R surjecting onto R_0, lift the g_i to $f_i \in R$, and put $S = R/(f_1, \ldots, f_d)$, again as a derived quotient. The setup is summarized by the following cartesian squares:

$$\begin{tikzcd}
X^{\Delta,\blacksquare} \arrow[r, "\Delta_f"] & X^{\Delta,\blacksquare} \times_{Y^{\Delta,\blacksquare}} X^{\Delta,\blacksquare} \\
\mathrm{Spa}(\Delta_{S_0}, \Delta_{S_0}) \arrow[u] \arrow[r, "\Delta_{f,R_0}"] \arrow[d] & \mathrm{Spa}(\Delta_{R_0}, \Delta_{R_0}) \arrow[u] \arrow[d] \\
\mathrm{Spa}(\Delta_S, \Delta_S) \arrow[r] & \mathrm{Spa}(\Delta_R, \Delta_R).
\end{tikzcd}$$

It is therefore enough to show that $\mathrm{Spa}(\Delta_S, \Delta_S) \longrightarrow \mathrm{Spa}(\Delta_R, \Delta_R)$ is affine proper. Indeed, by construction $\mathrm{Spa}(\Delta_{R_0}, \Delta_{R_0})$ is an effective epimorphism of analytic stacks, hence a universal $D^*_\blacksquare$-cover. After a further Zariski refinement, orient the prismatic ideal as $I = (\widetilde{d})$ and choose lifts $\widetilde{f}_i \in \Delta_R$ of $f_i \in R \simeq \Delta_R/I$. One then has

$$\Delta_S \simeq \Delta_R \left\{ \frac{\widetilde{f}_1}{\widetilde{d}}, \ldots, \frac{\widetilde{f}_d}{\widetilde{d}} \right\}^{\wedge}_{\delta,(p,\widetilde{d})}.$$

Lemma A.14 and [Kip26a, Example 2.1.13] then show that $\mathrm{Spa}(\Delta_S, \Delta_S) \longrightarrow \mathrm{Spa}(\Delta_R, \Delta_R)$ is affine proper.

Thus we have verified the hypotheses of [Mik25, Lemma 1.50] for Δ_f, and hence obtain a natural equivalence $R\Delta_{f,!} \simeq R\Delta_{f,*}$. It follows canonically that

$$\delta_f \simeq R\pi_{2,*} R\Delta_{f,!}\mathcal{O} \simeq R\pi_{2,*} R\Delta_{f,*}\mathcal{O} \simeq \mathcal{O},$$

where the last equivalence uses $\pi_2 \circ \Delta_f = \mathrm{id}$. Substituting this identification in the preceding equivalence gives $Rf_! \simeq Rf_*$. By the descent construction of [Mik25, Lemma 1.50], this equivalence is independent of the chosen atlas; its compatibility with arbitrary base change follows from [Mik25, Remark 1.49 and Corollary 1.31(2)].

Theorem 4.1.3(F) of [Kip26a] identifies the dualizing object with the d-fold Tate object. Kipp's homological shift $[n]$ is our cohomological shift $[-n]$, so his formula $\mathcal{O}\{d\}[-2d]$ becomes $\mathcal{O}\{d\}[2d]$. Strictness gives

$$(X, \mathcal{M}_X)^{\Delta,\blacksquare} \simeq X^{\Delta,\blacksquare} \times_{Y^{\Delta,\blacksquare}} (Y, \mathcal{M}_Y)^{\Delta,\blacksquare}.$$

The claimed result for $(X, \mathcal{M}_X)^{\Delta,\blacksquare}$ follows from Lemma 6.2 and base-change compatibility of $Rf_! \simeq Rf_*$. For the Laurent stack $(X, \mathcal{M}_X)^{\mathcal{L},\blacksquare}$, the claim follows by base change. □

The following proposition bridges the gap between the stacky approach and the site-theoretic approach we have taken thus far.

Proposition 6.5. *For every log formal scheme* $(\mathrm{X}, \mathcal{M})$ *satisfying Assumption 3.1, there is a natural equivalence*

$$\mathrm{an}_{(\mathrm{X},\mathcal{M})} : \mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}) \xrightarrow{\sim} \mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})^{\mathcal{L},\blacksquare}).$$

Proof. For every orientable object $(\mathrm{B}, (d), \mathcal{N})^a$ of the strict log prismatic site, applying analytic realization (Definition 6.1) to the map of Lemma A.4 gives

$$\mathrm{Spa}(\mathrm{B}, \mathrm{B}) \longrightarrow (\mathrm{X}, \mathcal{M})^{\Delta,\blacksquare}.$$

The Hodge-Tate pullback is cut out by d, so Definition A.11 gives

$$\mathrm{D}_{\blacksquare}\big(\mathrm{Spa}(\mathrm{B}, \mathrm{B}) \times_{(\mathrm{X},\mathcal{M})^{\Delta,\blacksquare}} (\mathrm{X}, \mathcal{M})^{\mathcal{L},\blacksquare}\big) \simeq \mathrm{D}_{\blacksquare}(\mathrm{B}[1/d]_p^{\wedge}).$$

Indeed, the analytic localization on the left is computed by the pairs $((\mathrm{B}/^{\mathrm{L}}p^n)[1/d], \mathrm{B}/^{\mathrm{L}}p^n)$. Taking the limit of the compatible pullback functors defines

$$\mathrm{res}_{(\mathrm{X},\mathcal{M})} : \mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})^{\mathcal{L},\blacksquare}) \longrightarrow \mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}).$$

Both sides satisfy strict-étale descent. For the target this is Corollary A.18. For the source, for $(\mathrm{X}, \mathcal{M})^{\Delta,\blacksquare}$ by strictness this reduces to [Kip26a, Theorem 2.3.3(A),(B) and Construction 3.7.1] which give étale descent for $\mathrm{X}^{\Delta,\blacksquare}$ (i.e. étale covers are sent to effective epimorphisms of analytic stacks). The same holds for the Laurent stack $(\mathrm{X}, \mathcal{M})^{\mathcal{L},\blacksquare}$, using the analogous result for the Hodge-Tate stack from [Kip26a]. Descent for $\mathrm{D}_{\blacksquare}$ then proves the claim. Lemma 3.4 therefore reduces us to a framed log affine; choose the logarithmic q-de Rham Breuil-Kisin prism $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf}\,\mathrm{A}})^a$ attached to the framing, and let $\mathrm{A}^{(m)}$ be its Čech nerve in the strict log prismatic site. By rigidity of prisms, we also write I for its extension to every term of this nerve.

Proposition A.22 and Laurent base change give

$$\mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})^{\mathcal{L},\blacksquare}) \simeq \varprojlim_{[m]\in\Delta} \mathrm{D}_{\blacksquare}(\mathrm{A}^{(m)}[1/\mathrm{I}]_p^{\wedge}).$$

Theorem A.15 identifies this limit with $\mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})$. Both equivalences are induced by the augmentations, and their composite is $\mathrm{res}_{(\mathrm{X},\mathcal{M})}$. Thus restriction is an equivalence, with inverse $\mathrm{an}_{(\mathrm{X},\mathcal{M})}$. □

Lemma 6.6. *For every* $(\mathrm{X}, \mathcal{M})$ *satisfying Assumption 3.1, there is a functorial symmetric monoidal equivalence*

$$\mathrm{an}_{(\mathrm{X},\mathcal{M})} : \mathsf{Perf}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}) \xrightarrow{\sim} \mathsf{Perf}((\mathrm{X}, \mathcal{M})^{\mathcal{L},\blacksquare}).$$

It is compatible with pullback and $\underline{\mathrm{RHom}}$*. If* $(\mathrm{Y}, \mathcal{N})$ *also satisfies Assumption 3.1,* $f : (\mathrm{X}, \mathcal{M}) \to (\mathrm{Y}, \mathcal{N})$ *is strict log smooth and proper, and* $f_{\mathcal{L}}$ *is the induced map of Laurent analytic stacks, then the right adjoint* $\mathrm{R}f_{\Delta,*}$ *to pullback on crystals preserves perfect objects and*

$$\mathrm{an}_{(\mathrm{Y},\mathcal{N})} \mathrm{R}f_{\Delta,*} \simeq \mathrm{R}f_{\mathcal{L},*} \, \mathrm{an}_{(\mathrm{X},\mathcal{M})} \,.$$

Proof. All functors in the construction of Proposition 6.5 are symmetric monoidal. Thus its equivalence restricts to dualizable objects and is compatible with $\underline{\mathrm{RHom}}$. Its compatibility with pullback gives

$$\begin{tikzcd} \mathrm{D}_{\blacksquare}((\mathrm{Y},\mathcal{N})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}) \arrow[r, "\mathrm{an}_{(\mathrm{Y},\mathcal{N})}"] \arrow[d, "\mathrm{L}f^*"] & \mathrm{D}_{\blacksquare}((\mathrm{Y},\mathcal{N})^{\mathcal{L},\blacksquare}) \arrow[d, "\mathrm{L}f^*_{\mathcal{L}}"] \\ \mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}) \arrow[r, "\mathrm{an}_{(\mathrm{X},\mathcal{M})}"] & \mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})^{\mathcal{L},\blacksquare}). \end{tikzcd}$$

Taking right adjoints gives the asserted comparison of pushforwards on the ambient categories. Lemma 6.4 gives $\mathrm{R}f_{\mathcal{L},!} \simeq \mathrm{R}f_{\mathcal{L},*}$, so Lemma 6.3 shows that $\mathrm{R}f_{\mathcal{L},*}$ preserves perfect objects. The comparison and the equivalence $\mathrm{an}_{(\mathrm{Y},\mathcal{N})}$ give the same assertion for $\mathrm{R}f_{\Delta,*}$. □

We also obtain Poincaré duality.

Corollary 6.7. *Let* $f : (\mathrm{X}, \mathcal{M}) \to (\mathrm{Y}, \mathcal{N})$ *be as in Lemma 6.4, and suppose both source and target satisfy Assumption 3.1. Transport* $f^!$ *on perfect objects from the map* $(\mathrm{X}, \mathcal{M})^{\mathcal{L},\blacksquare} \to (\mathrm{Y}, \mathcal{N})^{\mathcal{L},\blacksquare}$ *through Lemma 6.6. Then* $f^!\mathcal{G} \simeq f^*\mathcal{G}\{d\}[2d]$ *for* $\mathcal{G} \in \mathsf{Perf}((\mathrm{Y},\mathcal{N})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$*, the right adjoint to pullback on crystals preserves perfect complexes, and*

$$\mathrm{R}f_*\mathcal{E} \simeq \underline{\mathrm{RHom}}_{\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}}\big(\mathrm{R}f_*(\mathcal{E}^{\vee}\{d\}), \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}[-2d]\big)$$

for every $\mathcal{E} \in \mathsf{Perf}((\mathrm{X},\mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$*.*

Proof. Transporting Lemma 6.4 through Lemma 6.6 gives $f^!\mathcal{G} \simeq f^*\mathcal{G}\{d\}[2d]$. Lemma 6.3, together with the pushforward comparison in Lemma 6.6, gives

$$(\mathrm{R}f_*\mathcal{E})^{\vee} \simeq \mathrm{R}f_*(\mathcal{E}^{\vee}\{d\})[2d].$$

Dualizing, this equivalence gives the claim. □

We also note that the Poincaré pairing is compatible with Frobenius. The identification $f^!\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p} \simeq \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}\{d\}[2d]$ is Frobenius equivariant since it is upgraded to an identification of F-gauges in [Kip26a, Theorem 4.1.3(F)].

6.2. **Pushforward compatibility.** Let $f : (\mathrm{X}, \mathcal{M}_{\mathrm{X}}) \to (\mathrm{Y}, \mathcal{M}_{\mathrm{Y}})$ be a strict log smooth proper morphism of pure relative dimension d between log p-adic formal schemes whose adic generic fibers have pure dimensions d_{X} and d_{Y}. We assume $(\mathrm{X}, \mathrm{M}_{\mathrm{X}})$ and $(\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})$ satisfy Assumption 3.1 over one of the bases $\mathrm{Spf}\, \mathrm{W}(k)[\zeta_{p^\ell}]$ covered in Corollary 5.29; closed strata are equipped with their pullback log structures.

Let $d = d_{\mathrm{X}} - d_{\mathrm{Y}}$. In such a setup, define

$$\mathrm{R}f_+(-) := \mathrm{R}f_*(-)\{d\}[d].$$

We will show that $\mathrm{R}f_+$ induces a functor

$$\mathrm{R}f_+ : \mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathcal{M}_{\mathrm{X}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^\wedge_p) \to \mathrm{D}_{\mathrm{fgu}}((\mathrm{Y}, \mathcal{M}_{\mathrm{Y}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^\wedge_p).$$

After this, the pushforward compatibility result is formal.

We first need some preliminaries on relative stacks. Let $(\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})$ be a framed log affine with associated logarithmic q-de Rham prism (A, I), so $\mathrm{Y} = \mathrm{Spf}(\mathrm{A}/\mathrm{I})$, and let $(\mathrm{X}, \mathrm{M}_{\mathrm{X}}) \to (\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})$ be strict smooth. By Proposition A.22, the map $\mathrm{Spa}(\mathrm{A}, \mathrm{A}) \to (\mathrm{Y}, \mathcal{M}_{\mathrm{Y}})^{\Delta,\blacksquare}$ is a representable proper effective epimorphism. Define $(\mathrm{X}/\mathrm{A})^{\Delta,\blacksquare}$ via the cartesian square

$$\begin{tikzcd} (\mathrm{X}/\mathrm{A})^{\Delta,\blacksquare} \arrow[r] \arrow[d, "f_{\mathrm{A}}"] & (\mathrm{X}, \mathcal{M}_{\mathrm{X}})^{\Delta,\blacksquare} \arrow[d] \\ \mathrm{Spa}(\mathrm{A}, \mathrm{A}) \arrow[r, "\rho_{\mathrm{A}}"] & (\mathrm{Y}, \mathcal{M}_{\mathrm{Y}})^{\Delta,\blacksquare} \end{tikzcd}$$

where the bottom map is the effective epimorphism of Proposition A.22. We define $(\mathrm{X}/\mathrm{A})^{\mathcal{L},\blacksquare}$ by further base change over $\mathrm{Spa}(\mathrm{A}, \mathrm{A})$ to

$$(\mathrm{Spf}(\mathrm{A}/\mathrm{I})/\mathrm{A})^{\mathcal{L},\blacksquare} = \mathrm{colim}_n\, \mathrm{Spa}((\mathrm{A}/p^n)[1/\mathrm{I}], \mathrm{A}/p^n).$$

Equivalently, we replace all terms in the cartesian square with their Laurent versions. The method of Proposition 6.5 identifies $\mathsf{Perf}((\mathrm{X}/\mathrm{A})^{\mathcal{L},\blacksquare})$ with $\mathsf{Perf}((\mathrm{X}/\mathrm{A})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^\wedge_p)$, by using the Čech presentation.

There is a natural Frobenius φ on $(\mathrm{X}/\mathrm{A})^{\mathcal{L},\blacksquare}$ over φ_{A} on $(\mathrm{Spf}(\mathrm{A}/\mathrm{I})/\mathrm{A})^{\mathcal{L},\blacksquare}$. The superscript (φ) denotes base change along this φ_{A}, also for charts and their coefficient rings. If f_{A} denotes the structure map, relative Frobenius is the induced map $\mathrm{Fr} = (\varphi, f_{\mathrm{A}}) : (\mathrm{X}/\mathrm{A})^{\mathcal{L},\blacksquare} \to (\mathrm{X}/\mathrm{A})^{\mathcal{L},\blacksquare,(\varphi)}$.

Lemma 6.8. *In the above setting, relative Frobenius induces a fully faithful functor*

$$\mathrm{Fr}^* : \mathsf{Perf}((\mathrm{X}/\mathrm{A})^{\mathcal{L},\blacksquare,(\varphi)}) \longrightarrow \mathsf{Perf}((\mathrm{X}/\mathrm{A})^{\mathcal{L},\blacksquare}).$$

Proof. The projection formula immediately reduces the claim to proving that the unit $\mathcal{O} \to \mathrm{RFr}_*\mathcal{O}$ is an equivalence.

We may also work strict-étale locally by descent of the source and target: thus we assume that X is a framed log affine relative to $(\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})$ with relative coordinates $\mathrm{T}_1, \ldots, \mathrm{T}_d$. More precisely, we have

$$(\mathrm{X}, \mathrm{M}_{\mathrm{X}}) \to (\mathrm{Y}, \mathrm{M}_{\mathrm{Y}}) \times_{\mathrm{Spf}\, \mathrm{W}(k)[\zeta_{p^\ell}]} \widehat{\mathbf{G}}_m^d \to \mathbb{T}_r^{n+d}$$

where $\widehat{\mathbf{G}}_m^d$ has a trivial log structure. Thus $(\mathrm{X}, \mathrm{M}_{\mathrm{X}})$ inherits a framing, and has a corresponding q-de Rham prism with ring B^+ and Laurentized ring $\mathrm{B} = \mathrm{B}^+[1/(q-1)]_p^\wedge$. A straightforward variant of Proposition 3.12 gives a proper morphism $\rho : [\mathrm{AnSpec}\, \mathrm{B}/\Gamma] \to (\mathrm{X}/\mathrm{A})^{\mathcal{L},\blacksquare}$ with a fully faithful pullback where $\Gamma = \mathbf{Z}_p^d$ fixes A and $\gamma_i(\mathrm{T}_j) = q^{p^\ell \delta_{ij}} \mathrm{T}_j$.

As in the strategy of Proposition 3.12, to verify $\mathcal{O} \to \mathrm{RFr}_* \mathcal{O}$ is an isomorphism we may base change by a proper effective epimorphism. One considers the cartesian square

$$\begin{tikzcd}
{[\mathrm{AnSpec}\, \mathrm{C}/\Gamma]} \arrow[r] \arrow[dd] & {[\mathrm{AnSpec}\, \mathrm{B}/\Gamma]} \arrow[d, "\rho"] \\
 & (\mathrm{X}/\mathrm{A})^{\mathcal{L},\blacksquare} \arrow[d, "\mathrm{Fr}"] \\
\mathrm{AnSpec}\, \mathrm{B}^{(\varphi)} \arrow[r] & (\mathrm{X}/\mathrm{A})^{\mathcal{L},\blacksquare,(\varphi)}
\end{tikzcd}$$

Using Proposition A.22 and the definition of the relative stack the bottom horizontal map is also a proper effective epimorphism, and ρ is as well. We make two reductions: first, that $\mathcal{O} \to \mathrm{R}\rho_* \mathcal{O}$ is an equivalence, so it suffices to check $\mathcal{O} \to \mathrm{R}(\mathrm{Fr} \circ \rho)_* \mathcal{O}$ is an equivalence. Second, as in Proposition 3.12 we may use proper base change to reduce to the claim that the canonical map

$$\mathrm{B}^{(\varphi)} \to \mathrm{R}\Gamma_\blacksquare(\Gamma, \mathrm{C})$$

is an equivalence. Indeed, one can check Fr is affine proper using [Kip26a, Example 2.1.13] (one may check integrality modulo p where it is clear). Thus we must make a single calculation similar to Lemma 3.11.

Using Lemma A.6 and finite limit compatibility in Definition 6.1 we may identify C with the laurentization of

$$\mathrm{C}^+ \simeq \mathrm{B}^+[u_1^{\pm 1}, \ldots, u_d^{\pm 1}] \left\{ \frac{u_1 - 1}{[p]_{q^{p^\ell}}}, \ldots, \frac{u_d - 1}{[p]_{q^{p^\ell}}} \right\}_{\delta,(p,q-1)}^\wedge .$$

Here $\delta(u_i) = 0$ and $u_i = \mathrm{T}_i^p / \mathrm{T}_i^{(\varphi)}$, where $\mathrm{T}_i^{(\varphi)}$ belongs to the fixed target copy of $\mathrm{B}^{+,(\varphi)}$; thus $\gamma_i(u_j) = q^{p^{\ell+1}\delta_{ij}} u_j$. We regard B^+ as a module over B^φ, where it decomposes naturally as

$$\mathrm{B}^+ \simeq \bigoplus_{\beta \in \{0,\ldots,p-1\}^d} \mathrm{B}^{+,(\varphi)} \mathrm{T}^\beta .$$

The ring C is p-completely nuclear over $\mathrm{B}^{(\varphi)}$, so Lemma 3.9 and (1) compute its cohomology by the Koszul complex of $\gamma_1 - 1, \dots, \gamma_d - 1$. We use the toric η-basis from Definition 3.7 directly, as in our situation $\Gamma = \mathbf{Z}_p^d$ and this is easier to work with.

Explicitly, put $\eta_{i,m} = \prod_{j=0}^{m-1}(u_i - q^{p^{\ell+1}j})/([p]_{q^{p^\ell}}^m [m]_{q^{p^{\ell+1}}}!)$ and $\eta_\alpha = \prod_i \eta_{i,\alpha_i}$. Using Lemma 3.8, with q replaced by q^{p^ℓ}, we obtain

$$\mathrm{C}^+ \simeq \widehat{\bigoplus_{\substack{\beta \in \{0,\dots,p-1\}^d \\ \alpha \in \mathbf{N}^d}}} \mathrm{B}^{+,(\varphi)} \mathrm{T}^\beta \eta_\alpha.$$

where the completion is $(p, q-1)$-adic. This decomposition is Γ-equivariant after Laurentizing.

One checks $(\gamma_i - 1)\eta_\alpha = (q^{p^\ell} - 1)u_i \eta_{\alpha'}$, where α' is the same multi-index but with the ith position lowered by one, and negative indices mean zero. Since $q^{p^\ell} - 1$ becomes a unit after Laurentization and p-completion, we may divide the Koszul differentials by it. For $\beta = 0$ the normalized operators are unit multiples of the lowering operators; thus the $\beta = 0$ summand yields a copy of $\mathrm{B}^{(\varphi)}$ in degree zero. When $\beta \neq 0$, choose i with $\beta_i \neq 0$. Since $\gamma_i \cdot \mathrm{T}^\beta = q^{p^\ell \beta_i} \mathrm{T}^\beta$, the normalized formula is

$$\frac{\gamma_i - 1}{q^{p^\ell} - 1}(\mathrm{T}^\beta \eta_\alpha) = \mathrm{T}^\beta \left([\beta_i]_{q^{p^\ell}} \eta_\alpha + q^{p^\ell \beta_i} u_i \eta_{\alpha'} \right).$$

On the integral lattice this becomes modulo $(p, q-1)$ multiplication by β_i plus the lowering operator, which is invertible because the latter is locally nilpotent on the reduced direct sum. Completeness lifts this inverse, so every nonzero β-summand is acyclic. Using the Koszul complex it follows that $\mathrm{B}^{(\varphi)} \xrightarrow{\sim} \mathrm{R}\Gamma_{\blacksquare}(\Gamma, \mathrm{C})$, as desired. □

The previous proof indeed shows more, but we will only need the statement for Perf in what follows.

Proposition 6.9. *Let $f : (\mathrm{X}, \mathcal{M}_\mathrm{X}) \to (\mathrm{Y}, \mathcal{M}_\mathrm{Y})$ be a strict log smooth proper morphism between log formal schemes satisfying Assumption 3.1 over $\mathrm{W}(k)[\zeta_{p^\ell}]$, with $\ell \geq 0$. For every $\mathcal{E} \in \mathsf{Perf}((\mathrm{X}, \mathcal{M}_\mathrm{X})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^\wedge)$, the canonical comparison is an equivalence*

$$\varphi^* \mathrm{R} f_* \mathcal{E} \xrightarrow{\sim} \mathrm{R} f_* \varphi^* \mathcal{E}.$$

Proof. By Proposition 6.5 and proper base change from Lemma 6.4 we may reduce to the case where $(\mathrm{Y}, \mathcal{M}_\mathrm{Y})$ is a framed log affine; note here we also use strict-étale descent from Corollary A.18.

We may evaluate the canonical comparison morphism on a logarithmic q-de Rham chart $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf}\,\mathrm{A}})$ of $(\mathrm{Y}, \mathcal{M}_\mathrm{Y}) = (\mathrm{Spf}\,\mathrm{A}/\mathrm{I}, \mathrm{M}_{\mathrm{Spf}\,\mathrm{A}/\mathrm{I}})$. This evaluation is conservative, so we need only check the claim there.Strictness makes X smooth over $\mathrm{Spf}\,\mathrm{A}/\mathrm{I}$. Writing f_A for the structure map $(\mathrm{X}/\mathrm{A})^{\mathcal{L},\blacksquare} \to \mathrm{AnSpec}\,\mathrm{A}[1/\mathrm{I}]^{\wedge}_p = (\mathrm{Spf}\,\mathrm{A}/\mathrm{I}/\mathrm{A})^{\mathcal{L},\blacksquare}$, we have a commutative diagram

$$\begin{tikzcd}
(\mathrm{X}/\mathrm{A})^{\mathcal{L},\blacksquare} \arrow[r, "\mathrm{Fr}"] \arrow[dr, "f_{\mathrm{A}}"'] & (\mathrm{X}/\mathrm{A})^{\mathcal{L},\blacksquare,(\varphi)} \arrow[r, "\widetilde{\varphi}_{\mathrm{A}}"] \arrow[d, "f^{(\varphi)}_{\mathrm{A}}"] & (\mathrm{X}/\mathrm{A})^{\mathcal{L},\blacksquare} \arrow[d, "f_{\mathrm{A}}"] \\
& \mathrm{AnSpec}\,\mathrm{A}[1/\mathrm{I}]^{\wedge}_p \arrow[r, "\varphi_{\mathrm{A}}"] & \mathrm{AnSpec}\,\mathrm{A}[1/\mathrm{I}]^{\wedge}_p
\end{tikzcd}$$

whose right square is cartesian, so $\widetilde{\varphi}_{\mathrm{A}}$ is the base change of φ_{A} along f_{A}, and $\widetilde{\varphi}_{\mathrm{A}} \circ \mathrm{Fr} = \varphi$. Put $\mathcal{E}^{(\varphi)} = \widetilde{\varphi}^*_{\mathrm{A}}\mathcal{E}$. Proper base change and Lemma 6.8 give

$$\begin{aligned}
\varphi^*_{\mathrm{A}}\mathrm{R}f_{\mathrm{A},*}\mathcal{E} &\simeq \mathrm{R}f^{(\varphi)}_{\mathrm{A},*}\mathcal{E}^{(\varphi)} \\
&\xrightarrow{\sim} \mathrm{R}f^{(\varphi)}_{\mathrm{A},*}\mathrm{RFr}_*\mathrm{Fr}^*\mathcal{E}^{(\varphi)} \simeq \mathrm{R}f_{\mathrm{A},*}\varphi^*\mathcal{E}.
\end{aligned}$$

Conservativity finishes the proof. □

Theorem 6.10. *Let $f : (\mathrm{X}, \mathrm{M}_{\mathrm{X}}) \to (\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})$ be strict log smooth proper, where the source and target satisfy Assumption 3.1 over $\mathrm{Spf}\,\mathrm{W}(k)[\zeta_{p^\ell}]$. Then $\mathrm{R}f_+$ induces a functor*

$$\mathrm{R}f_+ : \mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathrm{M}_{\mathrm{X}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p) \to \mathrm{D}_{\mathrm{fgu}}((\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p).$$

Proof. Since the finitely generated unit category is stable under shifts and twists we may test the assertion for $\mathrm{R}f_*$. We may also test the assertion modulo p by derived Nakayama, and further reduce to the case where the p-torsion input $\mathcal{E}$ lies in the heart. Then by Corollary 4.13, there is a coherent root $\mathcal{E}_0$ of $\mathcal{E}$. But then since $\mathrm{R}f_* \simeq \mathrm{R}f_!$ canonically in this situation on the Laurent stacks and Proposition 6.5, the pushforward commutes with colimits. In particular,

$$\mathrm{R}f_*(\mathrm{colim}_n\,\varphi^{*n}\mathcal{E}_0) \simeq \mathrm{colim}_n\,\mathrm{R}f_*\varphi^{*n}\mathcal{E}_0.$$

By Proposition 6.9 we see this is $\mathrm{colim}_n\,\varphi^{*n}\mathrm{R}f_*\mathcal{E}_0$. Using Lemma 6.6, we see $\mathrm{R}f_*\mathcal{E}_0$ is perfect, which proves the result is finitely generated unit.

□

Theorem 6.11. *Let $f : (\mathrm{X}, \mathrm{M}_{\mathrm{X}}) \to (\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})$ be a strict log smooth proper morphism of pure relative dimension d where both satisfy Assumption 3.1 over $\mathrm{Spf}\,\mathrm{W}(k)[\zeta_{p^\ell}]$ whose log adic generic fibers are pure of dimensions d_{X} and d_{Y} respectively. Let D_{X} and D_{Y} be horizontal normal crossings divisors inducing M_{X} and M_{Y}. There is a commutative diagram*

$$\begin{tikzcd}
\mathrm{D}_{\mathrm{fgu}}((\mathrm{X}, \mathrm{M}_{\mathrm{X}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)^{\mathrm{op}} \arrow[r, "\mathrm{T}_{\text{ét},\mathrm{X}}[d_{\mathrm{X}}]"] \arrow[d, "\mathrm{R}f_*\{d\}[d]"] & \mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}_{\mathrm{X}}}((\mathrm{X}, \mathrm{M}_{\mathrm{X}})_{\eta}, \mathbf{Z}_p) \arrow[d, "\mathrm{R}f_{\eta,*}"] \\
\mathrm{D}_{\mathrm{fgu}}((\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)^{\mathrm{op}} \arrow[r, "\mathrm{T}_{\text{ét},\mathrm{Y}}[d_{\mathrm{Y}}]"] & \mathrm{D}^{(b)}_{\mathrm{cons},\mathrm{D}_{\mathrm{Y}}}((\mathrm{Y}, \mathrm{M}_{\mathrm{Y}})_{\eta}, \mathbf{Z}_p).
\end{tikzcd}$$

Proof. Theorem 6.10 and Corollary 4.15 allow us to restrict the relevant functors to the finitely generated unit categories. For $\mathcal{E}$ on X and $\mathcal{G}$ on Y, transporting Lemma 6.4 through Proposition 6.5 and using the projection formula give

$$\begin{aligned}\mathrm{RHom}(\mathrm{R}f_+\mathcal{E}[d], \mathcal{G}) &\simeq \mathrm{RHom}(\mathrm{R}f_*(\mathcal{E}\{d\}[2d]), \mathcal{G}) \\ &\simeq \mathrm{RHom}(\mathcal{E}\{d\}[2d], f^!\mathcal{G}) \simeq \mathrm{RHom}(\mathcal{E}, \mathrm{L}f^*\mathcal{G}).\end{aligned}$$

Thus $\mathrm{R}f_+[d]$ is left adjoint to $\mathrm{L}f^*$. Frobenius base change and the Frobenius-normalized trace make the units and counits in this calculation Frobenius compatible, so the adjunction restricts to the unit categories. The functor $\mathrm{R}f_{\eta,*}$ preserves p-Kummer constructible complexes, so the adjunction $\mathrm{L}f_\eta^* \leftrightarrows \mathrm{R}f_{\eta,*}$ restricts to these categories. Indeed, proper smooth pushforward preserves lissity on each logarithmic stratum. If g is a saturated log perfectoid test object from Definition 4.19 and f' is the base change of f, then proper base change and strictness identify $g^*\mathrm{R}f_{\eta,*}\mathbb{K}$ with $\epsilon^*\mathrm{R}f'_{\mathrm{pro\acute{e}t},*}\mathbb{L}$ whenever the pullback of $\mathbb{K}$ to the source of f' is $\epsilon^*\mathbb{L}$.

Pullback compatibility of $\mathrm{T}_{\acute{e}t}$ carries the opposite of the first adjunction to an adjunction with left adjoint $\mathrm{L}f_\eta^*$. Uniqueness of right adjoints therefore gives

$$\mathrm{T}_{\acute{e}t,Y}(\mathrm{R}f_+\mathcal{E}[d]) \simeq \mathrm{R}f_{\eta,*}\mathrm{T}_{\acute{e}t,X}(\mathcal{E}).$$

On framed strict-étale charts this is the local (φ, Γ) comparison, and both sides satisfy strict-étale descent; hence the equivalence is the global comparison. Since $\mathrm{T}_{\acute{e}t}$ is contravariant and $d_X = d_Y + d$, this is the asserted commutative diagram. □

Appendix A. Log prismatic stacks

A.1. **The log prismatic stack.** In this appendix by formal stacks we mean (accessible) fpqc sheaves in groupoids on animated p-nilpotent rings.

We denote by Log the p-completion of Olsson's stack of fine log structures [Ols03, Theorem 1.1]. For every fine monoid P, the toric stack

$$\mathcal{A}_P = [\mathrm{Spf}\, \mathbf{Z}_p[P]/\,\mathrm{Spf}\, \mathbf{Z}_p[P^{gp}]]$$

classifies fine log structures equipped with a map from P which fpqc locally lifts to a chart by [Ols03, Proposition 5.14], and these chart stacks cover Log by [Ols03, Corollary 5.25]. One may check that these, and in fact Log, are formal δ-stacks in the sense of [AKN26, Construction 10.20]. We let Log^{Δ} be its prismatization.

The general philosophy behind the log prismatic stack is easy to motivate. The data of a log formal scheme X with a fine log structure amounts to a map $X \to \mathrm{Log}$. Moreover in this interpretation many logarithmic cohomology theories $\mathrm{H}_{\log}(-)$ can often be produced by taking a relative version $\mathrm{H}(X/\mathrm{Log})$ using the ordinary cohomology theory $\mathrm{H}(-)$. For example logarithmic de Rham cohomology can be recovered this way. In [AKN26] they define prismatic stacks relative to δ-rings. More generally, if $X \to \mathcal{S}$ is a morphism to a formal δ-stack, we write

$$(X/\mathcal{S})^{\Delta} := X^{\Delta} \times_{\mathcal{S}^{\Delta}} (\mathcal{S} \times \mathbf{Z}_p^{\Delta})$$

for its *relative prismatization.* The point we explain next is what the map $\mathbf{Z}_p^{\Delta} \times \mathcal{S} \to \mathcal{S}^{\Delta}$ is.

We will use the same construction with $\mathcal{S} = \mathrm{Log}$, which we explain in more detail. We must specify a map $\mathrm{Log} \times \mathbf{Z}_p^{\Delta} \to \mathrm{Log}^{\Delta}$. Let R be a p-nilpotent ring and let $\mathcal{M}_{\mathrm{R}} \in \mathrm{Log}(\mathrm{R})$. The Witt vector adjunction and the δ-stack structure on Log give

$$\mathrm{Log}(\mathrm{R}) \simeq \mathrm{Map}_{\delta}(\mathrm{Spf\,W(R)}, \mathrm{Log}) \longrightarrow \mathrm{Map}(\mathrm{Spf\,W(R)}, \mathrm{Log}).$$

Let $\mathcal{M}_{\mathrm{W(R)}}$ be the resulting log structure. If $\mathrm{P} \to \mathcal{M}_{\mathrm{R}}$ is a fine chart with structure map $\alpha : \mathrm{P} \to \mathrm{R}$, then locally

$$\mathcal{M}_{\mathrm{W(R)}} = \left(\mathrm{P} \xrightarrow{m \mapsto [\alpha(m)]} \mathrm{W(R)}\right)^a.$$

We then define the map as

$$\pi_{\mathrm{Log}}(\mathcal{M}_{\mathrm{R}}, \mathrm{I} \to \mathrm{W(R)}) = \left(\mathrm{I} \to \mathrm{W(R)}, \mathcal{M}_{\mathrm{W(R)}}|_{\mathrm{Spec(W(R)/^{L}I)}}\right).$$

Then fpqc sheafification yields $\pi_{\mathrm{Log}} : \mathrm{Log} \times \mathbf{Z}_p^{\Delta} \to \mathrm{Log}^{\Delta}$.

Definition A.1. Let $(\mathrm{X}, \mathcal{M})$ be a bounded fs log formal scheme. Then there is a map $\mathrm{X} \to \mathrm{Log}$ induced by the log structure. We define $(\mathrm{X}, \mathcal{M})^{\Delta}$ to be the fiber product

$$\begin{tikzcd} (\mathrm{X}, \mathcal{M})^{\Delta} \arrow[r] \arrow[d] & \mathrm{X}^{\Delta} \arrow[d] \\ \mathrm{Log} \times \mathbf{Z}_p^{\Delta} \arrow[r, "\pi_{\mathrm{Log}}"] & \mathrm{Log}^{\Delta}. \end{tikzcd}$$

This is the log prismatic stack introduced by Olsson in [Ols23].[8]

To make arguments easier, and still cover the general scope needed for applications, we work in the situation of Assumption 3.1 for a finite extension $K/\mathbf{Q}_p$. Strict-étale locally, every such log formal scheme is a framed log affine, for which there is a logarithmic Breuil-Kisin atlas. We define this terminology as follows.

Definition A.2. Let $(\mathrm{X}, \mathcal{M})$ satisfy Assumption 3.1 over $\mathcal{O}_K$. A logarithmic Breuil-Kisin prism for $(\mathrm{X}, \mathcal{M})$ is a log prism $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf\,A}})^a$ such that

$$(\mathrm{Spf\,A/I}, \mathrm{M}_{\mathrm{Spf\,A/I}}) \to (\mathrm{X}, \mathcal{M})$$

is a strict open immersion of log p-adic formal schemes.

A logarithmic Breuil-Kisin atlas for $(\mathrm{X}, \mathcal{M})$ means this is an isomorphism. We recall that an ordinary Breuil-Kisin prism for a p-adic formal scheme X is a bounded prism (A, I) such that $\mathrm{Spf\,A/I} \to \mathrm{X}$ is an open immersion. It is an atlas if this map is an isomorphism. If $(\mathrm{X}, \mathcal{M})$ is a framed log affine, $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf\,A}})^a$ is the logarithmic Breuil-Kisin prism associated with its framing by Lemma A.3, and $(\mathrm{B}, (d), \mathcal{N})^a$ is a strict test prism, then Lemma A.6 constructs the coproduct and [Kos22, Proposition 4.13] makes

$$\mathrm{B} \longrightarrow \mathrm{A} \coprod_{*} \mathrm{B}$$

[8] We note that the definition of this stack was also considered by Koshikawa.

completely faithfully flat. Thus the sheaf represented by the atlas covers the final object of the strict log prismatic site for the fpqc topology.

Lemma A.3. *Let* $(\mathrm{X}, \mathcal{M}) = (\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ *be a framed log affine over* $\mathcal{O}_K$. *The construction below associates to its framing a logarithmic Breuil-Kisin prism* $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf}\,\mathrm{A}})^a$ *which is an atlas for* $(\mathrm{X}, \mathcal{M})$.

Proof. Write $\mathrm{R} \simeq \mathrm{R}_0 \widehat{\otimes}_{\mathrm{W}(k)} \mathcal{O}_K$ as in the framing. Choose a uniformizer π of $\mathcal{O}_K$ with Eisenstein polynomial $\mathrm{E}(u)$, and give $\mathrm{S} = \mathrm{W}(k)[\![u]\!]$ the Frobenius lift $u \mapsto u^p$; write E for $\mathrm{E}(u) \in \mathrm{S}$. This makes $(\mathrm{S}, (\mathrm{E}))$ a Breuil-Kisin prism for $\mathcal{O}_K$. When $\mathcal{O}_K = \mathrm{W}(k)[\zeta_p]$, we instead take $(\mathrm{S}, (\mathrm{E})) = (\mathrm{W}(k)[\![q-1]\!], ([p]_q))$ with $q \mapsto q^p$. Give

$$\mathrm{W}(k)\langle \mathrm{T}_1, \ldots, \mathrm{T}_r, \mathrm{T}_{r+1}^{\pm 1}, \ldots, \mathrm{T}_n^{\pm 1} \rangle$$

the Frobenius lift $\mathrm{T}_i \mapsto \mathrm{T}_i^p$. Since R_0 is p-completely étale over this ring, the Frobenius lift extends uniquely to R_0. The completed base change

$$\mathrm{A} = (\mathrm{R}_0 \otimes_{\mathrm{W}(k)} \mathrm{S})^{\wedge}_{(p,\mathrm{E})}$$

with prismatic ideal E and the chart $\mathbf{N}^r \to \mathrm{A}$ sending the ith generator to T_i and $\delta_{\log}(e_i) = 0$, is therefore a bounded fine log prism. The framing gives a strict identification

$$(\mathrm{Spf}\,\mathrm{A}/\mathrm{I}, \mathrm{M}_{\mathrm{Spf}\,\mathrm{A}/\mathrm{I}}) \simeq (\mathrm{X}, \mathcal{M}).$$

□

A.2. **Quasicoherent sheaves on the log prismatic stack.** We show that quasicoherent sheaves on $(\mathrm{X}, \mathcal{M})^{\mathbb{A}}$ recover log prismatic crystals. This ends up being slightly more subtle than the argument for $\mathrm{X}^{\mathbb{A}}$, because we must generally avoid using $\mathbb{A}_{\mathrm{R}}$ for R quasiregular semiperfect. The basic issue is that in the log setting these will generally not come with fine log structures, so they will not admit maps to Log.

We first show strict log prisms give points of Olsson's log prismatic stack.

Lemma A.4. *Let* $(\mathrm{X}, \mathcal{M})$ *satisfy Assumption 3.1 over* $\mathcal{O}_K$, *and let* $(\mathrm{A}, \mathrm{I}, \mathcal{N})^a$ *be a bounded log prism with fine log structure and a strict map* $(\mathrm{Spf}\,\mathrm{A}/\mathrm{I}, \mathcal{N}|_{\mathrm{Spf}\,\mathrm{A}/\mathrm{I}}) \longrightarrow (\mathrm{X}, \mathcal{M})$. *Then there is a natural map*

$$\rho_{\mathrm{X},\mathrm{A}} : \mathrm{Spf}\,\mathrm{A} \to (\mathrm{X}, \mathcal{M})^{\mathbb{A}}.$$

If $(\mathrm{X}, \mathcal{M})$ *is a framed log affine and* $(\mathrm{A}, \mathrm{I}, \mathcal{N})^a$ *is the logarithmic Breuil-Kisin prism associated with its framing, then* $\rho_{\mathrm{X},\mathrm{A}}$ *is an fpqc epimorphism.*

Proof. Construction 8.14 of [BL22] gives $\mathrm{Spf}\,\mathrm{A} \to \mathrm{X}^{\mathbb{A}}$. By the fiber product definition of $(\mathrm{X}, \mathcal{M})^{\mathbb{A}}$, it remains to complete the diagram

$$\begin{tikzcd} \operatorname{Spf} A \arrow[r] \arrow[d, dashed] & X^{\Delta} \arrow[d] \\ \mathbf{Z}_p^{\Delta} \times \operatorname{Log} \arrow[r, "\pi_{\operatorname{Log}}"] & \operatorname{Log}^{\Delta} \end{tikzcd}$$

The log structur on the prism supplies the dashed arrow; unwinding definitions and using that we have a log prism one may check the composites to $\operatorname{Log}^{\Delta}$ agree so we get the desired map.

Now suppose that $(X, \mathcal{M})$ is framed and A is the logarithmic Breuil-Kisin prism associated with its framing. After forgetting the log structure, $\rho_A : \operatorname{Spf} A \to X^{\Delta}$ is an fpqc epimorphism. It follows

$$\operatorname{Spf} A \times_{X^{\Delta}} (X, \mathcal{M})^{\Delta} \to (X, \mathcal{M})^{\Delta}$$

is an fpqc epimorphism. By étale descent of X^{Δ}, the stack $(X, \mathcal{M})^{\Delta}$ satisfies strict-étale descent. Thus when working locally we may choose a chart $P = \mathbf{N}^r$ for the log structure on Spf A and we may replace Log with a toric stack $\mathcal{A}_P = [\operatorname{Spf} \mathbf{Z}_p[P]/\operatorname{Spf} \mathbf{Z}_p[P^{gp}]]$ in the fiber product defining $(X, \mathcal{M})^{\Delta}$. Lemma A.21 gives a fine log prism B_P and an fpqc epimorphism to $\operatorname{Spf} A \times_{\mathcal{A}_P^{\Delta}} (\mathbf{Z}_p^{\Delta} \times \operatorname{Spf} \mathbf{Z}_p[P])$. Since $\operatorname{Spf} A \to X^{\Delta}$ and $\operatorname{Spf} \mathbf{Z}_p[P] \to \mathcal{A}_P$ are fpqc epimorphisms[9], one obtains an fpqc epimorphism

$$\rho_{X,B_P} : \operatorname{Spf} B_P \to (X, \mathcal{M})^{\Delta}.$$

We may check whether $\rho_{X,A} : \operatorname{Spf} A \to (X, \mathcal{M})^{\Delta}$ is an fpqc epimorphism after base change along one, and indeed we obtain

$$\begin{tikzcd} \operatorname{Spf}(A \coprod_* B_P) \arrow[r] \arrow[d] & \operatorname{Spf} B_P \arrow[d] \\ \operatorname{Spf} A \arrow[r, "\rho_{X,A}"] & (X, \mathcal{M})^{\Delta} \end{tikzcd}$$

using Lemma A.6 (which only uses the first part of this lemma). Here $(A \coprod_* B_P)$ is the coproduct in the log prismatic site $(X, \mathcal{M})_{\Delta}$. But one can directly check that

$$\operatorname{Spf}(A \coprod_* B_P) \to \operatorname{Spf} B_P$$

is fpqc, e.g. by [Kos22, Proposition 4.13]. This proves the claim. □

Using these fpqc epimorphisms, we next want to relate fiber products $\operatorname{Spf} A \times_{(X,\mathcal{M})^{\Delta}} \operatorname{Spf} B$ to coproducts in the category of log prisms over $(X, \mathcal{M})$. However since [Kos22, Proposition 4.13] works with the relative rather than absolute site there is not a direct reference. We use a similar method to construct the envelope in our particular case.

[9]Even better, both are flat covers of formal stacks. The latter is clear, while the former follows by noting $A \to A_{\mathrm{perf}}$ is p-completely faithfully flat (using Kunz's theorem) and applying the proof of [Pen25, Proposition 2.19].

Lemma A.5. *Let* $(\mathrm{X}, \mathcal{M})$ *be a framed log affine over* $\mathcal{O}_K$*, let* $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf\,A}})^a$ *be the logarithmic Breuil-Kisin prism associated with its framing, and let* $(\mathrm{B}, (d), \mathcal{N})^a$ *be an orientable bounded strict object of* $(\mathrm{X}, \mathcal{M})_{\mathbb{\Delta}}$*. The coproduct* $\mathrm{A} \coprod_* \mathrm{B}$ *exists.*

Proof. Write $\mathrm{R} = \mathrm{R}_0 \widehat{\otimes}_{\mathrm{W}(k)} \mathcal{O}_K$, with the strict p-completely étale framing

$$\big(\mathrm{W}(k)\langle \mathrm{T}_1, \ldots, \mathrm{T}_r, \mathrm{T}_{r+1}^{\pm 1}, \ldots, \mathrm{T}_n^{\pm 1}\rangle, \mathbf{N}^r\big) \longrightarrow (\mathrm{R}_0, \mathbf{N}^r).$$

In the notation of Lemma A.3, put $\mathrm{A}_0 = \mathrm{R}_0[u]$ and $\mathrm{E} = \mathrm{E}(u)$ for the Breuil-Kisin prism, or $\mathrm{A}_0 = \mathrm{R}_0[q]$ and $\mathrm{E} = [p]_q$ for the cyclotomic prism. Thus $\mathrm{A} = (\mathrm{A}_0)^\wedge_{(p,\mathrm{E})}$ and $\mathrm{I} = (\mathrm{E})$; the polynomial variable and the T_i have $\delta = 0$. Put $\bar{\mathrm{B}} = \mathrm{B}/d$, and give B its unique δ-map from $\mathrm{W}(k)$ lifting the structural map to $\bar{\mathrm{B}}$. By [Kos22, Lemma 3.8], the pullbacks of $e_i \in \mathbf{N}^r$ lift to sections $m_i \in \mathcal{N}$ forming a chart. Put $b_i = \alpha(m_i)$.

We must construct a bounded log prism C with maps from A and B such that, for every bounded test log prism T over $(\mathrm{X}, \mathcal{M})$,

$$\mathrm{Hom}_{(\mathrm{X},\mathcal{M})_{\mathbb{\Delta}}}(\mathrm{C}, \mathrm{T}) \simeq \mathrm{Hom}_{(\mathrm{X},\mathcal{M})_{\mathbb{\Delta}}}(\mathrm{A}, \mathrm{T}) \times \mathrm{Hom}_{(\mathrm{X},\mathcal{M})_{\mathbb{\Delta}}}(\mathrm{B}, \mathrm{T}).$$

For a pair $f : \mathrm{A} \to \mathrm{T}$ and $g : \mathrm{B} \to \mathrm{T}$, rigidity identifies the prismatic ideal of T with $d\mathrm{T}$. The log sections $g(m_i)$ and $f(e_i)$ have the same reduction. Lemma 3.8 of [Kos22] therefore gives a unique unit $\lambda_i \in 1 + d\mathrm{T}$ such that $g(m_i) = \lambda_i f(e_i)$. We adjoin variables v_i representing these units.

Form

$$\mathrm{D}_0^{\mathrm{ex}} = \big((\mathrm{A}_0 \otimes_{\mathrm{W}(k)} \mathrm{B})[v_1^{\pm 1}, \ldots, v_r^{\pm 1}]/(b_i - v_i \mathrm{T}_i)_{i=1}^r\big)^\wedge_{(p,d)}.$$

Give it the log structure pulled back from B and set $\delta(v_i) = v_i^p \delta_{\log}(m_i)$. The relations $b_i = v_i \mathrm{T}_i$ are compatible with δ, and $e_i \mapsto v_i^{-1} m_i$ has $\delta_{\log} = 0$. This is the exactification of [Kos22, Construction 2.17]. The maps $\mathrm{A}_0 \to \mathrm{R} \to \bar{\mathrm{B}}$, $\mathrm{B} \to \bar{\mathrm{B}}$, and $v_i \mapsto 1$ define an augmentation $q_0 : \mathrm{D}_0^{\mathrm{ex}} \to \bar{\mathrm{B}}$.

The framing and the substitutions $\mathrm{T}_i = b_i/v_i$ for $i \leq r$ make $\mathrm{D}_0^{\mathrm{ex}}$ (p,d)-completely smooth over B of relative dimension $n+1$. Modulo d, the augmentation defines a section, so locally $\ker(q_0) = (d, x_1, \ldots, x_{n+1})$, with the x_j a (p,d)-completely regular sequence relative to B. Propositions 3.2, 3.6, and 3.7 of [Kos22] give its bounded log prismatic envelope, locally

$$\mathrm{C} = (\mathrm{D}_0^{\mathrm{ex}}\{x_1/d, \ldots, x_{n+1}/d\}_\delta)^\wedge_{(p,d)}.$$

The map $\mathrm{B} \to \mathrm{C}$ makes its reduction strict over $(\mathrm{X}, \mathcal{M})$. Since $\mathrm{E} \in \ker(q_0)$ maps into $d\mathrm{C}$, the map $\mathrm{A}_0 \to \mathrm{C}$ extends to a log prism map $\mathrm{A} \to \mathrm{C}$.

For the pair (f, g) above, the restrictions to $\mathrm{W}(k)$ agree. Moreover, logarithmic δ-compatibility gives $\delta(\lambda_i) = \lambda_i^p g(\delta_{\log}(m_i))$. Thus $f|_{\mathrm{A}_0}$, g, and $v_i \mapsto \lambda_i$ define a unique $\delta_{\log}$-map $\mathrm{D}_0^{\mathrm{ex}} \to \mathrm{T}$ reducing through q_0. The envelope property extends it uniquely to $\mathrm{C} \to \mathrm{T}$. Conversely, a map from C recovers this pair by restriction, proving the required universal property. ∎

For use below, put $\mathrm{D}^{\mathrm{ex}} = (\mathrm{D}_0^{\mathrm{ex}})^{\wedge}_{(p,d,\mathrm{E})}$. Its augmentation $q : \mathrm{D}^{\mathrm{ex}} \to \bar{\mathrm{B}}$ has the same log prismatic envelope: every map used above sends E into the target's prismatic ideal, and therefore extends uniquely across this completion. Now that we know coproducts exist, we show they can also be computed using the log prismatic stack.

Lemma A.6. *Let $(\mathrm{X}, \mathcal{M})$ be a framed log affine over $\mathcal{O}_K$, let $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf\,A}})^a$ be the logarithmic Breuil-Kisin prism associated with its framing, and let $(\mathrm{B}, (d), \mathcal{N})^a$ be an orientable bounded strict object of $(\mathrm{X}, \mathcal{M})_{\Delta}$. The coproduct $\mathrm{A} \coprod_* \mathrm{B}$ exists, and*

$$\mathrm{Spf}(\mathrm{A} \coprod_* \mathrm{B}) \simeq \mathrm{Spf\,A} \times_{(\mathrm{X},\mathcal{M})^{\Delta}} \mathrm{Spf\,B}.$$

If (A, I) is an ordinary Breuil-Kisin atlas for X obtained from a smooth lift over $\mathrm{W}(k)[\![u]\!]$ and $(\mathrm{B}, (d))$ is an orientable bounded object of X_{Δ}, the analogous assertion holds without log structures.

Proof. Put $\bar{\mathrm{A}} = \mathrm{A}/\mathrm{I}$ and $\bar{\mathrm{B}} = \mathrm{B}/d$. Let $\mathrm{P} = \mathbf{N}^r$ be the chart supplied by the framing and use the resulting factorization $\mathrm{X} \to \mathcal{A}_{\mathrm{P}} \to \mathrm{Log}$. Since $\mathcal{A}_{\mathrm{P}} \to \mathrm{Log}$ is étale, the definition of $(\mathrm{X}, \mathcal{M})^{\Delta}$ may be computed relative to $\mathcal{A}_{\mathrm{P}}$.

The maps from Spf A and Spf B to $(\mathrm{X}, \mathcal{M})^{\Delta}$ are induced by the two commutative squares

$$\begin{array}{ccccc} \mathrm{Spf}\,\bar{\mathrm{A}} & \longrightarrow & \mathrm{X} & \longleftarrow & \mathrm{Spf}\,\bar{\mathrm{B}} \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Spf\,A} & \longrightarrow & \mathcal{A}_{\mathrm{P}} & \longleftarrow & \mathrm{Spf\,B}. \end{array}$$

Indeed as $(\mathrm{Spf}\,\bar{\mathrm{A}}/\,\mathrm{Spf\,A})^{\Delta}$ and $(\mathrm{Spf}\,\bar{\mathrm{B}}/\,\mathrm{Spf\,B})^{\Delta}$ are simply Spf A and Spf B, applying relative prismatization to the two squares gives the maps to $(\mathrm{X}/\mathcal{A}_{\mathrm{P}})^{\Delta} \simeq (\mathrm{X}, \mathcal{M})^{\Delta}$ by its functoriality in δ-pairs, allowing δ-stacks. We note $(\mathrm{X}/\mathcal{S})^{\Delta}$ also has finite limit compatibility for limits in the category of pairs $(\mathrm{X}, \mathcal{S})$ where $\mathcal{S}$ is a δ-stack. Regarding the previous diagram as a diagram of pairs occuring in its columns, one can use this compatibility to identify

$$\mathrm{Spf\,A} \times_{(\mathrm{X},\mathcal{M})^{\Delta}} \mathrm{Spf\,B} \simeq (\bar{\mathrm{A}}/\mathrm{A})^{\Delta} \times_{(\mathrm{X}/\mathcal{A}_{\mathrm{P}})^{\Delta}} (\bar{\mathrm{B}}/\mathrm{B})^{\Delta} \simeq \left(\frac{\mathrm{Spf}\,\bar{\mathrm{A}} \times^{\mathrm{R}}_{\mathrm{X}} \mathrm{Spf}\,\bar{\mathrm{B}}}{\mathrm{Spf\,A} \times^{\mathrm{R}}_{\mathcal{A}_{\mathrm{P}}} \mathrm{Spf\,B}}\right)^{\Delta}.$$

Since $\mathrm{Spf}\,\bar{\mathrm{A}} \simeq \mathrm{X}$, equivalently

$$\mathrm{Spf\,A} \times_{(\mathrm{X},\mathcal{M})^{\Delta}} \mathrm{Spf\,B} \simeq \big(\mathrm{Spf}\,\bar{\mathrm{B}}/\big(\mathrm{Spf\,A} \times^{\mathrm{R}}_{\mathcal{A}_{\mathrm{P}}} \mathrm{Spf\,B}\big)\big)^{\Delta}.$$

We identify the right side using the log prismatic envelope constructed in Lemma A.5, retaining its notation E, m_i, b_i, and v_i. The completed derived exactification is

$$\mathrm{D}^{\mathrm{ex}} := \big((\mathrm{A} \otimes^{\mathrm{L}}_{\mathrm{W}(k)} \mathrm{B})[v_1^{\pm 1}, \ldots, v_r^{\pm 1}]/^{\mathrm{L}}(b_i - v_i \mathrm{T}_i)_{i=1}^r\big)^{\wedge}_{(p,\mathrm{E},d)},$$

where v_i is the ratio between the two images of the ith chart generator. Its envelope is constructed from the (p, d)-completely smooth lift $\mathrm{D}_0^{\mathrm{ex}}$ in the preceding proof. Considering $\big(\mathrm{Spf\,A} \times^{\mathrm{R}}_{\mathcal{A}_{\mathrm{P}}} \mathrm{Spf\,B}\big)$ (which is concentrated in degree zero using smoothness of $\mathrm{X} \to \mathrm{Spf}\,\mathcal{O}_K \times$

$\mathcal{A}_{\mathrm{P}}$), this identifies with the variant of D^{ex} where the tensor product is over $\mathbf{Z}_p$. However the map $\operatorname{Spf}\bar{\mathrm{B}} \to \left(\operatorname{Spf}\mathrm{A} \times^{\mathrm{R}}_{\mathcal{A}_{\mathrm{P}}} \operatorname{Spf}\mathrm{B}\right)$ factors through the component D^{ex} of the tensor product, so the distinction is not relevant. Its prelog structure agrees with the exactification of [Kos22, Construction 2.17].

Write $q : \mathrm{D}^{\mathrm{ex}} \to \bar{\mathrm{B}}$ for the natural map. The previous discussion identifies

$$\operatorname{Spf}\mathrm{A} \times_{(\mathrm{X},\mathcal{M})^{\Delta}} \operatorname{Spf}\mathrm{B} \simeq \operatorname{Spf}\Delta_{\bar{\mathrm{B}}/\mathrm{D}^{\mathrm{ex}}}$$

as by [BL22, Theorem 7.17], the relative prismatization is $\operatorname{Spf}\Delta_{\bar{\mathrm{B}}/\mathrm{D}^{\mathrm{ex}}}$. Locally, $\ker(q)$ is generated by d and a sequence which modulo d is p-completely Koszul regular. Since $(\mathrm{B},(d))$ is bounded, it follows that $(\mathrm{D}^{\mathrm{ex}},(d))$ is a bounded prelog prism and that q satisfies the hypotheses of [BS22, Example 7.9]. That result identifies $\mathrm{C} = \Delta_{\bar{\mathrm{B}}/\mathrm{D}^{\mathrm{ex}}}$ with the corresponding prismatic envelope. By [Kos22, Propositions 3.2, 3.6, and 3.7], this envelope is bounded, and the log structure associated to the extension of the exactified prelog structure makes it the log prismatic envelope of q over $(\mathrm{B},(d),\mathcal{N})^a$. This is the envelope constructed in Lemma A.5, so $\mathrm{C} \simeq \mathrm{A}\coprod_* \mathrm{B}$.

The ordinary assertion follows by a similar argument. □

Proposition A.7. *Let* $(\mathrm{X},\mathcal{M}) = (\operatorname{Spf}\mathrm{R},\mathcal{M})$ *be a framed log affine over* $\mathcal{O}_K$, *and let*

$$\rho_{\mathrm{X},\mathrm{A}} : \operatorname{Spf}\mathrm{A} \to (\mathrm{X},\mathcal{M})^{\Delta}$$

be the fpqc epimorphism of Lemma A.4 associated with the logarithmic Breuil-Kisin prism of its framing. Then the Čech nerve of $\operatorname{Spf}\mathrm{A}$ *identifies with* $\operatorname{Spf}\mathrm{A}^{(n)}$ *where* $\mathrm{A}^{(n)}$ *are the iterated self-coproducts of* $(\mathrm{A},\mathrm{I},\mathrm{M}_{\operatorname{Spf}\mathrm{A}})^a$ *in the category of log prisms. In particular*

$$(\mathrm{X},\mathcal{M})^{\Delta} \simeq \left|\operatorname{Spf}\mathrm{A}^{(\bullet)}\right|$$

in formal stacks.

Proof. Lemma A.6, applied with $\mathrm{B}=\mathrm{A}$, identifies the degree one term of the Čech nerve with $\operatorname{Spf}(\mathrm{A}\coprod_* \mathrm{A})$. The higher terms are the iterated coproducts $\operatorname{Spf}\mathrm{A}^{(n)}$, for both the site-theoretic Čech nerve and the Čech nerve of $\rho_{\mathrm{X},\mathrm{A}} : \operatorname{Spf}\mathrm{A} \to (\mathrm{X},\mathcal{M})^{\Delta}$. Using Lemma A.4, this is an fpqc epimorphism so the claim follows. □

Proposition A.8. *Let* $(\mathrm{X},\mathcal{M})$ *satisfy Assumption 3.1 over* $\mathcal{O}_K$. *There is a natural symmetric monoidal equivalence*

$$\iota_{(\mathrm{X},\mathcal{M})} : \mathrm{D}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}) \xrightarrow{\sim} \mathrm{D}_{\mathrm{qc}}((\mathrm{X},\mathcal{M})^{\Delta}).$$

It is compatible with pullback along strict morphisms between pairs satisfying Assumption 3.1 and restricts to an equivalence on perfect objects. The same assertions hold after base change along an algebra object on $\mathbf{Z}_p^{\Delta}$.

Proof. We first note that the assertion can be checked strict-étale locally. Indeed for the crystal side by [Ino25, Lemma 5.9], crystals may be computed on the strict subsite; on a framed log affine, using the logarithmic Breuil-Kisin prism $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf}\,\mathrm{A}})^a$ associated with its framing yields

$$\lim_{[n]\in\Delta} \mathrm{D}_{(p,\mathrm{I}^{(n)})}(\mathrm{A}^{(n)}) \simeq \mathrm{D}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta})$$

where $\mathrm{A}^{(n)}$ is the logarithmic Čech nerve. The proof of Corollary A.18 gives strict-étale descent of the crystal side.

The same holds for the stack $(\mathrm{X}, \mathcal{M})^{\Delta}$ by étale descent for X^{Δ} in X [BL22]. We may therefore assume $(\mathrm{X}, \mathcal{M}) = (\mathrm{Spf}\,\mathrm{R}, \mathcal{M})$ is a framed log affine. The logarithmic Breuil-Kisin prism A associated with its framing covers the log prismatic stack by Lemma A.4.

Now we can use Proposition A.7 to identify

$$\mathrm{D}_{\mathrm{qc}}((\mathrm{X}, \mathcal{M})^{\Delta}) \simeq \lim_{[n]\in\Delta} \mathrm{D}_{\mathrm{qc}}(\mathrm{Spf}\,\mathrm{A}^{(n)}) \simeq \mathrm{D}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}).$$

Indeed, $\rho_{\mathrm{X},\mathrm{A}} : \mathrm{Spf}\,\mathrm{A} \to (\mathrm{X}, \mathcal{M})^{\Delta}$ is an fpqc epimorphism and we have identified its Čech nerve with $\mathrm{Spf}\,\mathrm{A}^{(n)}$, and thus D_{qc} satisfies descent for such maps. See the discussion after [Pen25, Definition 2.4] for more details. □

For a strict morphism f, we write $\mathrm{R}f_{\Delta,*}$ and $\mathrm{R}f_*$ for the right adjoints to pullback on the two categories of Proposition A.8, respectively.

Proposition A.9. *Let $f : (\mathrm{X}, \mathcal{M}) \to (\mathrm{Y}, \mathcal{N})$ be a strict morphism between pairs satisfying Assumption 3.1 over $\mathcal{O}_K$. For every $\mathcal{E} \in \mathrm{D}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta})$, there is a natural equivalence*

$$\iota_{(\mathrm{Y},\mathcal{N})} \mathrm{R}f_{\Delta,*}\mathcal{E} \simeq \mathrm{R}f_* \iota_{(\mathrm{X},\mathcal{M})}\mathcal{E}.$$

Proof. Compatibility of Proposition A.8 with strict pullback gives a commutative diagram

$$\begin{tikzcd} \mathrm{D}((\mathrm{Y}, \mathcal{N})_{\Delta}, \mathcal{O}_{\Delta}) \arrow[r, "\iota_{(\mathrm{Y},\mathcal{N})}"] \arrow[d, "f^*"] & \mathrm{D}_{\mathrm{qc}}((\mathrm{Y}, \mathcal{N})^{\Delta}) \arrow[d, "(f^{\Delta})^*"] \\ \mathrm{D}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}) \arrow[r, "\iota_{(\mathrm{X},\mathcal{M})}"] & \mathrm{D}_{\mathrm{qc}}((\mathrm{X}, \mathcal{M})^{\Delta}). \end{tikzcd}$$

Both vertical pullback functors preserve colimits. Taking right adjoints of this square gives the asserted equivalence. □

Remark A.10. Laurent F-crystals in [KY25, IKY26] use the saturated log prismatic site. By [Ino25, Lemma 5.9], the strict subsite is cofinal in the full log prismatic site. If $(\mathrm{X}, \mathcal{M})$ is fs and saturated, a strict object has the same characteristic monoid as the pullback of $\overline{\mathcal{M}}$, and is therefore saturated. The strictification in [Ino25, Lemma 5.9] has this same characteristic monoid, so the strict subsite is also cofinal in the saturated log prismatic site. Thus the limits defining the crystal categories on the full and saturated sites agree.

We suppress this distinction for $\mathrm{D}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta})$ and its perfect and Laurent variants, but make no comparison between arbitrary sheaves on the underlying ringed topoi.

A.3. **Solid Laurent crystals.** In this subsection, we define a solid generalization of

$$\mathsf{Perf}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}).$$

We always use the analytic structure $\mathrm{Spa}(\mathrm{A},\mathrm{A})$.

Definition A.11. Let $(\mathrm{A},(d))$ be an orientable prism. Associated to the derived (p,d)-complete ring A is the analytic adic ring (A,A) in the sense of [AM24]. We define

$$\mathrm{D}_{\blacksquare}(\mathrm{A}[1/d]^{\wedge}_{p}) := \varprojlim_{n} \mathsf{Mod}_{(\mathrm{A}/^{\mathrm{L}}p^{n})[1/d]}\mathrm{D}_{\blacksquare}((\mathrm{A},\mathrm{A})),$$

where $(\mathrm{A}/^{\mathrm{L}}p^{n})[1/d]$ is regarded as an algebra object of $\mathrm{D}_{\blacksquare}((\mathrm{A},\mathrm{A}))$. Equivalently, this is $\mathrm{D}_{\blacksquare}$ of $\mathrm{colim}_{n}\,\mathrm{Spa}((\mathrm{A}/^{\mathrm{L}}p^{n})[1/d],\mathrm{A}/^{\mathrm{L}}p^{n})$, where we use Spa to denote the analytic stack attached to a Huber pair as in [Kip26a, Example 2.1.5].

Then

$$\mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}) := \lim_{(\mathrm{A},(d),\mathrm{M}_{\mathrm{Spf}\,\mathrm{A}})^{a}\in(\mathrm{X},\mathcal{M})_{\Delta}} \mathrm{D}_{\blacksquare}(\mathrm{A}[1/d]^{\wedge}_{p}).$$

The definition is independent of the generator d, and one may (by ind-Zariski localization) assume that the prismatic ideal is principal.

In what follows we repeatedly use the following formal fact: if $\mathcal{C} \simeq \varprojlim_{i}\mathcal{C}_{i}$ is a limit of presentable symmetric monoidal stable ∞-categories and an algebra $\mathrm{S}\in\mathcal{C}$ has images S_{i}, then

$$\mathrm{Mod}_{\mathrm{S}}(\mathcal{C}) \simeq \varprojlim_{i}\mathrm{Mod}_{\mathrm{S}_{i}}(\mathcal{C}_{i}).$$

Indeed, algebra objects and their module objects are computed in limits.

We define perfect Laurent crystals by dualizability:

$$\mathsf{Perf}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}) := \mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})^{\mathrm{dual}}.$$

Lemma A.12. *Let $\mathrm{A}\to\mathrm{B}$ be a faithfully flat map of connective animated rings. If $\pi_0\mathrm{B}$ is countably presented as a $\pi_0\mathrm{A}$-module, then $\mathrm{A}\to\mathrm{B}$ is descendable of index at most 2.*

Proof. See [Mat16, Proposition 3.31 and the proof of Corollary 3.33]. □

Lemma A.13. *Let $(\mathrm{X},\mathcal{M})$ be a framed log affine over $\mathcal{O}_{K}$, let $(\mathrm{A},\mathrm{I},\mathrm{M}_{\mathrm{Spf}\,\mathrm{A}})^{a}$ be the logarithmic Breuil-Kisin prism associated with its framing, and let $(\mathrm{B},(d),\mathcal{N})^{a}$ be an orientable bounded strict object of the same log prismatic site. Put*

$$\mathrm{C} = \mathrm{A}\coprod_{*}\mathrm{B}.$$

Then C *exists, and*

$$\mathrm{B} \longrightarrow \mathrm{C}$$

is completely descendable for the (p, d)*-adic topologies in the sense of* [Kip26b, Definition 2.4.1.7], *and the induced algebra is descendable in* $\mathrm{D}_{\blacksquare}((\mathrm{B}, \mathrm{B}))$.

Moreover, $\mathrm{Spa}(\mathrm{C}, \mathrm{C}) \to \mathrm{Spa}(\mathrm{B}, \mathrm{B})$ *is a proper descendable cover. The same assertions hold when* X *is smooth over* $\mathcal{O}_K$*,* (A, I) *is an ordinary Breuil-Kisin atlas obtained from a smooth lift over* $\mathrm{W}(k)[\![u]\!]$*,* $(\mathrm{B}, (d))$ *is an orientable bounded object of* X_{Δ}*, and* $\mathrm{C} = \mathrm{A} \coprod_* \mathrm{B}$.

Proof. We prove the logarithmic assertion. The ordinary assertion is proved identically (in particular a degenerate case of the argument).

There are two main points in the argument. First, the reductions modulo (p^k, d^k) are descendable of index ≤ 2. Second, the nullhomotopies witnessing this need not lift compatibly to the derived adic limit. We show that the resulting $\varprojlim^1$ obstruction disappears after taking its tensor square, giving complete descendability of index ≤ 4.

By the construction in Lemma A.5, Zariski locally we have

$$\mathrm{C} \simeq (\mathrm{D}^{\mathrm{ex}}\{x_1/d, \ldots, x_m/d\}_\delta)^\wedge_{(p,d)}\,, \tag{6}$$

where $\ker(\mathrm{D}^{\mathrm{ex}} \to \bar{\mathrm{B}}) = (d, x_1, \ldots, x_m)$. The regularity established there and [Mao24, Example 5.43, Proposition 5.52, and Remark 5.54] imply that $\mathrm{B} \to \mathrm{C}$ is (p, d)-completely faithfully flat. In particular, its derived reductions $\mathrm{B}/(p^n, d^k) \to \mathrm{C}/(p^n, d^k)$ are faithfully flat for all $n, k \geq 1$. One may also easily check $\pi_0(\mathrm{C}/(p^n, d^k))$ is countably presented over $\pi_0(\mathrm{B}/(p^n, d^k))$ since we adjoin finitely many variables as a δ ring, so Lemma A.12 proves the first assertion (namely that the reductions modulo (p^k, d^k) are descendable of index ≤ 2).

Rigidity of prisms makes $\mathrm{B} \to \mathrm{C}$ adic. Let $\mathrm{J} = \mathrm{fib}(\mathrm{B} \to \mathrm{C})$ in the derived (p, d)-complete category. The assertion that the map is descendable of index ≤ 2 modulo (p^k, d^k) says that the canonical map $u : \mathrm{J}\widehat{\otimes}_{\mathrm{B}}\mathrm{J} \to \mathrm{B}$ becomes null modulo (p^k, d^k) for every k. If $\mathrm{J}_k = \mathrm{J} \otimes^{\mathrm{L}}_{\mathrm{B}} \mathrm{B}/(p^k, d^k)$, derived completeness gives

$$\mathrm{Map}_{\mathrm{B}}(\mathrm{J}\widehat{\otimes}_{\mathrm{B}}\mathrm{J}, \mathrm{B}) \simeq \mathrm{R}\varprojlim_k \mathrm{Map}_{\mathrm{B}/(p^k,d^k)}(\mathrm{J}_k^{\otimes 2}, \mathrm{B}/(p^k, d^k)).$$

Write X_k for the mapping spectrum on the right. The Milnor exact sequence gives

$$0 \longrightarrow \mathrm{R}^1 \lim_k \pi_1 \mathrm{X}_k \longrightarrow \pi_0 \mathrm{R} \lim_k \mathrm{X}_k \longrightarrow \lim_k \pi_0 \mathrm{X}_k \longrightarrow 0.$$

The reduction of u is null for every k, so its image in $\varprojlim_k \pi_0 \mathrm{X}_k$ vanishes. Hence $[u]$ lies in $\mathrm{Fil}^1 = \varprojlim^1_k \pi_1 \mathrm{X}_k$. Tensoring maps gives compatible pairings

$$\mathrm{X}_k \otimes \mathrm{X}_k \longrightarrow \mathrm{Map}_{\mathrm{B}/(p^k,d^k)}(\mathrm{J}_k^{\otimes 4}, \mathrm{B}/(p^k, d^k)).$$

The induced pairing of derived limit spectral sequences carries $\mathrm{Fil}^a \otimes \mathrm{Fil}^b$ into Fil^{a+b}. Therefore $[u] \in \mathrm{Fil}^1$ implies $[u\widehat{\otimes}u] \in \mathrm{Fil}^2 = 0$ since $\varprojlim^s_k = 0$ for $s \geq 2$ on a countable tower.

Thus the fourth tensor power of the fiber map $\mathrm{J} \to \mathrm{B}$ is null. The criterion in [Mat18, Proposition 2.32] makes C descendable in the derived (p, d)-complete category.

This shows complete descendability in the sense of [Kip26b, Definition 2.4.1.7]. Under $\mathrm{D}(\mathrm{Spf\,B}) \simeq \mathrm{D}_{(p,d)\text{-comp}}(\mathrm{B})$, the lax symmetric monoidal analytification functor used in the proof of [Kip26a, Theorem 2.3.3(A)] preserves descendable algebras. Hence the algebra induced by C in $\mathrm{D}_{\blacksquare}((\mathrm{B}, \mathrm{B}))$ is descendable.

It remains to prove properness. Example 2.1.13 of [Kip26a] says that a map $\mathrm{Spa}(\mathrm{C}, \mathrm{C}) \to \mathrm{Spa}(\mathrm{B}, \mathrm{B})$ is proper if and only if C is integral over the subring generated by B and topologically nilpotent elements of C. In the present adic situation, it is therefore enough to prove that

$$\pi_0(\mathrm{B}/(p, d)) \longrightarrow \pi_0(\mathrm{C}/(p, d))$$

is integral. Indeed, if a monic polynomial f over $\pi_0\mathrm{B}$ annihilates c modulo (p, d), then $f(t) - f(c)$ is a monic polynomial over $\pi_0\mathrm{B} + (p, d)\pi_0\mathrm{C}$ annihilating c.

Applying Lemma A.14 to (6), using $\mathrm{D}^{\mathrm{ex}}/(d, x_1, \ldots, x_m) \simeq \mathrm{B}/d$, gives the required integrality. Its algebra is descendable by the first part of the proof, so it is a cover by [Kip26a, Example 2.1.12]. The same argument applies to the ordinary case. □

Lemma A.14. *Let* A *be an animated* δ*-ring, let* $d \in \pi_0\mathrm{A}$ *be distinguished, and let* $\{f_i\}$ *be any (arbitrarily large) family of elements of* $\pi_0\mathrm{A}$*. Suppose* A *is further derived* (p, d) *complete. Then the morphism*

$$\mathrm{Spf}\,(\mathrm{A}\{f_i/d\}_\delta)^\wedge_{(p,d)} \to \mathrm{Spf\,A}$$

induced by the structure map is integral in the sense of [Kip26a, 2.2.5]. *In other words, on rings it is true in the ordinary sense modulo* (p, d).

Proof. As we discuss integrality on formal schemes, we must show

$$\pi_0\mathrm{A}/(p, d) \to \pi_0\left(\mathrm{A}\{f_i/d\}_\delta\right)^\wedge_{(p,d)}/(p, d)$$

is integral. Here the reductions are both derived. It suffices to check all of the iterates $\delta^n(f_i/d)$ satisfy monic polynomials in $\mathrm{A}/(p, d)[\mathrm{T}]$ modulo (p, d). We will show that $\delta^n(f_i/d)$ satisfies a monic polynomial in $\pi_0\mathrm{A}/(p, d)[(f_i/d), \delta(f_i/d), \ldots, \delta^{n-1}(f_i/d)]$.

The key identity for us will be

$$0 = \delta^n(d(f_i/d) - f_i) = \varphi^n(d)\delta^n(f_i/d) + \mathrm{F}_n((f_i/d), \delta(f_i/d), \ldots, \delta^{n-1}(f_i/d))$$

where F_n is a polynomial with coefficients in A. Moreover, for $n \geq 1$, modulo (p, d), F_n has degree p in $\delta^{n-1}(f_i/d)$, with leading coefficient $\delta(\varphi^{n-1}(d))$. We will drop the argument of F_n from here on.

The identity holds for $n = 0$ with $\mathrm{F}_0 = -f_i$. Applying δ to the relation, one sees that $\delta^{n+1}(d(f_i/d) - f_i)$ equals

$$\delta(\varphi^n(d)\delta^n(f_i/d)) + \delta(\mathrm{F}_n) - \sum_{j=1}^{p-1} \frac{1}{p}\binom{p}{j}(\varphi^n(d)\delta^n(f_i/d))^j \mathrm{F}_n^{p-j}.$$

One computes $\delta(\varphi^n(d)\delta^n(f_i/d)) = \varphi^{n+1}(d)\delta^{n+1}(f_i/d) + \delta(\varphi^n(d))(\delta^n(f_i/d))^p$ as well. Collecting the remaining terms to form F_{n+1} gives the first part of the claim. For the second part, if we work modulo (p,d), the previous display becomes $\delta(\varphi^n(d)\delta^n(f_i/d)) + \delta(\mathrm{F}_n)$, as the factor we subtract has all terms divisible by $\varphi^n(d) \in (p,d)$. Since F_n involves only the preceding iterates, $\delta(\mathrm{F}_n)$ has degree at most one in $\delta^n(f_i/d)$ modulo p. Thus the degree-p term of F_{n+1} has the asserted coefficient.

To deduce the desired claim for $\delta^n(f_i/d)$, one then reduces this relation for the index $n+1$ modulo (p,d): this kills the term $\varphi^{n+1}(d)\delta^{n+1}(f_i/d)$, leaving only the remaining terms $\mathrm{F}_{n+1}((f_i/d), \delta(f_i/d), \ldots, \delta^n(f_i/d))$. We know the leading coefficient in $\delta^n(f_i/d)$ for F_{n+1} modulo (p,d) was $\delta(\varphi^n(d))$ attached to $(\delta^n(f_i/d))^p$, but φ and δ commute so this is a unit (since d is distinguished; i.e. $\delta(d)$ is a unit). Thus after we divide by this unit, we obtain a monic polynomial in $\pi_0\mathrm{A}/(p,d)[(f_i/d), \delta(f_i/d), \ldots, \delta^{n-1}(f_i/d)]$ that $\delta^n(f_i/d)$ satisfies modulo (p,d). The claim follows. $\square$

Theorem A.15. *Let $(\mathrm{X}, \mathcal{M})$ be a framed log affine over $\mathcal{O}_K$, and let $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf\,A}})^a$ be the logarithmic Breuil-Kisin prism associated with its framing. If $(\mathrm{A}^{(m)}, \mathrm{I}, \mathrm{M}^{(m)})^a$ is the degree m term of its Čech nerve in the strict log prismatic site, then*

$$\mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p) \simeq \varprojlim_{[m]\in\Delta} \mathrm{D}_{\blacksquare}(\mathrm{A}^{(m)}[1/\mathrm{I}]^{\wedge}_p).$$

Proof. By [Ino25, Lemma 5.9], it suffices to work on the strict log prismatic subsite, where orientable objects form a Zariski basis. For an orientable strict test prism $(\mathrm{B}, (d), \mathcal{N})^a$, Lemma A.6 identifies the pullback of the atlas with

$$\mathrm{B} \longrightarrow \mathrm{A} \coprod_{*} \mathrm{B}.$$

The complete descendability result of Lemma A.13 gives

$$\mathrm{D}_{\blacksquare}(\mathrm{B}[1/d]^{\wedge}_p) \simeq \varprojlim_{[m]\in\Delta} \mathrm{D}_{\blacksquare}((\mathrm{A}^{(m)} \coprod_{*} \mathrm{B})[1/d]^{\wedge}_p)$$

because modulo every p^n, Laurent base change identifies the analytic Čech algebra with $((\mathrm{A}^{(m)} \coprod_* \mathrm{B})/^{\mathrm{L}}p^n)[1/d]$, and forming module objects commutes with limits in symmetric monoidal presentable categories. This shows the assertion that $(\mathrm{B}, (d), \mathrm{M}_{\mathrm{Spf\,B}})^a \mapsto \mathrm{D}_{\blacksquare}(\mathrm{B}[1/d]^{\wedge}_p)$ satisfies descent for $h_{\mathrm{A}} \to *$ in the log prismatic topos of $(\mathrm{X}, \mathcal{M})_{\Delta}$; here we use h for the Yoneda embedding on a log prism. Indeed, it suffices to check this on all representable sheaves, which we have just done. Thus, taking global sections one obtains

$$\mathrm{D}_{\blacksquare}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p) \simeq \varprojlim_{[m]\in\Delta} \mathrm{D}_{\blacksquare}(\mathrm{A}^{(m)}[1/\mathrm{I}]^{\wedge}_p).$$

$\square$

Lemma A.16. *Let* $(\mathrm{A},(d))$ *be an orientable transversal prism, i.e.* A/d *is* p*-torsion-free. There are functorial equivalences*

$$\mathrm{D}_{\blacksquare}(\mathrm{A}[1/d]_p^{\wedge})^{\mathrm{dual}} \simeq \varprojlim_n \mathsf{Perf}((\mathrm{A}/p^n)[1/d]) \simeq \mathsf{Perf}(\mathrm{A}[1/d]_p^{\wedge}).$$

In the setting of Theorem A.15, these give

$$\mathsf{Perf}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}) \simeq \varprojlim_{[m]\in\Delta} \mathsf{Perf}(\mathrm{A}^{(m)}[1/\mathrm{I}]_p^{\wedge}).$$

Proof. Transversality implies that A is p-torsion-free, so $\mathrm{A}/^{\mathrm{L}}p^n \simeq \mathrm{A}/p^n$. The finite-level Laurent analytic category in Definition A.11 is therefore the category associated with the ordinary Tate-Huber pair $((\mathrm{A}/p^n)[1/d],\mathrm{A}/p^n)$. By [Sch26b, Theorem 14.10], for a general Huber pair the dualizable objects in its solid module category are precisely the perfect complexes over its underlying ring. Applying this result at each level and using that taking dualizable objects commutes with limits of symmetric monoidal categories proves the first equivalence. Since $\mathrm{A}[1/d]$ is p-torsion-free, its derived p-completion is concentrated in degree zero and p-torsion-free; p-adic effectivity for perfect complexes gives the second equivalence.

For the final assertion, the logarithmic Breuil-Kisin atlas is transversal, and complete flatness of its Čech terms over the atlas preserves transversality. Thus all reductions $\mathrm{A}^{(m)}/^{\mathrm{L}}p^n$ are underived. Apply the first assertion termwise to the symmetric monoidal descent equivalence of Theorem A.15, and again take dualizable objects. □

Lemma A.17. *Let* $(\mathrm{Spf}\,\mathrm{R},\mathcal{M})$ *be a framed log affine over* $\mathrm{Spf}\,\mathrm{W}(k)[\zeta_p]$*. Let* $((\mathrm{A}^+)^{(m)},[p]_q,\mathcal{M}^{(m)})^a$ *be the degree* m *term of the log* q*-de Rham Čech nerve in* $(\mathrm{Spf}\,\mathrm{R},\mathcal{M})_{\Delta}$*, and put* $\mathrm{A}^{(m)} = (\mathrm{A}^+)^{(m)}[1/[p]_q]_p^{\wedge}$*. Then*

$$\mathrm{D}_{\blacksquare}((\mathrm{Spf}\,\mathrm{R},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge}) \simeq \varprojlim_{[m]\in\Delta} \mathrm{D}_{\blacksquare}(\mathrm{A}^{(m)}).$$

Moreover, the Γ *action on the degree zero term gives a conservative evaluation functor to* $\mathrm{D}_{\blacksquare}(\mathrm{A})^{h\Gamma}$.

Proof. For the cyclotomic choice of base prism in Lemma A.3, the log q-de Rham prism is the logarithmic Breuil-Kisin atlas. Theorem A.15 therefore gives the asserted Čech description. The Γ action arises from log prism automorphisms, and the crystal property therefore gives the action on degree zero evaluation. The analytic maps $\mathrm{A}^{(s)} \to \mathrm{C}(\Gamma^s,\mathrm{A})$ constructed in Lemma 3.8 identify the coface maps and show that this action is continuous. The projection from the Čech limit to its degree zero term is conservative. □

Corollary A.18. *On log formal schemes satisfying Assumption 3.1 over* $\mathcal{O}_K$*, the assignment*

$$(\mathrm{X},\mathcal{M}) \longmapsto \mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})$$

is a stack for the strict-étale topology. The same holds for $\mathsf{Perf}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]_p^{\wedge})$.

Proof. Framed log affines form a basis for the strict-étale topology by Lemma 3.4. It is therefore enough to consider an affine strict-étale cover $U \to X$ by framed log affines with X framed. Give every term $U^{(n)}$ of its Čech nerve the framing induced from X. Let A_X be the logarithmic Breuil-Kisin prism associated with the framing of X. The unique completely étale lifting property identifies its pullback to each $U^{(n)}$ with the logarithmic Breuil-Kisin prism $A_{U^{(n)}}$ associated with the induced framing; see [Kos22, Lemma 4.12]. Write $A_X^{(m)}$ and $A_{U^{(n)}}^{(m)}$ for the Laurent coefficient rings in degree m of their prismatic Čech nerves. They form the (augmented) bicosimplicial diagram

$$\begin{tikzcd}
D_{\blacksquare}((X,\mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}) \arrow[r] \arrow[d] & D_{\blacksquare}(A_X^{(0)}) \arrow[r, shift left] \arrow[r, shift right] \arrow[d] & D_{\blacksquare}(A_X^{(1)}) \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] \arrow[d] & \cdots \\
D_{\blacksquare}((U,\mathcal{M}|_U)_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}) \arrow[r] \arrow[d, shift left] \arrow[d, shift right] & D_{\blacksquare}(A_U^{(0)}) \arrow[r, shift left] \arrow[r, shift right] \arrow[d, shift left] \arrow[d, shift right] & D_{\blacksquare}(A_U^{(1)}) \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] \arrow[d, shift left] \arrow[d, shift right] & \cdots \\
D_{\blacksquare}((U^{(1)},\mathcal{M}|_{U^{(1)}})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}) \arrow[r] \arrow[d, shift left=2] \arrow[d] \arrow[d, shift right=2] & D_{\blacksquare}(A_{U^{(1)}}^{(0)}) \arrow[r, shift left] \arrow[r, shift right] \arrow[d, shift left=2] \arrow[d] \arrow[d, shift right=2] & D_{\blacksquare}(A_{U^{(1)}}^{(1)}) \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] \arrow[d, shift left=2] \arrow[d] \arrow[d, shift right=2] & \cdots \\
\vdots & \vdots & \vdots &
\end{tikzcd}$$

For fixed n, Theorem A.15 identifies the limit of the nth row with the Laurent crystal category of $U^{(n)}$. For fixed m, compatibility with strict-étale localization identifies the mth column with the Čech nerve of a completely faithfully étale map. Such maps are completely descendable, so the limit of this column is $D_{\blacksquare}(A_X^{(m)})$. Since limits commute with limits, we obtain

$$\begin{aligned}
\varprojlim_{[n]\in\Delta} D_{\blacksquare}((U^{(n)},\mathcal{M}|_{U^{(n)}})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}) &\simeq \varprojlim_{[n]\in\Delta} \varprojlim_{[m]\in\Delta} D_{\blacksquare}(A_{U^{(n)}}^{(m)}) \\
&\simeq \varprojlim_{[m]\in\Delta} \varprojlim_{[n]\in\Delta} D_{\blacksquare}(A_{U^{(n)}}^{(m)}) \\
&\simeq \varprojlim_{[m]\in\Delta} D_{\blacksquare}(A_X^{(m)}) \\
&\simeq D_{\blacksquare}((X,\mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge}).
\end{aligned}$$

All equivalences are induced by the augmentations, so their composite is the strict-étale descent equivalence.

The descent equivalence is symmetric monoidal, and dualizable objects are detected termwise in limits. It therefore restricts to the asserted descent equivalence for $\mathsf{Perf}((X,\mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/I_{\Delta}]_p^{\wedge})$. ◻

The canonical t-structure on $\mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$ is defined by taking the connective objects to be the crystals whose evaluations on every prism are connective, and taking the coconnective objects to be the right orthogonal of the connective objects shifted by one. Extension of scalars preserves connective objects, so this defines an accessible t-structure.

Corollary A.19. *Let $(\mathrm{X},\mathcal{M})$ satisfy Assumption 3.1 over $\mathcal{O}_K$. The canonical t-structure on $\mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$ restricts to $\mathsf{Perf}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$.*

Proof. Using a covering by framed log affines via Lemma 3.4, we may use their associated logarithmic Breuil-Kisin prisms $(\mathrm{A},\mathrm{I},\mathrm{M}_{\mathrm{Spf\,A}})^{a}$ from Lemma A.3 to produce a covering family (by Theorem A.15). The argument ends up being slightly technical since there is a subtlety that it is not clear the maps in the Breuil-Kisin Čech nerve are t-exact for $\mathrm{D}_{\blacksquare}(-)$.

First we simply check that there exists a t-structure on $\mathsf{Perf}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$ regarded as ordinary Laurent crystals, defined via orthogonality from the Breuil-Kisin presentation of $\mathsf{Perf}^{\leq 0}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$: we ask for the connective objects to be those which are connective on each Čech term $\mathrm{A}^{(n)}[1/\mathrm{I}]^{\wedge}_{p}$. This follows by the abstract argument of [Bha22, Footnote 70], noting that we actually only need the maps $\mathrm{A}[1/\mathrm{I}]^{\wedge}_{p}\to\mathrm{A}^{(n)}[1/\mathrm{I}]^{\wedge}_{p}$ to be flat (which they are in the ordinary sense since A is Noetherian, using that (p,I)-complete flatness upgrades to ordinary flatness with a Noetherian source). By Lemma A.16 this also yields a t-structure on the dualizable objects in $\mathrm{D}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$ as dualizability is detected termwise in a limit.

One may check $\mathcal{E}\in\mathsf{Perf}^{\leq 0}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$ is also in $\mathrm{D}^{\leq 0}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$. We must check all evaluations on log prisms are connective, as we define the canonical t-structure on $\mathrm{D}^{\leq 0}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$ this way by orthogonality. By construction of $\mathsf{Perf}^{\leq 0}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$ we see the evaluation on $(\mathrm{A}\coprod_{*}\mathrm{B})[1/\mathrm{I}]^{\wedge}$ is connective (as it is obtained by pullback from the value on $(\mathrm{A},\mathrm{I},\mathrm{M}_{\mathrm{Spf\,A}})^{a}$, which by definition is connective). In Lemma A.13 we argued $(\mathrm{A}\coprod_{*}\mathrm{B})[1/\mathrm{I}]^{\wedge}$ is a descendable algebra in $\mathrm{D}_{\blacksquare}(\mathrm{B}[1/\mathrm{J}]^{\wedge}_{p})$. If M is the evaluation of $\mathcal{E}$ in $\mathrm{D}_{\blacksquare}(\mathrm{B}[1/\mathrm{J}]^{\wedge}_{p})$, descendability says that M is a retract of a finite truncation of the Čech/Amitsur complex of $((\mathrm{A}\coprod_{*}\mathrm{B})[1/\mathrm{I}]^{\wedge})^{\bullet}\otimes^{\mathrm{L},\blacksquare}\mathrm{M}$ in $\mathrm{D}_{\blacksquare}(\mathrm{B}[1/\mathrm{J}]^{\wedge}_{p})$, where all terms are connective. Similarly using perfectness of M the same assertion applies to $\mathrm{M}^{\vee}$, deducing that it is coconnective – the coconnective objects are closed under finite limits, so $\mathrm{M}^{\vee}$ is coconnective. Passing back to M it is connective.

One may also check that $\mathcal{E}\in\mathsf{Perf}^{\geq 0}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$ is also in $\mathrm{D}^{\geq 0}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$ for its canonical t-structure. Indeed this is defined as the right orthogonal of

$$\mathrm{D}^{\leq -1}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p}).$$

For $\mathcal{E}'\in\mathrm{D}^{\leq -1}_{\blacksquare}((\mathrm{X},\mathcal{M})_{\Delta},\mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_{p})$, every evaluation on Čech terms $\mathrm{A}^{(n)}[1/\mathrm{I}]^{\wedge}_{p}$ lies in the ≤ -1 part of the solid t-structure. Writing $\mathcal{E}_n$ and $\mathcal{E}'_n$ for these evaluations, we get

$$\mathrm{Map}_{\mathrm{D}_{\blacksquare}(\mathrm{A}^{(n)}[1/\mathrm{I}]^{\wedge}_{p})}(\mathcal{E}'_n,\mathcal{E}_n)=0$$

using that $\mathcal{E}_n$ are in the ≥ 0 part (using that it is perfect here). But mapping complexes are computed by the limit over the Čech nerve. Thus the first claim of the corollary follows. □

We denote its heart by $\mathsf{Coh}((\mathrm{X}, \mathcal{M})_{\Delta}, \mathcal{O}_{\Delta}[1/\mathrm{I}_{\Delta}]^{\wedge}_p)$. Corollary A.18 gives strict-étale descent for its heart.

A.4. **Analytifications.** We compare Breuil-Kisin presentations with the analytification of prismatic stacks.

Proposition A.20. *Let* X *be an affine smooth* p*-adic formal scheme over* $\mathcal{O}_K$*, and let* (A, I) *be an ordinary Breuil-Kisin atlas obtained from a smooth lift over* $\mathrm{W}(k)[\![u]\!]$*. The induced map*

$$\mathrm{Spa}(\mathrm{A}, \mathrm{A}) \longrightarrow \mathrm{X}^{\Delta, \blacksquare}$$

is a representable proper effective epimorphism. If $\mathrm{A}^{(n)}$ *denotes the* n*th iterated self-coproduct in the category of prisms over* X*, then*

$$\mathrm{X}^{\Delta, \blacksquare} \simeq \left|\mathrm{Spa}(\mathrm{A}^{(\bullet)}, \mathrm{A}^{(\bullet)})\right| .$$

Proof. Choose a classical regular semiperfectoid cover of X in the infinite-root étale topology. Construction 3.7.1 and Corollary 3.6.3 of [Kip26a] show that

$$\coprod_i \mathrm{Spa}(\mathrm{B}_i, \mathrm{B}_i) \longrightarrow \mathrm{X}^{\Delta, \blacksquare}$$

is an effective epimorphism; after a Zariski refinement, the B_i are orientable bounded prisms over X. It therefore suffices to test the first assertion after pullback to each $\mathrm{Spa}(\mathrm{B}_i, \mathrm{B}_i)$. Lemma A.6 and finite limit preservation of analytic realization (Definition 6.1) identify this pullback with

$$\mathrm{Spa}(\mathrm{A} \coprod_* \mathrm{B}_i, \mathrm{A} \coprod_* \mathrm{B}_i) \longrightarrow \mathrm{Spa}(\mathrm{B}_i, \mathrm{B}_i),$$

which is a proper descendable cover by Lemma A.13. Representability is local on the target, as is properness by [Kip26a, Remark 2.3.1]; effective epimorphisms are local on the target in any ∞-topos. This proves the first assertion. Applying Lemma A.6 iteratively with $\mathrm{B} = \mathrm{A}$ identifies the Čech terms with $\mathrm{Spa}(\mathrm{A}^{(n)}, \mathrm{A}^{(n)})$. The resulting Čech nerve realizes to $\mathrm{X}^{\Delta, \blacksquare}$ because $\mathrm{Spa}(\mathrm{A}, \mathrm{A}) \to \mathrm{X}^{\Delta, \blacksquare}$ is an effective epimorphism. □

For a fine monoid P, let $\mathcal{A}_\mathrm{P} = [\mathrm{Spf}\, \mathbf{Z}_p[\mathrm{P}] / \mathrm{Spf}\, \mathbf{Z}_p[\mathrm{P}^{\mathrm{gp}}]]$ be its Artin cone. Its toric atlas is $\mathrm{Spf}\, \mathbf{Z}_p[\mathrm{P}] \to \mathcal{A}_\mathrm{P}$. The universal log structure induces an étale map $\mathcal{A}_\mathrm{P} \to \mathrm{Log}$ by [Ols03, Corollary 5.24].

Lemma A.21. *Let* $(\mathrm{X}, \mathcal{M})$ *be a framed log affine over* $\mathcal{O}_K$*, let* $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf}\, \mathrm{A}})^a$ *be the logarithmic Breuil-Kisin prism associated with its framing, and let* $\mathrm{P} = \mathbf{N}^r$ *be its framing chart. There is an orientable bounded fine log prism* $(\mathrm{B}_\mathrm{P}, \mathrm{I}, \mathcal{N}_\mathrm{P})^a$ *with a strict map*

$$(\mathrm{Spf}(\mathrm{B}_\mathrm{P}/\mathrm{I}), \mathcal{N}_\mathrm{P}|_{\mathrm{Spf}(\mathrm{B}_\mathrm{P}/\mathrm{I})}) \longrightarrow (\mathrm{X}, \mathcal{M})$$

and a commutative diagram

$$\begin{tikzcd} \operatorname{Spf} \mathrm{B}_{\mathrm{P}} \arrow[r, "\mathrm{fpqc}"] \arrow[dr, "\rho_{\mathrm{X},\mathrm{B}_{\mathrm{P}}}"'] & \operatorname{Spf} \mathrm{A} \times_{\mathcal{A}_{\mathrm{P}}^{\mathbb{A}}} (\mathbf{Z}_p^{\mathbb{A}} \times \operatorname{Spf} \mathbf{Z}_p[\mathrm{P}]) \arrow[d] \\ & (\mathrm{X}, \mathcal{M})^{\mathbb{A}}. \end{tikzcd}$$

The right vertical arrow is the natural map. After applying the analytic realization functor of Definition 6.1, the induced map

$$\operatorname{Spa}(\mathrm{B}_{\mathrm{P}}, \mathrm{B}_{\mathrm{P}}) \longrightarrow \operatorname{Spa}(\mathrm{A}, \mathrm{A}) \times_{\mathcal{A}_{\mathrm{P}}^{\mathbb{A},\blacksquare}} (\mathbf{Z}_p^{\mathbb{A}} \times \operatorname{Spf} \mathbf{Z}_p[\mathrm{P}])^{\blacksquare}$$

is a proper effective epimorphism.

Proof. Put

$$\mathrm{X}_{\mathrm{P}} = \operatorname{Spf}(\mathrm{A}/\mathrm{I}) \times_{\mathcal{A}_{\mathrm{P}}} \operatorname{Spf} \mathbf{Z}_p[\mathrm{P}].$$

We equip X_{P} with the pulled-back log structure. Let $\mathrm{T}_i \in \mathrm{A}$ be the images of the framing chart generators. This chart trivializes the torus torsor $\mathrm{X}_{\mathrm{P}} \to \operatorname{Spf}(\mathrm{A}/\mathrm{I})$, giving

$$\mathrm{X}_{\mathrm{P}} \simeq \operatorname{Spf}\left((\mathrm{A}/\mathrm{I})[u_1^{\pm 1}, \ldots, u_r^{\pm 1}]_p^{\wedge}\right).$$

Choose the units u_i so that the toric chart sends e_i to $u_i \bar{\mathrm{T}}_i$. Put $\mathrm{T}_{\mathrm{P}} = \mathrm{A} \widehat{\otimes}_{\mathbf{Z}_p} \mathbf{Z}_p[\mathrm{P}]$, with toric coordinates x_i and $\delta(x_i) = 0$. Relative base change and finite limit preservation give

$$\operatorname{Spf} \mathrm{A} \times_{\mathcal{A}_{\mathrm{P}}^{\mathbb{A}}} (\mathbf{Z}_p^{\mathbb{A}} \times \operatorname{Spf} \mathbf{Z}_p[\mathrm{P}]) \simeq (\mathrm{X}_{\mathrm{P}}/\mathrm{A})^{\mathbb{A}} \times_{(\operatorname{Spf}(\mathrm{T}_{\mathrm{P}}/\mathrm{I})/\mathrm{A})^{\mathbb{A}}} \operatorname{Spf} \mathrm{T}_{\mathrm{P}}.$$

Here we use [BL22, Constructions 3.11, 5.2 and Remarks 8.9, 8.11].

Set

$$\mathrm{C}_{\mathrm{P}} = \mathrm{A}[u_1^{\pm 1}, \ldots, u_r^{\pm 1}]_{(p,\mathrm{I})}^{\wedge},$$

with $\varphi(u_i) = u_i^p$. This is an ordinary Breuil-Kisin atlas of $\mathrm{X}_{\mathrm{P}}/\mathrm{A}$. The relative form of Proposition A.20, proved by the same argument, shows that

$$\operatorname{Spf} \mathrm{C}_{\mathrm{P}} \longrightarrow (\mathrm{X}_{\mathrm{P}}/\mathrm{A})^{\mathbb{A}}$$

is an fpqc epimorphism and becomes a representable proper effective epimorphism after applying the functor $(-)^{\blacksquare}$ in Definition 6.1. Indeed, the relative form of [BL22, Proposition 7.5] allows us to work fpqc-locally on an orientable bounded relative test prism. The relative ordinary case of Lemma A.6 identifies the pullback with the corresponding coproduct; complete faithful flatness follows from [Kos22, Proposition 4.13] with trivial log structures, while the properness argument of Lemma A.13 applies verbatim. Representability and properness are local on the target.

Its pullback to the stack in the statement is

$$\operatorname{Spf} \mathrm{C}_{\mathrm{P}} \times_{(\operatorname{Spf}(\mathrm{T}_{\mathrm{P}}/\mathrm{I})/\mathrm{A})^{\mathbb{A}}} \operatorname{Spf} \mathrm{T}_{\mathrm{P}}.$$

Here $(\mathrm{T}_{\mathrm{P}}, \mathrm{I})$ is the ordinary relative Breuil-Kisin atlas of $\operatorname{Spf}(\mathrm{T}_{\mathrm{P}}/\mathrm{I})$, and $(\mathrm{C}_{\mathrm{P}}, \mathrm{I})$ is an ordinary relative test prism over it. The relative ordinary case of Lemma A.6 therefore identifies this fiber product with the formal spectrum of their ordinary prismatic coproduct B_{P}, an orientable bounded prism. The resulting map is an fpqc epimorphism and becomes a proper effective epimorphism after analytic realization, by base change.

Give $\mathrm{B_P}$ the log structure $\mathcal{N}_\mathrm{P}$ pulled back from $\mathrm{T_P}$, with chart $e_i \mapsto x_i$ and $\delta_{\log}(e_i) = 0$. The reduction diagram of the ordinary coproduct gives $\bar{x}_i = \bar{u}_i \bar{\mathrm{T}}_i$, so its reduction maps strictly to $\mathrm{X_P}$, and hence to $(\mathrm{X}, \mathcal{M})$. The chart generator on X is identified with $\bar{u}_i^{-1} e_i$. To prove commutativity of the triangle, consider

$$\begin{tikzcd} \mathrm{Spf}(\mathrm{B_P}/\mathrm{I}) \arrow[r] \arrow[d] & \mathrm{X_P} \arrow[r] \arrow[d] & \mathrm{X} \arrow[d] \\ \mathrm{Spf}\,\mathrm{B_P} \arrow[r] & \mathrm{Spf}\,\mathrm{T_P} \arrow[r] & \mathcal{A}_\mathrm{P}. \end{tikzcd}$$

The left square is the reduction square of the coproduct, and the right square carries the tautological isomorphism defining $\mathrm{X_P}$, with ratios $\bar{u}_i$. The bottom arrows respect the δ-structures. Applying relative prismatization to this diagram shows that the triangle in the statement commutes: the outer square induces $\rho_{\mathrm{X},\mathrm{B_P}}$ for the chosen log structure and strict reduction, as in the proof of Lemma A.6. □

Proposition A.22. *Let $(\mathrm{X}, \mathcal{M})$ be a framed log affine over $\mathcal{O}_K$, and let $(\mathrm{A}, \mathrm{I}, \mathrm{M}_{\mathrm{Spf\,A}})^a$ be the logarithmic Breuil-Kisin prism associated with its framing. The induced map*

$$\mathrm{Spa}(\mathrm{A}, \mathrm{A}) \longrightarrow (\mathrm{X}, \mathcal{M})^{\Delta,\blacksquare}$$

is a representable proper effective epimorphism. If $\mathrm{A}^{(n)}$ denotes the nth iterated self-coproduct in $(\mathrm{X}, \mathcal{M})_{\Delta}$, then

$$(\mathrm{X}, \mathcal{M})^{\Delta,\blacksquare} \simeq \left|\mathrm{Spa}(\mathrm{A}^{(\bullet)}, \mathrm{A}^{(\bullet)})\right|.$$

Proof. Let $\mathrm{P} = \mathbf{N}^r$ be the framing chart; its reduction is a chart for $\mathcal{M}$ on $\mathrm{X} = \mathrm{Spf\,A/I}$. This induces a map $\mathrm{Spf\,A} \to \mathcal{A}_\mathrm{P}^{\Delta} \to \mathrm{Log}^{\Delta}$. By Definition A.1, the fiber product $\mathrm{Spf\,A} \times_{\mathrm{X}^{\Delta}} (\mathrm{X}, \mathcal{M})^{\Delta}$ is $\mathrm{Spf\,A} \times_{\mathrm{Log}^{\Delta}} (\mathbf{Z}_p^{\Delta} \times \mathrm{Log})$. Since $\mathcal{A}_\mathrm{P} \to \mathrm{Log}$ is étale, the étale Artin-cone comparison of [BL22, Remark 3.9 and Construction 3.11] gives

$$\mathrm{Spf\,A} \times_{\mathrm{Log}^{\Delta}} (\mathbf{Z}_p^{\Delta} \times \mathrm{Log}) \simeq \mathrm{Spf\,A} \times_{\mathcal{A}_\mathrm{P}^{\Delta}} (\mathbf{Z}_p^{\Delta} \times \mathcal{A}_\mathrm{P}).$$

The toric atlas $\mathrm{Spf}\,\mathbf{Z}_p[\mathrm{P}] \to \mathcal{A}_\mathrm{P}$ and Lemma A.21 give an effective epimorphism

$$\mathrm{Spa}(\mathrm{B_P}, \mathrm{B_P}) \longrightarrow \mathrm{Spa}(\mathrm{A}, \mathrm{A}) \times_{\mathrm{X}^{\Delta,\blacksquare}} (\mathrm{X}, \mathcal{M})^{\Delta,\blacksquare}.$$

The projection from the fiber product on the right to $(\mathrm{X}, \mathcal{M})^{\Delta,\blacksquare}$ is the base change of $\mathrm{Spa}(\mathrm{A}, \mathrm{A}) \to \mathrm{X}^{\Delta,\blacksquare}$, so it is an effective epimorphism by Proposition A.20. Thus the analytic realization of $\rho_{\mathrm{X},\mathrm{B_P}}$ is an effective epimorphism. Lemma A.6 identifies the pullback of $\mathrm{Spa}(\mathrm{A}, \mathrm{A})$ along this map with

$$\mathrm{Spa}(\mathrm{A} \coprod_* \mathrm{B_P}, \mathrm{A} \coprod_* \mathrm{B_P}) \longrightarrow \mathrm{Spa}(\mathrm{B_P}, \mathrm{B_P}).$$

This is a proper descendable cover by Lemma A.13. Representability is local on the target, as is properness by [Kip26a, Remark 2.3.1]; the covering property is local in any ∞-topos. This proves the first assertion. Lemma A.6 and finite limit preservation of $(-)^{\blacksquare}$ (as in Definition 6.1) identify the nth Čech term with $\mathrm{Spa}(\mathrm{A}^{(n)}, \mathrm{A}^{(n)})$. Since the map from $\mathrm{Spa}(\mathrm{A}, \mathrm{A})$ is an effective epimorphism, its Čech nerve realizes to $(\mathrm{X}, \mathcal{M})^{\Delta,\blacksquare}$. □

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